3. 2026 Assessment of the Sablefish Stock in Alaska

2026 Center for Independent Experts (CIE) Review Document

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May 2026

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This report may be cited as:

Goethel, D.R. and Cheng, M.L.H. 2026. Assessment of the sablefish stock in Alaska: 2026 Center for Independent Experts (CIE) Review Document. North Pacific Fishery Management Council, Anchorage, AK.

The last operational full assessment for sablefish was in 2021 and a full description of the model can be found in the 2021 SAFE document:
https://www.fisheries.noaa.gov/resource/data/2021-assessment-sablefish-stock-alaska

The last update assessment was performed in 2024:
https://www.npfmc.org/wp-content/PDFdocuments/SAFE/2024/Sablefish.pdf

Changes that were planned (but not used operationally due to the US government shutdown) for the full assessment in 2025 (Model *25.12*) are described in the September NPFMC Plan Team document:
https://meetings.npfmc.org/CommentReview/DownloadFile?p=02344a6f-99f3-4239-990e-e5dd9393e1a7.pdf&fileName=Sablefish%20proposed%20model%20changes.pdf

The following appendices can be found at the CIE GitHub repository:

Appendix 3A. The Spatial Population over Regional Components (SPoRC) Model: Technical Description of the Assessment Model for Sablefish (here)
Appendix 3B. Model Building and Bridging Exercise for the Sablefish Single Region Models (Bridging from Model *25.12* to *26.1_1Reg*) (here)
Appendix 3C. Development of a Fleets-as-Areas (FAA) Stock Assessment Model for Alaska Sablefish (*Model 26.2_FAA*) (here)
Appendix 3D. Results of Spatial Stock Assessment Models for Alaska Sablefish (here)
Appendix 3E. Evaluating Alternative Assessment Structures for Alaska Sablefish Under Survey Design Changes via Closed-Loop Simulation (here)

Executive Summary

Summary of Changes to the Assessment

In 2024, there was an update assessment for sablefish (*Anoplopoma fimbria*) using Model *23.5*. A full assessment was scheduled for 2025 with model changes implemented (Model *25.12*) and reviewed at the North Pacific Fisheries Management Council's (NPFMC) September Groundfish Plan Team meeting. However, an operational assessment was not performed in 2025 due to the US government shutdown. For the 2026 Center for Independent Expert's (CIE) review of the sablefish assessment, two models are being provided as potential candidates for the operational assessment in 2026 and beyond (results from the continuity Model *25.12_1Reg_Cont* are available as part of the model building exercise; Appendix 3B).

Model *26.1_1Reg* builds from Model *25.12*, but implements updates to parametrization (e.g., estimating sex-specific natural mortality, *M*, and switching to a logistic selectivity function for the trawl fishery fleet) and data inputs (e.g., removing catch-per-unit effort, CPUE, data, removing length compositions for fleets with age compositions, updating growth and weight estimates, and updating how the NOAA longline survey index is calculated; Appendix 3B). Additionally, a spatially-implicit fleets-as-areas (FAA) model was developed (Model *26.2_FAA*) to fit the longline survey data at a regional scale, such that extrapolation of the resulting indices would no longer be necessary (Appendix 3C). Because large portions of the sablefish resource (~50%) are not being surveyed in a given year (i.e., the survey now covers the Gulf of Alaska, GOA, in odd years and the Bering Sea, BS, and Aleutian Islands, AI, in even years), the longline survey extrapolation approach is not ideal. Model *26.2_FAA* was built from Model *26.1_1Reg* with incremental complexity in fleet structure of the longline survey, which resulted in three total NOAA longline survey fleets (instead of just one aggregate fleet as in Model *26.1_1Reg*). Each regional fleet (BS, AI, and GOA) was modeled with its own estimated selectivity function and catchability parameter, q, by fitting to regionally disaggregated indices and age composition data, thereby, eliminating the need to extrapolate indices or aggregate composition data across regions. Two spatially explicit assessments were also developed that integrated mark-recapture data and estimated movement among either 3 regions (matching the regional structure of the longline survey in the FAA model; Model *26.3a_Spat_3Reg*) or 5 regions (matching the management structure across Alaska; Model *26.3_Spat_5Reg*; Appendix 3D). However, neither spatial model is being put forward for operational management due to model stability concerns and the need for further model refinement.

A closed loop simulation analysis was also undertaken to understand performance of the various model structures being explored (i.e., single region, FAA, and three region spatial models; Appendix 3E). The simulation analysis utilized a spatially explicit simulation operating model conditioned on the five region spatial assessment. The results of the simulation experiment helped guide model building, particularly the structure of the FAA model, and provided insight on expected assessment performance in terms of potential bias and stability. Moreover, it also identified expected management outcomes via the closed loop feedback and implemented $F_{40\%}$ sloped harvest control rule (HCR) currently utilized for sablefish management.

Units

Throughout the document mass (i.e., for spawning stock biomass, SSB, biomass, and catch) is reported in metric tons (abbreviated as either mt or t; equivalent to a tonne) or kilotons (kt; equivalent to 1,000 metric tons). Recruitment (age-2) and abundance are in millions of fish. Fishing and natural mortality are instantaneous rates ($year^{-1}$).

Changes to the Input Data

Newly available data since the 2024 assessment that were used in the assessment models for 2026 (*26.1_1Reg and 26.2_FAA)* included (Table 3.3):

- Length data from the trawl fisheries for 2024.
- Age data from the fixed gear fishery for 2024.
- Finalized catch for 2024 and 2025.
- The NOAA longline survey abundance indices for 2025.
- Updated length- and weight-at-age estimates using survey age-length-weight data through 2023.
- For the single region NOAA longline survey index (used in Model *26.1_1Reg*), the index was constructed using a 'carryover' approach for non-surveyed regions (i.e., for a given region the Relative Population Numbers, RPN – the survey index, from the previous time the region was surveyed was used for off-year survey RPNs) instead of extrapolating off-year survey RPN values for the BS or AI based on RPN change in the GOA (this traditional approach is no longer feasible given the change in survey design in 2025 because the GOA is no longer surveyed every year).
- The input sample size (ISS) for composition data utilized a relative scaling within gears/surveys to maintain information on sampling across years (previously a static ISS was used for each gear).

Data that was removed from the proposed assessment models for 2026, include:

- The fixed gear fishery CPUE index, including the historical Japanese CPUE (1964 – 1981) and the domestic CPUE (1995 – 2022).
- The fixed gear fishery length composition data from 1990 – 1998.
- The NOAA Longline Survey length composition data from 1990 – 1995.
- For the NOAA Longline Survey index, extrapolated relative population numbers (RPNs) from stations in the western AI were removed from the index.

Changes to the Assessment Methodology

For Model *26.1_1Reg*, changes from the previously reviewed Model *25.12*, included:

- Sex-specific *M* was estimated.
- Logistic selectivity was assumed for the trawl fishery (instead of domed).
- The fixed gear fishery early/pre-IFQ (1960 -- 1994) time block was removed.
- All selectivity parameter linkages were removed.
- The selectivity parameter priors were updated.

For Model *26.2_FAA*, changes from Model *26.1_1Reg*, included:

- A spatially-implicit FAA structure was implemented for the NOAA Longline Survey fleets, which disaggregated the longline survey into three separate regional fleets (BS, AI, and GOA), each of which had an independently estimated set of selectivity parameters and *q* values that were estimated by fitting the corresponding disaggregated indices (without any carryover or extrapolation approaches applied) and age compositions.

Summary of Results

In 2025, the NOAA Longline Survey was conducted using the new survey design where only the GOA was sampled. The indices continued to increase, though rates of increase in the GOA RPNs were not as large as in recent years (Figures 3.7 – 3.8). Due to biological and socioeconomic drivers, catch has remains below acceptable biological catch (ABC) with the proportion of the quota utilized less than 50% in 2024 and 2025.

Two models are presented, where an FAA model was identified as a potential suitable structure for sablefish, as it could disaggregate NOAA Longline Survey data by region and avoid the need for 'carryover' extrapolation of the NOAA Longline Survey index as is required in the single region assessment (Model *26.1_1Reg*; see Appendix 3B). Ultimately, the choice of model structure for sablefish is a tradeoff between model stability, assessment complexity, and assumed data representativeness (See Table Below). Although

an FAA model would forego the need for data extrapolation, it may be more unstable and might overinterpret signals from sparse composition data (e.g., if the noise to signal ratio is high).

For both models, fit to the available data retain similar issues, where NOAA Longline Survey indices are generally well fit at the expense of composition data, primarily due to the Francis reweighting approach of 'right-weighting' the data to ensure adequate fit to the indices. Although residual patterns in the composition data remain, the aggregated fits continue to appear adequate given associated model tension between fitting the NOAA Longline Survey indices versus age compositions (a long-standing issue with the sablefish assessment; Goethel and Cheng, 2024). Both models have trouble reconciling the larger year classes and population growth suggested by the age composition data in recent years compared with the more tempered, but still extremely rapid, population growth observed in the survey RPNs since ~2018. No major diagnostic issues were identified for either model based on retrospective analysis, profile likelihoods, MCMC, or jitter analysis. Thus, both Model *26.1_1Reg* and Model *26.2_FAA* appear to be robust and stable models and adequate for interpreting sablefish population dynamics.

Population trends align across models with both indicating strong 2014, 2016, and 2017 year classes, and a reduction towards average recruitment since 2020 (Figure 3.20). Recent recruitment trends in Model *26.2_FAA* differ moderately from those in Model *26.1_1Reg*. For instance, the time series average recruitment has declined, while the interpretation of recent dynamics has altered resulting in a larger estimate of the 2014 and 2017 year classes and a smaller 2016 year class (Figure 3.20). Alterations in recruitment estimates are due to differences in recruitment signals across the regional NOAA Longline Survey age compositions, where the AI provides a mixture of recruitment signals (strong 2014, 2016, and 2017 year classes; Figure 3.44) that balances those in the BS (strong 2014 and 2016 and very strong 2017 year classes; Figure 3.41) and GOA (strong 2014 and very strong 2016 year classes; Figure 3.36).

Although Model *26.2_FAA* assumes a moderate downward scaling of SSB and catch advice, general population dynamics and stock status remain consistent with Model *26.1_1Reg* (Figure 3.21). Interestingly, an exploratory run with the FAA model, where extrapolated regional carryover indices were constructed and fit to better emulate the aggregate index used in Model *26.1_1Reg*, led to a strong upward scaling of SSB, particularly since the start of the NOAA Longline Survey in the mid-1990s (Appendix 3C, Figure 3C.12). Thus, there may be inherent risk with using extrapolated survey data in the assessment, which can potentially result in artificially inflating SSB and catch advice. Moreover, the phase-plane for Model *26.2_FAA* suggests that overfishing may have been occurring on the sablefish population for much of the 2000s, until recent recruitment events began entering the population around 2021 (and ABC utilization rates declined; Figure 3.85). Therefore, the FAA model suggests that the single region model may be more risk prone, if population scale is more in line with those estimated in Model *26.2_FAA*.

Overall, the sablefish age structure continues to gradually expand as the recent large year classes (e.g., 2014, 2016, and 2017) begin to enter ages that are nearing full maturity. For instance, the 2014 year class is now 12 years old, while the larger 2016 year class is 10 years of age. It is expected that these cohorts will continue to survive in large numbers and help expand the age structure over the next 5 – 10 years.

Sablefish are managed under Tier 3 of the NPFMC harvest control rule (HCR), which aims to maintain the population at $B_{40\%}$. For both models, female spawning biomass in 2026 is projected to be ~$B_{67\%}$ placing sablefish in NPFMC Tier 3a. Spawning biomass is projected to increase rapidly in the near-term, and the maximum permissible value of F_{ABC} under Tier 3a is 0.08 with an F_{OFL} of 0.09 and 0.10 for the single region and FAA models, respectively. Due to the scaling differences among models, the projected ABC and OFL levels differ by ~10,000 mt. For Model *26.1_1Reg*, 2026 ABC and OFL are 41,000 mt and 48,000 mt, respectively. For Model *26.2_FAA*, 2026 ABC and OFL are 32,000 mt and 38,000 mt, respectively. For both models, projections indicate that the Alaskan sablefish stock is not subject to overfishing, not overfished, and not approaching an overfished condition.

ES1. Summary of tradeoffs between using a single region (1Reg), Fleets-as-Areas (FAA), and muti-region (spatial) models for sablefish.

Model	*Benefits*	*Issues*	*MSE Conclusions*	*Tradeoffs*
Single Region Model (1Reg)	High sample sizes. Low model complexity. Intuitive and transparent results. High inertia towards moving away from status quo assessment methodology.	Requires fitting extrapolated data for the Longline Survey Index. Implicit historical over-weighting of GOA composition data during aggregation or high interannual variability recently (GOA then BS/AI). Little signal from BS/AI compositions, because GOA given higher weight (RPNs or catch) in data aggregation. Ignores spatial structure.	Moderate SSB bias and moderate instability due to the survey extrapolation and aggregated composition data but impacts on resulting management advice may be negligible.	Higher sample sizes and improved stability but requires fitting the extrapolated longline survey index and primarily represents the recruitment signals from the GOA due to data aggregation assumptions.
Fleets-as-Areas (FAA)	Fitting non-extrapolated Longline Survey index. Fitting disaggregated survey compositions. Better accounts for signals from BS/AI compositions on recent year class strength. Implicitly accounts for ontogenetic movement dynamics	Low sample sizes and sparse data for BS/AI compositions. Increased instability from added complexity (estimating extra q and selectivity parameters for 'regional' survey fleets and/or extra F and selectivity parameters for fishery fleets). Non-intuitive estimates that are difficult to interpret (e.g., due to using multiple fleets that are interacting with a single underlying population – not multiple populations/regions). Selectivity needs to be flexible to address ontogenetic dynamics.	Low SSB bias and instability with improved performance over other structures when survey coverage decreases, but adequate flexibility in selectivity parametrizations is likely needed to ensure good performance.	Able to fit the regional longline survey indices without aggregation and likely provides more robust outputs when survey coverage decreases, but sample size limitations may prevent adequate selectivity flexibility to address ontogenetic availability issues due to model instability.
Spatial (3 or 5 Region)	Fit data at the regional scale it was collected. Better account for signals from each region. More intuitive than FAA. Explicitly account for ontogenetic movement dynamics.	Increased sample size limitations. Increased parametrization decisions related to spatial processes. Lack of best practices for fitting tagging data (e.g., dealing with tag mixing issues and data weighting). Increased parameter correlation, particularly between regional recruitment parameters and movement. Diagnostics can be difficult to interpret. Increased instability.	High instability due to overparameterization and low sample sizes but best represents localized dynamics.	Best represents spatial dynamics but suffers from high model instability due to data sparsity that requires extremely careful parametrization, while appropriate handling of tagging data is difficult and potentially subjective.

Summary Table

Quantity/Status	Specified *last* SAFE for Model *23.5*:		Estimated *this* SAFE for Model *26.1_1Reg*:		Estimated *this* SAFE for Model *26.2_FAA*:	
	2025	2026*	2026	2027**	2026	2027^
M Female (estimated)[#]	0.11		0.092		0.088	
M Male (estimated)	--		0.090		0.087	
Average Recruitment (millions of fish)	26		21		18	
Tier	3a		3a		3a	
Projected Total Biomass (Age 2+; kt)	705	696	628	611	469	453
Projected Female Spawning Biomass (kt)	220	241	237	253	205	211
$B_{100\%}$ (kt)	303	303	348	348	306	306
$B_{40\%}$ (kt)	121	121	139	139	122	122
$B_{35\%}$ (kt)	106	106	122	122	107	107
F_{OFL}	0.10	0.10	0.09	0.09	0.10	0.10
$maxF_{ABC}$	0.09	0.09	0.08	0.08	0.08	0.08
F_{ABC}	0.09	0.09	0.08	0.08	0.08	0.08
OFL (kt)	59	58	48	46	38	35
max ABC (kt)	50	50	41	40	32	30
ABC (kt)	50	50	41	40	32	30
Status	As determined *last* year for:		As determined *this* year for:		As determined *this* year for:	
	2023	2024	2024	2025	2024	2025
Overfishing	No	n/a	No	n/a	No	n/a
Overfished	n/a	No	n/a	No	n/a	No
Approaching overfished	n/a	No	n/a	No	n/a	No

*The 2024 SAFE projections of future SSB and biomass were based on specified catch of 25,000 mt in 2025 (a yield ratio of 0.49 was assumed based on a 2024 extrapolated catch of 23,000 mt and an ABC of 47,000 mt). Specified catch projections are utilized in response to management requests for a more accurate two-year projection. SSB and biomass are less than presented when the maximum ABC is removed in 2025.

**The 2026 SAFE Model *26.1_1Reg* projections were based on specified catch of 20,000 mt in 2026 (a yield ratio of 0.49 was assumed based on a 2025 catch of 23,200 mt and an ABC of 47,600 mt).

^The 2026 SAFE Model *26.2_FAA* projections were based on specified catch of 16,000 mt in 2026 (a yield ratio of 0.49 was assumed based on a 2025 catch of 23,200 mt and an ABC of 47,600 mt).

[#]A single female natural mortality parameter is estimated in Model *23.5* and is used for both sexes.

Spatial Catch Apportionment

A five-year moving average of the NOAA Longline Survey proportions of biomass in each region are used to apportion catch to management area, as a means to ensure regional catch is in relative proportion to regional biomass. The apportionment values are updated yearly as new survey data are collected.

Accounting for Whale Depredation

For the final recommended ABC (ABC_w) or OFL (OFL_w), sperm and killer whale depredation in the fixed gear fishery is accounted for by reducing the maximum ABC by the last estimate of the three-year average of depredation estimates by area. Beginning in 2023, whale depredation estimates have been held constant at the 2022 levels, per SSC recommendations. By accounting for whale depredation in the assessment and associated catch projections, the approach does not create additional regulations or burden for in-season management. The ABC is reduced by ~170 mt and the OFL by ~200 mt annually to account for depredation.

Introduction

For a full description of the sablefish resource and fishery dynamics, see Goethel et al. (2021; available at https://www.fisheries.noaa.gov/resource/data/2021-assessment-sablefish-stock-alaska).

Sablefish are found from northern Mexico to the Gulf of Alaska (GOA), westward to the Aleutian Islands (AI), northward in the Bering Sea (BS), and into the northwestern Pacific Ocean off the Siberian coast of Russia and the Kuril Islands in Japan (Wolotira et al. 1993). Genetic analysis has found no population sub-structure throughout their range along the US West Coast to Alaska and suggested that observed differences in growth and maturation rates may be due to phenotypic plasticity or are environmentally driven (Jasonowicz et al., 2017; Kapur et al., 2020; Timm et al., 2024). Sablefish are assessed as a single, homogeneous population in Federal waters off Alaska, because of their high movement rates and panmictic stock structure. However, management is by discrete regions to distribute exploitation throughout their wide geographical range, including the Bering Sea (BS), Aleutian Islands (AI), Western GOA (WG), Central GOA (CG), and Eastern GOA (EG) (composed of West Yakutat, WY, and Southeast Outside, SEO; Figure 3.1).

Adult sablefish live at depths greater than 200 m and are highly mobile, though, exact movement patterns and drivers are not well understood (Hanselman et al., 2015). Recently, results of a spatially explicit tag-integrated assessment model (Cheng et al., 2025d) demonstrated support for historical hypotheses of a clockwise ontogenetic movement pattern (Heifetz and Fujioka, 1991), wherein immature fish tend to move towards and reside in western region (WG, BS, AI) and then move eastward as they mature with high residency of older fish in the CG and EG. Spawning occurs in winter at depths of 300 - 500 m near the edges of the continental slope (Mason et al. 1983). Juvenile sablefish spend their first two to three years on the continental shelf of the GOA and the southeast BS. After their second summer, they begin moving offshore to deeper water, typically reaching their adult habitat, the upper continental slope, at 4 to 5 years of age. This corresponds to the age range when sablefish start becoming reproductively viable (Mason et al. 1983, Rodgveller et al. 2016).

Drivers of recruitment dynamics for sablefish are not well understood, but, unlike other Alaska groundfish species, they appear to have responded positively to recent large marine heatwaves in the North Pacific potentially due to their rapid growth (Callahan et al., 2021). Although young, immature sablefish are found throughout Alaska federal waters, recent extremely large year classes have been more concentrated in western regions, especially the eastern Bering Sea (EBS) shelf (Cheng et al., 2025d). Thus, the EBS may act as a nursery ground during periods of high recruitment and may represent recolonization of a pre-industrial fishing population center (Sasaki, 1985). For further information on sablefish life history dynamics and ecosystem drivers see the Sablefish Ecosystem and Socioeconomic Profile (ESP; the last full ESP is available here in Appendix 3C and the most recent ESP update is available here in Appendix 3D).

Fishery

Sablefish have been exploited since the end of the 19th century with rapid fishery expansion in the 1960s when Japanese longliners began operations in the EBS (Table 3.1, Figures 3.1 -- 3.4). As the fishing grounds in the EBS were preempted by expanding Japanese trawl fisheries, the Japanese longline fleet expanded to the AI and the GOA in the 1970s (Figure 3.1). Heavy fishing by foreign vessels resulted in a peak in total catch at 53,000 t in 1972, which led to a substantial population decline and implementation of fishery regulations in Alaska. The U.S. longline fishery began expanding in 1982 in the GOA and by 1988 the U.S. harvested all sablefish taken in Alaska, primarily by hook-and-line gear in the eastern and central GOA. Over the last two decades, the EG and CG have constituted the bulk of sablefish landings (~4 – 7kt each), followed by the WG (~1 – 3kt), and BS and AI (0.2 – 2kt each). In the 2010s, the AI experienced slightly higher catch than the BS (Figure SM3C.12), which corresponded with a period of high catch rates of old,

large fish (e.g., from 2010 to 2016 ~20 – 70% of the AI age compositions were composed of fish older than age-30 leading to an unrealistically large plus group in the population level age compositions during that period; Figures 3.48 and SM3C.42). Catch and effort in the AI has decreased drastically since 2023 (likely due to poor sablefish markets and the high cost of fishing given the distance to fishing grounds in the AI). In 2025, there were no observed trips or biological sampling from the AI due to a combination of low fishing effort and the sampling design of observer deployment.

From a year round fishery in 1983, the fishing season shrank to 10 days in 1994, warranting the "derby" fishery label. In 1995, Individual Fishery Quotas (IFQ) were implemented for hook-and-line vessels along with an 8-month season. The IFQ Program is a catch share fishery that issued quota shares to individuals based on sablefish and halibut landings made from 1988 -- 1990. Since the implementation of IFQs, the number of longline vessels with sablefish IFQ experienced a substantial and anticipated decline from 616 in 1995 to 362 in 2011, which was expected as shareholders consolidated their holdings and fished fewer vessels to reduce costs (Fina, 2011). The season dates have varied by several weeks since 1995, but the monthly pattern has been from March to November with the majority of landings occurring in May to June. IFQ management increased fishery catch rates and decreased the harvest of immature fish after implementation (Sigler and Lunsford 2001).

In response to consistent sperm whale depredation on hook and line gear, the NPFMC passed a regulation in 2015 to allow pot fishing in the GOA starting in 2017. In 2017 and 2018, pot fishing made up a small proportion of the fixed gear catch (10% and 12%, respectively). However, since 2021 the majority of removals by the fixed gear fleet was taken by pot gear (Table 3.1, Figure 3.3). The transition to pot gear was extremely rapid and primarily driven by the increasing popularity of collapsible 'slinky' pots that are easier to store on smaller vessels while effectively eliminating whale depredation that has been rampant on hook and line gear (Figure 3.5; Peterson and Carothers, 2013; Peterson and Hanselman, 2017; Cheng et al., 2024a).

Sablefish are also caught during trawl fisheries for other species groups such as rockfish and deep-water flatfish, and more recently walleye pollock. Allocation of the Total Allowable Catch (TAC) by gear group varies by management region and influences the amount of catch in each region. Allocation percentages by area are: 80% to fixed gear and 20% to trawl in the Western and Central GOA; 95% to fixed gear and 5% to trawl in the Eastern GOA (i.e., 5% to trawl in WY, given trawling bans in the SEO management area); 50% to fixed gear and 50% to trawl in the eastern BS; and 75% to fixed gear and 25% to trawl gear in the Aleutian Islands. For much of the last twenty years, trawl gear has constituted around 10% of total sablefish catch, but this proportion increased rapidly starting around 2016 and reaching a peak of 39% in 2020 (Table 3.1; Figure 3.4). Recent increases in trawl fishery sablefish catch are predominantly from the BS due to the increased prevalence of small sablefish from recent strong year classes on the primary fishing grounds in the EBS (Tables 3.1 – 3.2, Figures 3.1 – 3.2). Although most sablefish catch in trawl gears are incidental, haul-specific targeting of sablefish has risen, particularly in the CG and WG, as part of other directed bottom trawl fisheries likely due to the increasing prevalence of sablefish. Over the last two decades, the majority of sablefish catch in trawl gears has occurred in the CG (0.5 – 2kt), but since 2018 catch in the BS has increased rapidly (~1 – 5kt) as has the catch in the AI (~0.5 – 1.25kt). The increase in the BS and AI is attributable to the rapid increase in small sablefish due to strong recruitment since 2016, which have typically been first encountered in the BS and AI (see Appendix 3D), particularly in the pelagic trawl fisheries on the EBS shelf (see Appendix 3E of the 2024 SAFE).

Five State of Alaska fisheries land sablefish outside the IFQ program, including those in Prince William Sound, Chatham Strait, and Clarence Strait with minor fisheries in the northern GOA and AI. The minor state fisheries were established by the State of Alaska in 1995 at the same time that the Federal Government established the IFQ fishery, primarily to provide open-access fisheries to fishermen who could not participate in the IFQ fishery. State catch from the northern GOA and AI minor fisheries are included in

the current assessment, because they are reported using the area code of the adjacent Federal waters in the Alaska Regional Office Catch Accounting System (CAS), the source of the catch data used in the assessment. Major state fisheries in the Northern Southeast Inside (NSEI) or Southern Southeast Inside (SSEI) waters are managed and assessed by the ADFG and catch associated with these fisheries are not included in the model.

Further details on the Alaskan sablefish fisheries can be found in the 2024 SAFE, Appendices 3D – 3F. Appendix 3D contains the sablefish Ecosystem and Socioeconomic Profile (ESP), which includes summaries of various fishery socioeconomic metrics. Appendix 3E describes the catch of small sablefish in the eastern Bering Sea in the various trawl fisheries. Appendix 3F summarizes sampling and observer coverage rates in the sablefish fisheries, including analysis of coverage rates by electronic monitoring (EM).

Management Measures and Bycatch

A summary of historical catch and management measures pertinent to sablefish in Alaska are provided in Table 3.2. As noted, the sablefish fixed gear fishery transitioned to IFQs in 1995 to avoid the growing derby-style fishery dynamics. Recently, the legalization of pot gear in the GOA in 2017 has had a monumental shift on gear usage, leading to a rapid transition from a predominantly hook-and-line fixed gear fishery to one composed primarily of longline pots since 2021 (Cheng et al., 2024a). Under current regulations, release of any sablefish by the sablefish IFQ fishery is prohibited, as long as there is remaining IFQ for persons onboard the fishing vessel. Following recent extreme recruitment of sablefish starting around 2014, markets for sablefish have become flooded causing socioeconomic downturns and resulting in upwards of 50% of the Alaska sablefish quota being left unharvested in recent years (Goethel et al., 2025). The NPFMC is currently exploring options to loosen full retention requirements based on requests from stakeholders to be able to release small, low value fish (NPFMC, 2024; Goethel et al., 2025; https://shinyfin.psmfc.org/small_sablefish/).

Sablefish discards in groundfish target fisheries are dominated by trawl fisheries in recent years, primarily in the BSAI due to an influx of recent large year classes in the BS, but discards have declined substantially starting in 2023 (Table SM3.3). Bycatch of federally managed groundfish species in the sablefish target fishery has primarily been composed of shortspine thornyhead, arrowtooth flounder, halibut, skates, and shortraker and rougheye rockfish, with amounts of incidental catch increasing drastically (by upwards of three-fold) in 2024 for many flounder and rockfish species (Table SM3.4). In terms of non-target bycatch, grenadiers are consistently the primary species encountered in the target sablefish fishery (Table SM3.5). The predominant prohibited species catch (PSC) in both the BSAI and GOA sablefish target fisheries is golden king crab and tanner crab (Table SM3.6).

Data

A variety of data sources are included in the assessment and were updated for 2026 (as available; Table 3.3), including fishery catch (Figure 3.4; Table 3.1), fishery and survey compositional data (Tables 3.4 – 3.5), and various indices of abundance and biomass (Figures 3.6 – 3.11; Table 3.6). The final input data used for the assessment can be found on the sablefish SAFE GitHub site at https://github.com/dgoethel-noaa/2026_Sablefish_CIE/tree/main/Model%20Runs/_Final_Mods). All data is pulled directly from the Alaska Fisheries Information Network (AKFIN) database, which maintains and hosts fishery catch, observer, and survey data for the Alaska region. All data is QA/QC'd by the collection agency prior to uploading to the AKFIN database. For further information on data collection protocols, estimation procedures, and quality assurance measures see the associated documents and webpages (e.g., Alaska Regional Office Catch Accounting System for fishery catch, North Pacific Observer Program for fishery

biological samples, the NOAA Longline Survey for fishery independent indices and biological samples, and the AFSC Age and Growth Program for otolith age determination).

Data Used in the Assessment

Fixed Gear and Trawl Fishery Catch and Compositions

For all fishery data, data streams for two fleets are created: fixed gear and trawl gear (Figure 3.4). The fixed gear fleet represents a combination of the historic Japanese fleet and the current U.S. domestic fleet combining both hook-and-line and pot longline gear. Given the lack of reliable historic composition data to estimate selectivity, the Japanese fleet cannot be independently partitioned. Conversely, because the recent transition from hook-and-line to longline pot occurred so rapidly (<5 years) and the apparent similarity in selectivity between these gear types, combining hook and pot gear into a single fixed gear fleet has been deemed adequate (Cheng et al., 2024a,b). Similarly, given the lack of age data and limited length frequency sample sizes, all trawl gears (i.e., bottom and pelagic) are combined into a single trawl fleet.

Length, catch, and effort data were historically collected from the Japanese and U.S. fixed gear and trawl fisheries, and are now collected from U.S. fixed gear and trawl fisheries. The Japanese data were collected by fishermen trained by Japanese scientists. The U.S. fishery length and age data are collected by at-sea and plant observers. No otoliths were collected from the fisheries until 1999, because of the difficulty of obtaining representative samples from the fishery, and otoliths are only routinely collected from the directed fixed gear fishery (not the trawl fishery). Observer coverage, and resulting biological sampling, varies by region due to the level of fishing effort, sampling strata of associated vessels, and the amount of electronic monitoring (EM) in the region and sampling strata (e.g., EM coverage is much higher for fixed gear vessels in the CG and EG; see summary of fishery data collection in Appendix 3F of the 2024 Sablefish SAFE).

The catch used in this assessment (Table 3.1) represents total catch (landings plus bycatch or discards -- which are minimal in the fixed gear IFQ fishery -- assuming 100% mortality), and includes catch from minor state-managed fisheries (i.e., these are reported by federal statistical areas and directly incorporated into the NOAA catch accounting system) in the northern GOA and in the AI region (constituting about 1% of the average total catch). Additionally, all non-commercial sablefish removals (see Table 3.1) are integrated into the fixed gear fishery catch, including research catches and Alaska state sport fishery removals not associated with the NSEI or SSEI waters fisheries, as well as estimates of whale depredation from hook-and-line gear (see next section). Trawl fishery catch is summed across all trawl gear types (e.g., bottom and pelagic) with widely differing distributions and targeting. For instance, pelagic trawls primarily catch smaller sablefish as bycatch in the EBS pollock fishery, whereas bottom trawls catch larger sablefish across the Alaska region occasionally during targeted hauls.

Age compositions from the U.S. fixed gear fishery are available since 1999 with relatively consistent sample sizes (~1,200 otoliths aged yearly; Tables 3.4 and 3.5), and sex and length data recorded for each otolith. Aged sablefish otoliths are available the year following collection (i.e., there is a one year lag for use in the assessment), and the total number of aged otoliths varies slightly due to otolith damage or issues during the age reading process (Table 3.5). To ensure representative composition data for the entire population, a random sampling protocol (e.g., across size, sex, and haul) is first used to identify which fishery otoliths to age in a given year. The goal is to choose ~200 otoliths from each of the six management regions, where regional allocations are adjusted slightly when any given region does not have the requisite number of samples. Next, a weighted mean is used to determine the population aggregate, sex-specific age compositions to ensure a representative estimate of catch-at-age by sex. The regional age proportions are calculated across both sexes and weighted by total catch in numbers by region (to better reflect recruitment as opposed to older fish – as is the case when weighting by catch in biomass). Because the joint approach to fitting compositions is used for all models (Cheng et al., 2025c), a single vector of compositions that sums to one across both sexes and all ages is calculated each year, hence this single vector for each region

is weighted by total regional catch and not sex-specific catch. Finally, the weighted average (by catch in numbers) is taken across all regions to provide the final vector of population age compositions across both sexes. The two-stage protocol ensures adequate samples from across Alaska, such that differences in composition data by region are taken into account, while simultaneously avoiding over-weighting regions with lower catch. For models that disaggregate regions or fleets as regions (e.g., spatially explicit or implicit models, see Appendix 3C and 3D), weighted averages are no longer needed as the raw age compositions from a given region can be used directly (except when modeling the GOA as a single unit, then the same weighted average approach is used but just for regions within the GOA). Length compositions from the fixed gear fishery are collected but not used in the assessment models.

Only length data is available from the trawl fishery and sample sizes are generally much lower than in the directed fisheries, representing a mixture of lower sablefish catch and different observer protocols in non-directed fisheries (Tables 3.4 and 3.5). To obtain representative catch-at-length, the same approach using regional catch (in numbers) weighted averages as the fixed gear fishery age compositions is used to construct a single vector of population level trawl length compositions across sexes. As with fixed gear fishery age compositions, trawl fishery length compositions are available to the assessment on a one year lag from data collection. Only years with at least 300 length samples are used in the assessment.

Whale Depredation

Whale depredation on hook-and-line gear has been pervasive in the fishery, so a general additive modeling (GAM) approach for estimating fixed gear fishery whale depredation was developed (Peterson and Hanselman, 2017) and the model estimates were updated in 2022 (Goethel et al., 2022). The two-step modeling approach estimates total fishery depredation based on observer and high-resolution catch data (i.e., in 1/3° by 1/3° bins). First, the impact on depredated sets (i.e., a CPUE reduction factor) is estimated using a Generalized Additive Mixed Model (GAMM). Next, a Generalized Additive Model (GAM) with a zero-inflated Poisson distribution is used to determine the proportion of sets in an area depredated by whales. The total depredation is then determined based on the number of sets depredated per grid, the reduction in catch per depredated set, and the total catch per grid. Estimated depredation has generally been below 1,000 t per annum, often composing less than 1% of the total catch in recent years. Despite low overall impact relative to total catch, depredation varies by area and species with orca depredation higher in western regions (primarily the WG) and sperm whale depredation more significant in the CG and EG (Figure 3.5). Whale depredation estimates are added to the fixed gear catch and accounted for in associated catch projections (i.e., in the final whale-depredated catch recommendation values). Since 2023, fishery whale depredation values have been held constant at 2022 levels (131 t), per SSC recommendations (i.e., the depredation model is no longer updated annually).

NOAA and Japanese Longline Survey Indices and Compositions

Japan and the U.S. conducted a cooperative fixed station Alaska-wide longline survey for sablefish annually from 1978 (the fist year was experimental and not included in indices) to 1994 (herein termed the 'Japanese Longline Survey'; Sasaki 1985; Sigler and Fujioka 1988). In 1988, NOAA instigated a domestic longline survey for sablefish (herein termed the 'NOAA Longline Survey') that has been conducted annually in the GOA with the aim of continuing the Japanese Longline Survey methodology, and was expanded to biennial sampling of the AI starting in 1996 and biennial sampling of the eastern BS starting in 1997. Catch, effort, age (since 1996 and sporadically on the Japanese survey), length, and weight (since 1996) have been collected annually. Relative population abundance in numbers (RPNs) are computed using survey catch rates scaled to management area sizes, except for initial survey years (i.e., the RPN time series begins in 1990; Kimura and Zenger, 1997).

The NOAA Longline Survey operates on a cost-recovery basis, where the sale of the catch offsets the cost of the survey. Due to market conditions and vessel availability, the longline survey was cancelled in 2024. Starting in 2025, a new survey contract led to a shortened schedule, resulting in an unforeseen change in experimental design. The GOA is now sampled in odd-numbered years and the BS and AI are sampled in even-numbered years. Details on the survey operations and recent trends can be found in Siwicke and Malecha (2026).

Longline gear is unable to estimate absolute population numbers or biomass, but it can provide a relative index, in this case a relative population number (RPN). Catch per unit effort (CPUE, $\hat{\theta}_{i,j,k}$) from station (i), depth strata (j), and geographic area (k) is calculated as

$$\hat{\theta}_{i,j,k} = \frac{n_fish_{i,j,k}}{n_skate_{i,j,k}},$$

where $n_fish_{i,j,k}$ is the number of fish caught and $n_skate_{i,j,k}$ is the number of skates of gear at station i, depth stratum j, and geographic area k, after filtering out skates of gear with > 5 ineffective hooks and adjusting the number of fish to a standard 45-hook skate. These are used to get an estimate of the mean CPUE for depth stratum j and geographic area k

$$\mathrm{E}[\bar{\theta}_{j,k}] = \frac{\sum_i \hat{\theta}_{i,j,k}}{n_{j,k}},$$

where $n_{j,k}$ is the number of station-level CPUE observations within depth stratum j and geographic area k. The associated sample variance of CPUE by depth stratum j and geographic area k is

$$\hat{V}[\bar{\theta}_{j,k}] = \frac{\sum_i (\hat{\theta}_{i,j,k} - \bar{\theta}_{j,k})^2}{n_{j,k}}.$$

Each depth stratum within a geographic region has an area weight ($w_{j,k}$) which is used to scale the CPUE to an RPN ($\bar{\psi}_{j,k}$) for each depth stratum j and area k

$$\mathrm{E}[\bar{\psi}_{j,k}] = \bar{\theta}_{j,k} w_{j,k},$$

These RPNs can then be summed across depth strata and geographic areas to provide regional or management area RPNs for areas sampled. Area sizes were refined in 2014 (Echave et al., 2013). The total RPN variance of an area ($\hat{V}[\hat{\psi}_k]$) combines sample variance of CPUE and area sizes with covariance of CPUE among depth stratum:

$$\hat{V}[\hat{\psi}_k] = \sum_j w_{j,k}^2 \hat{V}[\bar{\theta}_{j,k}] + 2 \sum_j \sum_{m \neq j} cov[\bar{\theta}_{j,k}, \bar{\theta}_{m,k}],$$

where j and m are different depth strata in the same region. These variances can be summed with the associated RPNs to regional or management areas.

To enable development of a consistent abundance index for use in a single region assessment model for sablefish, an extrapolation is necessary for unsurveyed regions in a given year. The extrapolation ensures that the resulting index represents the entire Alaska-wide population in a given year, avoiding annual fluctuations due to differences in survey coverage in even and odd years. By creating a single consistently scaled index, the assessment model can assume a time-invariant catchability (q) scalar instead of trying to estimate time-varying catchability and implicitly losing information on population trends and scale. In past

assessments, for years when the BS and AI were being biennially sampled, estimates for RPNs in off-years (y) for a given region were extrapolated from the previous year RPN in that region based on an assumption of a proportional change as the annually sampled GOA:

$$RPN_{AI,y} = RPN_{AI,y-1}\frac{RPN_{GOA,y}-RPN_{GOA,y-1}}{RPN_{GOA,y-1}},$$
$$RPN_{BS,y} = RPN_{BS,y-1}\frac{RPN_{GOA,y} - RPN_{GOA,y-1}}{RPN_{GOA,y-1}}.$$

Extrapolating the BS or AI based on the GOA trends generally aligns with the assumption of a single population across Alaska, which is supported by the high mobility of sablefish and cross-region movement (see tagging section below). However, it ignores potential depth-based shifts in demographics that may differ among these large regions, as well as recent differences in RPN trends for the AI and BS which have increased more rapidly than the GOA (Figures 3.8 – 3.9).

With the 2025 survey redesign, the GOA extrapolation approach is no longer feasible as the GOA is only surveyed in odd years. Thus, a new approach was implemented termed the 'carryover' method, where holdover RPN values from the last time a given region (k) was surveyed (y-1) are used for interim years (y) between survey sampling events:

$$RPN_{k,y} = RPN_{k,y-1}$$

The carryover approach ensures a consistent scaling similar to the previous GOA extrapolation method, but relies solely on an assumption of steady-state regional dynamics. For use in the assessment model, the approach likely leads to conservative catch recommendations during periods of rapid population growth and moderate overestimation of catch during periods of population decline (Goethel and Cheng, 2025; See Appendix 3E). For 2026, the carryover method was implemented for all survey years for the NOAA Longline Survey RPN index (i.e., for the single region assessment models; Figures 3.6 – 3.7). Ultimately, the population trends between the GOA and carryover extrapolation approaches remain similar, but the GOA extrapolation approach predicts more rapid recent growth with higher interannual variability (Figure 3.7).

Due to concerns with either extrapolation approach, mainly, basing index trends on extrapolated data and potentially inflating RPNs, regional NOAA Longline Survey indices were also constructed for 2026 (i.e., for the fleets-as-areas and research-oriented spatial models) based solely on the observed data in a given year (i.e., without any extrapolation). Regional indices were constructed for the BS, AI, and GOA by summing RPNs from stations within the associated regions. Thus, the regional indices represent only the 'observed' survey data and, when summed across regions, are less than the single region extrapolated indices by the amount of the extrapolation in a given year (Figure 3.7).

Historically, the NOAA Longline Survey indices have also utilized an extrapolation within the AI region to account for a decrease in survey coverage of western AI stations between the Japanese and NOAA Longline Surveys. Mainly, the NOAA Longline survey has not sampled the northwestern or southwestern AI, given logistical constraints and associated costs with sampling remote regions of Alaska. Thus, extrapolations have been used to estimate the unsampled western AI RPNs in even-numbered years based on ratios between the western and eastern AI from the Japanese survey from 1983 to 1994. The northwestern AI is extrapolated from the observed northeastern AI RPNs based on a fixed ratio of 0.1820, while the southwestern AI is extrapolated from the southeastern AI RPNs based on a fixed ratio of 0.6581. The western AI extrapolation implicitly assumes a proportional, time-invariant linkage in regional population trends between the northwest or southwest AI with the northeast or southeast AI, respectively. However,

the extrapolation methods do not account for numerous quantitative and experimental design changes that have occurred in the 30+ years since they were calculated (e.g., changes in area sizes). Therefore, given the low reliability of these extrapolation methods, the lack of observed data from the western AI, and the rapid changes in Alaska-wide sablefish population dynamics, it was deemed appropriate to remove the western AI extrapolated stations from the AI RPNs for 2026.

Finally, whale depredation on the NOAA Longline Survey is accounted for by removing stations or adjusting area-RPNs depending on the form of depredation. Sets impacted by killer whale depredation are removed from calculations, because of the significant and variable impacts killer whales can have on catch rates. For sets impacted by sperm whale depredation, which is more difficult to detect, catch rates are inflated (i.e., an inflation factor of 1.18 or 1/0.85) based on depredation estimates from an area-wide Generalized Linear Mixed Model (GLMM) using year, depth strata, station, management area, and total number of effective hooks as explanatory variables (Hanselman et al., 2018a). The standard error and covariance of this estimate is included in the total variance of the RPNs. Because sperm whale depredation only occurs on a subset of the stations, the overall increase in the RPN index is modest, ranging from 1 - 5 % over time (Figure 3.10). The correction by area is minimal, but generally most impactful in EG stations.

The lowest RPN values in the NOAA Longline Survey time series occurred in 2015, but have increased rapidly with new time series highs continually being reached each year (Figures 3.6 – 3.7). However, population growth appears to be leveling off, at least in the GOA based on the 2025 GOA index (Figure 3.8). On a station by station basis, catch rates were generally higher in both the EG and WG in 2025 compared to 2023 but lower in the CG (Figure 3.11).

Since 1996, a random sample of otoliths has been collected and aged (~1,200 per year; Tables 3.4 – 3.5) from the NOAA Longline Survey, where survey protocols require otoliths be taken if a sablefish is caught on the first hook of a skate (Siwicke and Malecha, 2026). A similar two stage method as is used for the fixed gear fishery age compositions is used for the Japanese and NOAA Longline Survey age compositions, where a random selection of otoliths are aged, then regional compositions are weighted by regional RPNs to determine a weighted average population level survey age compositions. For the random sampling, there is a goal of aging ~900 otoliths from RPN stations (i.e., those that are used in the assessment) and 300 from gully (not included in the RPN index) stations. Of the 900 otoliths from RPN stations, there was an annual target of 600 from the GOA and 300 from either the BS or AI (depending on which region was surveyed). Historically, the number of GOA RPN otoliths that were aged was ~700 due to insufficient otoliths from other strata (Table 3.5). With survey changes in 2025, new targets include ~900 aged otoliths from the GOA and 300 from gully (non-RPN) stations in odd years, then 600 each from the BS and AI in even years. As noted, the regional age compositions are weighted by RPNs in each region to obtain a representative estimate of survey catch-at-age. For models that fit to regional data, the weighted average only needs to be applied within regions with multiple areas (i.e., the GOA). Age compositions are available to the assessment on a one year lag from data collection. Length data are collected from the NOAA Longline Survey, but not used in the assessment model.

From 1979 - 1994 otolith collections from the Japanese Longline Survey were length-stratified and only aged sporadically (approximately biennially; Table 3.4). Length compositions were collected annually and used in the assessment to enable stable estimation of survey selectivity parameters. As with other composition data sources, the regional Japanese Longline Survey age and length compositions were weighted by regional RPNs to achieve a representative population level survey catch-at-age.

Biological Inputs

Because the assessment model is age-structured, but length data from the Japanese Longline Survey and trawl fishery must be fit, a size-age transition matrix must be constructed based on mean length estimates

(with uncertainty) from a growth curve. Two time blocks (pre- and post-1996) of growth are assumed in the assessment to account for potential changes in growth between historical and recent periods, and address changes in data collection protocols on the NOAA Longline Survey. For instance, before 1996 otolith collection was length-stratified and no weights were collected, while random otolith collection and weight sampling began in 1996. Age-length pairs from the Japanese Longline Survey are utilized to estimate pre-1996 growth parameters, based on bias-corrected age-length pairs (Echave, 2021). Thus, these growth parameter estimates remain unchanged from previous assessments (see Goethel et al., 2021). For the post-1996 time block, growth schedules (length-at-age and weight-at-age) were re-estimated using the NOAA Longline Survey age and length data from 1996 through 2023, with estimation assuming a von Bertalanffy growth function. Estimation of growth followed the methods established and approved in previous assessments (Echave, 2021). Sablefish growth exhibits strong sexual dimorphism with males reaching a much smaller maximum size, despite similar growth over the first ~5 years of life (Figure 3.12). In general, updated growth estimates were largely consistent with prior values (Table SM3.7). However, growth rates among younger individuals declined across both sexes, likely attributable to the influence of decreased growth rates of recent cohorts due to density dependence from extreme recruitment events (Cheng et al. 2025a). Size-age transition matrices were constructed using the re-estimated growth curves with normal error fit to the standard deviations of the collected lengths-at-age, which generally aligned across time blocks and with previous estimates (Figure 3.13).

Given the larger maximum sizes achieved by females, the resulting maximum weights achieved are also much higher (Figure 3.14). Not surprisingly, the associated weight-at-age for females is higher, especially as males approach their asymptotic maximum size starting around age-15 (Figure 3.15). Sablefish maturity schedules were developed based on histological data using an age-based logistic regression (Table SM3.7; see Williams and Rodgveller, 2021).

Sablefish are difficult to age, especially those older than eight years (Kimura and Lyons 1991). To compensate, an ageing error matrix based on known-age otoliths collected from tagged young-of-the-year fish is used (Heifetz et al. 1999; Hanselman et al. 2012). The ageing error matrix is directly incorporated into the assessment to account for uncertainty in the ageing process.

Sablefish first appear on the upper continental slope, where the longline survey and longline fishery occur, at age-2, with a fork length of about 45 cm. A higher proportion of young fish are susceptible to trawl gear compared to fixed gear. Trawl fisheries usually occur on the continental shelf and shelf break inhabited by younger fish. The transition towards pot gear in the fixed gear fishery may allow a higher proportion of small fish to be selected, but the location of fixed gear fishing activities (e.g., size-based targeting of more valuable larger fish) and the optional use of escape rings may limit the number of small sablefish retained by pots. Given these considerations, recruitment in the assessment occurs at age-2, which is when fish primarily become susceptible to the various gears. Based on analysis of NOAA Longline Survey sex-specific age data, a 50:50 sex ratio is assumed at the age of recruitment (i.e., age-2).

Sablefish are a relatively long-lived species with a maximum observed age in Alaska of 94 years (Kimura et al., 1998). However, relatively few fish older than age-31 are seen within the age composition data, aside from a few years of high catch rates of particularly old fish in the AI during the 2010s (Figures 3.48 and SM3C.42). Thus, based on the age composition data, a maximum age of 31 (modeled as a plus group) is assumed. Given the complex and sex-specific population dynamics, estimating natural mortality (M) within the assessment is difficult. Thus, a prior for M is required. The M prior was updated in 2025 [$\sim N(\ln(0.085), 0.1^2)$] based on estimates of natural mortality from previous tagging models (0.0852 from Hanselman et al., 2018b), while also considering realistic average longevity values and estimates of sablefish natural mortality from assessments on the US west coast and British Columbia (See Goethel and Cheng, 2025). In terms of longevity, a mean of 0.085 corresponds to a maximum age of ~65 years, which was deemed appropriate, on average, because there are only 6 fish >70 years in the survey age compositions. The US

west coast trawl survey observes slightly more older sablefish, and the associated longevity based prior in the west coast sablefish assessment was based on a maximum age of 75 resulting in a prior of ~$N($0.072, 0.31^2). The final estimate of natural mortality from the west coast sablefish assessment was 0.088 (Wetzel et al., 2025). For British Columbia, sex-specific natural mortality is estimated as part of the management procedure process, where female M is ~0.094 and male M is ~0.05 (DFO, 2023).

Data Input Changes for 2026

Newly available data since the 2024 assessment that were used in the assessment models for 2026 (*26.1_1Reg and 26.2_FAA)* included (Table 3.3):

- Length data from the trawl fisheries for 2024.
- Age data from the fixed gear fishery for 2024.
- Finalized catch for 2024 and 2025.
- The NOAA longline survey abundance indices for 2025.
- Updated inputs of length- and weight-at-age estimated using all available survey age-length-weight data through 2023.
- For the aggregate NOAA longline survey index (used in Model *26.1_1Reg*), the index was constructed using the 'carryover' approach for non-surveyed regions.

Data that was removed from the assessment models for 2026, include:

- The fixed gear fishery CPUE index, including the historical Japanese CPUE (1964 – 1981) and the domestic CPUE (1995 – 2022).
- The fixed gear fishery length composition data from 1990 – 1998.
- The NOAA longline survey length composition data from 1990 – 1995.
- For the NOAA Longline Survey index, extrapolated relative population numbers (RPNs) from stations in the western AI were removed from the index.

Available Data Not Used in the Assessment

NOAA Gulf of Alaska Trawl Survey

Since 1984, NOAA has conducted triennial or biennial multispecies trawl surveys in the GOA on the upper continental slope to 500 m depth. GOA trawl survey biomass indices were not used in the assessment model prior to 2007, because they were not considered good indicators of the relative biomass of adult sablefish since the full range of sablefish habitat is not sampled and sablefish may also outswim the net. Beginning in 2007, trawl survey indices were integrated into the assessment, primarily as an index of juvenile sablefish biomass. However, recent evidence suggests that the NOAA Longline Survey provides an adequate indicator of incoming recruitment strength, while signals regarding recent cohort strength in the longline survey data have countered those observed in the trawl survey, and the indices themselves strongly diverged in 2023 (Figure 3.6). Thus, the trawl survey index was again removed from the assessment in 2025 (See Goethel and Cheng, 2025). The GOA trawl survey index was at its lowest level of the time series in 2013, more than quadrupled until 2021, then declined precipitously by ~50% in 2023 and remained relatively stable in 2025 (Figure 3.6; Table 3.6). Lengths are sampled during each survey year (Table 3.4). AI and BS trawl survey biomass estimates are also available, but not used in the assessment model given their relatively low sablefish biomass estimates, high sampling error, and relatively short time series, especially in the BS.

Fixed Gear Fishery Catch-per-Unit Effort Index

Based on Japanese longline fishery catch rate data, a nominal index of historical CPUE was previously included in the assessment from 1964 – 1981 (Table 3.5). For the domestic fixed gear fishery, records of catch weight and effort for vessels that target sablefish are collected by observers and also reported by vessel captains in voluntary and required logbooks. A combined gear (i.e., including data from both hook-and-line and pot gear) standardized fixed gear CPUE index was developed (Cheng et al., 2023) and used in previous assessments (Table 3.5; Goethel and Cheng, 2024). However, the logbook data needed to produce the standardized index is no longer available to NOAA, and the last year of the CPUE index is 2022. Although the CPUE index generally matches trends in the NOAA Longline Survey RPNs, the CPUE index generally lags RPNs by several years (likely due to the fishery targeting larger fish and CPUE better representing mature biomass, whereas the RPNs better reflect total biomass; Figure 3.6). Given that the CPUE index can no longer be updated and the likely issues with standardizing for management and gear changes, the CPUE index was removed from the assessment in 2026 (See Appendix 3B). Likewise, the Japanese CPUE index was removed, since it cannot be standardized and given the lack of information on the provenance of the underlying data.

NOAA Longline Survey and Fixed Gear Fishery Length Compositions

Extensive length data from the NOAA Longline survey and fixed gear fishery are available, where survey length data are available in the year of collection while fishery length data is available on a one year lag. Length data were randomly collected for all years since 1990 for both the survey and fishery. Given the difficulties in adequately addressing assumptions for fitting length data in an age-structured assessment (e.g., accounting for time-varying growth and seasonal growth dynamics), length data has been removed for all data sources that otherwise have age data from which selectivity can be estimated (See Appendix 3B; Goethel and Cheng, 2025).

Tagging Data

The spatial distribution and movement of sablefish have been studied by the Alaska Fisheries Science Center (AFSC) through various forms of tagging data, including traditional anchor tags, electronic archival tags, and pop off satellite tags. Tagging occurs on the NOAA Longline Survey and approximately 5% of the longline survey catch of sablefish are tagged and released each year, which generally equals about 3,000 – 3,500 fish per year. Since 1972, approximately 413,000 sablefish have been tagged in Alaska waters, of which over 38,500 have been recovered. The AFSC sablefish tag program has one of the longest time series of groundfish tag data in the nation. These data have been used to examine movement patterns, evaluate areal apportionment strategies of annual catch quotas, validate ageing methods, examine growth and mortality, and develop spatially explicit tagging and assessment models. Additionally, the AFSC conducts a GOA juvenile tagging program that has released 42,625 juvenile sablefish with conventional tags and 1,082 electronic archival tags have been implanted, with 2,630 conventional tag returns and 41 electronic tag returns. Because of the known-age (age-1) of juvenile sablefish, this tagging program provides valuable information that differs from the tagging of adults on the NOAA Longline Survey. Tagging of known-age juveniles before they leave coastal areas offers an opportunity to document age-specific movement, that is, recoveries of known-age fish provides information on the age at which fish become available to the fisheries (Maloney and Sigler 2008). Recoveries of known-age fish can also provide evaluation of ageing methods, such as otolith reading, enabling the development of ageing error matrices (Heifetz et al. 1999; Hanselman et al. 2012).

Recovered tags from both programs are either physically returned to the Auke Bay Laboratory with a prepaid business reply envelope or are reported with the tag reporting application that can be downloaded for free to a mobile or tablet device. Typical information given by the respondent and collected is: (1) tag

number, (2) date of capture, (3) location, (4) size of fish, (5) sex, (6) depth of capture and (7) gear type. In recent years, as many as 500 to 1,000 tag recoveries have been submitted per year. The number of individuals returning tags varies widely. Many fishers and processors associate the tagging programs with the NMFS observer program and utilize observers onboard fishing vessels and in processing plants to collect and return all tags from that particular vessel or plant. The usual number of non-NOAA people participating ranges between 300 and 400 annually.

Existing data sets are used to match recovery and release information for each tag submitted. With each tag recovery, a small reward (hat) is offered along with a brief letter describing the history of the fish in terms of movement and growth. Additionally, an annual tag reward lottery drawing is held in which all tags recovered in the past year are entered into a drawing for a monetary reward. These incentives are to encourage participation and improve tag recovery rates.

Further information on the tagging program and the data can be found in the most recent report to industry on the sablefish tag program (Echave et al., 2013) and at the AFSC groundfish tag website found at https://apps-afsc.fisheries.noaa.gov/maps/tagmap/tagmap-v2/combined.php. The tagging data has been analyzed extensively via tagging studies (e.g., Maloney and Sigler, 2008; Hanselman et al., 2015; Kapur et al., 2025) and is integrated directly into the spatially explicit model recently developed for sablefish (Cheng et al., 2025d; See Appendix 3D).

Population Trends in Nearby Regions Not Incorporated in the Assessment Model

Alaska Northern Southeast and Southern Southeast Inside Waters

Sablefish in the Northern Southeast Inside (NSEI) Subdistrict waters and Southern Southeast Inside (SSEI) Subdistrict waters of Alaska are treated as separate stocks from the federal population, but some migration into and out of Alaska federal and state waters has been confirmed with tagging studies (Hanselman et al. 2015). NSEI sablefish continue to demonstrate similar population trends as the federal stock with strong recent recruitment observed, particularly the 2016-year class, which is now over 50% mature and comprises approximately 32% of the biomass (Figure 3.16). The NSEI longline survey CPUE has remained substantially above the long term mean since 2020 with minimal variation over the last three years. However, estimates from the 2023 stock assessment suggest sablefish spawning stock biomass remains at lower levels than estimated during the 1980s and 1990s. In SSEI waters, the longline survey CPUE showed a 40% increase from 2022 to 2023 (Figure 3.16). The SSEI pot fishery CPUE increased 10% from 2022 to 2023 while the SSEI longline fishery CPUE decreased in recent years, most likely due to a substantial shift from longline gear to pot gear. Alaska state fisheries are seeing a similar rapid transition from hook-and-line gear to pot gear as the federal fishery.

Canada

The estimated biomass trend for the British Columbia (BC) stock of sablefish is similar to that in Alaska federal waters, with strong increases in the mid-2010s due to above average recruitment in 2016 – 2017. Spawning stock biomass has shown consistent increases in recent years (Figure 3.17). The survey index value in 2023 was 10% less than the 2022 value, but still in line with the above-average levels seen since 2018. Total allowable catches (TACs) for the BC Sablefish stock are determined using a surplus production model implemented as part of a management procedure approach chosen through management strategy evaluation (Kendra Holt, pers. comm.).

United States West Coast (Washington, Oregon, and California)

The United States West Coast stock of sablefish was most recently assessed by NOAA's Northwest Fisheries Science Center in 2025. At the beginning of 2025, spawning output (i.e., the biomass of mature

females) was estimated at 74,409 metric tons (mt), corresponding to a stock status of 33.9% relative to unfished conditions. The population declined rapidly from the mid-1970s to levels below the minimum stock size threshold (MSST) by the mid-1990s. From 2000 to 2020, spawning output remained relatively stable but below the MSST. Since 2020, however, the population has increased steadily, reaching its highest spawning output since 1989 (Figure 3.18). This recent increase is driven by several above-average recruitment years since 2013, including strong cohorts in 2013, 2016, 2020, 2021, and 2023 (Figure 3.18). Recruitment estimates for 2024 remain highly uncertain at the time of the assessment. Stock size and status are projected to continue increasing over the next several years, with spawning output expected to surpass the management target ($B_{40\%}$) by 2027, even under projected catches that are considerably higher than the recent average (5,651 mt per year over the last five years).

Pacific Sablefish Transboundary Assessment Team (PSTAT)

Concurrent sablefish trends seen in Alaska, Canada, and the West Coast highlights the need to better understand the contribution to Alaska sablefish productivity from other areas and vice versa. A Pacific Sablefish Transboundary Assessment Team (PSTAT) consisting of scientists from the U.S. (west coast and Alaska regions, including both federal and state scientists) and Canada has been working to better understand the dynamics, population trends, and biology of sablefish across the eastern Pacific Ocean (https://www.pacificsablefishscience.org/). The PSTAT has engaged in a multi-year research endeavor that produced three key products: 1) a tagging model estimating movement across the northeastern Pacific (Kapur et al., 2025); 2) a spatially structured, data-conditioned operating model (OM), and 3) the development of an MSE for the transboundary sablefish stock, with seven management strategies explored (Kapur et al., 2024). No future meetings are scheduled due to lack of future funding for PSTAT work.

Analytic approach

Model Structure

The assessment models are implemented in the generalized R-TMB based Spatial Populations over Regional Components (SPoRC) stock assessment R package (Cheng et al., 2026; https://chengmatt.github.io/SPoRC/index.html), which utilizes the integrated analysis paradigm (Maunder and Punt, 2013). Two assessment model configurations are presented with identical parametrizations and assumptions, aside from the treatment of the NOAA Longline Survey data. Model *26.1_1Reg* utilizes a single aggregate index based on the carryover extrapolation method and aggregates associated age compositions Alaska-wide. Conversely, Model *26.2_FAA* utilizes a spatially implicit fleets-as-areas (FAA) configuration, where the NOAA Longline Survey is disaggregated into BS, AI, and GOA fleets with each fleet having a unique index and set of age compositions. Separate catchability and selectivity parameters are then estimated for each regional survey fleet. The following describes the general assessment structure as it applies to both model configurations and highlights areas of discrepancy in terms of survey dynamics. A complete description of the general SPoRC assessment model structure as it pertains to the sablefish application is available in Appendix 3A. Associated R code for implementing each model as well as data inputs and model outputs and figures are available from the GitHub repo (Model *26.1_1Reg*, Model *26.2_FAA*). Information on model bridging from previous AD Model Builder applications to SPoRC as well as recent assessment model changes and rationale is available in the 2025 model development document (Goethel and Cheng., 2025). Bridging from Model *25.12* to Model *26.1_1Reg* is provided in Appendix 3B, while development of Model *26.2_FAA* is provided in Appendix 3C. A research-oriented spatially explicit model has also been developed and is described in Appendix 3D, but not discussed further here.

Spatial Structure

The assessment models both assume a single Alaska-wide sablefish population. For Model *26.1_1Reg*, a single region panmictic configuration is used where spatial homogeneity is assumed across Alaska, with all data sets fit Alaska-wide. In Model *26.2_FAA*, the FAA configuration uses selectivity and catchability assumptions to implicitly account for spatial heterogeneity. Similar to the single region assessment, the FAA model assumes that the 'regional' survey trends in indices and compositions represent the dynamics from a single underlying population, where survey-specific catchability (q) and selectivity estimates are used to implicitly account for varying age-specific availability (e.g., due to the survey footprint being smaller than the distribution of the population of inference or caused by ontogenetic or seasonal migration patterns; Berger et al., 2012; Waterhouse et al., 2014). Interpreting the dynamics of FAA models can be less intuitive, because each fleet is still operating on a single, homogeneous population, such that variability in indices and compositions can only be interpreted as changes in availability to a given fleet. Thus, when signals in the data (e.g., survey indices or compositions) differ for fundamental population processes (e.g., recruitment) across space, the model must attempt to rectify these differences primarily through selectivity differences, potentially leading to non-intuitive predictions (e.g., predicted survey abundance trends). The FAA model foregoes the need for using any extrapolated data and can fit regionally disaggregated data, but may overinterpret signals from sparse composition data (e.g., if the noise to signal ratio is high). In particular, given the need for a sex-specific model for sablefish, data sparsity is of concern as compositions are partitioned by sex and region.

Age and Recruitment Dynamics

An age structured statistical catch-at-age (SCAA) framework was utilized, which tracks population numbers-at-age by sex. Sex-specific modeling was required to address sexually dimorphic growth and associated selectivity (Cheng et al., 2025b). Equilibrium age structure in 1960 (model start year) was initialized assuming exponential decay from the mean recruitment based on natural mortality and an assumption that fixed gear fishing mortality is at 10% of the mean fishing mortality. Estimated age deviations were then applied to the equilibrium age structure to define the final initial age structure.

Recruitment at age-2 is estimated based on lognormal deviations about a mean recruitment parameter. As described in Methot and Taylor (2011), a bias correction term was implemented in the stock-recruit relationship and recruitment penalty term in the objective function to ensure that recruitment estimates were mean unbiased. The correction term was scaled linearly from 0 in data poor years (when there was little information to inform recruitment) to 0.9 in recent data rich years (when there was extensive information to inform recruitment), but then declined near the terminal year to account for lack of information on the most recent year classes. The values of the bias correction terms were iteratively tuned as described in Methot and Taylor (2011). For sablefish, the break points for the bias correction term included a start year of 1968 (i.e., start of the increasing linear ramp), a max of 0.9 starting in 1984, and a subsequent decline (i.e., start of the declining linear ramp) starting in 2023 to account for limited information in the compositional data for recent recruitment events. To ensure adequate flexibility to estimate extreme, spasmodic recruitment events the recruitment variance term (σ_R) was fixed at 0.9 for the recent period (since 1975), while historical recruitment was constrained with a value of 0.4 to aid model stability in years with no composition data.

The population was then projected forward from the initial age structure and annual age-2 recruitment assuming age-based dynamics that tracked fish from age-2 to age-31+, where the terminal age was a plus group. Exponential cohort decay was modeled via total instantaneous mortality composed of natural and fishing mortality.

Demographic Parameters

Primary demographic parameters were estimated outside the model and treated as fixed inputs, including maturity- (single time block), length- (two time blocks, pre- and post-1996), and weight-at-age (single time block; see Table SM3C.7 for age-based values for all inputs). To allow fitting length data directly (i.e., in the trawl fishery and Japanese Longline Survey), predicted age compositions were converted to length compositions using input size-age transition matrices. Because sablefish can be difficult to age, an ageing error matrix was directly integrated into the assessment based on known-age otoliths (Hanselman et al., 2012). Sex-specific natural mortality is estimated with a prior [$\sim N(\ln(0.085), 0.1^2)$].

Fishery Dynamics

The model assumes two primary fishing fleets (i.e., the directed fixed gear fishery, including both pot and hook-and-line gear, and the combined trawl gear fishery) each with independent dynamics, including unique fishing mortality and fishery selectivity parameters. Each fleet had an estimated mean fishing mortality parameter and annual lognormal deviations. Selectivity was estimated by sex and fishery using a two-parameter (i.e., age at 50% selectivity, $a_{50\%}$, and the slope of the curve, k) logistic selectivity function estimated with priors [$\sim N(\ln(4),1.5^2)$, estimated in log-space]. Two selectivity time blocks were assumed for the fixed-gear fishery (1960 – 2015 and 2016 – present), roughly corresponding to the transition point between gear types from a primarily hook-and-line fishery to a primarily pot fishery (Cheng et al., 2024a,b). The trawl fishery had a single time block.

Survey Structure

Two fishery-independent survey fleets were modeled, including the NOAA Longline Survey and the Japanese Longline Survey. Each survey fleet was modeled by estimating a single catchability parameter and sex-specific logistic survey parameters (parametrized in the same way as the fishery selectivity functions). A single selectivity and catchability time block was assumed for the Japanese Longline Survey, while the NOAA Longline Survey was modeled with two selectivity time blocks (1990 – 2015 and 2016 -- present), to address apparent changes in age-specific availability by depth and region in recent years and a single catchability time block. For Model *26.2_FAA*, the NOAA Longline Survey indices were disaggregated as regional (BS, AI, and GOA) fleets. Each fleet was modeled with unique, estimated catchability and sex-specific logistic selectivity parameters (with two selectivity time blocks).

Observation Models and Estimated Parameters

Each of the data sources (i.e., fishery catch by fleet, fixed gear fishery age compositions, trawl fishery length compositions, NOAA and Japanese Longline Survey indices, NOAA Longline Survey age compositions, and Japanese Longline Survey age and length compositions; Figure 3.19; Table 3.3) are fit directly in the model. Indices and catch utilized a lognormal error structure with yearly standard errors (SEs; See Table 3.6 for the yearly SE values). For the compositional data, a multinomial error distribution was assumed. To account for interannual variability in composition data, a scaled input sample size (ISS) was utilized that proportionately scaled the ISS within a fleet based on the actual number of samples collected. Relative ISS values were assigned using a linear scaling based on the minimum and maximum samples taken across all years and the fleet-specific maximum ISS (see the maximum ISS values for each in Table 3.5) and a minimum ISS of 20. Generally, to better represent expected information content and representativeness of the data sources, age data was given a relatively higher maximum ISS than length data, survey data was given a higher value than fishery data, and the NOAA Longline Survey a higher value than the Japanese Longline Survey data.

A statistical maximum likelihood estimation (MLE) framework enabled estimation of key parameters, using

a penalized likelihood approach that included normal penalty functions [i.e., recruitment deviations from mean recruitment, $\sim N(0, \sigma_R^2)$, and fishing mortality deviations from mean fishing mortality, $\sim N(0, 0.75^2)$)] and lognormal priors (i.e., natural mortality and selectivity parameters). In total, 253 parameters were estimated for Model *26.1_1Reg* and 271 parameters for Model *26.2_FAA*, which represented the addition of 2 catchability parameters and 16 survey selectivity parameters to enable fitting the regional NOAA Longline Survey data in the FAA model (Table 3.7).

To ensure appropriate data weighting and avoid overfitting compositional data, the two-stage Francis iterative reweighting method was applied to 'right-weight' the compositional data and account for correlation among composition bins (Francis, 2011, 2017). In Stage 1, the model was run with equal input compositional data weights (i.e., all sources of age and length composition data fit in the model were given a weight, λ, of 1.0; exploratory runs demonstrated that final weights were insensitive to initial weights). Then, the compositional data weights were adjusted following Method TA1.8 and weighting assumption T3.4 of Francis (2011, Appendix Table A1, therein; i.e., using the assumption of a multinomial distribution and accounting for correlations among ages or length bins). In Stage 2, the model was then rerun with the new weights. The weights were iteratively adjusted until the difference between the current weights and the revised weights were minimal (i.e., the weights converged; for sablefish this usually took less than 10 iterations). The final 'effective' sample sizes after Francis reweighting were determined by multiplying the ISS by the data weights (See Table 3.8 for mean SE, ISS, and ESS for each data source fit in the model). It is worth noting that self-weighting Dirichlet multinomial likelihoods for the composition data were explored (See 'Sensitivity Runs' in Appendices 3B and 3C) to circumvent the need for data reweighting procedures, but fits to indices were greatly degraded and the approach was not pursued further.

Definitions

Recruitment was the estimated number of age-2 sablefish (in millions of fish). Total biomass (in kilotons) was the abundance of all sablefish age-2 and older multiplied by sex-specific input weight-at-age. Spawning stock biomass (SSB) was the mass of mature (based on input age-based maturity) females (in kilotons). Summary fishing mortality was fully selected F, which is the instantaneous mortality at the age of maximum fishery selectivity, summed across fleets.

Model Alternatives and Justification

As noted, two models were developed, which differed in their treatment of the NOAA Longline Survey data. Model *26.2_FAA* disaggregated survey data by region and avoided the need for index 'carryover' extrapolations as in Model *26.1_1Reg*. Ultimately, the choice of model structure for sablefish is a tradeoff between model stability, assessment complexity, and assumed data representativeness (See Appendix 3C, Table 3C.1). Although the FAA model would forego the need for data extrapolation, it may be more unstable and might overinterpret signals from sparse composition data (e.g., if the noise to signal ratio is high).

For Model *26.2_FAA*, changes from Model *26.1_1Reg*, included:

- A spatially-implicit FAA structure was implemented for the NOAA Longline Survey fleets, which disaggregated the longline survey into three regional fleets (BS, AI, and GOA), each of which had an independently estimated set of selectivity parameters and q values that were estimated by fitting the corresponding disaggregated indices (without any carryover or extrapolation approaches applied) and age compositions.

Model Bridging and Final Model Development

An extensive model bridging and development exercise was undertaken in 2026, which involved first implementing Model *25.12_1Reg_Cont* with updated data through 2025 and implementing Francis reweighting. Next, a single region model bridging exercise was implemented using incremental stepwise progression from Model *25.12_1Reg* to Model *26.1_1Reg* (the final single region model; See Appendix 3B). For Model *26.1_1Reg*, changes from the previously reviewed Model *25.12*, included:

- Sex-specific *M* was estimated.
- Logistic selectivity was assumed for the trawl fishery (instead of domed).
- The fixed gear fishery early/pre-IFQ (1960 -- 1994) time block was removed.
- All selectivity parameter linkages were removed.
- The selectivity parameter priors were updated.

Next, a FAA model development exercise was undertaken to develop a suitable model of intermediate complexity that did not rely on any extrapolated survey data, which was informed by an extensive, spatially explicit closed loop simulation analysis (Appendix 3E). Model building started with Model *26.1_1Reg* and added incremental complexity in fleet structure until adequate data fits and diagnostics were achieved (See Appendix 3C), which resulted in *Model 26.2_FAA*.

Model Evaluation

Model convergence was identified through successful inversion of the Hessian matrix, a maximum gradient component of less than 1e-4, reasonable estimates of uncertainty (SE), and limited parameter correlation. A jitter analysis was implemented to identify if altering the starting parameters led to an improved fit to the observed data, which would indicate that the model did not find a global minimum and suggest model instability. For the jitter analysis, the model was rerun 50 times with new starting parameter values. These new parameter values were determined by altering the starting values specified for the original model using an additive random deviation selected from a normal distribution $\sim N(0, 0.1)$.

Model performance and goodness of fit were determined by examining joint negative log-likelihood components, visual fits to the data (including indices of abundance and age and length composition data), runs tests for index data, and residual analysis of both Pearson and one step ahead (OSA) residuals (to identify potential age or time trends in residuals; for OSA residuals the first age or length bin was dropped). Evaluating compositional model fits using Pearson residuals has the potential to result in spurious conclusions regarding lack of fit and/or independence, because correlated observations are retained. OSA residuals, which decorrelate observations arising from multinomial distributions, are increasingly being recommended as best practice for interpreting stock assessment model fits (Trijoulet et al., 2023). The overall interpretation of OSA and Pearson residuals is similar: large magnitude (e.g., greater than three) or systematic trends in residuals could indicate model misspecification. Finally, time series estimates of population trends were explored to ensure these were consistent with observations from the resource and general expert judgement regarding temporal dynamics.

Uncertainty

Two forms of uncertainty were evaluated, including the Hessian approximations from the MLEs and the posterior distributions from Markov Chain Monte Carlo (MCMC) simulations. For MCMCs, the recruitment bias adjustment was turned off and priors were implemented on most parameters. The SparseNUTS package was utilized for MCMC with 20,000 iterations, 150 warmups (based on defaults of SparseNUTS), and 4 chains. Key diagnostics were reported including maximum Rhat and the minimum effective sample size.

Model Diagnostic Analyses

Retrospective Analysis

Retrospective analysis is the examination of the consistency among successive estimates of the same parameters obtained as new data are sequentially removed from a model. A seven-year retrospective analysis (i.e., terminal years 2018 through 2025) was undertaken to examine whether any significant time trends in estimates of spawning biomass or recruitment were present. Because the recent fishery and survey selectivity time blocks begin in 2016, peels before 2017 preclude informative retrospective analysis due to model instability when attempting to estimate selectivity parameters with only one year of data. 'Mohn's rho' (ρ), a measure of overall retrospective bias, was calculated as the mean of the relative 'bias' across all retrospective peels.

Historical Assessment Retrospective Analysis

An historical assessment retrospective analysis addresses consistency across successive stock assessment applications with the actual data available in the terminal model year. An 'all model' historical retrospective was undertaken, which compared the actual assessment outputs from the model used as the basis of management advice for a given SAFE year (i.e., going back to the 2018 SAFE; this included successive applications of models *16.5*, *21.12*, *23.5,* and *25.12*).

Profile Likelihoods

Likelihood profiles allow exploration of how the likelihood response surface varies for different values of a given parameter and highlights potential tension between data sources for estimation of key parameters. A likelihood profile was developed for the primary scaling parameters, mean recruitment and natural mortality.

Model Sensitivity Runs

Extensive model sensitivity runs were performed for both Model *26.1_1Reg* (Appendix 3B) and Model *26.2_FAA* (Appendix 3C) as part of the model bridging and model development process. See the associated appendices for results of the sensitivity model runs. No further model runs are presented here.

Results

Model Bridging and Final Model

Each model building step for the single region model helped improve model stability as diagnosed via various convergence diagnostics (e.g., maximum gradient values, number and degree of parameter correlations, jitter analysis, and profile likelihoods; see Appendix 3B). Model *26.1_1Reg* demonstrates more stable selectivity estimation, which has been an ongoing concern for the sablefish assessment, often requiring linking parameters across sexes. Moreover, improving biological realism by estimating sex-specific *M* helped stabilize estimates and move them towards more realistic, lower values. Similarly, removal of length compositions (for fleets where age compositions exist) further reduced model tension, providing better alignment in scaling parameters among the age composition data and the indices. By improving model parametrization, reducing model tension from uncertain data sources, and improving biological realism, the resulting Model *26.1_1Reg* appears to provide a more robust assessment of the sablefish population compared to Model *25.12_1Reg_Cont*. Results from the updated continuity model (Model *25.12_1Reg_Cont*) with data through 2025, suggest that Model *26.1_1Reg*, and Model *26.2_FAA* provided similar interpretations of the population trends and recruitment dynamics (Figures 3.20 – 3.21). In particular, the continuity and final single region models did not demonstrate any major differences, aside

from Model *26.1_1Reg* demonstrating slightly more rapid declines in SSB historically and moderately increased rebuilding recently (Figure 3.21). Model *26.2_FAA* demonstrated moderate differences in interpretation of recent recruitment, while providing a strong downward scaling in SSB (See Model Evaluation and Appendix 3C).

Data Reweighting

As expected, Francis reweighting significantly downweighted the composition data to ensure adequate fit to the indices (Table 3.8). Not surprisingly, the NOAA Longline Survey age compositions retained the highest final ESS, likely because these data most closely reflect the trends in the indices. However, the significant downweighting of the compositions further exemplifies the internal model tension between fitting the trends represented in the NOAA Longline Survey index compared to age compositions. Despite similar signals of rapid population growth since 2015, the model wants to interpret the magnitude of the recent year classes as much larger when closely fitting the compositions compared to the index. The fit to the mean age or length clearly demonstrates the difficulty fitting the rapid transition to a smaller, younger population since 2015. The mean age in the fishery has dropped from nearly 15 to less than 5 in ~7 years with a similar drop in the NOAA Longline Survey from a mean age of 10 to 5 in ~5 years (Figure 3.22). Moreover, reduction in mean age in the NOAA Longline Survey appears strongly driven by the BS and AI, as seen in the fit to mean age from the regional age compositions in Model *26.2_FAA* (Figure 3.23). Given that the AI age compositions receive moderate upweighting relative to other age composition data in the FAA model (Table 3.3), it is likely that the AI compositions better align with the trends in the indices, such that the Francis reweighting upweights the AI because it allows simultaneously good fit to the indices. The resulting fit to the mean composition bins after reweighting do not drastically change in the FAA compared to the single region model (Figures 3.22 – 3.23). However, disaggregation of the NOAA Longline Survey age compositions better elucidates the rapid reduction in mean age in recent years, particularly in the BS and AI, as well as the comparatively higher variation in mean age in these regions compared to the GOA.

Jitter Analysis

Jitter analyses for Models *26.1_1Reg* and *26.2_FAA* both indicated robust solutions (Figures 3.24 – 3.27), with negligible variation in objective function values across runs. Moreover, all jitter runs demonstrated small parameter gradients and positive definite Hessian matrices. Because no jitter run improved upon the original solution, both models were retained for all subsequent analyses. In the future, jitter analyses will apply larger parameter perturbations to better confirm solution robustness.

Model Evaluation

Both models had a positive-definite Hessian, small maximum gradient component (2.91 e -10 for Model *26.1_1Reg* and 7.64 e -10 for Model *26.2_FAA*), and no major parameter correlation, indicating (when also accounting for the jitter analysis results) that both models converged to a robust solution. However, a number of parameters demonstrate high uncertainty, including historical recruitment deviations defining the initial age structure (given lack of compositional data at the start of the model), recent recruitment deviations (due to elevated uncertainty associated with limited data on incoming year classes), and a handful of selectivity parameters (likely caused by the number of model partitions resulting from sex-specific dynamics; Tables SM3.8 – SM3.9). The objective function values by data source indicated that the compositional data, particularly the NOAA longline survey ages, and catch data had the largest absolute contributions (Figures 3.28 – 3.29).

Relative performance of each model are provided in Appendix 3C (Table 3C.2), and can be summarized as:

26.1_1Reg (full results here):

- Requires extrapolating (using the carryover method) ~50% of the NOAA Longline Survey index RPNs in a given year (to maintain a consistent scale for the single region assessment) for non-surveyed regions.
- Results in a stable model that provides good fits to the NOAA Longline Survey 'carryover' index.
- Demonstrates mediocre fits to age compositions.
- GOA composition data are implicitly upweighted during data aggregation due to higher sampling and catch, which likely under-represents recruitment signals from the BS and AI, resulting in an extremely large 2016 year class.

26.2_FAA (full results here):

- Model remains relatively stable and demonstrates adequate fits to the regionally disaggregated NOAA Longline Survey indices.
- Shows similar tension as the single region model when simultaneously trying to fit indices and age compositions, and demonstrates poor fit to the fixed gear fishery male age compositions.
- GOA NOAA Longline Survey selectivity becomes less steep, reflecting the need for the model to alias regional availability through selectivity estimation.
- Recruitment trends better reflect signals from the BS and AI as evident by increasing estimates of 2014 and 2017 year classes and decreases in the 2016 year class.
- Moderate downward scaling of SSB is due to decreasing estimates of M and R_0, likely reflecting the removal of extrapolated index data.

For both models, fit to the available data retain similar issues, where survey indices are generally well fit at the expense of composition data, primarily due to the Francis reweighting. Residual patterns in the composition data remain, but the aggregated fits appear adequate. Fundamentally, both models have trouble reconciling the much larger year classes and population growth suggested by the age composition data compared with the rapid, but more tempered, population growth observed in the survey RPNs. Both Model *26.1_1Reg* and Model *26.2_FAA* appear to provide robust and stable models for interpreting sablefish population dynamics. Although Model *26.2_FAA* assumes a moderate downward scaling of SSB and catch advice, general population dynamics and stock status remain consistent with Model *26.1_1Reg*. Interestingly, an exploratory run with the FAA model, where extrapolated regional carryover indices were fit to emulate the aggregate carryover index used in Model *26.1_1Reg*, led to a strong upward scaling of SSB, particularly since the start of the NOAA Longline Survey in the mid-1990s (Appendix 3C, Figure 3C.12). Thus, there may be inherent risk with using extrapolated survey data in the assessment, which might potentially result in artificially inflating SSB and catch advice.

Goodness of Fit

Fit to the catch data was good for both models, given the high assumed standard error (Figure 3.30). For *Model 26.1_1Reg*, the fit to both survey indices was also extremely good and the associated runs test did not indicate any significant non-randomness (Figures 3.31 and 3.33). The regionally disaggregated NOAA Longline Survey indices demonstrate general agreement on long-term population trends across regions, with all indicating rapid population growth since 2015 (Figure 3.32). Model *26.2_FAA* provided adequate fit to each index, given the slight discrepancies in signals. However, differences in the shape of the survey selectivity curves across regional fleets (i.e., much steeper in BS and AI compared to GOA; Figure 3.82) leads to a divergent trend in the last few years in the predicted indices. The BS and AI predicted indices begin to decline despite observed RPNs continuing to increase, while the GOA continues to increase and better fits the increasing observed GOA index (Figure 3.32). The reason for the divergence across regions is due to the tradeoff between survey selectivity, cohort mortality, and reductions in estimated year class sizes since the 2017 year class. Mainly, for the BS and AI survey indices, maximum selectivity occurs

around age-4, so a cohort's maximum contribution to RPNs will decline after this age due to ongoing mortality (i.e., the size of the cohort can only decline). Conversely, in the GOA, survey selectivity is maximum around age-20, so there is a tradeoff between slowly increasing selectivity and increasing cohort mortality across the lifespan, such that maximum contributions to RPNs likely occur at intermediate ages (but much later than in the BS and AI). Because the 2017 year class has reached full selection in the BS and AI surveys, it is the largest cohort in the time series, and subsequent cohorts have been much smaller, the model predicts a decline in BS and AI RPNs starting around 2022 (i.e., when the cohort reaches age-5). For the GOA, the predicted survey index continues to increase because the tipping point between mortality and selectivity has not yet been reached. To verify that this is true, a sensitivity run was undertaken with a steeper GOA survey selectivity, which resulted in the predicted GOA index increasing much more rapidly, leading to overestimation of the recent observed index and a decline in the predicted index after 2023 (Appendix 3C, Figure 3C.10). The index runs test indicated moderate non-randomness in the fit to the GOA NOAA Longline Survey index, particularly associated with the early data (prior to 2000), then underestimation from 2015 – 2020 and overestimation since 2020 (Figure 3.34).

For Model *26.1_1Reg*, fits to the composition data remain mediocre due to tension between fitting the index and age composition data. The poorer fits to the composition data can partially be attributed to the Francis reweighting approach (Francis, 2011), which prioritizes fit to the abundance indices. Fits to the NOAA Longline Survey age compositions were adequate, with a tendency to overestimate at age-2 and underestimate the peak of cohort abundance around ages 5-7, then slightly overestimate again between ages 10 through 20 (Figure 3.35). Fit to the mean proportions across the time series were generally good (Figure 3.37). Many of the same patterns observed from visual analysis of compositional data fits were observed in analysis of residuals (both Pearson and OSA residuals are provided for each compositional data source; Figures 3.37 and 3.39). Residual patterns indicate underestimation of recent large cohorts as they enter the population (e.g., since 2016; Figure 3.37), but the model has trouble fitting the massive recent cohorts (e.g., 2014, 2016, and 2017) when they begin to appear in the composition data at age-3 (Figure 3.35). The same patterns are observed in the fixed gear fishery age compositions (Figures 3.47), though recent recruitment events tend to be first observed at a slightly later age (e.g., age-4 or age-5), but fit to mean proportions is generally good. Moderate residual patterning exists, but the magnitude of residuals is relatively low (Figures 3.49 and 3.51). However, the predicted size of the plus group is greatly underestimated compared to the observed proportions (Figure 3.47), which has been an ongoing concern for the assessment (Goethel and Cheng, 2024). The large plus group is due an inordinate catch of large, old fish in the AI during the early 2010s (20 – 70% of catch from the AI was older than age-30 during these years; Appendix 3C, Figure SM3C.42), and fitting these data is not possible with realistic population dynamics (and application of a spatial model does not resolve the issue either, See Appendix 3D). Japanese Longline Survey age compositions are noisy but adequately fit (Figures 3.53, 3.55, and 3.57), while fit to the associated length compositions also appear adequate (Figures 3.59, 3.60, and 3.63). Trawl fishery length compositions are also relatively noisy and, at times, bimodal (representing the aggregation across diverse trawl fleets; Figure 3.66). However, fit to the mean proportions is reasonable and balances fitting the catch of small fish from the pelagic trawls with catch of larger females in bottom trawls (Figures 3.67 and 3.69).

Given the disaggregation by region of the NOAA Longline Survey age compositions in Model *26.2_FAA*, it is not possible to make a direct comparison to Model *26.1_1Reg* survey composition fits. However, because the GOA compositions are more highly sampled and receive higher relative weight in the aggregation method for the single region model, the GOA compositions in the FAA model are quite similar (Figures 3.35 – 3.36). Generally, the observed and predicted survey compositions from Model *26.1_1Reg* are similar to the fit to the GOA age composition fits in Model *26.2_FAA*, with generally similar residual patterns (Figures 3.37 – 3.40). A noticeable difference is that Model *26.2_FAA* strongly overestimates age-2 compositions across sexes but has slightly improved estimates for intermediate ages (e.g., ages 10 to 20; Figure 3.38). Fits to the BS and AI NOAA Longline Survey age compositions tend to be adequate for Model *26.2_FAA*, especially considering the relatively lower sample sizes and increased sparsity across ages

(Figures 3.41 and 3.44). The AI age data tends to be fit slightly better, given the relatively higher data weight (ESS). Residual patterns for all NOAA Longline Survey regional fleets tend to reflect the pervasive trend seen in Model *26.1_1Reg* of generally underestimating peak abundance at ages 4 – 6 and overestimating intermediate ages 10 – 20 (Figures 3.42 – 3.43 and 3.45 – 3.46).

Compared to Model *26.1_1Reg*, the fit to the fixed gear fishery male age compositions for Model *26.2_FAA* is noticeably worse, where expected values are much higher than observed for ages 4 – 6 (Figure 3.48). The pattern was present in Model *26.1_1Reg* for certain years (e.g., 2012 -- 2013) but has become much more exacerbated in Model *26.2_FAA*. Fixed gear fishery male selectivity in the first time block is estimated to be much steeper in Model *26.2_FAA* (Figure 3.80), which is likely driving the overestimation of young male fish. The relative downweighting of the fishery age compositions relative to the multiple NOAA Longline Survey age composition data sets is probably also a primary factor in the poorer fit to the fishery age compositions in Model *26.2_FAA*. Residual patterns remain similar to Model *26.1_1Reg* with generally low magnitude of residuals (Figures 3.49 – 3.52). No major changes in the fits to the Japanese Longline Survey (Figures 3.53 – 3.64) or trawl fishery compositions (Figures 3.65 – 3.70) were detected for Model *26.2_FAA* compared to Model *26.1_1Reg*.

Time Series Results

For both models, sablefish abundance and biomass dropped throughout much of the 1960s and 1970s (Table 3.9, Figures 3.71 – 3.72), as the population began to be heavily exploited, with catches peaking at 53,000 mt in 1972 (Table 3.1, Figure 3.1). The population recovered in the mid-1980s due to a series of strong year classes in the late 1970s and early 1980s (Figures 3.73 – 3.78). The population subsequently declined as these strong year classes were removed, with a slight rebound in the early 2000s due to a handful of above average year classes in the late 1990s, then declined to a time series low in 2015 (Figures 3.71 – 3.72). Associated with a series of extremely large recruitment events since 2014, age-2+ biomass has approximately tripled since 2015, which represents sablefish population levels on par with those at the time the fishery expanded in the 1960s. The population age structure is slowly expanding as recent year classes begin to enter more fully mature age classes, but the number of fish greater than age-10 still remains low compared to historical levels (Figures 3.73 – 3.74; see Tables SM3.10 – SM3.11 for numbers-at-age by sex). However, over the course of the next decade, it is expected that the large 2014, 2016, and 2017 year classes will continue to help expand the age structure. SSB typically lags behind biomass with less pronounced extremes and, aside from a ten-year upswing in the 1980s, had been on a general downward trajectory for the entire time series (Table 3.9; Figures 3.71 – 3.72). However, SSB has more than doubled from the time series low in 2017, but remains well below time series high levels observed in the 1960s.

Sablefish are characterized by highly spasmodic recruitment events with periodic year classes that are well above average, then prolonged periods of depressed recruitment (Figures 3.75 – 3.76). Moreover, no apparent relationship between SSB and subsequent recruitment exists, with many of the larger cohorts being spawned from relatively low SSB (Figures 3.77 – 3.78). However, the large recruitment events in the late 1970s and early 1980s were associated with a more balanced age structure, which may indicate a potential age diversity and spawning portfolio effect on recruitment success (e.g., Griffiths et al., 2023). The largest historical recruitment event was the 1978 year class followed by the 1981 year class, but below average recruitment events then occurred throughout much of the 1980s and early 1990s (Table 3.9, Figures 3.77 – 3.78). Since 2014, recruitment has been at or above the time series mean (Figures 3.75 – 3.76). The 2014, 2016, and 2017 year classes are the three of the largest cohorts in the time series. However, recruitment events since 2020 appear to have declined back towards the time series mean level (Figures 3.75 – 3.76). However, the size of recent strong recruitment events is uncertain, given that they are informed by limited age and length composition data. Recruitment trends in Model *26.2_FAA* do differ moderately from those in Model *26.1_1Reg*. For instance, the time series average recruitment has declined, while the interpretation of recent dynamics has altered slightly resulting in a larger estimate of the size of the 2014 and 2017 year

classes and a smaller 2016 year class (Figure 3.20). Alterations in recruitment estimates are attributable to subtle differences in recruitment signals across the regional NOAA Longline Survey age compositions, where the AI appears to provide a mixture of recruitment signals (strong 2014, 2016, and 2017 year classes; Figure 3.44) that balances those in the BS (strong 2014 and 2016 and very strong 2017 year classes; Figure 3.41) and GOA (strong 2014 and very strong 2016 year classes; Figure 3.36). However, there may also be a data-year interaction related to the timing of the AI survey in even years (potentially matching years when some of the large recent recruitment events have occurred; e.g., 2014 and 2016).

Estimates of natural mortality are generally lower for males than females and lower for Model *26.2_FAA* (i.e., 0.088 for females and 0.087 for males) than Model *26.1_1Reg* (i.e., 0.092 for females and 0.090 for males; Figure 3.79) The downward shift in *M* likely indicates improved model parametrization, given that previous assessment models tended to estimate unrealistically high *M* values (Goethel and Cheng, 2025). Generally, selectivity has shifted towards younger fish for the longline survey and fixed gear fishery over time (Figures 3.80 – 3.82). Males tend to be selected at an older age than females in all fleets, likely because they are smaller at a given age. Compared to the fixed gear, younger fish are more vulnerable and older fish are less vulnerable to trawl gear, reflecting the preponderance of pelagic trawl gear data (that primarily interacts with small sablefish on the EBS shelf) in the aggregate trawl fishery length compositions. For Model *26.2_FAA*, survey selectivity for the GOA becomes much less steep (Figure 3.82) in the recent time block. The GOA selectivity shape reflects the model assuming that the younger age classes (< age-20 and aside from age-2) were less available to the GOA survey, and is required by the model to avoid overestimating the GOA survey index (See Appendix 3C).

Fishing mortality was estimated to be high in the 1970s, relatively low in the early 1980s, then increased in the 1990s before undergoing a gradual decline throughout the 2000s (Figure 3.83). Over the last five years, fishing mortality has steadily declined and is on par with the low levels of the early 1980s. Fishing mortality from the trawl fleet is generally less than half that of the fishing mortality in the fixed gear fleet. The estimates of fixed gear fishing mortality have increased slightly in Model *26.2_FAA* compared to Model *26.1_1Reg*, associated with the relative increase in *M* (Figure 3.79) and downward scaling of overall population biomass (Figure 3.21). Recent management has generally constrained fishing mortality to below limit reference point values, while biomass is above limit reference point values (Figures 3.84 – 3.85). However, recent low fishing mortality is primarily associated with poor sablefish market conditions leading to low ABC utilization in recent years (~50% in 2024 and 2025). Interestingly, the phase-plane for Model *26.2_FAA* suggests that overfishing may have been occurring on the sablefish population for much of the 2000s, until recent recruitment events began entering the population around 2021 and ABC utilization rates declined (Figure 3.85). Therefore, the FAA model suggests that the single region model may be more risk prone, if population scale is more in line with those estimated in Model *26.2_FAA*.

Overall, disaggregating the NOAA Longline Survey indices and age compositions and using an FAA structure (Model *26.2_FAA*), led to moderate downward rescaling in SSB and average recruitment, primarily due to a decline in *M* estimates as well as other scaling parameters (Table 3.10). The result is a strong decline in $B_{40\%}$ and catch advice (i.e., Acceptable Biological Catch, ABC; See Summary Table). Generally, the SSB trends remain consistent, though stock rebuilding increases more rapidly since 2018 in Model *26.2_FAA* compared to Model *26.1_1Reg*. The primary difference between models is the interpretation of recent recruitment events, where Model *26.2_FAA* now predicts that the 2017 year class is much larger than the 2016 year class, and all subsequent recruitment has been considerably lower than in Model *26.1_1Reg* (Figures 3.75 – 3.76).

Uncertainty

For both Model *26.1_1Reg* and *26.2_FAA*, MCMC diagnostics indicated adequate performance (maximum Rhat of 1.008 and minimum effective sample size of 401 for each) to allow for model inference. The MLE

and MCMC estimates of parameters were generally similar, with some minor discrepancies among mean estimates for recent recruitment (Table 3.10). For primary scaling parameters (mean recruitment and catchability), the marginal posterior histograms from MCMC and asymptotic normal distributions generally align for both models (Figures 3.86 – 3.87), and no major pairwise correlations were noted aside from expected moderate correlation among primary scaling parameters (Figures 3.88 -- 3.89).

Comparison to Last Assessment Model

Generally, the population trends from the 2024 SAFE (Model *23.5*) are similar to those in both Model *26.1_1Reg* and Model *26.2_FAA*, though population scale and average recruitment have declined, particularly in Model *26.2_FAA* (Table 3.11; Figure 3.94). More importantly, model results for Model *26.1_1Reg* closely align with Model *25.12_1Reg_Cont* (the continuity model from 2025; Figures 3.20 – 3.21), despite the 2025 model not being used for provision of final management advice (due to the US government shutdown in 2025).

Model Diagnostic Analyses

Retrospective Analysis

The retrospective analysis indicated that the models are quite consistent, demonstrating a slight tendency to underestimate population scale and terminal SSB (Mohn's ρ = 0.0054 for Model *26.1_1Reg* and ρ = 0.046 for Model *26.2_FAA*; Figures 3.90 – 3.91). Variability exists in the estimates of recent year class strength, and Model *26.1_1Reg* tends to underestimate large year classes until about four years after it is first estimated (Figure 3.92). Similarly, for Model *26.2_FAA*, large recruitment year classes are often initially underestimated, but estimates tend to stabilize more quickly (Figure 3.93). However, the internal tension and differences in recruitment signals across age composition data sources (e.g., the different NOAA Longline Survey regional compositions) for Model *26.2_FAA* is apparent, given the switch between 2016 and 2017 being the dominant year class as more data is integrated over time (Figure 3.91). As noted, the GOA compositions tend to favor the 2016 year class, whereas the BS favors the 2017 year class and the AI demonstrates strong signals for both. Aside from conflicting signals within the age compositions themselves, there is likely also a year-data interaction, given the historical odd-even year survey design for the BS and AI regions, respectively. Therefore, the retrospective pattern partially reflects a data availability issue, where initial recruitment signals are probably dominated by patterns in the GOA, then shift slightly towards the BS and AI signals as those data become available. However, uncertainty in recruitment estimates and associated recruitment retrospective patterns for both models do not lead to retrospective patterns in SSB, because by the time a year class begins to significantly contribute to the SSB (e.g., by around age-5) the estimate of year class strength has already stabilized.

Historical Assessment Retrospective Analysis

Aside from Model *16.5* (i.e., used as the basis of management advice prior to 2021), which tended to severely overestimate recruitment and SSB growth, recent models have been highly consistent from one year to the next (i.e., based on the 'all model' historical retrospective; Figure 3.94). Model *26.2_FAA* demonstrates a significant downward scaling of population biomass, but trends remain similar to previous models, albeit with a moderately higher rate of population growth since 2018.

Profile Likelihoods

Likelihood profiles on key scaling parameters show similar trends across models, although those from Model *26.2_FAA* tend to be more stable and demonstrate a stronger minima for the age composition data (i.e., the primary source of information on recruitment and *M*; Figures 3.95 – 3.98). Not surprisingly, given

the final estimates (Table 3.10), the age composition data indicate lower values for M and R_0 in Model *26.2_FAA*.

Harvest Recommendations

Population Projections

A standard set of seven projections is required for each stock managed under Tiers 1, 2, or 3 by the NPFMC. Five of the seven standard scenarios support the alternative harvest strategies analyzed in the Alaska Groundfish Harvest Specifications Final Environmental Impact Statement. For all projections, yearly recruitment is simulated as an inverse Gaussian distribution parameterized using historical recruitment estimates (1979 – 2023; average recruitment was 21 million for Model *26.1_1Reg* and 18 million for Model *26.2_FAA*). A common random number seed is used across projection scenarios to ensure a common time series of recruitment across all 1,000 simulations. Projections start from the terminal year numbers-at-age and all projection parameters (e.g., selectivity, natural mortality, and biological inputs) are taken from the estimated or assumed values from the terminal year of the assessment. The relative fishing mortality among fishing fleets in the terminal year is used to assign future fleet-specific fishing mortality (associated with projection specific catch). The same assumptions are used to define biological reference points using spawning biomass per-recruit calculations. For NPFMC Tier 3 stocks, F_{ABC} is defined as $F_{40\%}$ and F_{OFL} is defined as $F_{35\%}$, except when reduced by the NPFMC threshold/sloped HCR due to SSB falling below SSB_{OFL} ($B_{35\%}$). For scenarios 1, 2, 6, and 7, the NPFMC threshold HCR is used to reduce specified fishing mortality when the biomass drops below the specified limit reference point ($B_{35\%}$; See Figure 3.84).

The seven projection scenarios are ("*max* F_{ABC}" refers to the maximum permissible value of F_{ABC}):

- *Scenario 1*: In all future years, F is set equal to *max* F_{ABC}.
- *Scenario 2*: In 2026 and 2027, F is set equal to the F associated with the specified catch, which is the whale corrected ABC multiplied by the fraction of the 2025 ABC that was harvested (i.e., a harvest ratio of 49% was calculated in 2025). For the remainder of the future years, maximum permissible ABC is used.
- *Scenario 3*: In all future years, F is set equal to 50% of max F_{ABC}.
- *Scenario 4*: In all future years, F is set equal to the 2020 – 2024 average F.
- *Scenario 5*: In all future years, F is set equal to zero.
- *Scenario 6*: In all future years, F is set equal to F_{OFL}. This scenario determines whether a stock is overfished. If the stock is expected to be 1) above its B_{MSY} level in 2025, or 2) above ½ of its B_{MSY} level in 2025 and above its B_{MSY} level in 2034 under this scenario, then the stock is not overfished.
- *Scenario 7*: In 2026 and 2027, F is set equal to *max* F_{ABC}, and in all subsequent years, F is set equal to F_{OFL}. This scenario determines whether a stock is approaching an overfished condition. If the stock is 1) above its B_{MSY} level in 2027, or 2) above 1/2 of its B_{MSY} level in 2027 and expected to be above its B_{MSY} level in 2037 under this scenario, then the stock is not approaching an overfished condition.

Spawning biomass, fishing mortality, and yield are tabulated for the seven standard projection scenarios (Tables 3.12 – 3.13). In Scenario 2 (Specified Catch), the specified catches are used to increase the accuracy of the short-term projections. For catch projections in the two years following the assessment terminal year, the ratio of the terminal year catch to terminal year ABC is used to determine the fraction of the ABC to be removed in each projection year. This method results in slightly higher future ABCs due to the lower initial removals in the initial projection years.

Status Determination

For the purpose of ABC and OFL reporting, the specified catch projections (Scenario 2) are used and reported in the executive summary. Under the Magnuson-Stevens Fishery Conservation and Management Reauthorization Act, the Secretary of Commerce is required to report on the status of each U.S. fishery with respect to overfishing. This report involves the answers to three questions: 1) Is the stock being subjected to overfishing? 2) Is the stock currently overfished? 3) Is the stock approaching an overfished condition?

The official catch estimate for the most recent complete year (2025) is 23,200 mt, which is less than the 2025 OFL of 58,500 mt. Therefore, the stock is not being subjected to overfishing. For Model *26.1_1Reg*, because 2025 SSB is at $B_{62\%}$ (i.e., above $B_{35\%}$), sablefish are not overfished. Similarly, given that the 2027 SSB is projected to be at $B_{73\%}$ (i.e., above $B_{35\%}$), sablefish are not approaching an overfished condition (Table 3.12). For Model *26.2_FAA*, because 2025 SSB is at $B_{63\%}$ (i.e., above $B_{35\%}$), sablefish are not overfished. Similarly, given that the 2027 SSB is projected to be at $B_{69\%}$ (i.e., above $B_{35\%}$), sablefish are not approaching an overfished condition (Table 3.12).

Thus, **overfishing is not occurring** on Alaskan sablefish and the stock **is not overfished nor is it approaching an overfished condition.**

Acceptable Biological Catches (ABCs) and Overfishing Limits (OFLs)

Sablefish are managed under Tier 3 of the NPFMC harvest control rule, which aims to maintain the population at $B_{40\%}$.

For Model *26.1_1Reg*, the updated point estimate of $B_{40\%}$ is 139,000 t. Since projected female spawning biomass (combined areas) for 2026 is 237,000 mt (equivalent to $B_{68\%}$), sablefish is in sub-tier "a" of Tier 3. The updated point estimates of $F_{40\%}$, and $F_{35\%}$ from this assessment are 0.08 and 0.09, respectively. Thus, the maximum permissible value of F_{ABC} under Tier 3a is 0.08, which translates into a 2026 ABC (combined areas, before whale adjustments) of 41,000 mt. The OFL fishing mortality rate is 0.09, which translates into a 2026 OFL (combined areas) of 48,000 mt.

For Model *26.2_FAA*, the updated point estimate of $B_{40\%}$ is 122,000 mt. Since projected female spawning biomass (combined areas) for 2026 is 205,000 mt (equivalent to $B_{67\%}$), sablefish is in sub-tier "a" of Tier 3. The updated point estimates of $F_{40\%}$, and $F_{35\%}$ from this assessment are 0.08 and 0.10, respectively. Thus, the maximum permissible value of F_{ABC} under Tier 3a is 0.08, which translates into a 2026 ABC (combined areas, before whale adjustments) of 32,000 mt. The OFL fishing mortality rate is 0.10, which translates into a 2026 OFL (combined areas) of 38,000 mt.

Area Apportionment of Harvests

An apportionment method that tracks regional biomass, or a proxy thereof, is likely the best defense against localized depletion or other conservation concerns (e.g., disproportionately targeting spawners in only a handful of regions or population strongholds). Based on a biological perspective, the five-year average survey apportionment method was recommended by the SSC in 2020, because it tracks biomass across management regions to the best of our current ability (i.e., by using estimates of regional biomass from the yearly longline survey that targets sablefish in primary adult habitat). Because a moving five-year average is used, the apportionment values change each year as new survey data is added into the calculation. Given that sablefish are assumed to be a single population across the GOA and BSAI, area-specific OFLs are not implemented. Explorations with the spatial model (Appendices 3D and 3E) concluded that the area apportionment based on survey biomass generally matched recommended regional harvest levels from the spatial assessment (Cheng et al., 2025d).

Data Gaps and Research Priorities

In terms of the assessment model and general knowledge of sablefish dynamics, the following research is being considered:

- Allow for time-varying growth (estimated internally or externally) to address the potential for declines in growth due to recent extreme year classes, and explore the incorporation of tagging data to inform growth processes.
- Continue to explore the use of self-weighting likelihoods in combination with time-varying selectivity formulations.
- Update the aging error matrix.
- Expand the survey age-structure (i.e., set the plus group to age-41+ or age-51+) to reduce the size of the plus group and better represent the age-structured dynamics.
- Explore use of bootstrapping methods to determine appropriate input sample sizes for compositional data using the methods of Hulson and Williams (2024).
- Pending NPFMC action on small sablefish release motions, explore methods for handling discards in the assessment model.
- Develop quantitative methods to inform recent recruitment based on analysis of environmental data and alternate data sets available in the ESP.
- Expand the spatial model to get a better understanding of juvenile distribution, spawning locations, and habitat utilization (e.g., using electronic and satellite tagging) through integration of a high-resolution tagging and movement model.
- Explore the use of data-weighting methods for tagging and composition data in the spatial model.
- Expand the spatial model into a full life cycle model that incorporates larval individual-based modeling outputs to inform connectivity during early life history stages and ecosystem drivers of settlement success.
- Continue work on the sablefish MSE tool to identify robust harvest control rules that account for age structure and fishery performance metrics.

Acknowledgments

The authors wish to acknowledge the many additional contributors that help make this sablefish assessment possible each year. The feedback we receive from industry participants is vital in aiding our understanding of patterns in the fishery and the stock. The careful data collection by fishery observers and survey biologists is greatly appreciated and form the backbone of all of our assessments. Finally, we thank our excellent AFSC Age and Growth program.

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Tables

Table 3.1. Alaska sablefish total catch (t). Eastern GOA includes West Yakutat (WY) and Southeast Outside (SEO). Catch is as of May 1, 2026 (from www.akfin.org). The values in this table are not adjusted for whale depredation. Abbreviations are: Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WGOA), Central Gulf of Alaska (CGOA), Eastern Gulf of Alaska (EGOA), West Yakutat (WY), Southeast Outside (SEO), non-commercial (Non-Comm.), Hook and Line (HAL).

		By Area							By Gear			
Year	Total	AI	BS	WGOA	CGOA	EGOA	WY	SEO	Non-Comm.	HAL	Trawl	Pot	% Trawl
1960	3,100	-	1,861	-	-	1,193	-	-	-	3,100	-	-	-
1961	16,100	-	15,627	-	-	451	-	-	-	16,100	-	-	-
1962	26,400	-	25,989	-	-	390	-	-	-	26,400	-	-	-
1963	16,900	664	13,706	266	1,324	941	-	-	-	10,600	6,300	-	37
1964	7,300	1,541	3,545	92	955	1,140	-	-	-	3,300	4,000	-	55
1965	8,700	1,249	4,838	764	1,449	433	-	-	-	900	7,800	-	90
1966	15,600	1,341	9,505	1,093	2,632	1,012	-	-	-	3,800	11,800	-	76
1967	19,200	1,652	11,698	523	1,955	3,368	-	-	-	3,900	15,300	-	80
1968	31,000	1,673	14,374	297	1,658	12,938	-	-	-	11,200	19,800	-	64
1969	36,800	1,673	16,009	836	4,214	14,099	-	-	-	15,400	21,400	-	58
1970	37,800	1,248	11,737	1,566	6,703	16,604	-	-	-	22,700	15,100	-	40
1971	43,500	2,936	15,106	2,047	6,996	16,382	-	-	-	22,900	20,600	-	47
1972	53,000	3,531	12,758	3,857	11,599	21,320	-	-	-	28,500	24,500	-	46
1973	36,900	2,902	5,957	3,962	9,629	14,439	-	-	-	23,200	13,700	-	37
1974	34,600	2,477	4,258	4,207	7,590	16,006	-	-	-	25,500	9,100	-	26
1975	29,900	1,747	2,766	4,240	6,566	14,659	-	-	-	23,300	6,600	-	22
1976	31,700	1,659	2,923	4,837	6,479	15,782	-	-	-	25,400	6,300	-	20
1977	21,403	1,897	2,718	2,968	4,270	9,543	-	-	3	18,900	2,500	-	12
1978	10,414	821	1,193	1,419	3,090	3,870	-	-	14	9,200	1,200	-	12
1979	12,031	782	1,376	999	3,189	5,391	-	-	131	10,400	1,500	-	12
1980	10,584	275	2,205	1,450	3,027	3,461	-	-	184	8,400	2,000	-	19
1981	12,838	533	2,605	1,595	3,425	4,425	-	-	238	11,000	1,600	-	12
1982	12,348	964	3,238	1,489	2,885	3,457	-	-	348	10,200	1,800	-	15
1983	12,082	684	2,712	1,496	2,970	3,818	-	-	282	10,200	1,600	-	13
1984	14,511	1,061	3,336	1,326	3,463	4,618	-	-	411	10,300	3,800	-	26
1985	15,076	1,551	2,454	2,152	4,209	4,098	-	-	576	13,000	1,500	-	10
1986	29,419	3,285	4,184	4,067	9,105	8,175	-	-	519	21,600	7,300	-	25
1987	35,666	4,112	4,904	4,141	11,505	10,500	-	-	466	27,600	7,600	-	21
1988	39,107	3,616	4,006	3,789	14,505	12,473	-	-	707	29,300	9,100	-	23
1989	35,564	3,704	1,516	4,533	13,224	11,852	-	-	764	27,500	7,300	-	21
1990	30,880	2,120	2,330	1,993	12,066	11,707	-	-	664	25,532	4,684	-	15
1991	27,092	2,190	1,209	1,931	11,178	9,938	4,069	5,869	645	23,349	3,097	-	11
1992	24,574	1,553	613	2,221	10,355	9,158	4,408	4,750	674	20,977	2,910	13	12
1993	26,099	2,078	669	740	11,955	9,976	4,620	5,356	682	22,912	2,506	-	10
1994	24,174	1,727	694	539	9,377	11,243	4,493	6,750	594	20,614	2,938	29	12
1995	21,080	1,119	930	1,747	7,673	9,223	3,872	5,352	388	18,062	2,613	18	12
1996	17,834	764	648	1,649	6,773	7,558	2,899	4,659	441	15,147	2,187	59	12
1997	14,951	781	552	1,374	6,234	5,666	1,930	3,735	344	12,975	1,632	1	11
1998	14,242	535	563	1,432	5,922	5,422	1,956	3,467	368	12,386	1,487	1	10
1999	13,977	683	675	1,488	5,874	4,867	1,709	3,159	390	11,566	1,985	37	14
2000	15,894	1,049	742	1,587	6,173	6,020	2,066	3,953	324	13,402	2,019	149	13
2001	14,435	1,074	864	1,588	5,518	5,021	1,737	3,284	370	12,057	1,783	225	12
2002	15,205	1,118	1,144	1,865	6,180	4,441	1,550	2,891	457	11,993	2,243	512	15
2003	16,797	1,118	1,012	2,118	6,994	5,170	1,822	3,347	386	13,671	2,060	680	12
2004	17,896	955	1,041	2,173	7,310	6,041	2,241	3,801	376	15,042	1,656	822	9
2005	16,951	1,481	1,070	1,930	6,706	5,399	1,824	3,575	366	13,741	1,556	1,288	9
2006	15,904	1,151	1,078	2,151	5,921	5,251	1,889	3,362	353	13,218	1,246	1,087	8
2007	16,284	1,169	1,182	2,101	6,004	5,502	2,074	3,429	326	13,087	1,235	1,636	8
2008	14,857	899	1,141	1,679	5,495	5,337	2,016	3,321	305	12,490	1,122	940	8
2009	13,364	1,100	916	1,423	4,967	4,656	1,831	2,825	302	11,370	1,057	635	8
2010	12,275	1,048	752	1,354	4,512	4,270	1,579	2,692	339	10,422	1,005	510	8
2011	13,328	1,027	707	1,395	4,922	4,936	1,902	3,034	341	11,251	1,180	556	9
2012	14,144	1,205	744	1,352	5,328	5,243	2,033	3,210	272	12,259	1,102	511	8
2013	13,851	1,082	635	1,358	5,187	5,349	2,102	3,246	240	12,134	1,037	439	7
2014	11,806	813	314	1,194	4,736	4,489	1,671	2,817	259	10,195	1,025	326	9
2015	11,179	422	210	998	4,626	4,677	1,866	2,811	246	9,721	1,090	122	10
2016	10,472	340	531	1,052	4,195	4,106	1,651	2,455	248	8,701	1,336	187	13
2017	12,552	588	1,150	1,181	4,838	4,510	1,694	2,816	285	8,464	2,272	1,531	18
2018	14,494	664	1,536	1,389	5,778	4,881	1,861	3,019	246	8,690	3,780	1,778	26
2019	16,912	663	3,162	1,533	6,280	4,915	1,802	3,113	360	8,268	5,154	3,130	30
2020	19,416	1,232	5,329	1,462	6,041	4,971	1,835	3,137	381	5,813	7,493	5,730	39
2021	21,748	1,578	4,169	1,994	7,325	6,201	2,329	3,872	481	4,644	4,853	11,771	22
2022	27,420	2,230	5,514	3,028	8,165	7,971	2,750	5,221	512	4,056	5,366	17,485	20
2023	25,493	2,488	6,132	2,789	6,457	7,151	2,453	4,698	498	3,310	5,602	16,103	22
2024	23,515	1,430	4,904	2,783	7,317	6,976	2,390	4,586	105	3,406	5,506	14,497	24
2025	23,641	948	3,909	3,136	7,708	7,546	2,293	5,247	394	2,814	4,105	16,321	18

Table 3.2. Summary of management measures with time series of catch, ABC, OFL, and TAC. All values are in metric tons (mt). Catch is as of May 1, 2026 (from www.akfin.org). Catch does not include non-commercial catch (i.e., as opposed to Table 3.1).

Year	Catch	OFL	ABC	TAC	Management measure
1980	10,400			18,000	Amendment 8 to the Gulf of Alaska Fishery Management Plan established the West and East Yakutat (now SEO) management areas for sablefish.
1981	12,600			19,300	
1982	12,000			17,300	
1983	11,800			14,500	
1984	14,100			14,800	
1985	14,500			13,500	Amendment 14 of the GOA FMP allocated sablefish quota by gear type: 80% to fixed gear and 20% to trawl gear in WGOA and CGOA and 95% fixed to 5% trawl in the EGOA.
1986	28,900			21,400	Pot fishing banned in Eastern GOA.
1987	35,200			27,700	Pot fishing banned in Central GOA.
1988	38,400		44,200	36,400	
1989	34,800		37,100	32,200	Pot fishing banned in Western GOA.
1990	30,200		33,400	33,200	Amendment 15 of the BSAI FMP allocated sablefish quota by gear type: 50% to fixed gear in and 50% to trawl in the EBS, and 75% fixed to 25% trawl in the Aleutian Islands.
1991	26,400		28,800	28,800	
1992	23,900	34,100	25,200	25,200	Pot fishing banned in Bering Sea (57 FR 37906).
1993	25,400	33,200	25,000	25,000	
1994	23,600	35,900	28,800	28,800	
1995	20,700	25,700	25,300	25,300	Amendment 20 to the Gulf of Alaska Fishery Management Plan and 15 to the Bering Sea/Aleutian Islands Fishery Management Plan established IFQ management for sablefish beginning in 1995. These amendments also allocated 20% of the fixed gear allocation of sablefish to a CDQ reserve for the Bering Sea and Aleutian Islands.
1996	17,400	22,800	19,600	19,400	Pot fishing ban repealed in Bering Sea except from June 1-30.
1997	14,600	45,600	17,200	16,800	Maximum retainable allowances for sablefish were revised in the Gulf of Alaska. The percentage depends on the basis species.
1998	13,900	27,800	16,800	16,800	
1999	13,600	24,700	15,900	15,400	
2000	15,600	21,500	17,200	17,200	
2001	14,100	20,700	16,900	16,900	
2002	14,700	26,100	17,300	17,300	
2003	16,400	28,900	20,900	20,900	
2004	17,500	30,800	23,000	22,600	
2005	16,600	25,400	21,000	21,000	
2006	15,600	25,300	21,000	20,700	
2007	16,000	23,700	20,100	20,100	
2008	14,600	21,300	18,000	18,000	Pot fishing ban repealed in Bering Sea for June 1-30 (74 FR 28733).
2009	13,100	19,000	16,100	16,100	
2010	11,900	18,000	15,200	15,200	
2011	13,000	19,000	16,000	16,000	
2012	13,900	20,400	17,200	17,200	
2013	13,600	19,200	16,200	16,200	
2014	11,500	16,200	13,700	13,700	
2015	10,900	16,100	13,700	13,700	NPFMC passes Amendment 101 to allow pot fishing in the GOA.
2016	10,200	13,400	11,800	11,800	Whale depredation accounted for in survey and fishery.
2017	12,300	15,400	13,100	13,100	Pot fishing begins in the GOA.
2018	14,200	29,500	15,000	15,000	
2019	16,600	32,800	15,100	15,100	
2020	19,000	50,500	22,000	18,300	OFL changed to Alaska-wide. TAC set below ABC based on AP recommendation.
2021	21,300	60,400	29,600	26,100	
2022	26,900	40,400	34,500	34,500	
2023	25,000	47,400	40,500	39,600	
2024	23,400	55,100	47,100	39,000	
2025	23,200	58,500	47,600	39,300	

Table 3.3. Data used in the 2026 assessment models. Years in **bold** are data new to this assessment.

Source	Data	Years
Fixed Gear Fisheries	Catch	1960 – **2025**
Trawl Fisheries	Catch	1960 – **2025**
Non-Commercial Catch	Catch	1977 – **2025**
U.S. Fixed Gear Fisheries	Age	1999 – **2024**
U.S. Trawl Fisheries	Length	1990,1991,1999, 2005 – **2024**
Japan-U.S. Cooperative Longline Survey	RPNs, Length	1979 - 1994
	Age	1981, 1985, 1987, 1989, 1991, 1993
NOAA Domestic Longline Survey	RPNs	1990 – 2023, **2025**
	Age	1996 – 2023

Table 3.4. Sample sizes for age and length data for all available data (not all data is used in the assessment). Japanese fishery data are from Sasaki (1985), U.S. fishery data are from the observer databases, and NOAA survey data are from AFSC databases.

	Age Samples			Length Samples				
Year	Japanese LL Survey	NOAA LL Survey	Fixed Gear Fishery	Japanese LL Survey	NOAA LL Survey	NOAA Trawl Survey	Fixed Gear Fishery	Trawl Gear Fishery
1979				18,673				
1980				26,568				
1981	490			32,789				
1982				62,241				
1983				65,579				
1984				98,077				
1985	1,294			120,453				
1986				121,834				
1987	1,211			98,246				
1988				108,695				
1989	2,069			105,575				
1990				71,858	86,960	4,954	32,936	1,204
1991	797			64,052	81,747		28,182	655
1992				73,075	90,725		20,929	
1993	1,718			74,402	82,033	7,015	21,943	
1994				69,020	62,714		11,914	
1995					80,652		17,735	
1996		957			72,720	4,299	14,416	
1997		1,126			73,694		20,330	
1998		912			48,094		8,932	
1999		791	1,141		62,812	3,968	28,070	447
2000		911	1,148		52,556		32,208	471
2001		889	1,003		64,448	3,501	30,315	422
2002		901	1,059		60,004		33,719	527
2003		872	1,185		62,972	4,839	36,077	463
2004		949	1,145		69,977		31,199	717
2005		976	1,114		62,514	4,643	36,213	2,541
2006		949	1,154		66,060		32,497	898
2007		905	1,115		59,425	3,690	29,854	2,142
2008		947	1,164		56,830		23,414	2,268
2009		963	1,126		55,637	3,455	24,674	1,897
2010		940	1,159		62,820		24,530	1,634
2011		954	1,190		73,063	2,064	22,659	1,877
2012		947	1,165		53,514		22,203	2,533
2013		957	1,157		43,230	1,206	16,093	2,674
2014		1,010	1,126		46,885		19,524	2,210
2015		1,098	1,176		44,150	2,023	20,056	2,320
2016		990	1,169		60,353		12,857	1,630
2017		975	1,190		61,616	2,860	12,345	2,625
2018		976	1,174		63,287		13,269	3,306
2019		946	1,140		84,705	7,555	13,537	2,620
2020		972	1,188		91,432		9,122	9,421
2021		963	1,183		76,162	8,208	15,762	7,681
2022		972	1,174		66,850		16,152	6,485
2023		916	991		61,037	2,795	17,993	6,732
2024			1,181				12,178	6,383
2025					93,164	2,249		

Table 3.5. The actual number of samples and associated Input Sample Size (ISS) by year for age and length composition data used in the assessment model. ISS is calculated using a relative linear scaling within each gear based on the maximum and minimum number of samples (see text). The single region model uses the values from the 'AK' row, while the fleets-as-areas (FAA) model uses the 'AK' row for all data sources except the NOAA Longline Survey (i.e., regional ISS are used for each regional fleet).

		Age Samples						Length Samples			
		Japanese LL Survey		NOAA LL Survey		Fixed Gear Fishery		Japanese LL Survey		Trawl Fishery	
Year	Reg	#	ISS	#	ISS	#	ISS	#	ISS	#	ISS
1979	AK							18,673	29		
1980	AK							26,568	32		
1981	AK	490	43					32,789	35		
1982	AK							62,241	49		
1983	AK							65,579	50		
1984	AK							98,077	65		
1985	AK	1,294	85					120,453	75		
1986	AK							121,834	75		
1987	AK	1,211	81					98,246	65		
1988	AK							108,695	70		
1989	AK	2,069	125					105,575	68		
1990	AK							71,858	53	1,204	28
1991	AK	797	59					64,052	49	655	24
1992	AK							73,075	53		
1993	AK	1,718	107					74,402	54		
1994	AK							69,020	52		
	AI			204	34						
1996	GOA			753	82						
	AK			957	172						
	BS			223	36						
1997	GOA			903	95						
	AK			1,126	200						
	AI			323	44						
1998	GOA			589	68						
	AK			912	165						
	BS			193	33						
1999	GOA			599	68						
	AK			791	144	1,141	145			447	22
	AI			281	41						
2000	GOA			630	71						
	AK			911	165	1,148	146			471	22
	BS			315	44						
2001	GOA			574	66						
	AK			889	161	1,003	129			422	22
	AI			248	38						
2002	GOA			653	73						
	AK			901	163	1,059	136			527	22
	BS			166	31						
2003	GOA			706	78						
	AK			872	158	1,185	150			463	22
	AI			282	41						
2004	GOA			667	74						
	AK			949	171	1,145	145			717	24
	BS			301	42						
2005	GOA			675	75						
	AK			905	164	1,114	142			2,541	40
	AI			295	42						
2006	GOA			654	73						
	AK			949	171	1,154	146			898	26
2007	BS			286	41						
	GOA			619	70						

	AK	905	164	1,115	142	2,142	37
	AI	265	39				
2008	GOA	682	76				
	AK	947	171	1,164	148	2,268	38
	BS	189	33				
2009	GOA	774	84				
	AK	963	173	1,126	143	1,897	35
	AI	156	30				
2010	GOA	784	85				
	AK	940	169	1,159	147	1,634	32
	BS	171	31				
2011	GOA	**783**	**85**				
	AK	954	172	1,190	150	1,877	34
	AI	242	37				
2012	GOA	705	78				
	AK	947	171	1,165	148	2,533	40
	BS	237	37				
2013	GOA	720	79				
	AK	957	172	1,157	147	2,674	41
	AI	198	33				
2014	GOA	812	87				
	AK	1,010	181	1,126	143	2,210	37
	BS	132	28				
2015	GOA	966	100				
	AK	1,098	196	1,176	149	2,320	38
	AI	304	43				
2016	GOA	686	76				
	AK	990	178	1,169	148	1,630	32
	BS	300	42				
2017	GOA	675	75				
	AK	975	175	1,190	150	2,625	41
	AI	297	42				
2018	GOA	679	75				
	AK	976	175	1,174	149	3,306	47
	BS	299	42				
2019	GOA	647	73				
	AK	946	170	1,140	145	2,620	41
	AI	296	42				
2020	GOA	676	75				
	AK	972	175	1,188	150	9,421	100
	BS	300	42				
2021	GOA	663	74				
	AK	963	173	1,183	150	7,681	85
	AI	298	42				
2022	GOA	674	75				
	AK	972	175	1,174	149	6,485	75
	BS	300	42				
2023	GOA	616	70				
	AK	916	165	991	128	6,732	77
2024	AK			1,181	149	6,383	74

Table 3.6. Sablefish abundance or biomass index values (only the NOAA Longline Survey and Japanese Longline Survey RPNs are used in the assessment model) along with associated standard errors (SEs). Relative population number (RPN; in 1000s of fish) equals CPUE in numbers weighted by respective strata areas. Relative population weight (RPW; in kilotons) equals CPUE in weight multiplied by strata areas. NMFS trawl survey biomass estimates (kilotons) are from the Gulf of Alaska only at depths < 500 m. The single region model uses the values from the 'AK' row, while the fleets-as-areas (FAA) model uses the 'AK' row for all data sources except the NOAA Longline Survey (i.e., regional RPNs are used for each regional fleet).

		Relative Population Numbers (RPNS)				Relative Population Weights (RPWs) or Biomass									
Year		NOAA LL Survey*		Japanese LL Survey		NOAA LL Survey*		Japanese LL Survey		NOAA Trawl Survey		Japanese Fishery CPUE		Fixed Gear Fishery CPUE	
	Reg	RPN	SE	RPN	SE	RPW	SE	RPW	SE	RPW	SE	RPW	SE	RPW	SE
1964	AK											1,452	0.1		
1965	AK											1,806	0.1		
1966	AK											2,462	0.1		
1967	AK											2,855	0.1		
1968	AK											2,336	0.1		
1969	AK											2,443	0.1		
1970	AK											2,912	0.1		
1971	AK											2,401	0.1		
1972	AK											2,247	0.1		
1973	AK											2,318	0.1		
1974	AK											2,295	0.1		
1975	AK											1,953	0.1		
1976	AK											1,780	0.1		
1977	AK											1,511	0.1		
1978	AK											942	0.1		
1979	AK			413	0.06			1,075	0.05			809	0.1		
1980	AK			387	0.07			968	0.05			1,040	0.1		
1981	AK			458	0.05			1,146	0.03			1,343	0.1		
1982	AK			613	0.06			1,572	0.03						
1983	AK			621	0.06			1,632	0.03						
1984	AK			685	0.07			1,804	0.04						
1985	AK			903	0.06			2,569	0.04						
1986	AK			838	0.06			2,456	0.04						
1987	AK			667	0.06			2,068	0.04						
1988	AK			707	0.06			2,088	0.03						
1989	AK			661	0.10			2,177	0.07						
1990	GOA	455	0.06			1,553	0.04								
	AK	513	0.06	449	0.08	1,710	0.04	1,454	0.06	214	0.17				
1991	GOA	457	0.06			1,625	0.04								
	AK	516	0.06	386	0.11	1,782	0.04	1,321	0.07						
1992	GOA	406	0.06			1,416	0.04								
	AK	465	0.05	402	0.06	1,573	0.03	1,390	0.04						
1993	GOA	454	0.07			1,573	0.05								
	AK	513	0.07	395	0.06	1,730	0.04	1,318	0.04	248	0.16				
1994	GOA	368	0.06			1,455	0.04								
	AK	427	0.06	366	0.09	1,612	0.04	1,288	0.06						
1995	GOA	377	0.07			1,384	0.04								
	AK	436	0.06			1,541	0.04							0.35	0.03
1996	AI	16	0.32			50	0.21								
	GOA	394	0.06			1,583	0.04								
	AK	452	0.06			1,739	0.04			144	0.17			0.34	0.03
1997	BS	42	0.17			157	0.11								
	GOA	379	0.07			1,396	0.05								

	AK	438	0.06	1,553	0.04			0.37	0.04
	AI	71	0.20	217	0.13				
1998	GOA	338	0.08	1,238	0.05				
	AK	452	0.07	1,612	0.05			0.33	0.03
	BS	47	0.19	125	0.13				
1999	GOA	371	0.07	1,338	0.05				
	AK	490	0.06	1,680	0.04	103	0.10	0.33	0.03
	AI	66	0.27	188	0.18				
2000	GOA	322	0.09	1,202	0.06				
	AK	435	0.08	1,514	0.05			0.33	0.03
	BS	85	0.14	255	0.09				
2001	GOA	349	0.08	1,252	0.05				
	AK	499	0.07	1,695	0.05			0.31	0.03
	AI	63	0.26	209	0.17				
2002	GOA	376	0.10	1,344	0.07				
	AK	524	0.08	1,807	0.05			0.32	0.03
	BS	70	0.28	219	0.18				
2003	GOA	367	0.09	1,236	0.06				
	AK	499	0.08	1,663	0.05	187	0.14	0.35	0.03
	AI	55	0.21	142	0.14				
2004	GOA	389	0.10	1,243	0.06				
	AK	514	0.08	1,605	0.06			0.33	0.03
	BS	103	0.20	260	0.13				
2005	GOA	366	0.10	1,202	0.07				
	AK	524	0.08	1,604	0.05	178	0.13	0.36	0.04
	AI	65	0.38	181	0.25				
2006	GOA	374	0.09	1,304	0.06				
	AK	542	0.08	1,744	0.06			0.30	0.03
	BS	98	0.20	286	0.13				
2007	GOA	329	0.09	1,120	0.06				
	AK	493	0.09	1,586	0.06	110	0.12	0.31	0.03
	AI	48	0.22	148	0.15				
2008	GOA	304	0.08	1,052	0.06				
	AK	451	0.08	1,487	0.05			0.32	0.03
	BS	21	0.20	71	0.13				
2009	GOA	323	0.11	1,108	0.07				
	AK	393	0.10	1,328	0.06	105	0.12	0.28	0.03
	AI	36	0.25	119	0.17				
2010	GOA	383	0.10	1,279	0.06				
	AK	440	0.09	1,470	0.06			0.26	0.03
	BS	33	0.17	91	0.12				
2011	GOA	461	0.10	1,406	0.06				
	AK	531	0.09	1,616	0.06	84	0.19	0.25	0.03
	AI	62	0.28	145	0.19				
2012	GOA	336	0.09	1,029	0.06				
	AK	431	0.08	1,266	0.05			0.27	0.03
	BS	94	0.19	254	0.12				
2013	GOA	261	0.09	870	0.06				
	AK	417	0.08	1,270	0.05	59	0.17	0.22	0.02
	AI	48	0.24	136	0.16				
2014	GOA	308	0.08	988	0.06				
	AK	450	0.07	1,379	0.05			0.21	0.02
	BS	48	0.23	135	0.15				
2015	GOA	279	0.09	894	0.06				
	AK	374	0.08	1,166	0.05	65	0.10	0.19	0.02

	AI	90	0.20	217	0.13				
2016	GOA	319	0.08	933	0.05				
	AK	457	0.07	1,286	0.05			0.17	0.02
	BS	107	0.35	181	0.22				
2017	GOA	329	0.07	923	0.05				
	AK	526	0.09	1,321	0.05	118	0.18	0.19	0.02
	AI	134	0.20	252	0.13				
2018	GOA	327	0.07	770	0.05				
	AK	568	0.09	1,204	0.05			0.17	0.02
	BS	218	0.27	426	0.17				
2019	GOA	442	0.08	952	0.05				
	AK	794	0.09	1,631	0.06	210	0.09	0.19	0.02
	AI	271	0.15	496	0.10				
2020	GOA	549	0.08	1,339	0.05				
	AK	1,038	0.08	2,262	0.05			0.20	0.02
	BS	406	0.10	916	0.06				
2021	GOA	535	0.07	1,311	0.05				
	AK	1,211	0.06	2,723	0.04	290	0.11	0.27	0.03
	AI	300	0.13	632	0.08				
2022	GOA	631	0.06	1,600	0.04				
	AK	1,337	0.05	3,147	0.03			0.37	0.04
	BS	442	0.17	962	0.11				
2023	GOA	657	0.09	1,548	0.06				
	AK	1,400	0.07	3,142	0.05	140	0.12		
2025	GOA	686	0.07	1750	0.05				
	AK	1,429	0.07	3,344	0.04	153	0.14		

*NOAA Longline Survey RPNs assume ‘carryover’ values for survey areas not sampled every year, including Aleutian Islands (1990 -- 1995 and subsequent odd numbered years), Bering Sea (1990 -- 1996 and subsequent even numbered years), and BS+AI (2025). There was no longline survey in 2024.

Table 3.7. Summary of the parameters estimated within the assessment model.

Parameter Name	***25.12_1Reg_Cont***	***26.1_1Reg***	***26.2_FAA***
Catchability	5	2	4
Mean recruitment	1	1	1
Natural Mortality	1	2	2
Initial Deviations	28	28	28
Recruitment Deviations	65	65	65
Mean Fishing Mortality	2	2	2
Fishing Mortality Deviations	129	129	129
Fishery Selectivity	14	12	12
Survey Selectivity	11	12	28
Total	256	253	271

Table 3.8. The assumed error terms for each data source used in the final assessment models. Fit to catch and survey data assumed lognormal error with an associated assumed standard error (SE). Fit to composition data (ages and lengths) assumed a multinomial distribution with associated sample sizes (input sample size, ISS, and final 'effective' sample size, ESS). Each model used Francis data reweighting to 'right-weight' the composition data relative to the indices (Francis, 2011), which resulted in a weighting factor (λ , Data Wt) used to scale the ISS to achieve the final ESS. Values represent the mean across all years of associated data, where the yearly SE can be found in Table 3.6 and the yearly ISS can be found in Table 3.5 (associated yearly ESS can be calculated by multiplying the ISS by the respective Data_Wt for the given data source).

Data Source	*26.1_1Reg*				*26.2_FAA*			
	SE	**ISS**	**Data Wt**	**ESS**	**SE**	**ISS**	**Data Wt**	**ESS**
Fixed Gear Catch	0.05	--	--	--	0.05	--	--	--
Trawl Catch	0.05	--	--	--	0.05	--	--	--
NOAA Longline Survey RPN BS	--	--	--	--	0.20	--	--	--
NOAA Longline Survey RPN AI	--	--	--	--	0.24	--	--	--
NOAA Longline Survey RPN GOA	--	--	--	--	0.08	--	--	--
NOAA Longline Survey RPN AK	0.07	--	--	--	--	--	--	--
Coop Survey RPN	0.07	--	--	--	0.07	--	--	--
Fixed Gear Age Composition	--	145	0.12	18	--	145	0.04	6
NOAA Longline Survey Age Composition BS	--	--	--	--	--	37	0.51	19
NOAA Longline Survey Age Composition AI	--	--	--	--	--	39	1.91	75
NOAA Longline Survey Age Composition GOA	--	--	--	--	--	77	0.29	23
NOAA Longline Survey Age Composition AK	--	171	0.28	47	--	--	--	--
Japanese Longline Survey Age Composition	--	83	0.17	14	--	83	0.18	15
Japanese Longline Survey Size Composition	--	55	0.34	19	--	55	0.33	18
Trawl Fishery Size Composition	--	45	0.25	12	--	45	0.14	7

Table 3.9. Estimates (MLE mean) of recruitment (millions of age-2 fish) and spawning biomass (kt) with lower and upper lower 95% MCMC credible intervals for Model *26.1_1Reg* and Model *26.2_FAA*.

	26.1_1Reg						*26.2_FAA*					
Year	**Recruits**	**2.5% CI**	**97.5% CI**	**SSB**	**2.5% CI**	**97.5% CI**	**Recruits**	**2.5% CI**	**97.5% CI**	**SSB**	**2.5% CI**	**97.5% CI**
1960	19	7	37	304	233	348	16	6	30	252	195	291
1961	19	7	37	304	233	346	16	6	31	252	195	291
1962	20	7	38	296	227	339	17	6	33	245	189	283
1963	20	8	41	284	215	325	18	7	35	232	178	270
1964	21	8	43	276	209	317	19	7	38	225	172	262
1965	22	8	45	275	208	315	19	7	40	224	171	260
1966	22	8	44	272	207	312	19	7	41	223	171	258
1967	21	8	43	267	204	306	19	7	40	218	167	252
1968	21	8	40	261	199	298	18	7	37	213	164	246
1969	20	7	38	250	191	286	17	7	34	203	156	234
1970	19	7	35	237	180	271	16	6	31	190	146	219
1971	18	7	32	223	169	257	15	6	28	177	136	205
1972	17	7	30	207	155	240	14	6	25	162	124	189
1973	16	7	28	186	137	218	13	5	23	142	108	168
1974	15	6	26	173	126	204	12	5	21	131	99	155
1975	7	1	17	161	116	191	6	1	13	120	90	142
1976	7	1	16	151	108	180	6	1	13	111	84	132
1977	7	1	16	140	99	167	6	1	13	101	75	120
1978	9	2	20	134	94	159	7	1	17	95	70	113
1979	20	3	45	132	93	155	16	3	48	94	70	110
1980	64	21	91	129	92	150	60	9	76	91	69	107
1981	14	2	47	126	91	145	11	2	42	90	69	103
1982	23	3	67	124	91	141	16	3	62	88	69	100
1983	63	9	86	127	94	142	49	4	67	92	72	101
1984	10	2	35	134	101	149	9	2	30	99	79	107
1985	10	2	28	145	111	160	9	2	26	109	88	117
1986	18	3	30	158	122	174	14	2	25	121	98	129
1987	7	2	18	165	128	183	6	1	15	126	102	135
1988	6	1	14	167	129	186	6	1	15	125	102	134
1989	10	2	20	162	125	182	10	2	19	118	96	128
1990	9	2	22	154	118	174	11	2	22	108	88	120
1991	22	8	32	144	109	164	15	4	26	98	79	109
1992	8	2	20	133	101	153	12	2	24	88	71	99
1993	19	6	30	122	92	142	12	4	20	79	64	90
1994	11	3	22	111	84	131	4	1	9	71	58	81
1995	9	2	18	103	78	122	8	2	13	64	53	74
1996	12	4	22	97	74	115	12	6	17	60	49	69
1997	16	5	25	94	72	112	8	2	15	57	47	66
1998	8	2	19	92	71	110	18	9	25	55	46	64
1999	26	12	37	91	70	108	10	3	18	53	44	62
2000	17	5	30	90	70	107	8	2	15	52	43	60
2001	14	3	29	88	69	106	13	4	20	50	42	58
2002	30	14	45	88	69	106	27	17	37	50	42	58
2003	14	4	27	90	70	108	12	4	20	50	42	58
2004	9	2	19	91	72	110	5	1	11	50	42	58
2005	10	3	18	94	74	113	7	2	11	51	43	59
2006	7	2	14	97	76	117	3	1	7	52	44	61
2007	9	3	16	100	79	120	10	3	14	54	46	62
2008	10	3	18	101	79	121	7	2	15	54	46	63

2009	15	5	25	100	79	121	16	6	24	53	44	62
2010	14	5	24	99	78	120	14	6	22	51	43	60
2011	9	2	17	97	76	117	7	2	14	49	41	58
2012	9	3	15	94	74	114	6	2	11	47	40	56
2013	5	1	11	92	72	111	6	2	11	47	39	55
2014	7	2	14	89	70	109	6	1	12	46	39	55
2015	12	4	23	88	69	107	9	3	18	47	39	56
2016	46	25	63	87	68	106	62	41	83	48	40	57
2017	18	4	39	86	67	105	29	11	51	49	41	58
2018	103	61	141	87	67	106	68	37	99	51	42	62
2019	76	30	119	90	70	110	112	72	148	57	47	70
2020	32	4	72	98	76	119	11	2	33	69	56	85
2021	33	5	73	113	87	138	11	2	26	88	71	108
2022	42	4	79	136	105	165	16	2	38	115	91	141
2023	33	3	64	163	125	197	10	1	29	144	113	176
2024	21	3	71	192	147	232	11	2	33	172	133	211
2025	21	14	23	219	167	263	17	12	18	194	149	238

Table 3.10. Key scaling parameter estimates along with their uncertainty, including 95% credible intervals from MCMC analysis for each model. Recruitment is in millions of fish. All parameter estimates are in ln-space.

Model	Parameter	Mean (MCMC)	Median (MCMC)	SD (MCMC)	97.5% CI	2.5% CI	Mean (MLE)	SD (MLE)
26.1_1Reg	ln_Mean_Recruit	2.87	2.86	0.13	2.66	3.09	3.03	0.13
	ln_M_Female	-2.41	-2.41	0.07	-2.52	-2.29	-2.39	0.07
	ln_M_Male	-2.43	-2.43	0.07	-2.55	-2.31	-2.41	0.07
	ln_q NOAA Longline Survey	1.90	1.91	0.10	1.73	2.06	1.90	0.10
	ln_q Japanese Longline Survey	1.73	1.73	0.10	1.56	1.89	1.71	0.09
26.2_FAA	ln_Mean_Recruit	2.67	2.67	0.11	2.49	2.86	2.82	0.11
	ln_M_Female	-2.44	-2.44	0.07	-2.57	-2.32	-2.43	0.07
	ln_M_Male	-2.46	-2.46	0.08	-2.59	-2.32	-2.44	0.08
	ln_q NOAA Longline Survey BS	0.53	0.53	0.10	0.36	0.68	0.55	0.10
	ln_q NOAA Longline Survey AI	0.31	0.32	0.10	0.15	0.47	0.33	0.10
	ln_q NOAA Longline Survey GOA	2.17	2.18	0.08	2.04	2.31	2.20	0.08
	ln_q Japanese Longline Survey	2.07	2.06	0.09	1.92	2.21	2.08	0.08

Table 3.11. Comparison of the 2024 SAFE model (*23.5*; last published sablefish assessment) estimates and the 2026 assessment outputs for recruitment and spawning stock biomass (SSB) including the % change from the 2024 SAFE estimates (Diff %).

	Recruits (millions)					SSB (kt)				
Year	2024 SAFE	2026 *26.1_1Reg*	Diff (%)	2026 *26.2_FAA*	Diff (%)	2024 SAFE	2026 *26.1_1Reg*	Diff (%)	2026 *26.2_FAA*	Diff (%)
1960	28.4	19	-33%	16	-44%	283.0	304	7%	252	-11%
1961	30.2	19	-37%	16	-47%	282.0	304	8%	252	-11%
1962	32.3	20	-38%	17	-47%	275.0	296	8%	245	-11%
1963	34.3	20	-42%	18	-48%	264.0	284	8%	232	-12%
1964	35.5	21	-41%	19	-46%	259.0	276	7%	225	-13%
1965	36.0	22	-39%	19	-47%	261.0	275	5%	224	-14%
1966	34.3	22	-36%	19	-45%	263.0	272	3%	223	-15%
1967	31.5	21	-33%	19	-40%	264.0	267	1%	218	-17%
1968	28.6	21	-27%	18	-37%	264.0	261	-1%	213	-19%
1969	26.0	20	-23%	17	-35%	259.0	250	-3%	203	-22%
1970	23.8	19	-20%	16	-33%	251.0	237	-6%	190	-24%
1971	21.8	18	-17%	15	-31%	241.0	223	-7%	177	-27%
1972	20.2	17	-16%	14	-31%	227.0	207	-9%	162	-29%
1973	19.1	16	-16%	13	-32%	207.0	186	-10%	142	-31%
1974	18.9	15	-21%	12	-37%	194.0	173	-11%	131	-32%
1975	8.8	7	-20%	6	-32%	180.0	161	-11%	120	-33%
1976	10.2	7	-31%	6	-41%	167.0	151	-10%	111	-34%
1977	10.8	7	-35%	6	-44%	153.0	140	-8%	101	-34%
1978	13.9	9	-35%	7	-50%	144.0	134	-7%	95	-34%
1979	83.0	20	-76%	16	-81%	141.0	132	-6%	94	-33%
1980	46.0	64	39%	60	30%	137.0	129	-6%	91	-34%
1981	23.0	14	-39%	11	-52%	136.0	126	-7%	90	-34%
1982	69.3	23	-67%	16	-77%	138.0	124	-10%	88	-36%
1983	39.7	63	59%	49	23%	147.0	127	-14%	92	-37%
1984	14.1	10	-29%	9	-36%	161.0	134	-17%	99	-39%
1985	15.5	10	-35%	9	-42%	179.0	145	-19%	109	-39%
1986	25.5	18	-29%	14	-45%	196.0	158	-19%	121	-38%
1987	9.8	7	-29%	6	-39%	205.0	165	-20%	126	-39%
1988	6.8	6	-12%	6	-12%	205.0	167	-19%	125	-39%
1989	7.7	10	30%	10	30%	197.0	162	-18%	118	-40%
1990	12.3	9	-27%	11	-11%	184.0	154	-16%	108	-41%
1991	23.7	22	-7%	15	-37%	170.0	144	-15%	98	-42%
1992	7.5	8	7%	12	60%	157.0	133	-15%	88	-44%
1993	25.5	19	-25%	12	-53%	144.0	122	-15%	79	-45%
1994	6.9	11	59%	4	-42%	130.0	111	-15%	71	-45%
1995	7.5	9	20%	8	7%	119.0	103	-13%	64	-46%
1996	12.6	12	-5%	12	-5%	112.0	97	-13%	60	-46%
1997	20.7	16	-23%	8	-61%	107.0	94	-12%	57	-47%
1998	9.5	8	-16%	18	89%	103.0	92	-11%	55	-47%
1999	37.2	26	-30%	10	-73%	99.0	91	-8%	53	-46%
2000	16.3	17	4%	8	-51%	96.0	90	-6%	52	-46%
2001	15.8	14	-11%	13	-18%	93.0	88	-5%	50	-46%
2002	43.7	30	-31%	27	-38%	93.0	88	-5%	50	-46%
2003	13.3	14	5%	12	-10%	94.0	90	-4%	50	-47%
2004	10.0	9	-10%	5	-50%	96.0	91	-5%	50	-48%
2005	12.1	10	-17%	7	-42%	98.0	94	-4%	51	-48%
2006	7.6	7	-8%	3	-61%	102.0	97	-5%	52	-49%
2007	10.6	9	-15%	10	-6%	104.0	100	-4%	54	-48%
2008	9.9	10	1%	7	-29%	105.0	101	-4%	54	-49%
2009	16.0	15	-6%	16	0%	103.0	100	-3%	53	-49%
2010	21.6	14	-35%	14	-35%	101.0	99	-2%	51	-50%
2011	10.2	9	-12%	7	-31%	97.0	97	0%	49	-49%
2012	11.6	9	-22%	6	-48%	93.0	94	1%	47	-49%
2013	4.5	5	11%	6	33%	89.0	92	3%	47	-47%

2014	7.6	7	-8%	6	-21%	86.0	89	3%	46	-47%
2015	14.3	12	-16%	9	-37%	85.0	88	4%	47	-45%
2016	49.9	46	-8%	62	24%	84.0	87	4%	48	-43%
2017	21.5	18	-16%	29	35%	83.0	86	4%	49	-41%
2018	95.9	103	7%	68	-29%	83.0	87	5%	51	-39%
2019	94.3	76	-19%	112	19%	86.0	90	5%	57	-34%
2020	42.7	32	-25%	11	-74%	95.0	98	3%	69	-27%
2021	82.4	33	-60%	11	-87%	110.0	113	3%	88	-20%
2022	25.1	42	67%	16	-36%	133.0	136	2%	115	-14%
2023	27.9	33	18%	10	-64%	161.0	163	1%	144	-11%
2024	--	21	--	11	--	191.0	192	1%	172	-10%
2025	--	--	--	--	--	--	219	--	194	--

Table 3.12. Sablefish spawning biomass (kilotons), fishing mortality, and yield (kilotons) for the seven projection harvest scenarios (columns) outlined in the 'Population Projections' section for Model *26.1_1Reg*. The 'Specified Catch' scenario uses the proportion of the ABC utilized in 2024 (~49%) to set the realized yield for 2026 and 2027. SSB and yield are in kilotons.

Year	Maximum Permissible F	Author's F (Specified Catches)	Half maximum F	5-year Average F	No Fishing	Overfished	Approaching Overfished
Spawning biomass (kt)							
2025	216	216	216	216	216	216	216
2026	237	237	237	237	237	237	237
2027	244	253	244	248	262	241	244
2028	245	264	246	254	283	239	245
2029	242	260	243	256	300	233	239
2030	236	253	237	254	313	224	230
2031	228	243	229	249	323	213	219
2032	219	232	219	242	330	202	207
2033	209	221	210	235	336	192	196
2034	201	210	201	228	339	182	185
2035	192	201	193	221	342	173	176
2036	185	192	186	215	344	166	168
2037	178	184	179	209	346	159	162
2038	173	178	175	205	347	154	156
Fishing mortality							
2025	0.05	0.05	0.05	0.05	0.05	0.05	0.05
2026	0.08	0.04	0.08	0.06	0.00	0.09	0.08
2027	0.08	0.04	0.08	0.06	0.00	0.09	0.08
2028	0.08	0.08	0.08	0.06	0.00	0.09	0.09
2029	0.08	0.08	0.08	0.06	0.00	0.09	0.09
2030	0.08	0.08	0.08	0.06	0.00	0.09	0.09
2031	0.08	0.08	0.08	0.06	0.00	0.09	0.09
2032	0.08	0.08	0.08	0.06	0.00	0.09	0.09
2033	0.08	0.08	0.08	0.06	0.00	0.09	0.09
2034	0.08	0.08	0.08	0.06	0.00	0.09	0.09
2035	0.08	0.08	0.08	0.06	0.00	0.09	0.09
2036	0.08	0.08	0.08	0.06	0.00	0.09	0.09
2037	0.08	0.08	0.08	0.06	0.00	0.09	0.09
2038	0.08	0.08	0.08	0.06	0.00	0.09	0.09
Yield (kt)							
2025	24	24	24	24	24	24	24
2026	41	20	40	31	0	48	41
2027	40	21	39	30	0	46	40
2028	38	41	38	30	0	44	45
2029	37	39	36	29	0	42	43
2030	36	38	35	28	0	40	41
2031	34	36	34	28	0	38	39
2032	33	34	32	27	0	36	37
2033	32	33	31	26	0	35	35
2034	31	32	30	25	0	33	34
2035	30	31	29	25	0	32	33
2036	29	30	28	24	0	31	32
2037	28	29	28	24	0	30	31
2038	27	28	27	24	0	30	30

Table 3.13. Sablefish spawning biomass (kilotons), fishing mortality, and yield (kilotons) for the seven projection harvest scenarios (columns) outlined in the 'Population Projections' section for Model *26.2_FAA*. The 'Specified Catch' scenario uses the proportion of the ABC utilized in 2024 (~49%) to set the realized yield for 2026 and 2027. SSB and yield are in kilotons.

Year	Maximum Permissible F	Author's F (Specified Catches)	Half maximum F	5-year Average F	No Fishing	Overfished	Approaching Overfished
Spawning biomass (kt)							
2025	193	193	193	193	193	193	193
2026	205	205	205	205	205	205	205
2027	203	211	200	205	219	200	203
2028	196	212	191	200	229	191	196
2029	188	202	180	193	236	179	185
2030	179	192	170	185	241	169	173
2031	171	182	160	178	246	159	163
2032	164	174	152	172	251	151	155
2033	158	166	145	166	256	144	147
2034	153	160	139	162	261	139	141
2035	149	154	134	158	265	134	136
2036	145	149	130	154	269	130	132
2037	141	145	127	151	272	126	128
2038	138	142	125	150	274	124	124
Fishing mortality							
2025	0.06	0.06	0.06	0.06	0.06	0.06	0.06
2026	0.08	0.04	0.09	0.07	0.00	0.10	0.08
2027	0.08	0.04	0.09	0.07	0.00	0.10	0.08
2028	0.08	0.08	0.09	0.07	0.00	0.10	0.10
2029	0.08	0.08	0.09	0.07	0.00	0.10	0.10
2030	0.08	0.08	0.09	0.07	0.00	0.10	0.10
2031	0.08	0.08	0.09	0.07	0.00	0.10	0.10
2032	0.08	0.08	0.09	0.07	0.00	0.10	0.10
2033	0.08	0.08	0.09	0.07	0.00	0.10	0.10
2034	0.08	0.08	0.09	0.07	0.00	0.10	0.10
2035	0.08	0.08	0.09	0.07	0.00	0.10	0.10
2036	0.08	0.08	0.09	0.07	0.00	0.10	0.10
2037	0.08	0.08	0.09	0.07	0.00	0.10	0.10
2038	0.08	0.08	0.09	0.07	0.00	0.10	0.10
Yield (kt)							
2025	24	24	24	24	24	24	24
2026	32	16	37	28	0	38	32
2027	30	16	35	27	0	35	30
2028	29	31	33	26	0	33	34
2029	28	30	31	25	0	32	33
2030	27	28	30	24	0	30	31
2031	26	27	28	24	0	29	30
2032	25	26	27	23	0	28	28
2033	25	25	26	23	0	27	27
2034	24	25	26	22	0	26	27
2035	23	24	25	22	0	26	26
2036	23	23	25	21	0	25	25
2037	22	23	24	21	0	24	25
2038	22	22	24	21	0	24	24

Figures

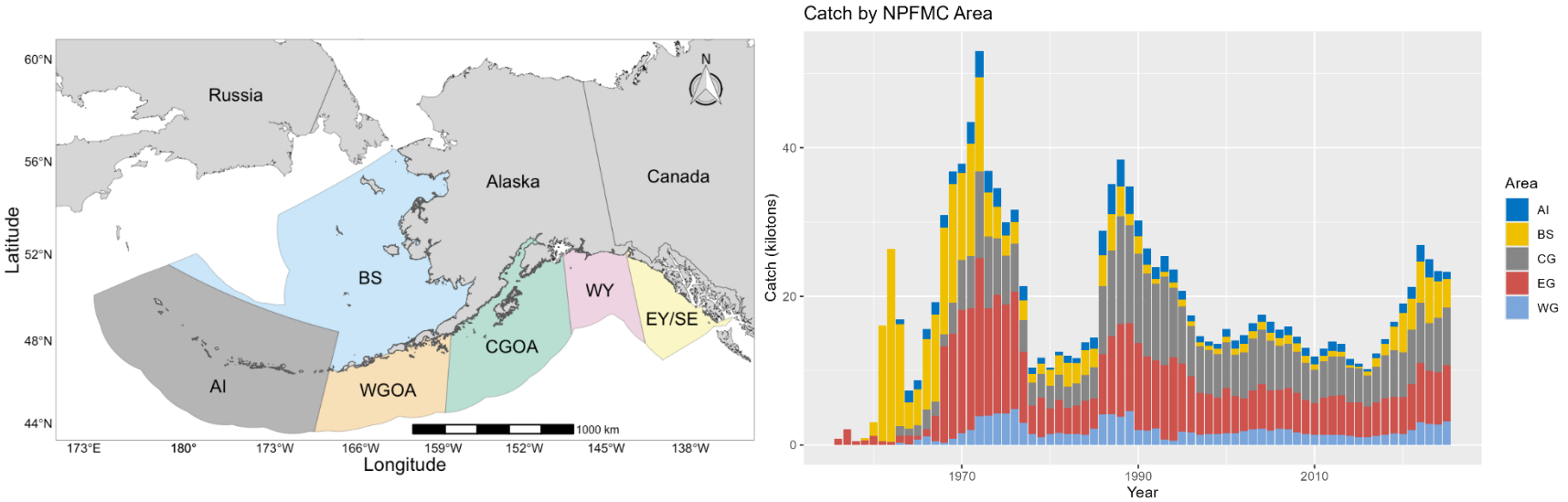


Figure 3.1. Sablefish North Pacific Fishery Management Council management areas and total catch (kt) summed across all fleets by management area.

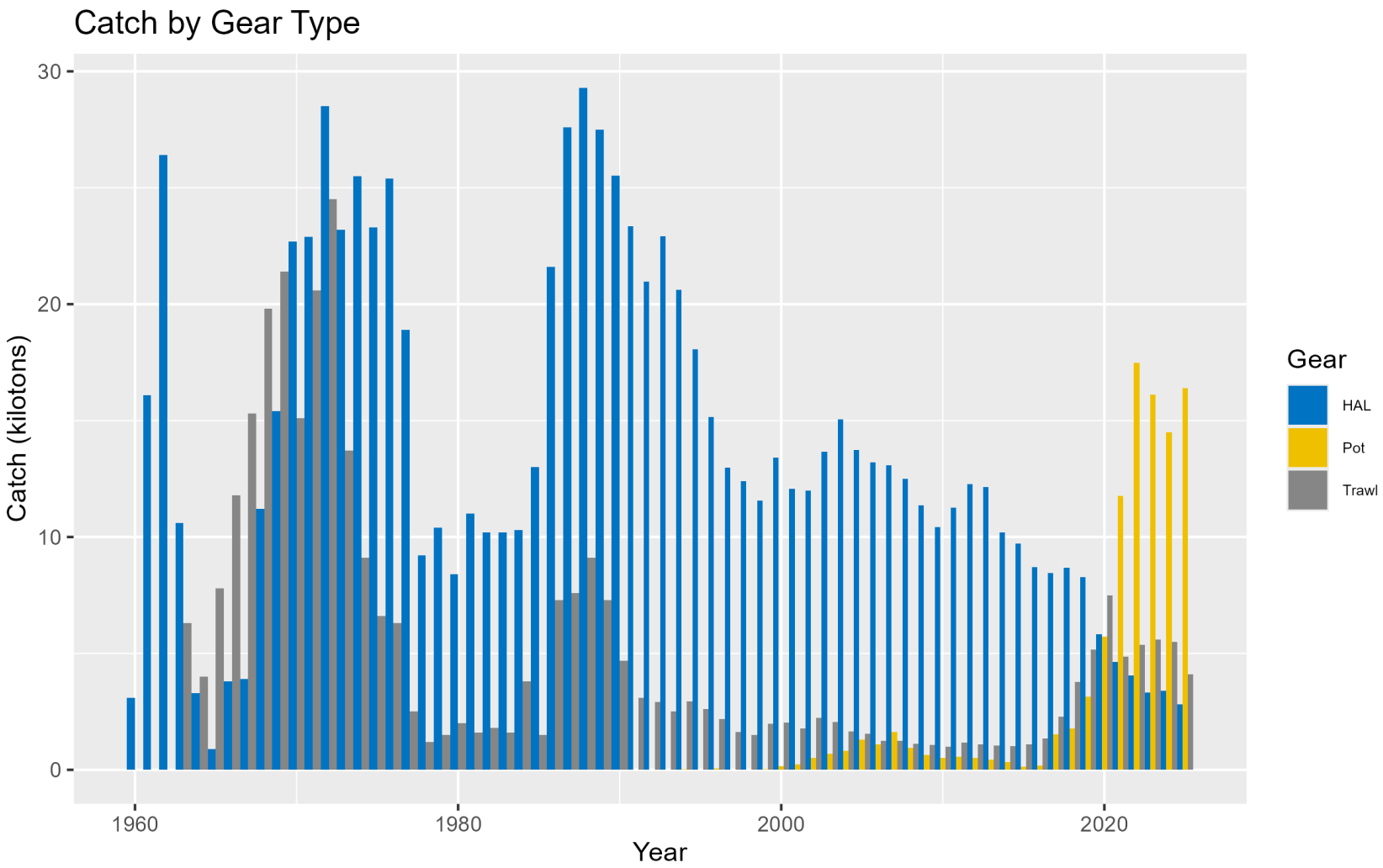


Figure 3.2. Sablefish catch (kt) by gear type. Note that hook and line (HAL) and pot gear catch are combined into a single 'fixed gear' fleet in the model.

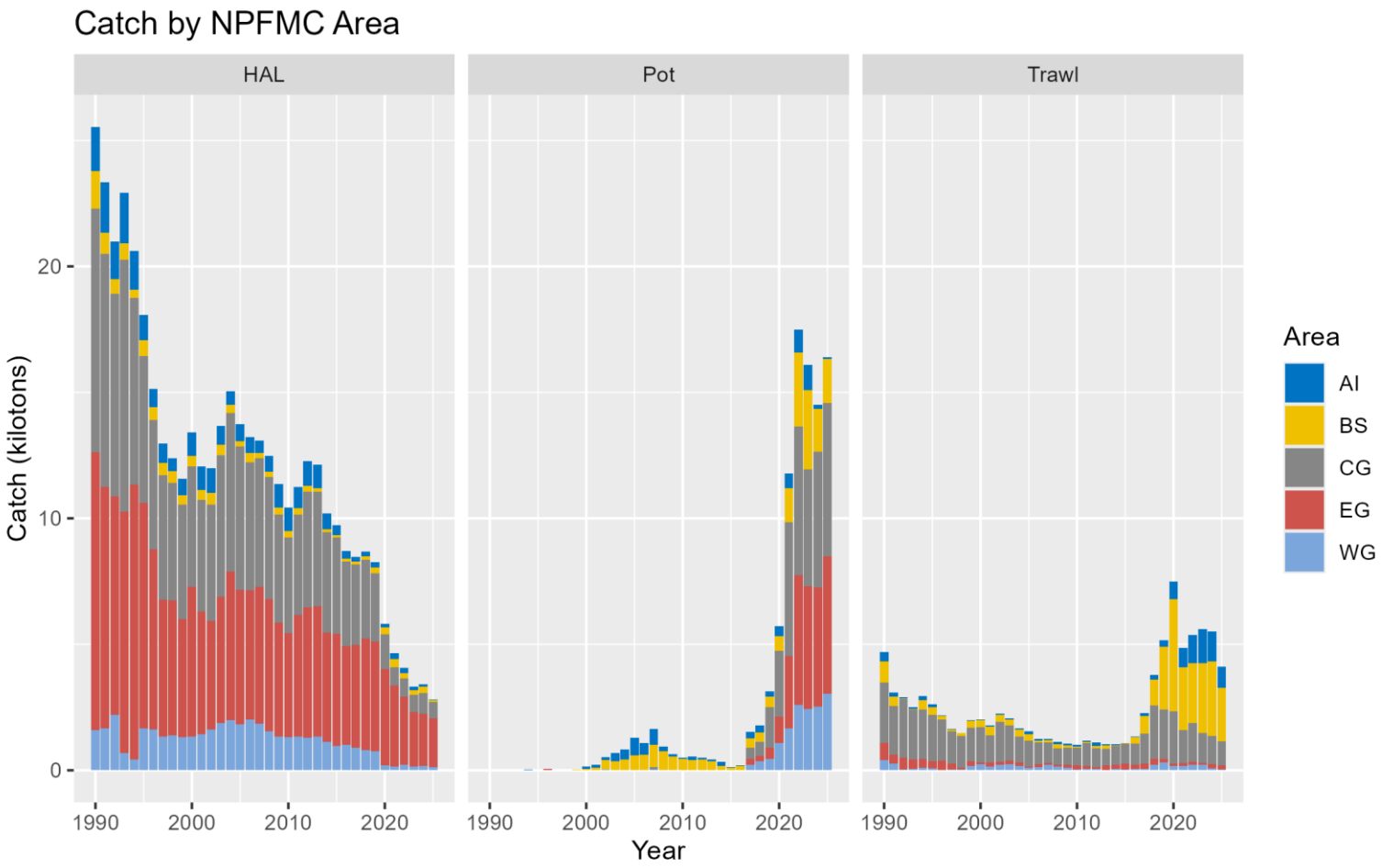


Figure 3.3. Sablefish catch (kt) by fleet (facet) and North Pacific Fishery Management Council area (colored bars).

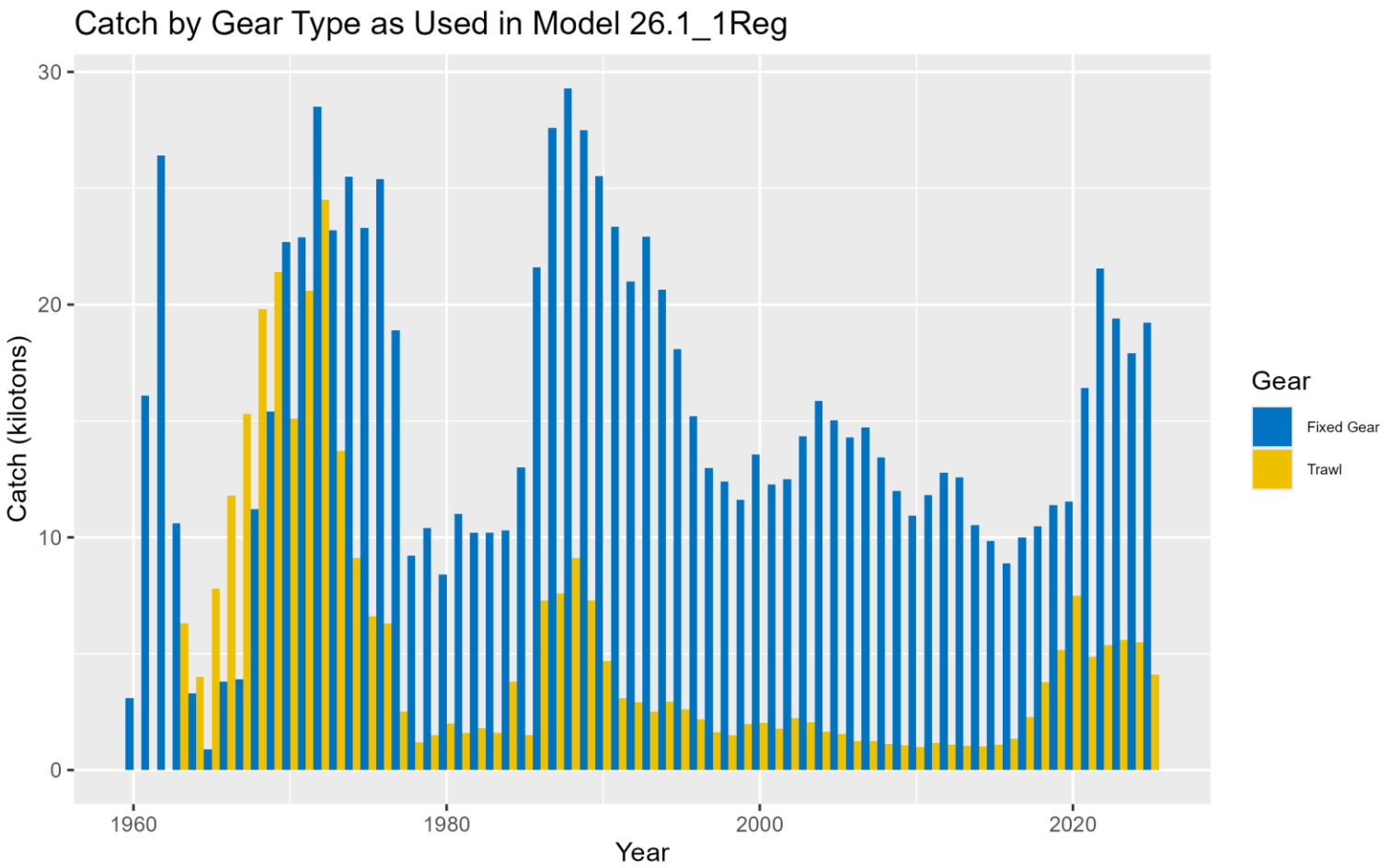


Figure 3.4. Sablefish total catch (kt) by fleet as used in the assessment models. Note the fixed gear fleet includes hook-and-line and pot gear.

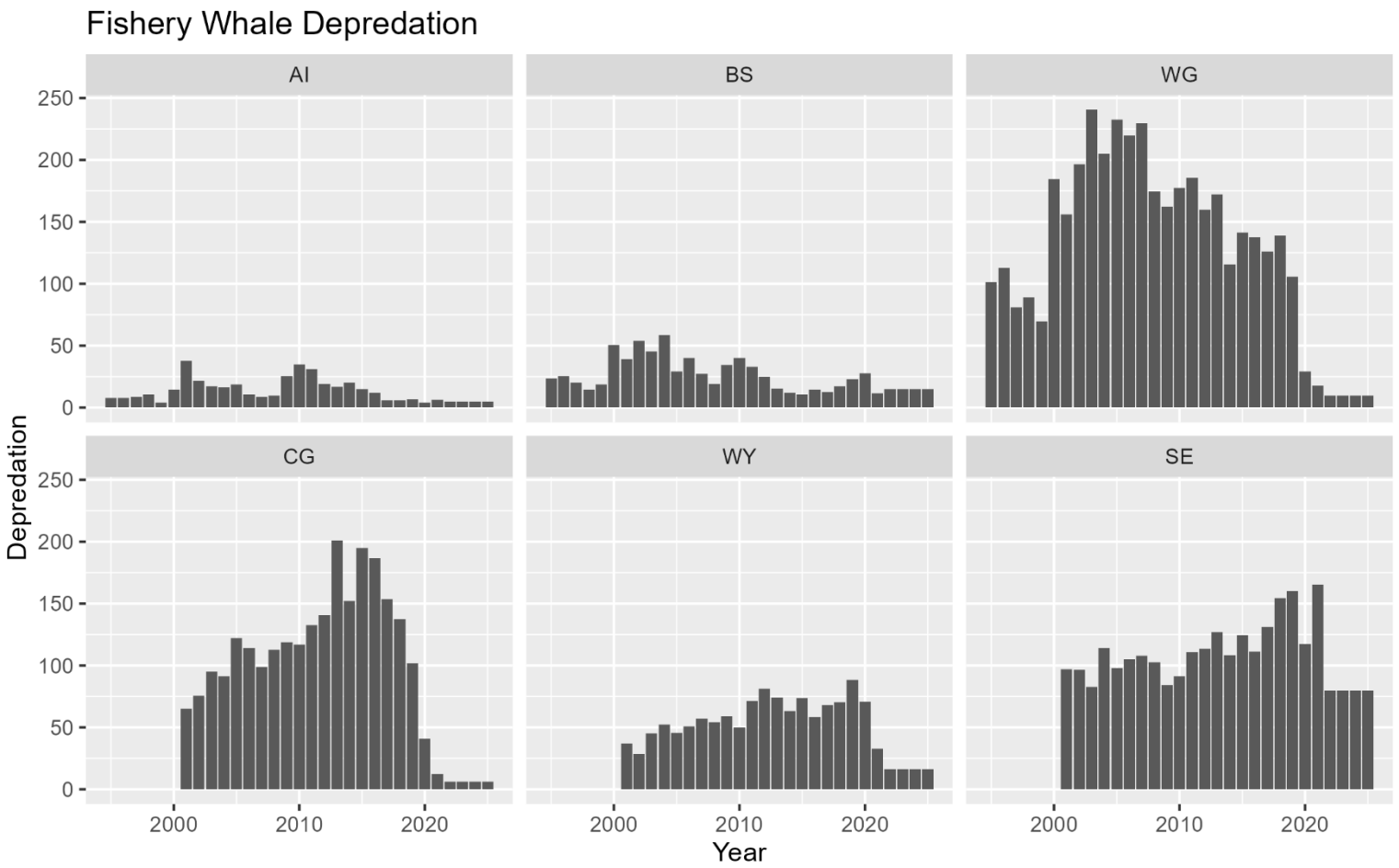


Figure 3.5. Estimated whale depredation in the sablefish fixed gear fishery. Depredation estimates reflect catch removals (metric tons, mt) by region (panel) due to orcas (top row) and sperm whales (bottom row). Starting in 2023 estimates were held constant at 2022 values, given the strong reduction in depredation due to the largescale switch to pot gear.

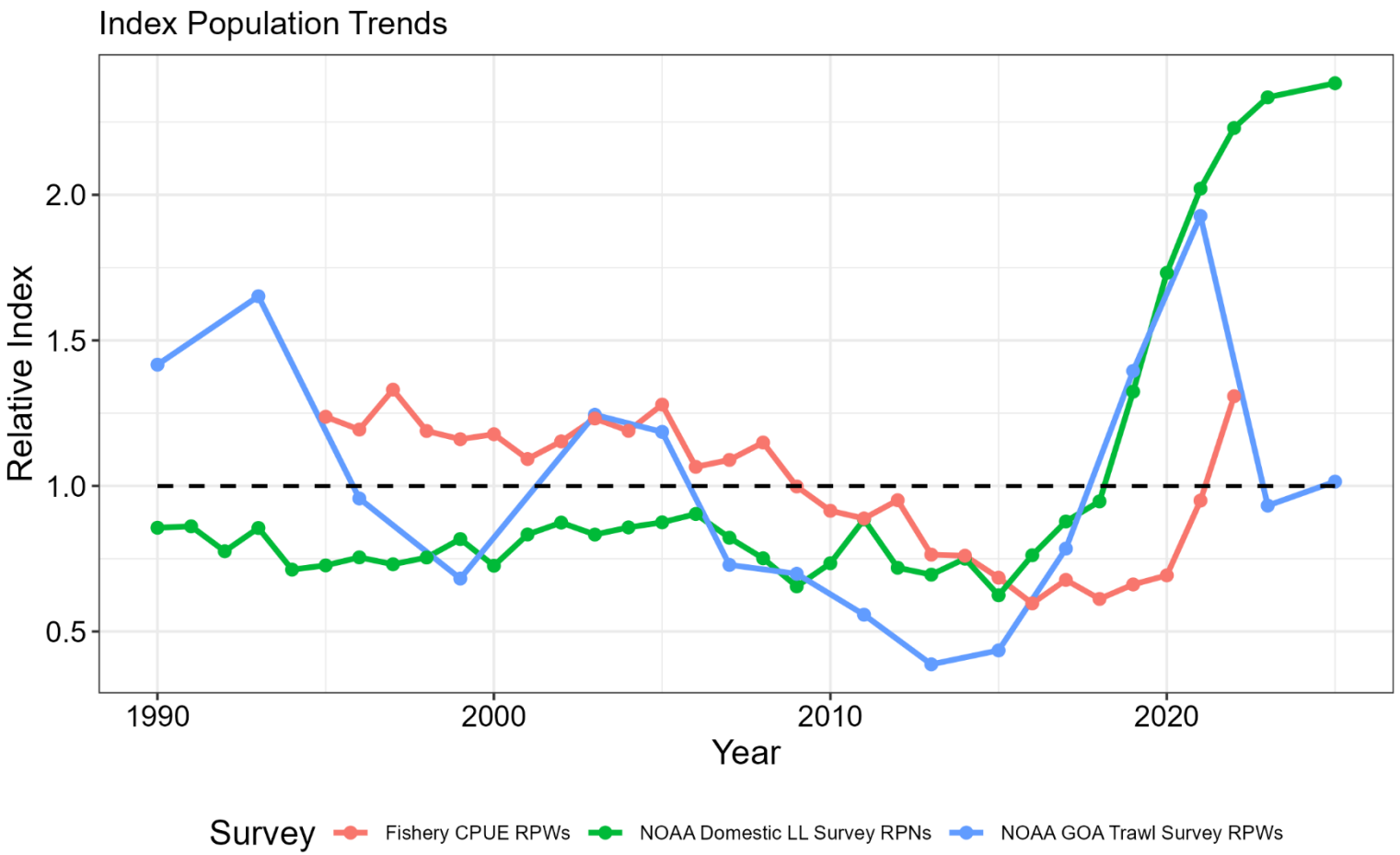


Figure 3.6. Comparison of the three abundance and biomass indices for sablefish, including the NOAA Longline Survey relative population numbers (RPNs), the fixed gear fishery standardized CPUE (in weight), and the NOAA Gulf of Alaska (stations < 500m depth) Trawl Survey relative population weights (RPWs). Each index is relativized to the associated mean value for the time series. Note that only the NOAA Longline Survey index is currently used in the assessment and represents the single region 'carryover' index. There was no NOAA Longline Survey in 2024.

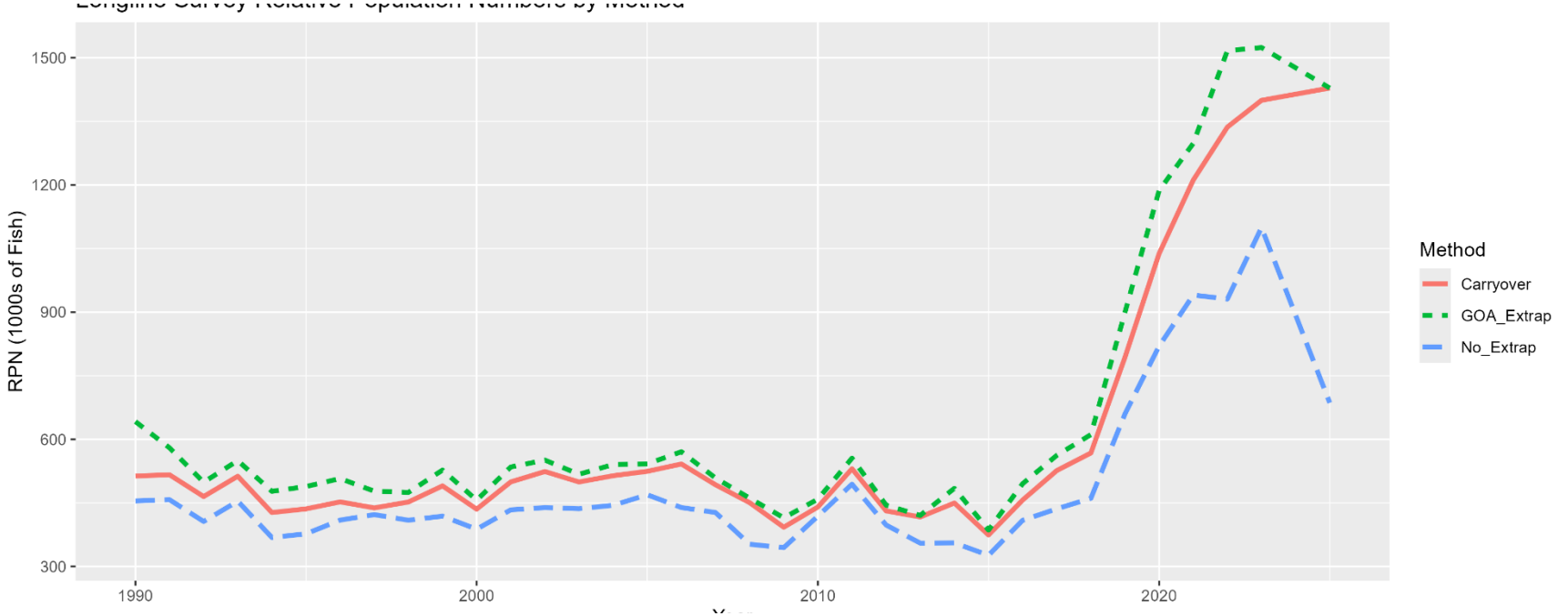


Figure 3.7. Comparison of the three versions of the NOAA Longline Survey abundance index for sablefish (relative population numbers, RPNs) demonstrating different extrapolation methods and station inclusion. The GOA_Extrap index (used in Model *25.12_1Reg_Cont*) represents the traditional approach of extrapolating the BS/AI in unsurveyed years based on the year-over-year growth in the GOA, then switching to the 'carryover' approach in 2025, and including the extrapolated western Aleutian Island (AI) stations in the index. The Carryover index (used in Model *26.1_1Reg*) represents the new method for all years, where holdover RPNs for unsurveyed regions are used based on the RPN from that region the last time it was surveyed. The No_Extrap index represents the sum of the area-specific RPNs without any extrapolation (i.e., including only RPNs from stations actually surveyed in a given year) and excluding western AI extrapolation stations – this index represents the sum of the regional indices as used in Model *26.2_FAA*.

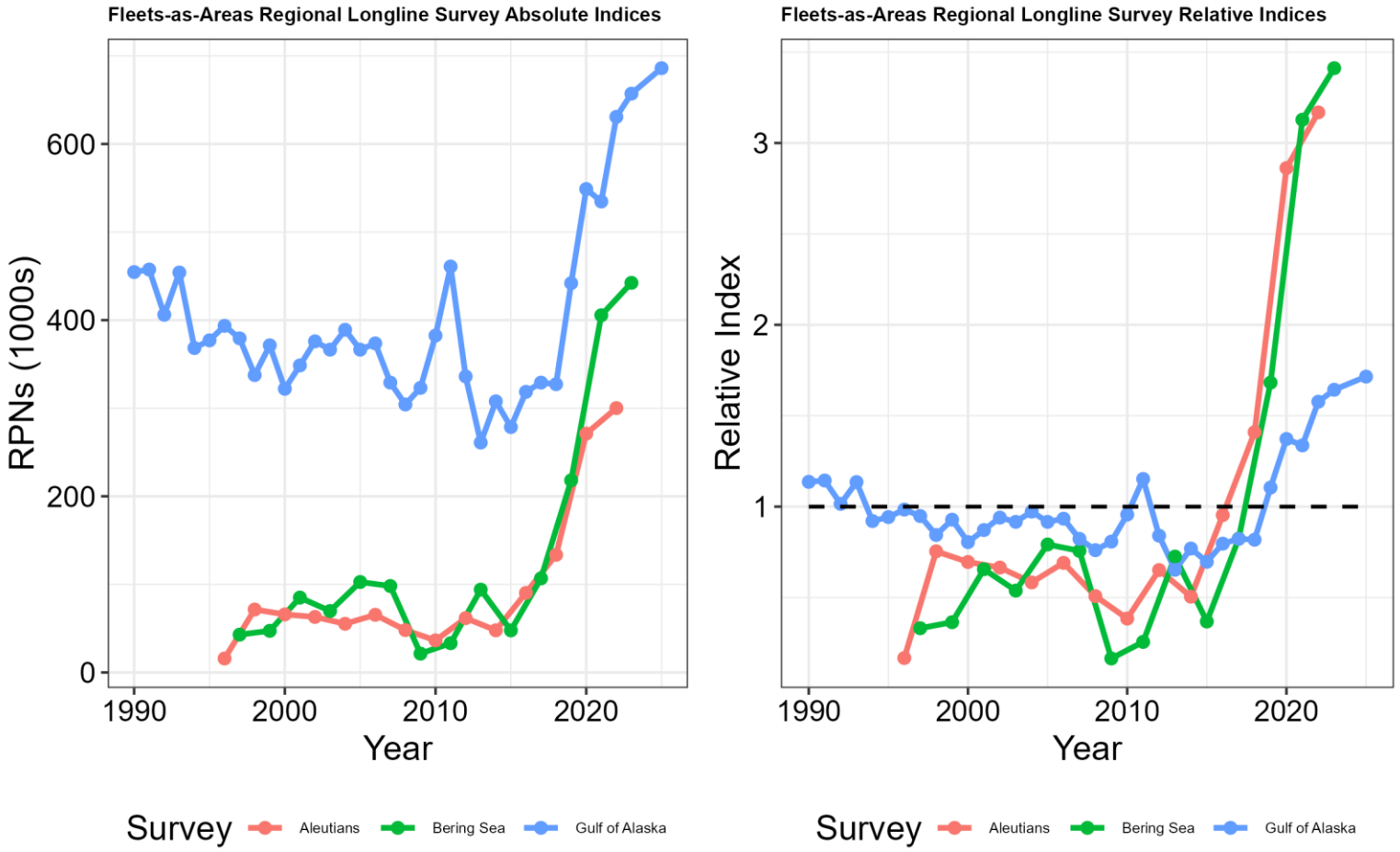


Figure 3.8. The regional NOAA Longline Survey indices as used in Model *26.2_FAA*, which do not include any extrapolation or western AI extrapolated stations. The left hand panel shows the absolute relative population numbers (RPNs, 1000s) for each region, whereas the right panel shows each index relativized to the associated mean value for the time series. Note that the Bering Sea was historically surveyed in odd years and the Aleutian Islands are surveyed in even years (i.e., sampling occurs every other year in these regions), there was no survey in 2024, and only the GOA was surveyed in 2025.

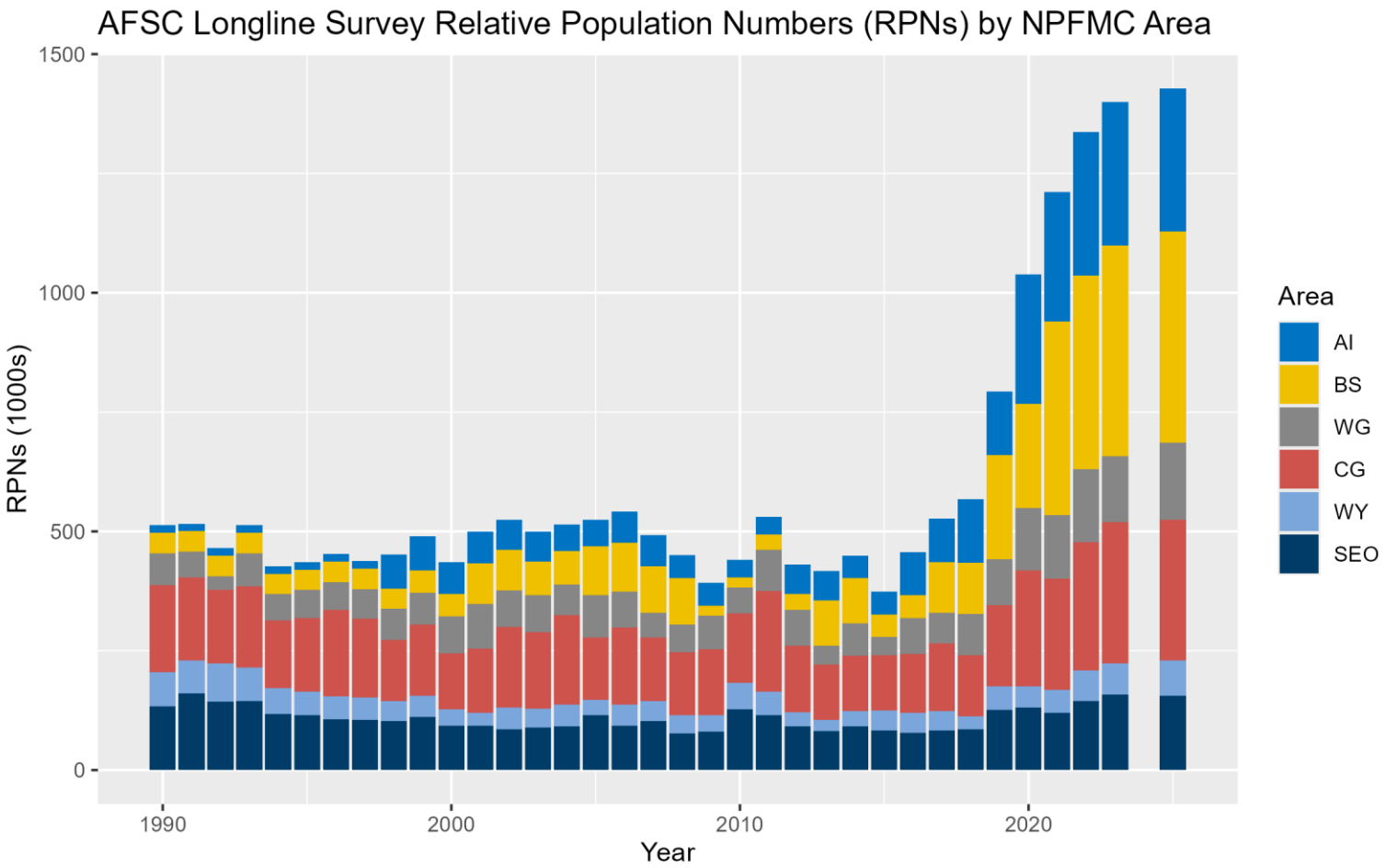


Figure 3.9. Relative abundance (relative population numbers, RPNs, in thousands) by region from the NOAA domestic longline survey utilizing the carryover method for non-surveyed regions (i.e., matching the combined index used in Model *26.1_1Reg*). Note that the Bering Sea is surveyed in odd years and the Aleutian Islands are surveyed in even years (i.e., sampling occurs every other year in these regions), there was no survey in 2024, and only the GOA was surveyed in 2025. The regional trends for non-surveyed regions are extrapolated by using holdover values from the last time the given region was surveyed.

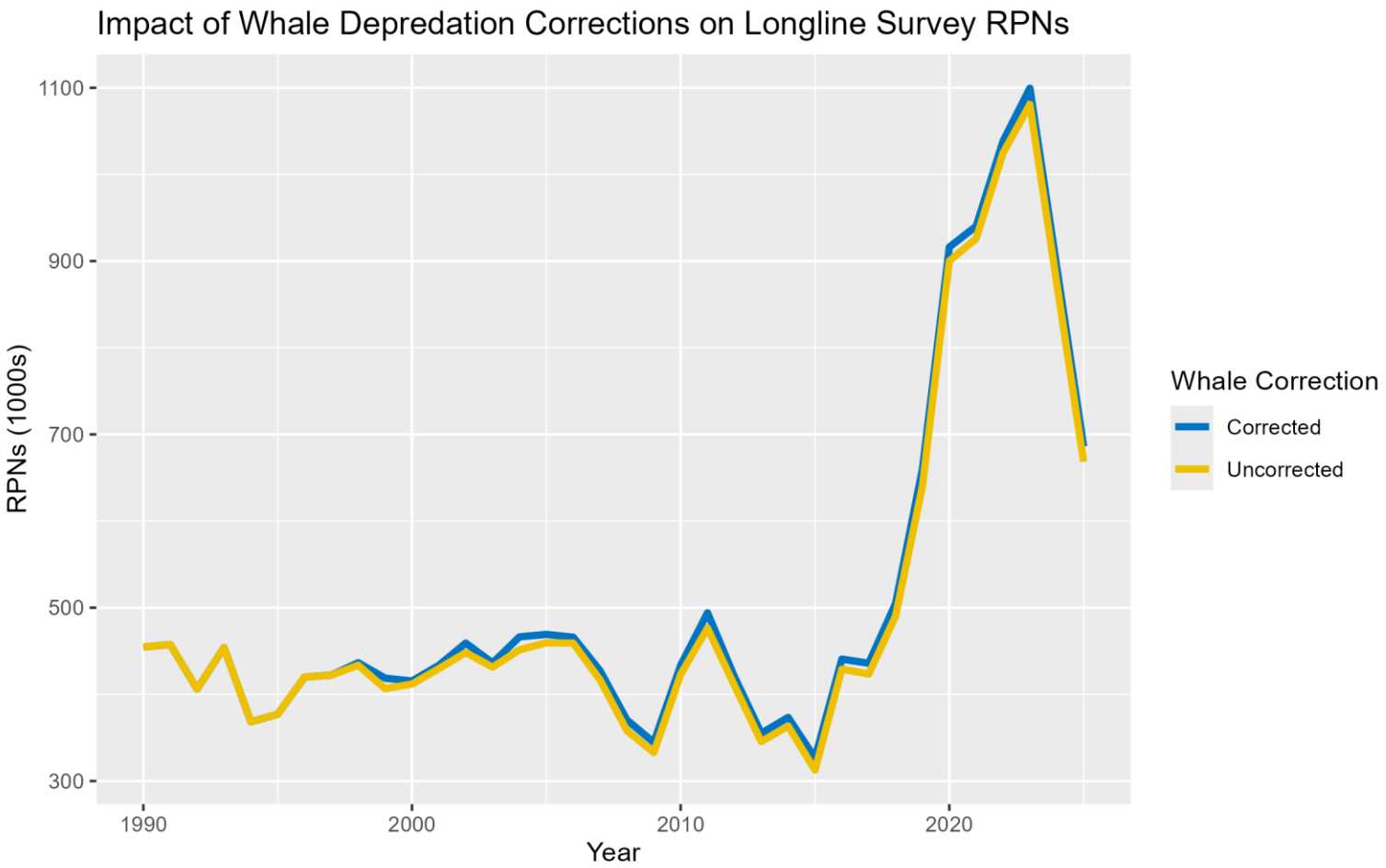


Figure 3.10. NOAA Longline Survey relative population numbers (RPNs, 1000s of fish) with (blue line) and without (yellow line) corrections for sperm whale depredation. Note that these are the summed regional RPN values that do not include extrapolations for areas not surveyed in the given year, thus, they are less than the final index by the 'carryover' RPN values.

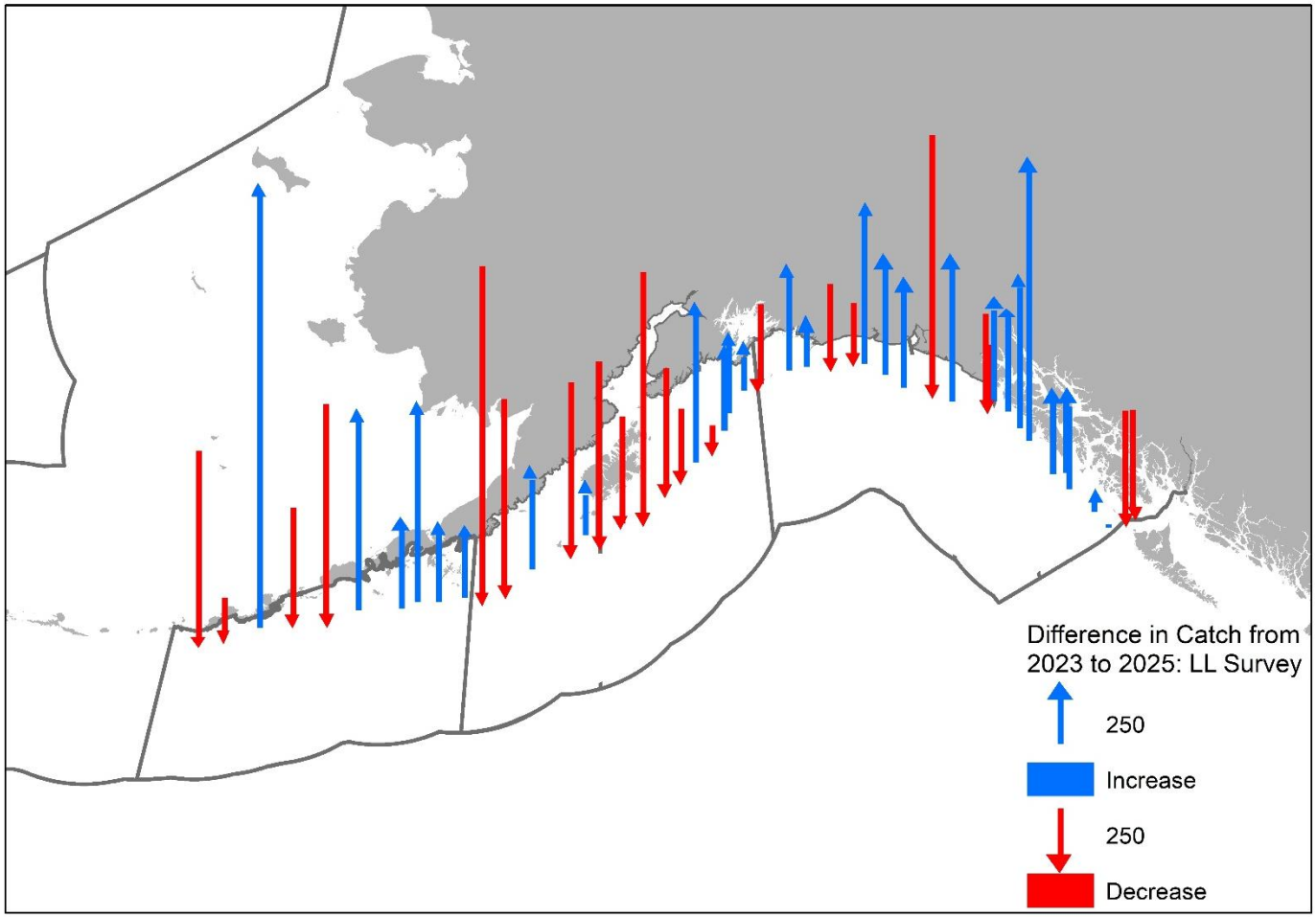


Figure 3.11. Comparison of the 2023 and 2025 NOAA Longline Survey catches in the Gulf of Alaska in terms of the difference in numbers of fish caught per station from 2023 to 2025. Numbers are not corrected for sperm whale depredation.

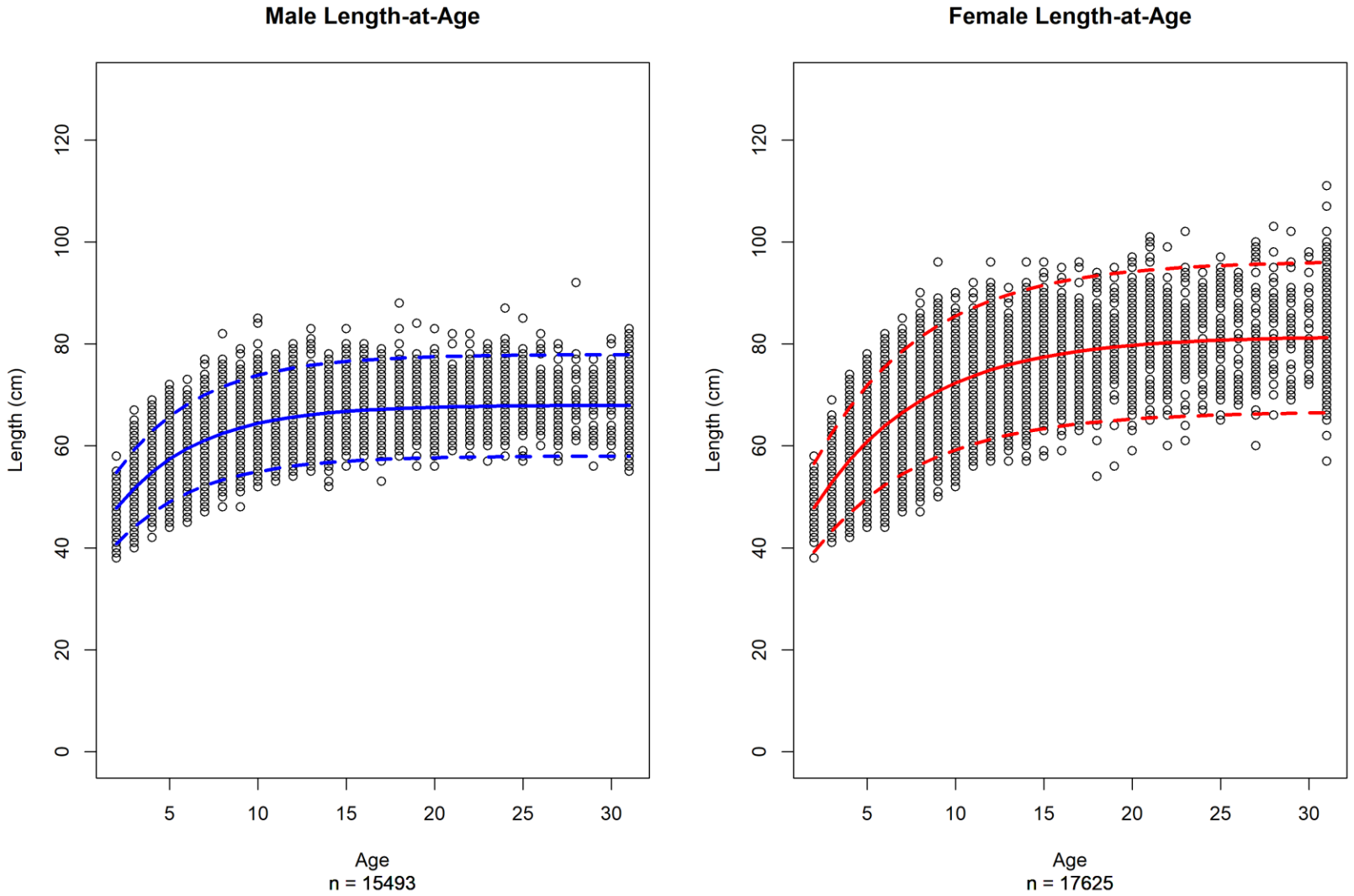


Figure 3.12. Observed (open circles) age-length pairs from the NOAA Longline Survey from 1996 – 2023 and predicted (colored lines) von Bertalanffy growth curves with associated 95% confidence intervals (dashed lines) for males (left panel) and females (right panel). Sex-specific sample sizes (*n*) are listed under each plot.

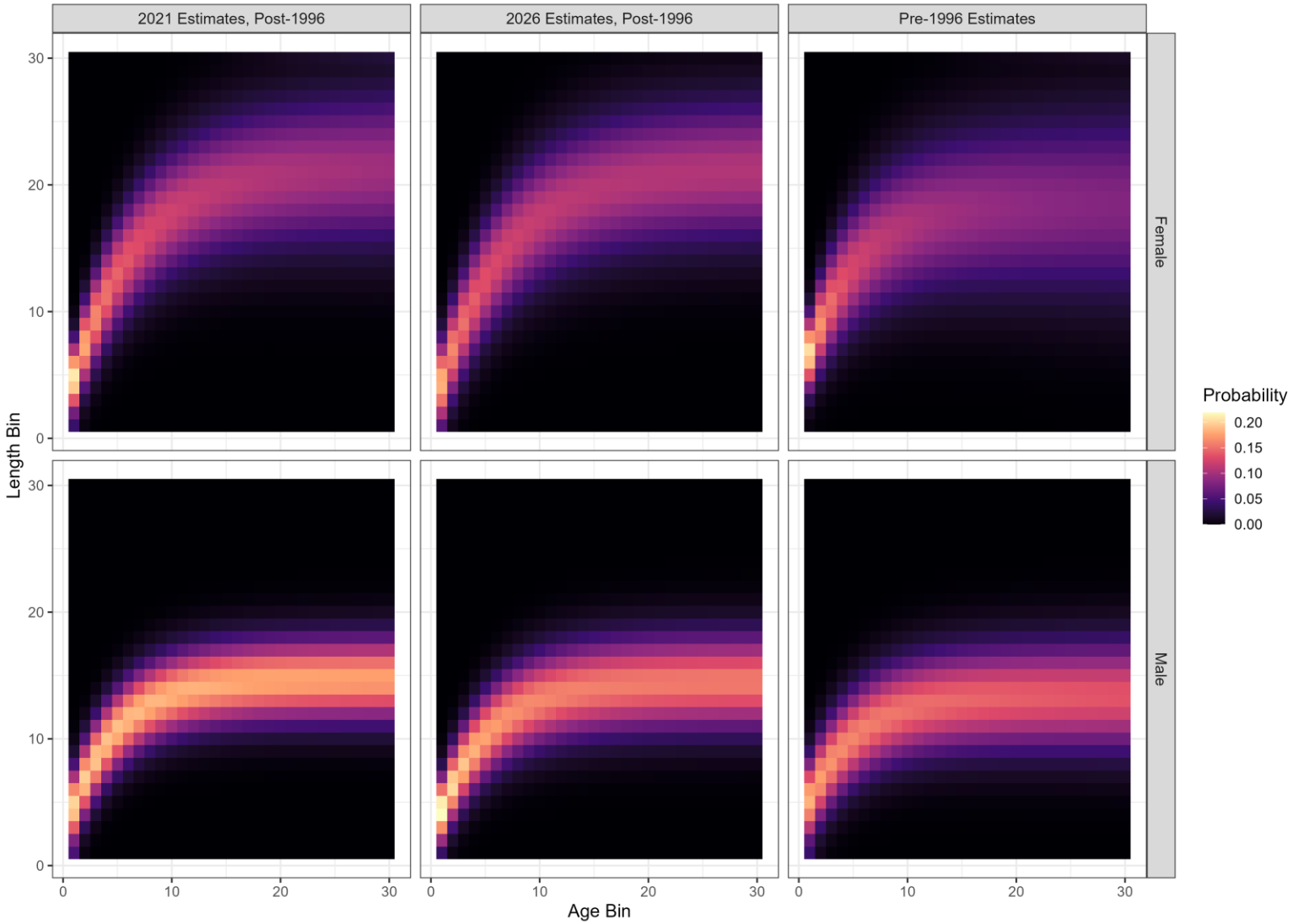


Figure 3.13. Comparison of size-age transition matrices from the previous SAFE (as estimated in 2021, 2021 Estimates) and as estimated in 2026 (2026 Estimates) with all NOAA Longline Survey data from 1996 – 2023, as well as the pre-1996 size-ag transitions estimated previously using the bias-corrected NOAA Longline Survey data prior to 1996 (and used in all models; see Echave, 2021).

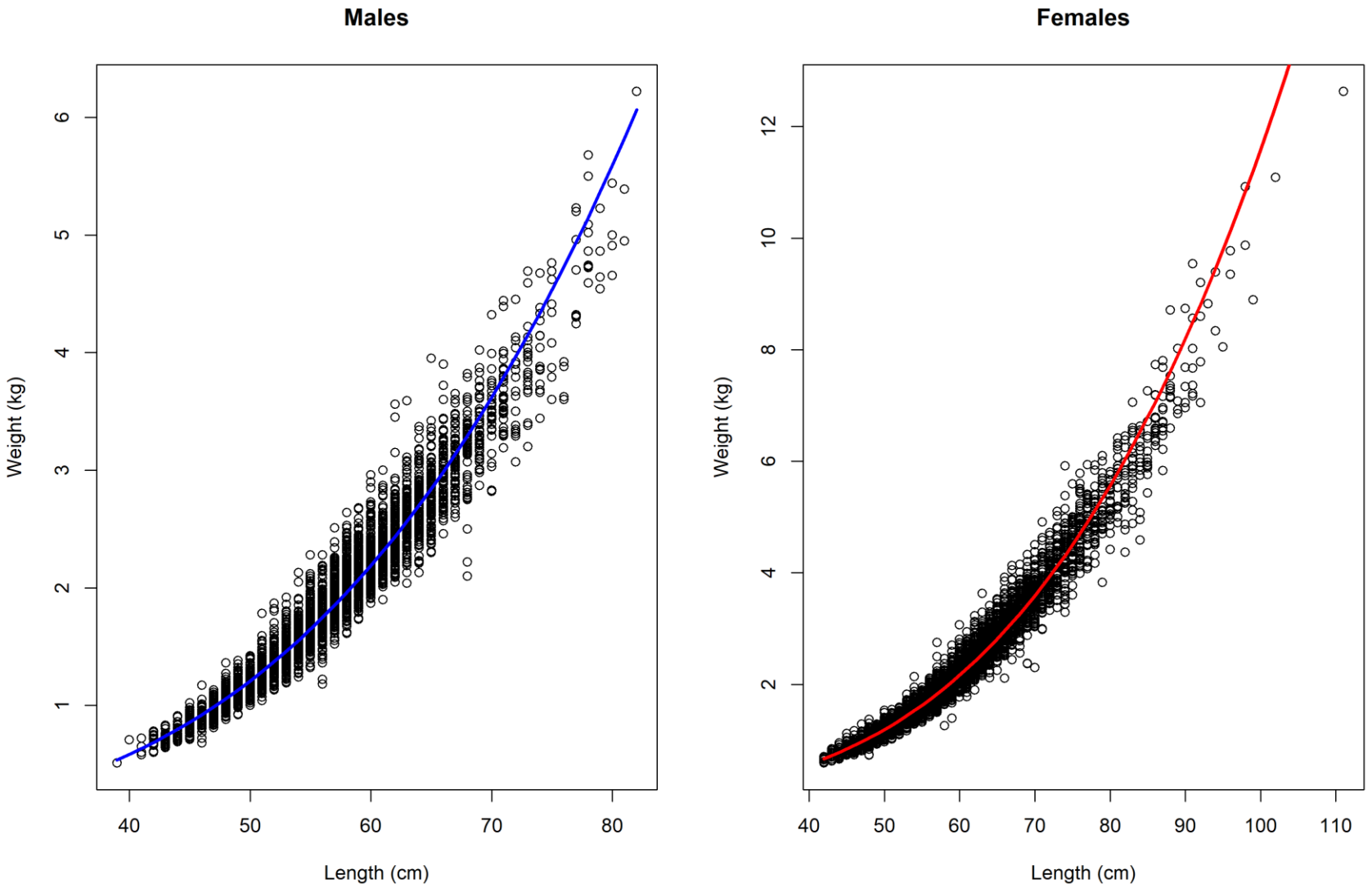


Figure 3.14. Observed (open circles) weight-length pairs from the NOAA Longline Survey from 1996 – 2023 and predicted (colored lines) weight-length curves for males (left panel) and females (right panel).

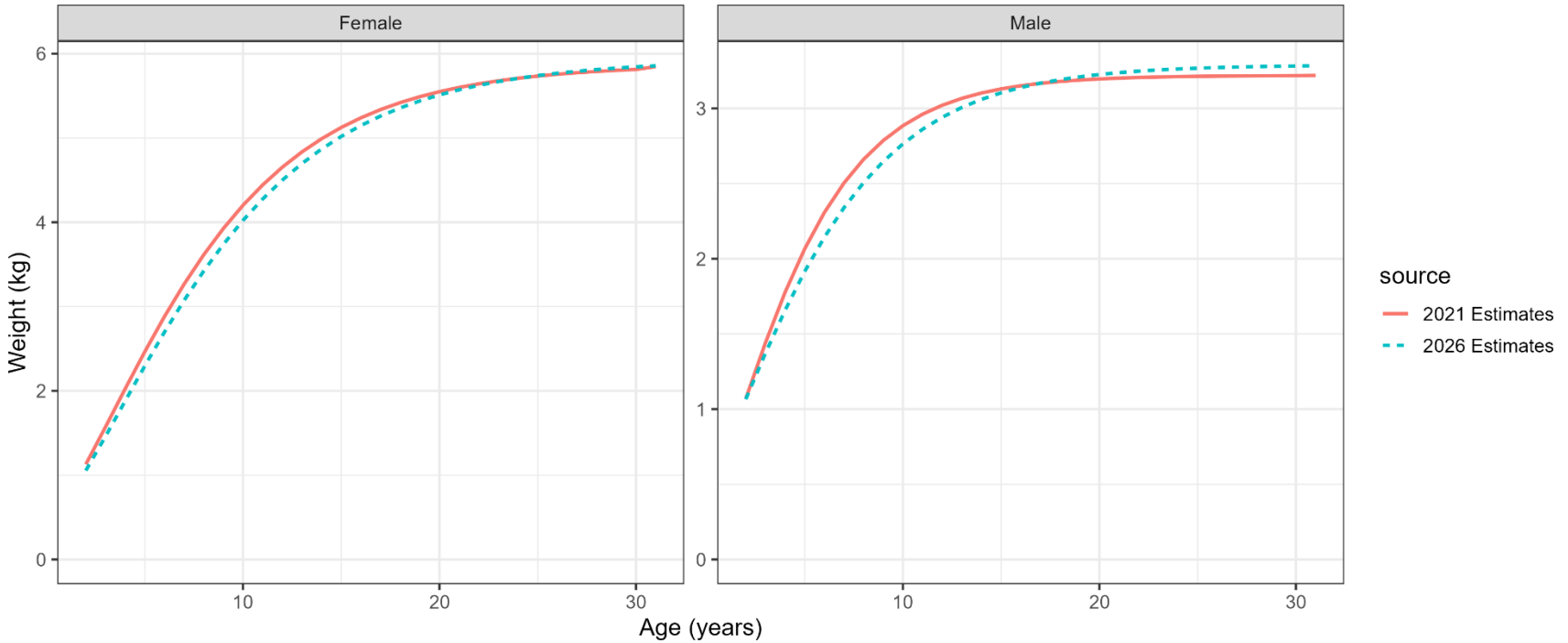


Figure 3.15. Comparison of weight-at-age estimates from the previous SAFE (as estimated in 2021, 2021 Estimates) and as estimated in 2026 (2026 Estimates) with all NOAA Longline Survey data from 1996 – 2023. The estimated weight-at-age is based on the estimate length-weight parameters and mean length from the von Bertalanffy growth curve estimates.

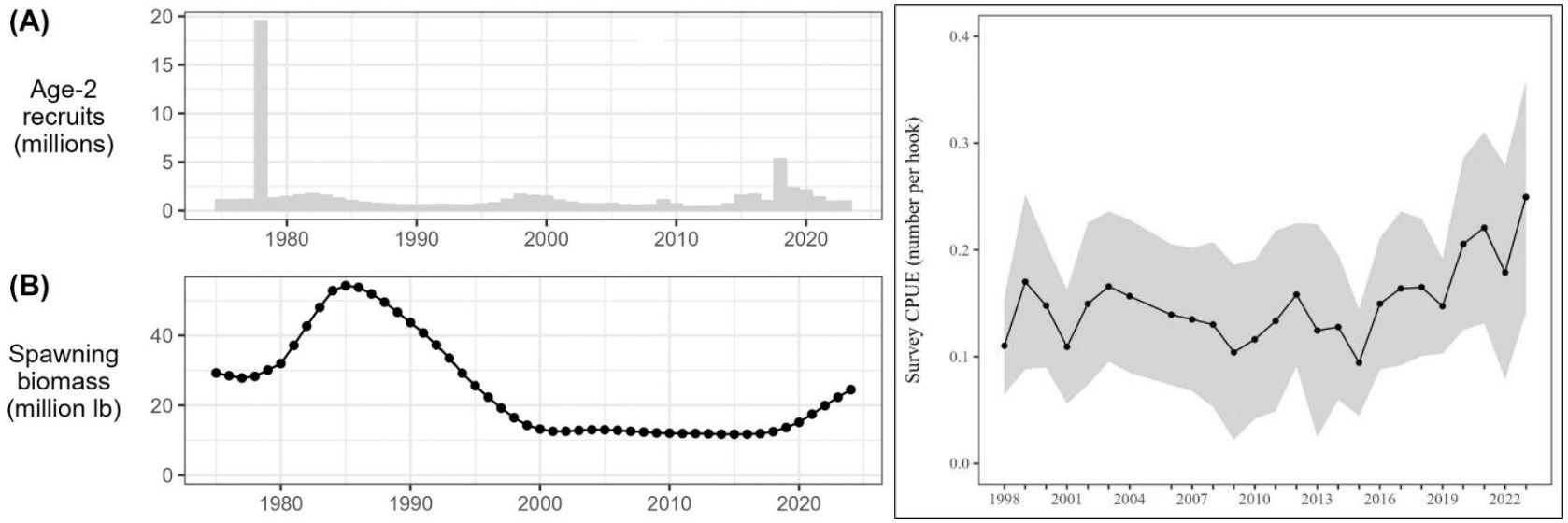


Figure 3.16. Model predictions for the ADFG Northern Southeast Inside (NSEI) sablefish stock assessment (left panels; reproduced here with permission from Rhea Ehresmann, pers. comm., https://github.com/commfish/seak_sablefish) of age-2 recruitment (millions; top) and female spawning stock biomass (million pounds; bottom). Southern Southeast Inside (SSEI) sablefish longline survey catch-per-unit-effort (CPUE) in individuals per hook (right panel) from 1998 to 2023 (except 2005; reproduced here with permission from Rhea Ehresmann, pers. Comm.).

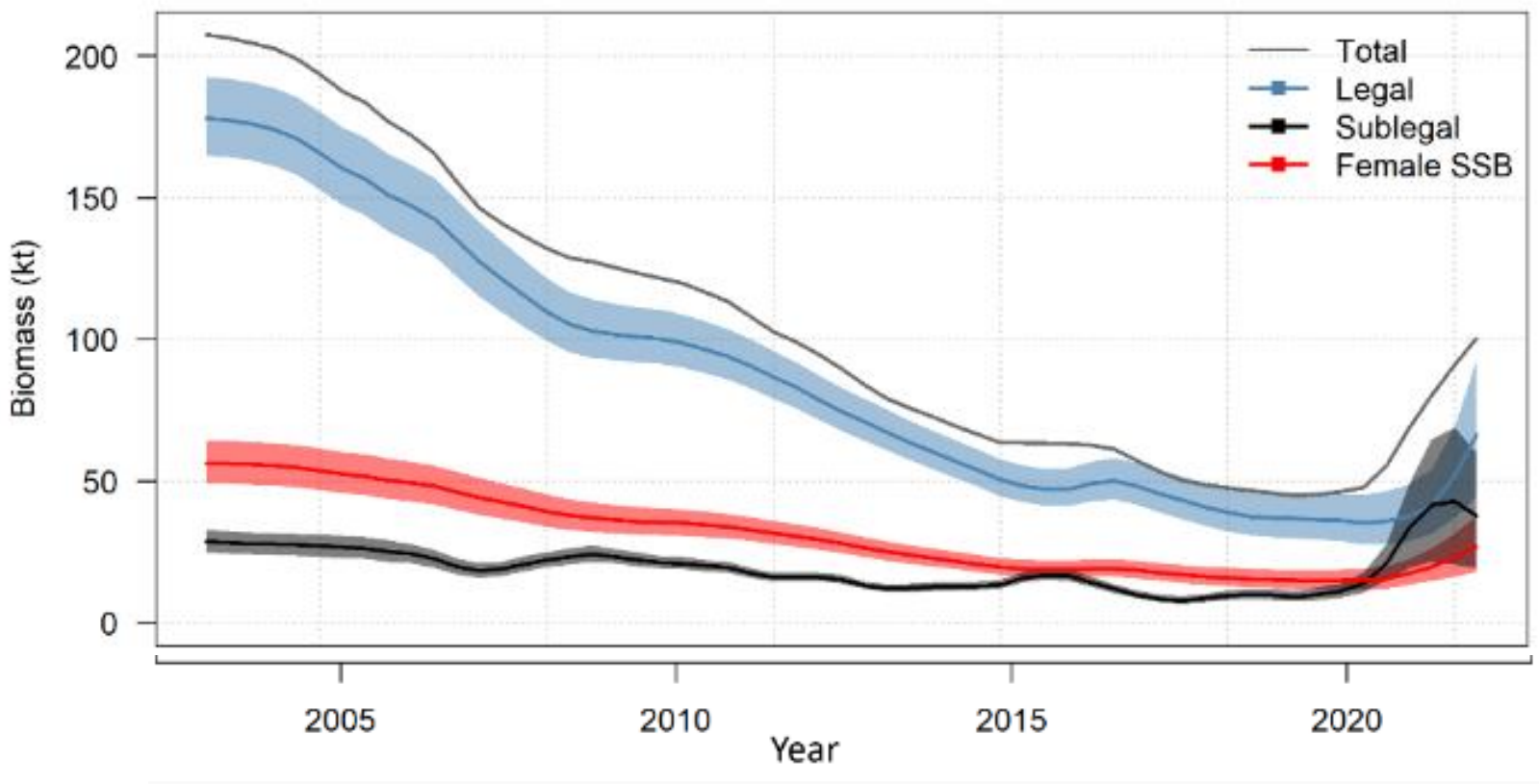


Figure 3.17. Time series of total biomass, legal-sized biomass, sub-legal-sized biomass, and female spawning biomass (SSB) estimates for the British Columbia stock of sablefish based on weighted averages over the five OM scenarios used in 2022. Note that total, legal, and sub-legal biomass estimates include both female and male fish, while SSB is shown for only female Sablefish (reproduced here with permission from Kendra Holt, DFO Canada, pers. comm.).

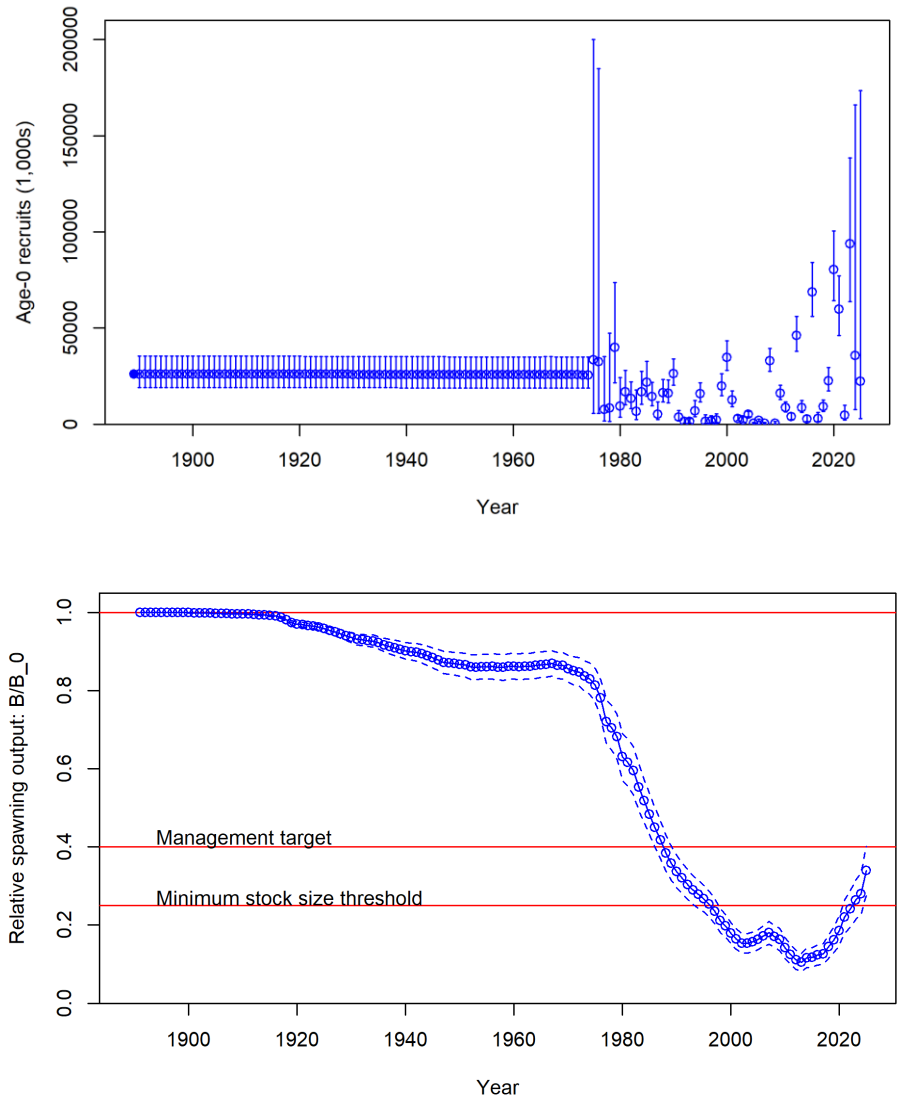


Figure 3.18. Time series (dots) of recruitment (1000s of age-0 fish, top panel) and relative spawning output (bottom panel) with associated 95% confidence intervals (bars or dashed lines) for west coast USA sablefish (reproduced here with permission from Wetzel et al. 2025).

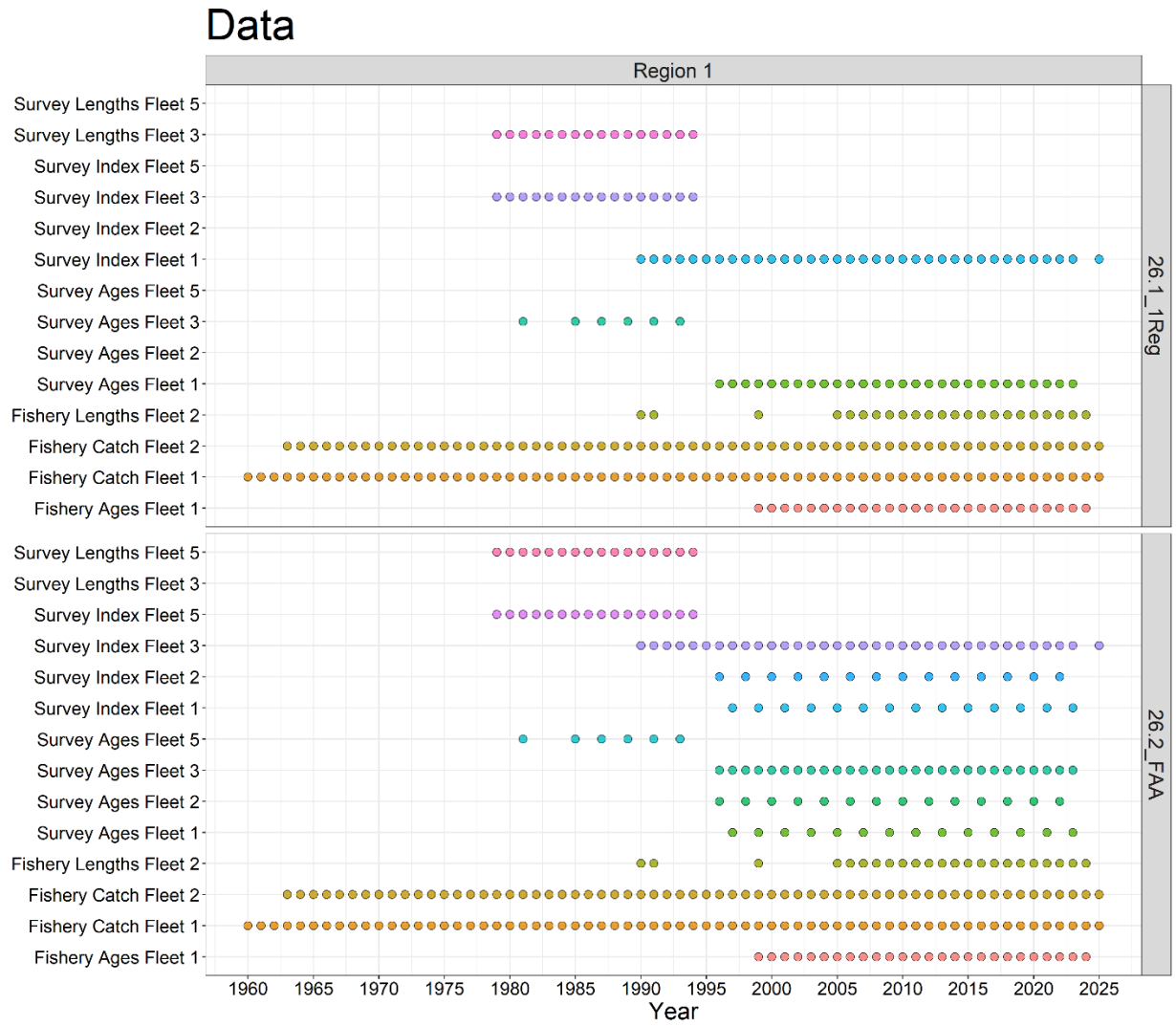


Figure 3.19. Data inputs used by each model. Fishery Fleet 1 is the fixed gear fleet and Fishery Fleet 2 is the trawl fleet. Note that survey fleet numbers vary across models. For Model *26.1_1Reg* Survey Fleet 1 is the NOAA Longline Survey and survey Fleet 3 is the Japanese Longline Survey. For Model *26.2_FAA* Survey Fleet 1 is the BS NOAA Longline Survey, Survey Fleet 2 is the AI NOAA Longline Survey, Survey Fleet 3 is the GOA NOAA Longline Survey, and Survey Fleet 5 is the Japanese Longline Survey.

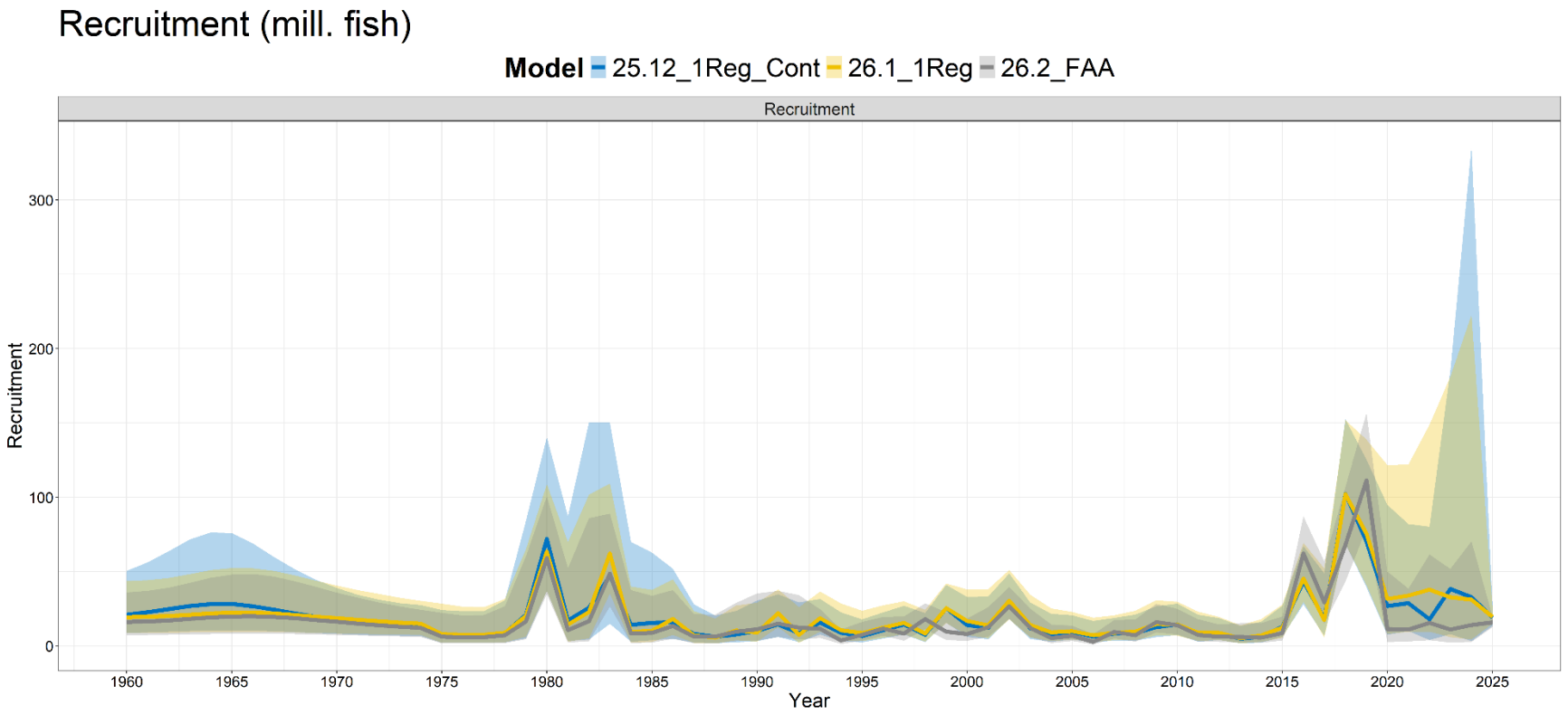


Figure 3.20. Results of the model bridging exercise in terms of recruitment (millions of age-2 fish, lines) with 95% confidence intervals (associated shaded regions). Model *25.12_1Reg_Cont* (blue line) is the continuity model updated with data through 2025. Model *26.1_1Reg* (yellow line) is the final single region model. Model *26.2_FAA* (grey line) is the final fleets-as-areas model. For a description of stepwise model building from Model *25.12_1Reg_Cont* to Model *26.1_1Reg* see Appendix 3B. For a description of the development of Model *26.2_FAA* (adding incremental complexity from Model *26.1_1Reg*) see Appendix 3C.

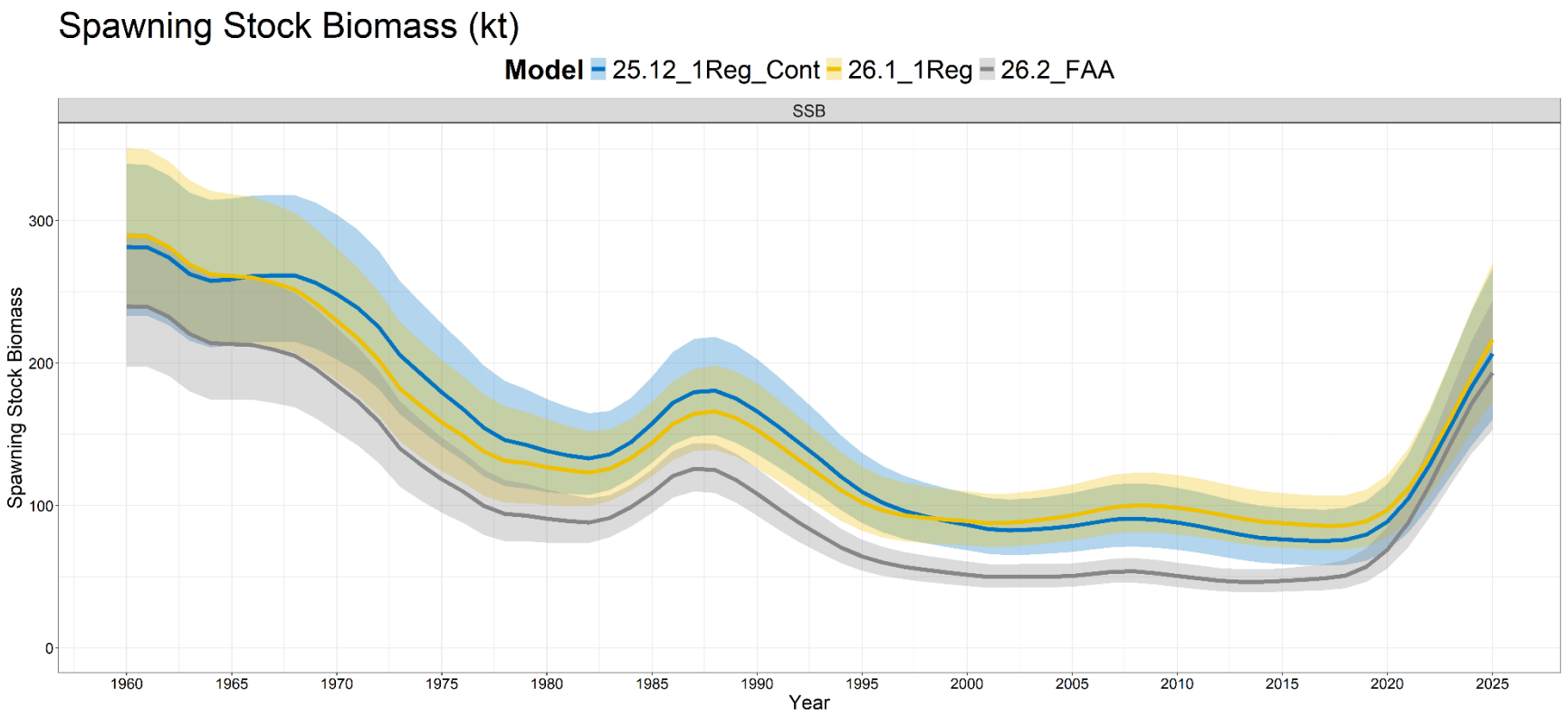


Figure 3.21. Results of the model bridging exercise in terms of spawning stock biomass (kt, lines) with 95% confidence intervals (associated shaded regions). Model *25.12_1Reg_Cont* (blue line) is the continuity model updated with data through 2025. Model *26.1_1Reg* (yellow line) is the final single region model. Model *26.2_FAA* (grey line) is the final fleets-as-areas model. For a description of stepwise model building from Model *25.12_1Reg_Cont* to Model *26.1_1Reg* see Appendix 3B. For a description of the development of Model *26.2_FAA* (adding incremental complexity from Model *26.1_1Reg*) see Appendix 3C.

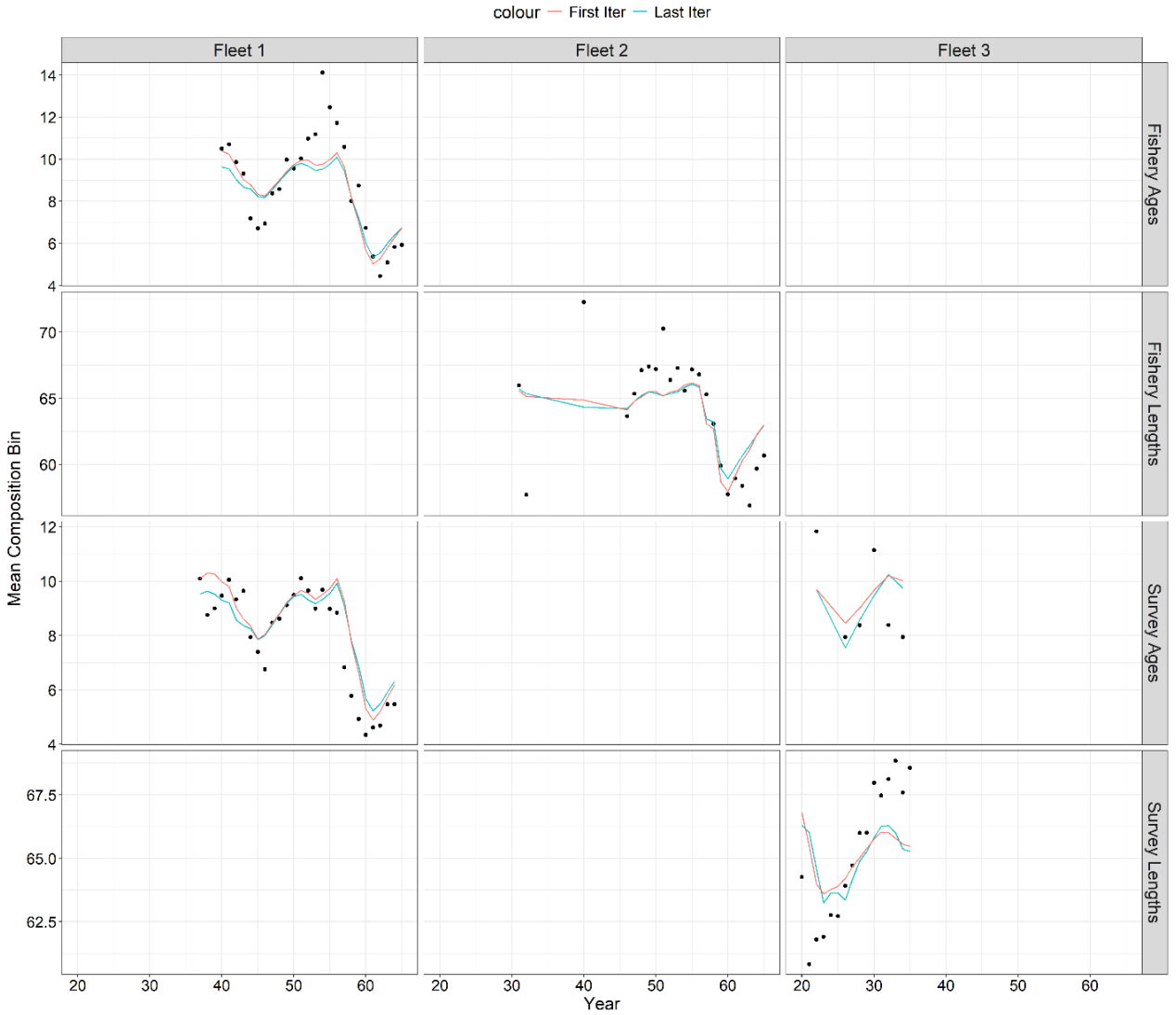


Figure 3.22. Fits to the mean age or mean length by year for each compositional data source at the start (using equal data weights, red line) and end (using final data weights, blue lines) of the Francis reweighting method (Francis, 2011) for Model *26.1_1Reg*. Fishery fleet 1 is the fixed gear fishery and fleet 2 is the trawl fishery. Survey fleet 1 is the NOAA Longline Survey and fleet 3 is the Japanese Longline Survey. Year 1 is 1960.

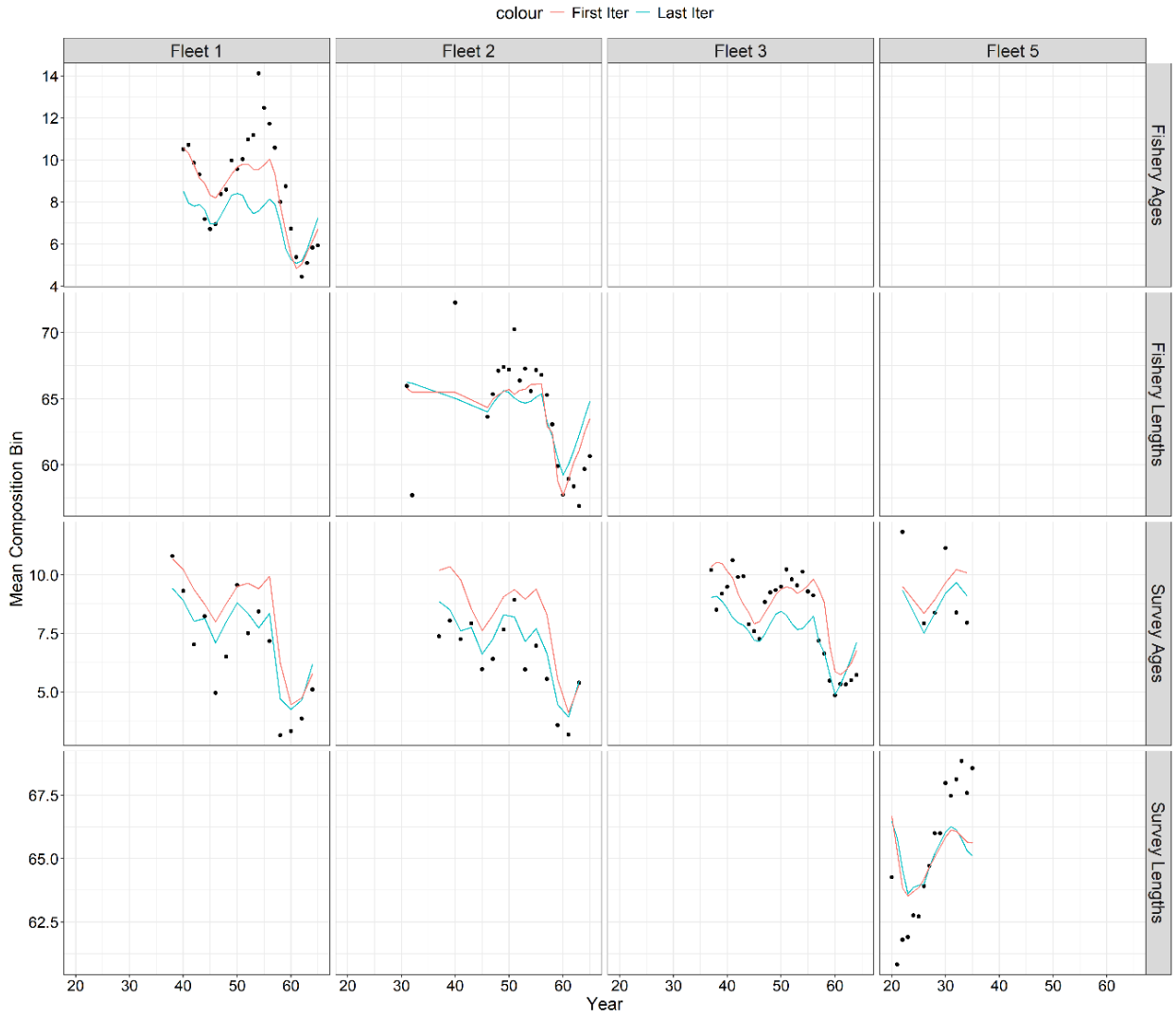


Figure 3.23. Fits to the mean age or mean length by year for each compositional data source at the start (using equal data weights, red line) and end (using final data weights, blue lines) of the Francis reweighting method (Francis, 2011) for Model *26.2_FAA*. Fishery fleet 1 is the fixed gear fishery and fleet 2 is the trawl fishery. Survey Fleet 1 is the BS NOAA Longline Survey, Survey Fleet 2 is the AI NOAA Longline Survey, Survey Fleet 3 is the GOA NOAA Longline Survey, and Survey Fleet 5 is the Japanese Longline Survey. Year 1 is 1960.

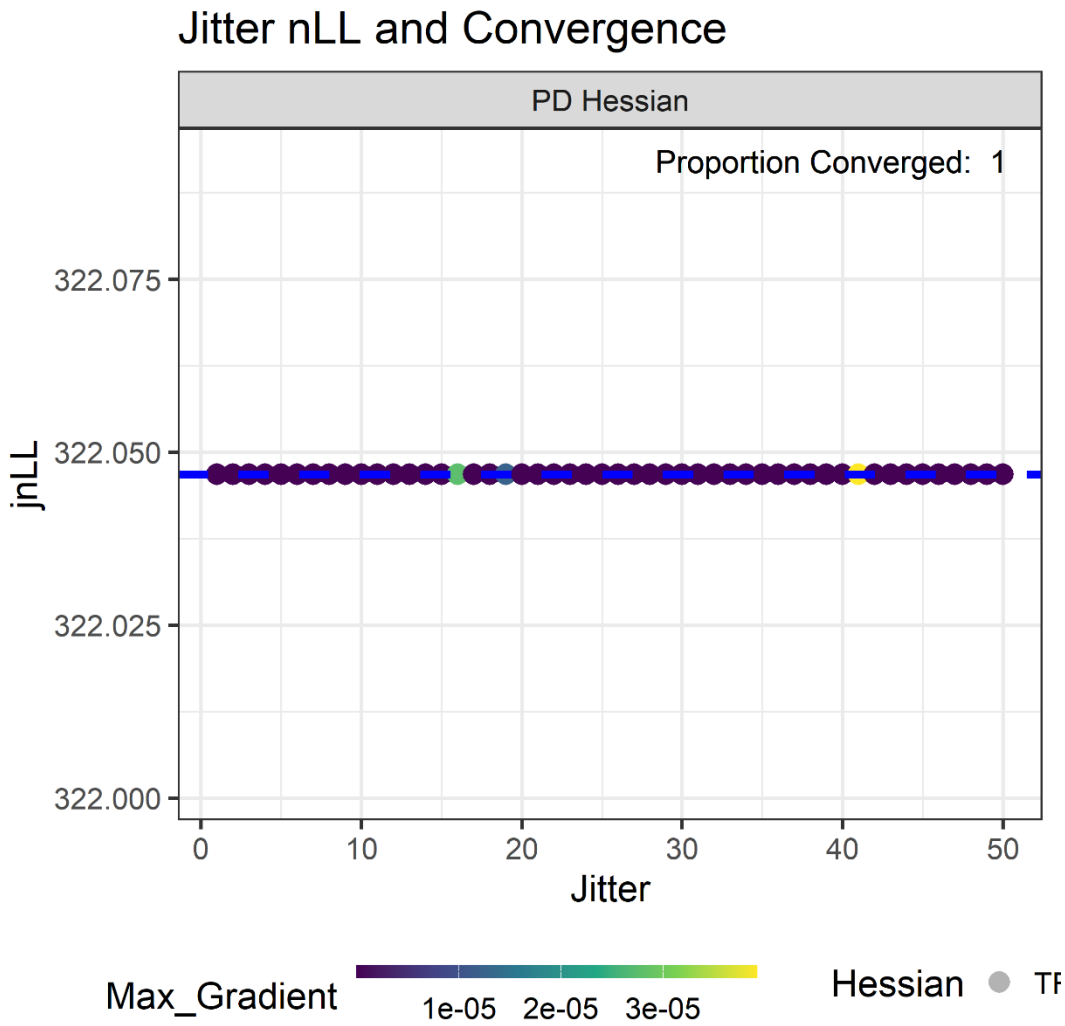


Figure 3.24. Results of the jitter analysis in terms of total joint negative log-likelihood (jnLL) values for Model *26.1_1Reg*. The starting parameter values were randomly jittered assuming ~$N(0,0.1^2)$. All 50 jitter runs are shown and the model with the smallest jnLL was used as the base model.

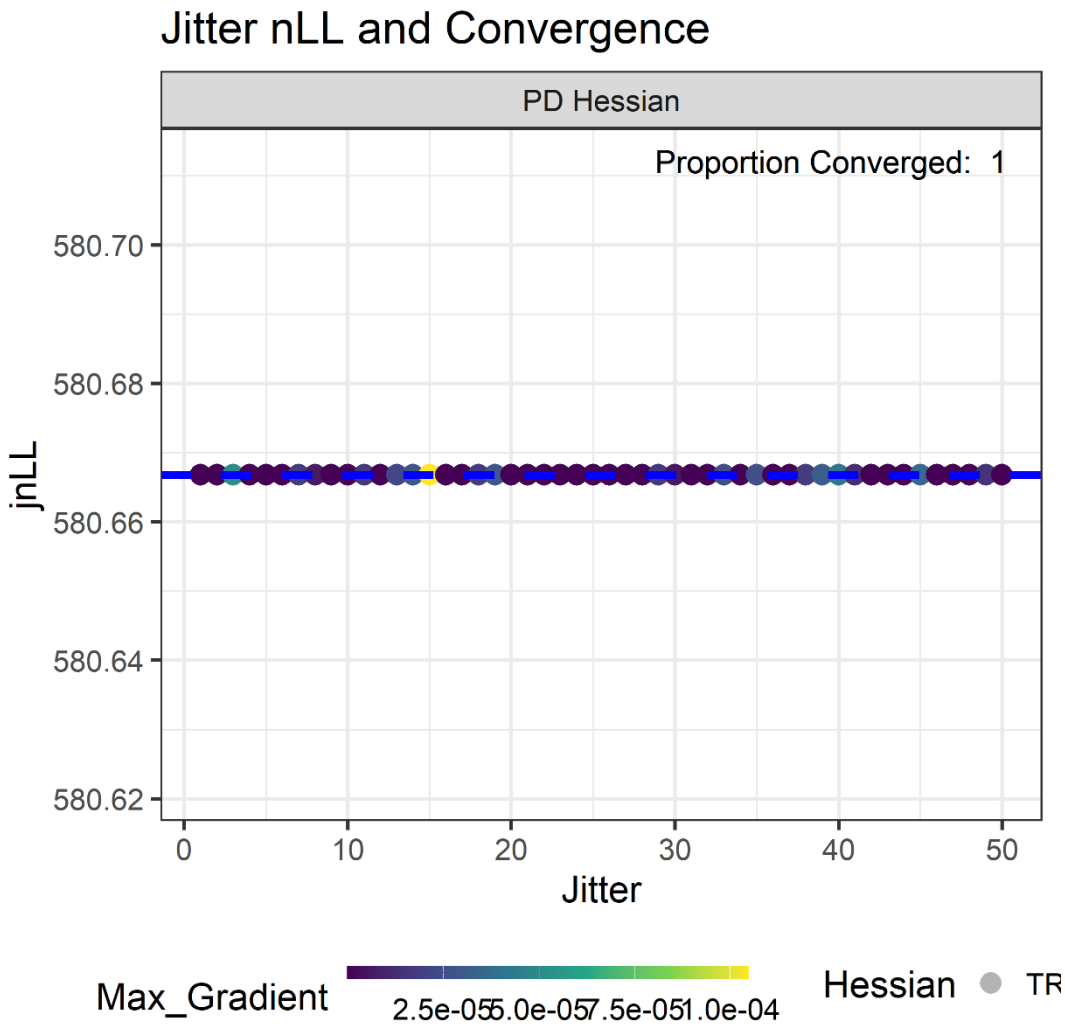


Figure 3.25. Results of the jitter analysis in terms of total joint negative log-likelihood (jnLL) values Model *26.2_FAA*. The starting parameter values were randomly jittered assuming ~$N(0,0.1^2)$. All 50 jitter runs are shown and the model with the smallest jnLL was used as the base model.

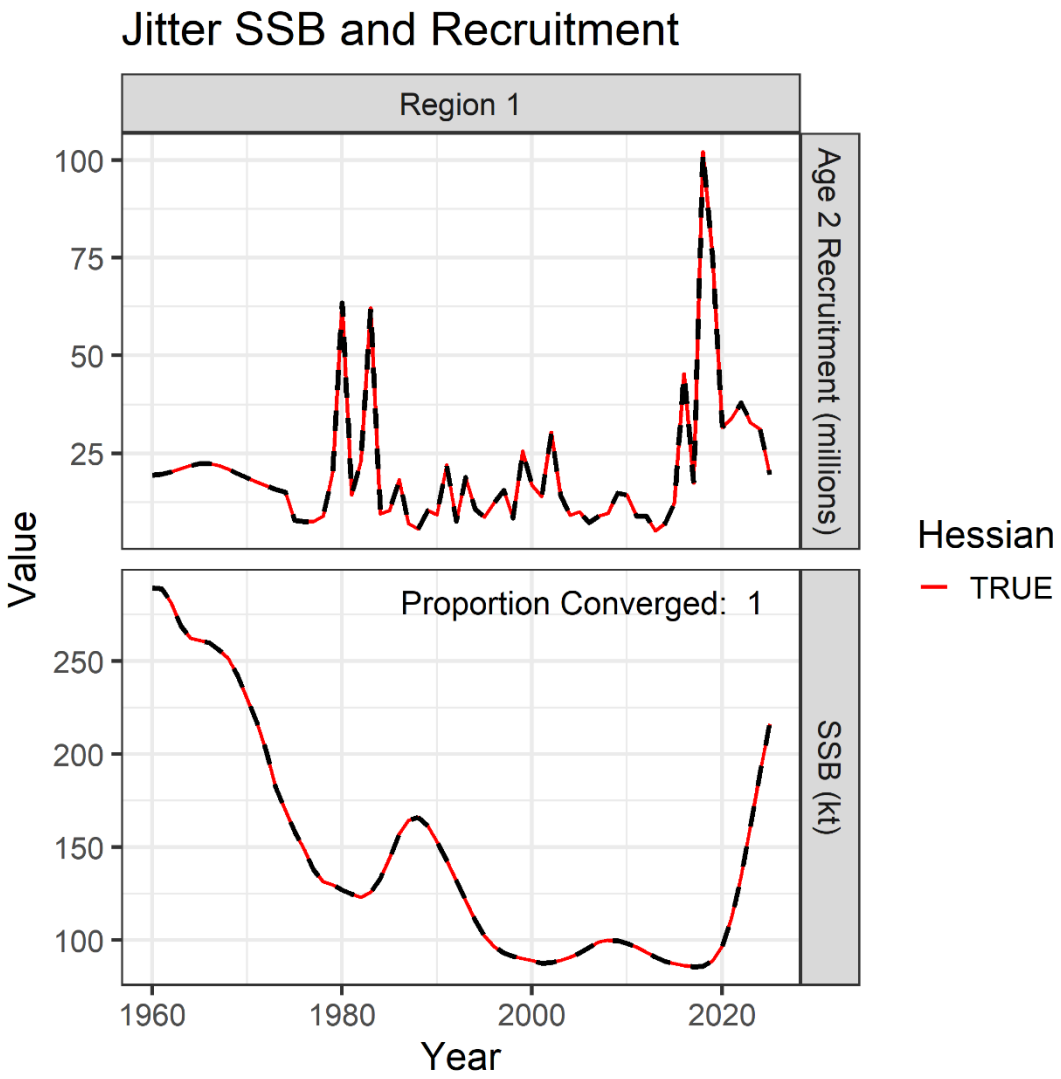


Figure 3.26. Results of the jitter analysis showing recruitment (top panel) and SSB (bottom panel) for each jitter iteration for Model *26.1_1Reg*. The starting parameter values were randomly jittered assuming ~$N(0,0.1^2)$ and all 50 jitter runs are shown.

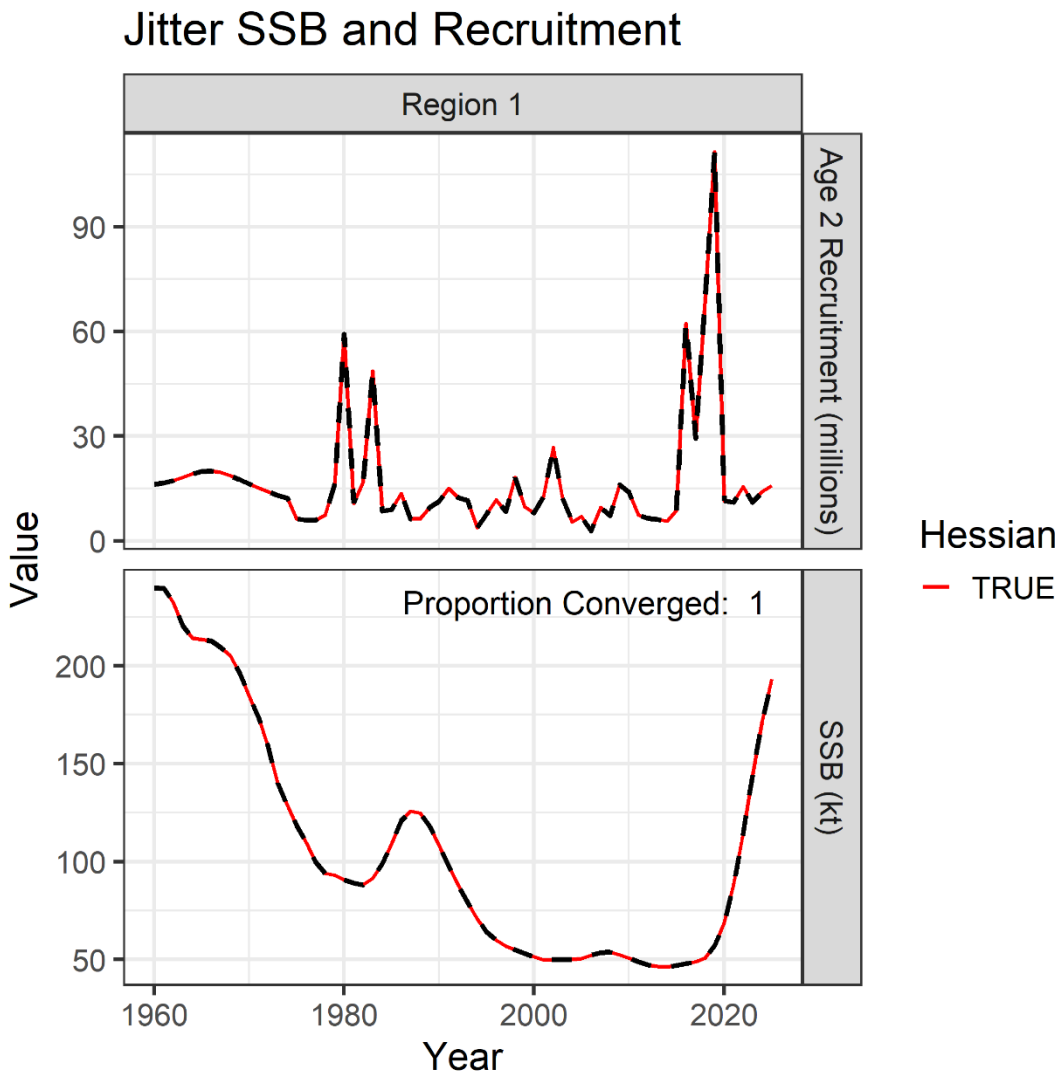


Figure 3.27. Results of the jitter analysis showing recruitment (top panel) and SSB (bottom panel) for each jitter iteration for Model *26.2_FAA*. The starting parameter values were randomly jittered assuming ~$N(0,0.1^2)$ and all 50 jitter runs are shown.

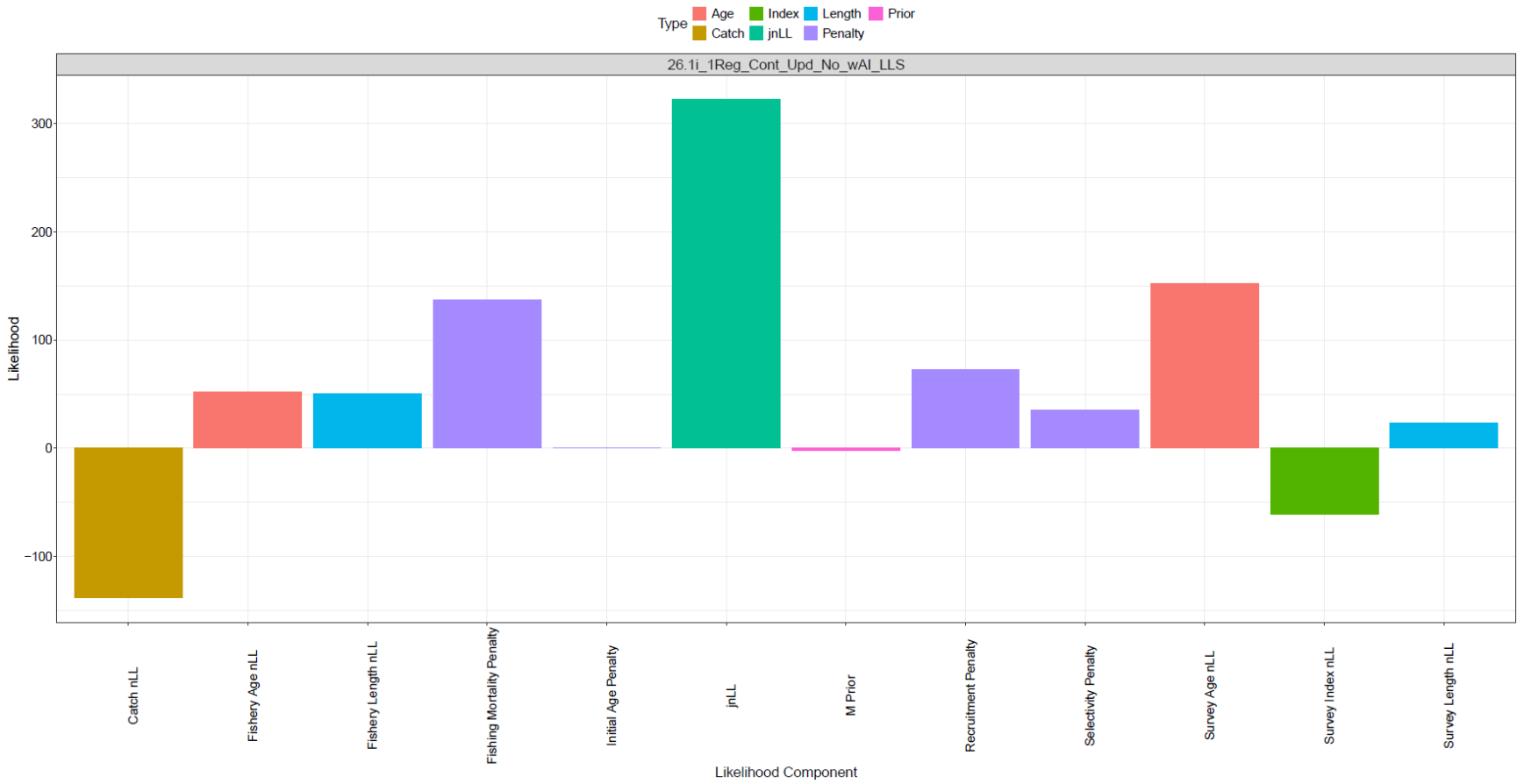


Figure 3.28. Joint negative log-likelihood (jnLL) components from Model *26.1_1Reg* where different colors represent the various data sets and priors or penalty terms. Note that due to differences in likelihood structures and data inputs (regional disaggregation of fleets) used in Model *26.1_1Reg* compared to Model *26.2_FAA*, Figures 3.25 and 3.26 are not directly comparable.

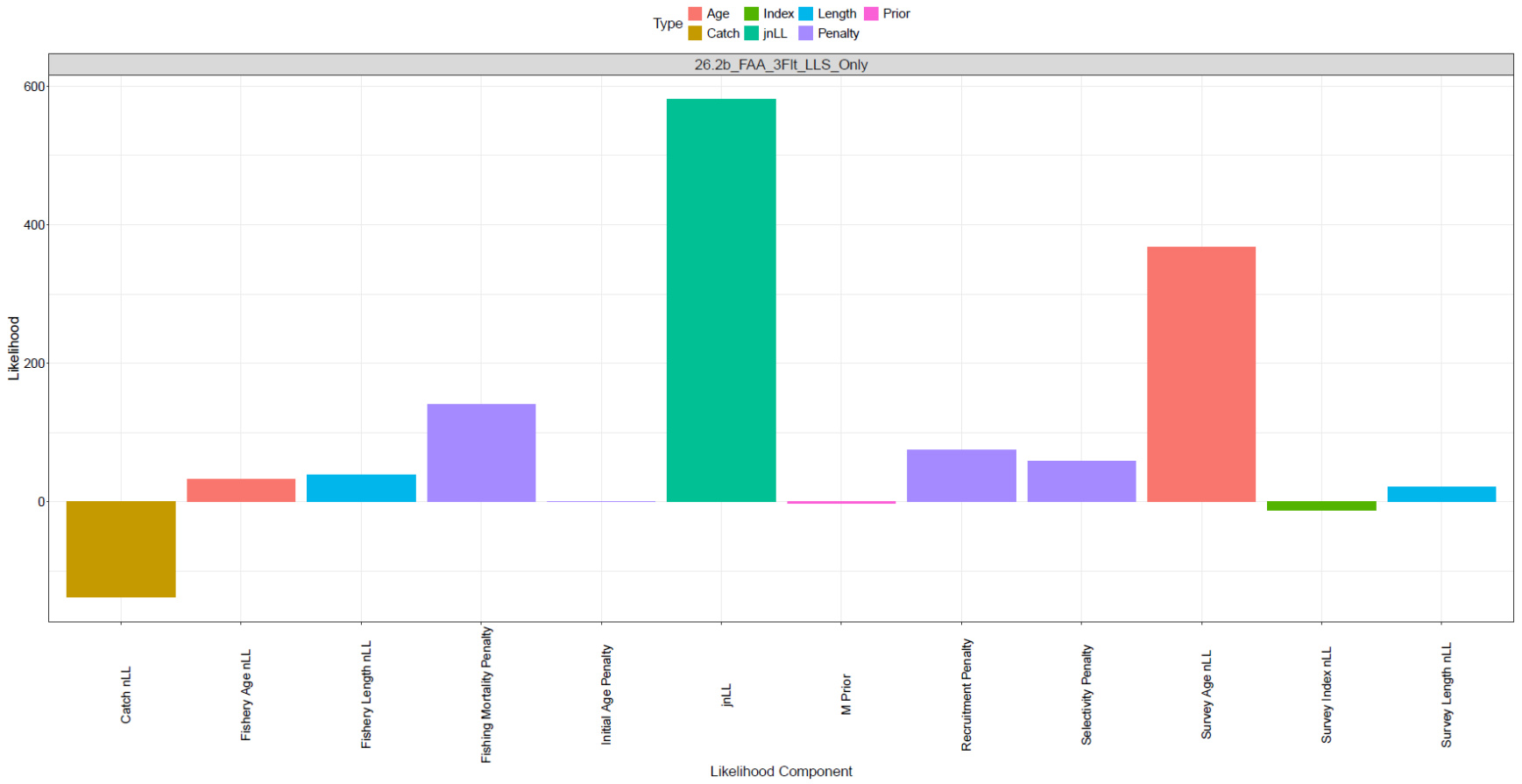


Figure 3.29. Joint negative log-likelihood (jnLL) components from Model *26.2_FAA* where different colors represent the various data sets and priors or penalty terms. Note that due to differences in likelihood structures and data inputs (regional disaggregation of fleets) used in Model *26.1_1Reg* compared to Model *26.2_FAA*, Figures 3.28 and 3.29 are not directly comparable.

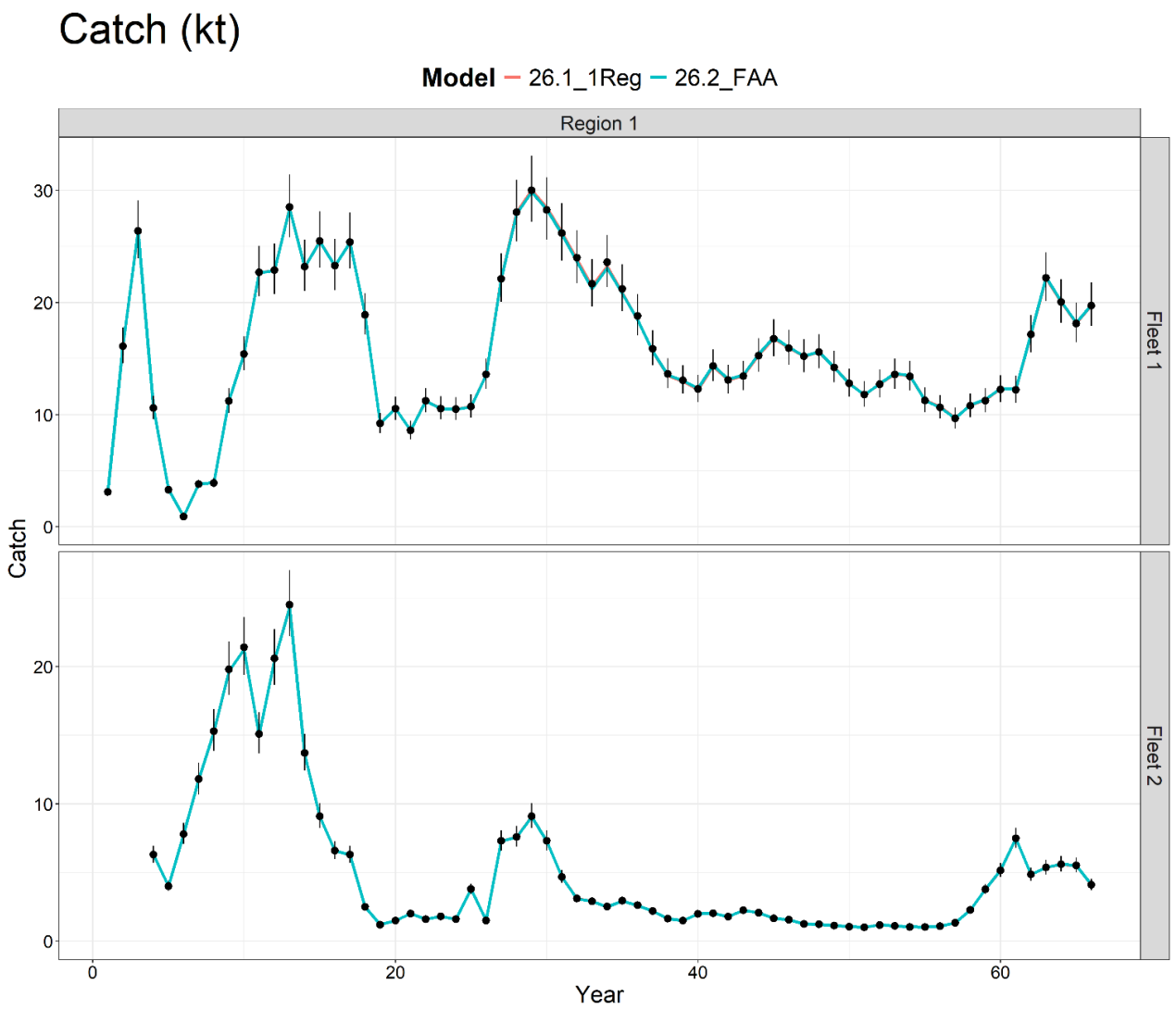


Figure 3.30. Fit to the catch data (black dots with 95% confidence intervals) by fleet (fleet 1 is the fixed gear fishery and fleet 2 is the trawl fleet) for Model *26.1_1Reg* (red line) and Model *26.2_FAA* (blue line). Note the varying y-axis scales.

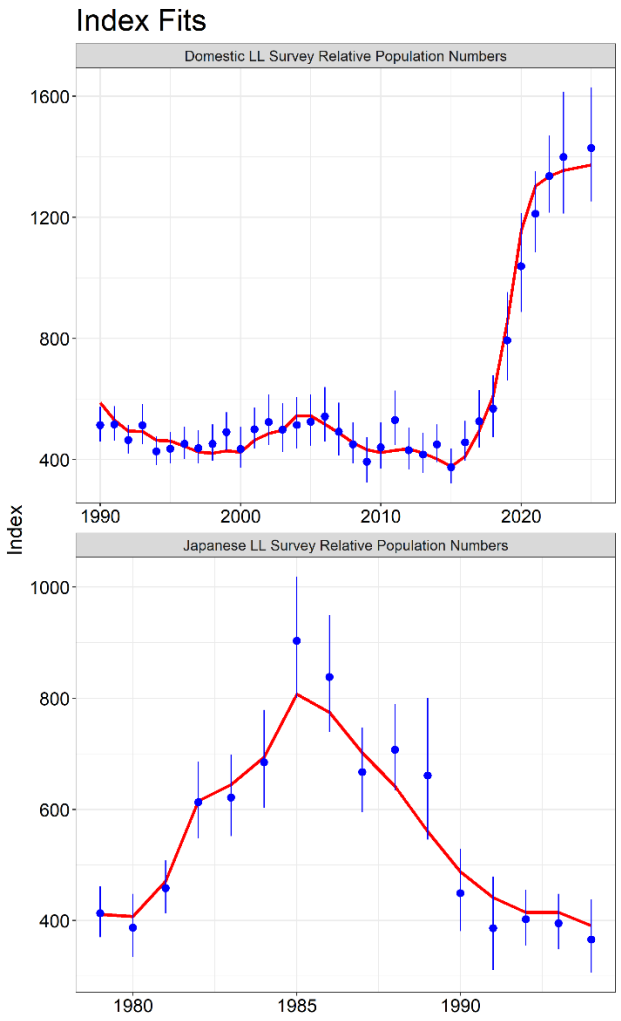


Figure 3.31. Model *26.1_1Reg* fit to indices, including the NOAA Longline Survey RPNs (top, 1000s of fish) and Japanese Longline Survey RPNs (bottom, 1000s of fish). Red lines are model predicted values, and blue dots are observed values (with 95% confidence intervals based on the assumed standard errors).

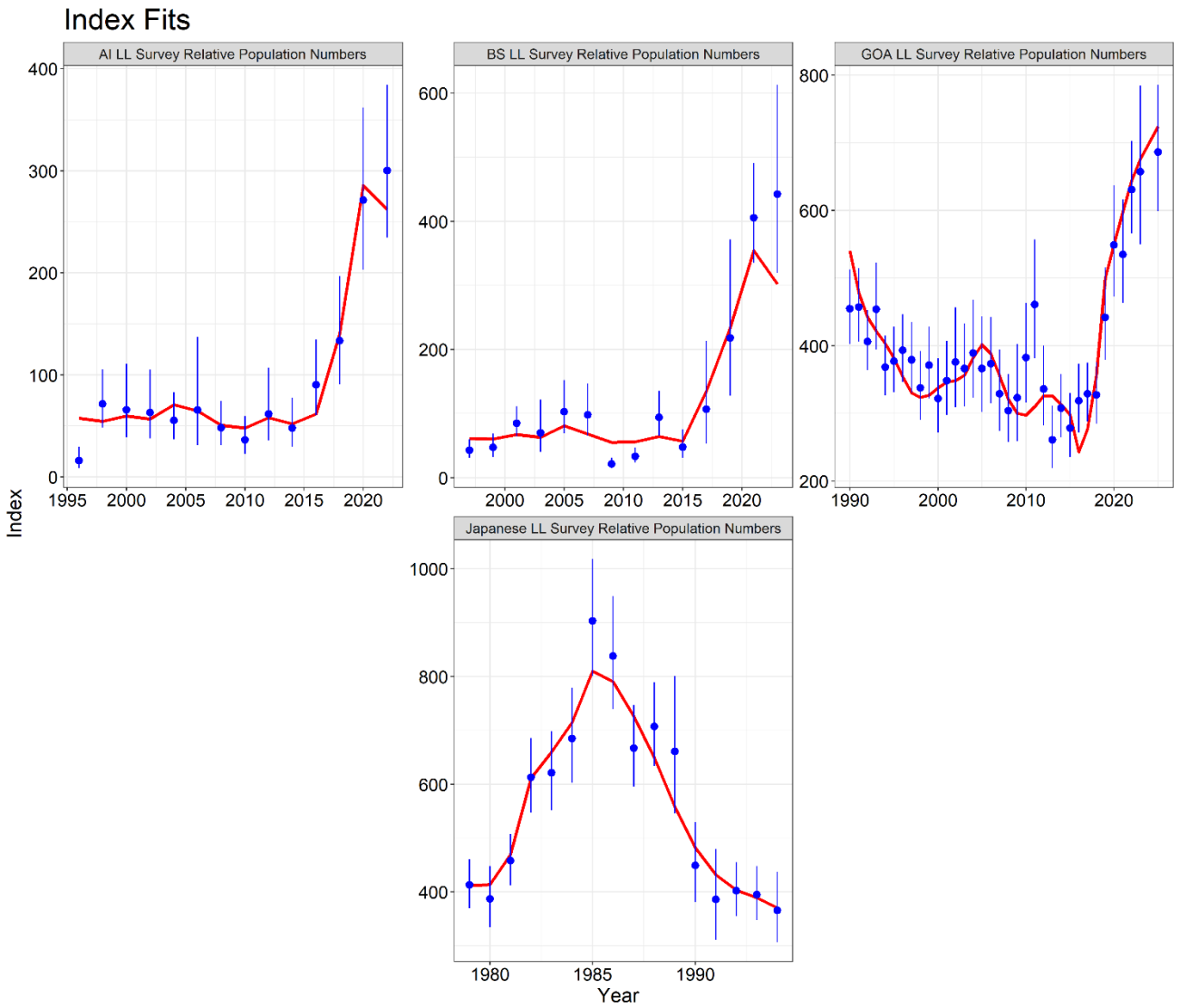


Figure 3.32. Model *26.2_FAA* fit to indices, including the NOAA Longline Survey RPNs (top, 1000s of fish) by region (columns) and Japanese Longline Survey RPNs (bottom, 1000s of fish). Red lines are model predicted values, and blue dots are observed values (with 95% confidence intervals based on the assumed standard errors).

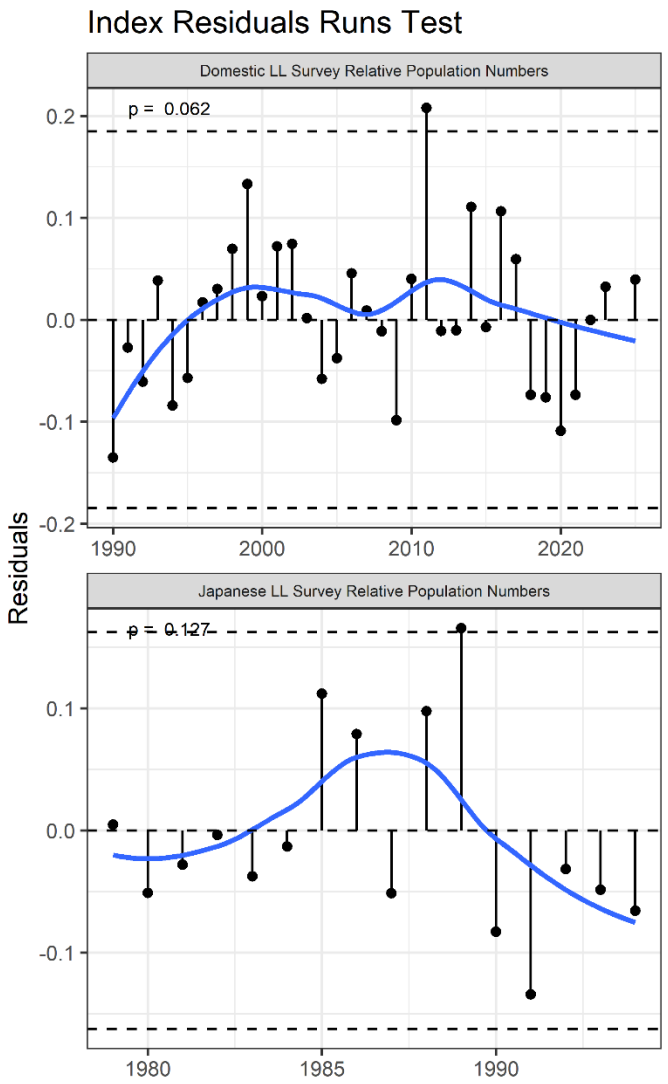


Figure 3.33. Results of an index residuals runs test for Model *26.1_1Reg* , where the black dots are the residual values, the blue lines represent a loess smoother across years, and the dashed lines represent the 95% confidence intervals.

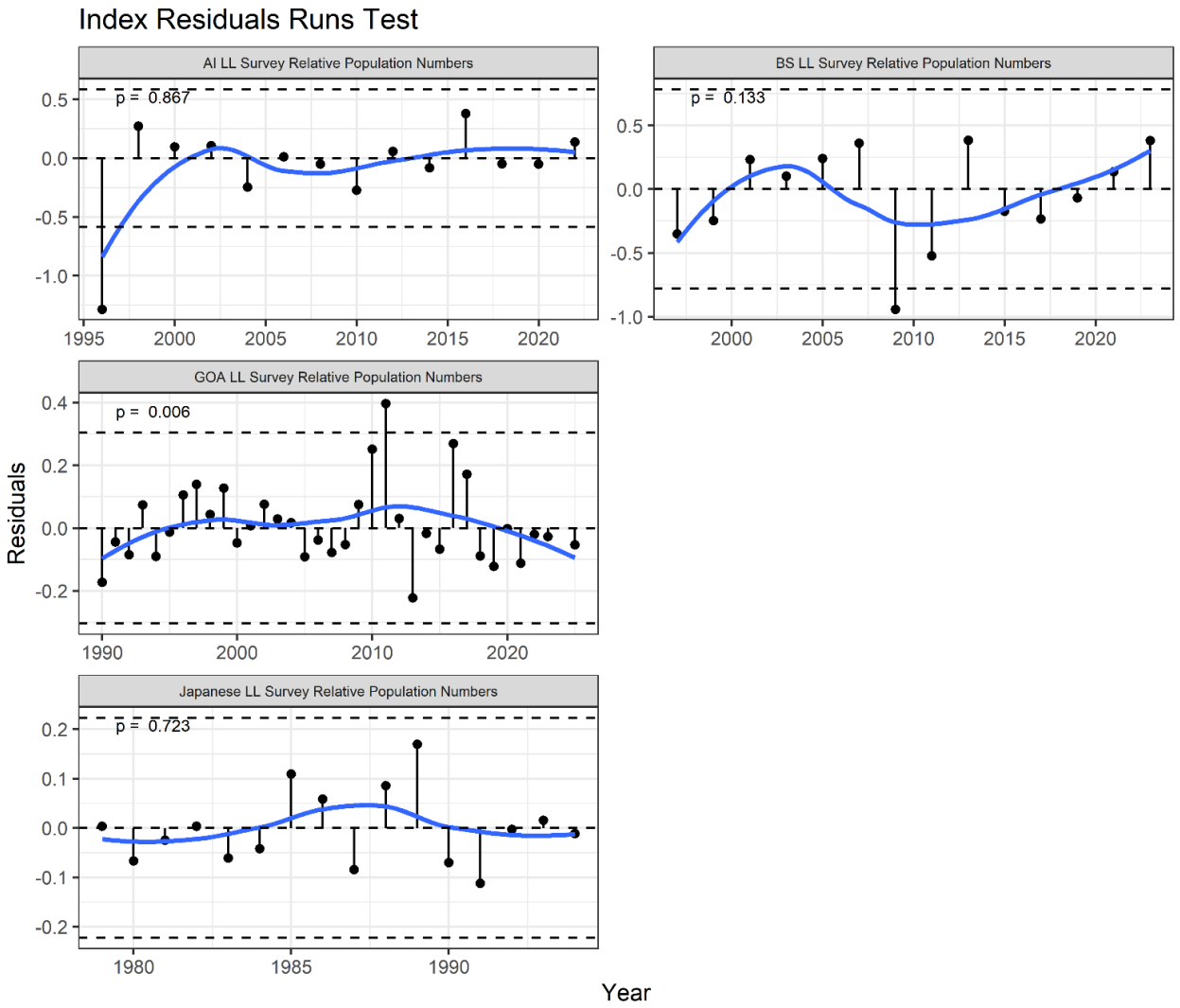


Figure 3.34. Results of an index residuals runs test for Model *26.2_FAA* , where the black dots are the residual values, the blue lines represent a loess smoother across years, and the dashed lines represent the 95% confidence intervals.

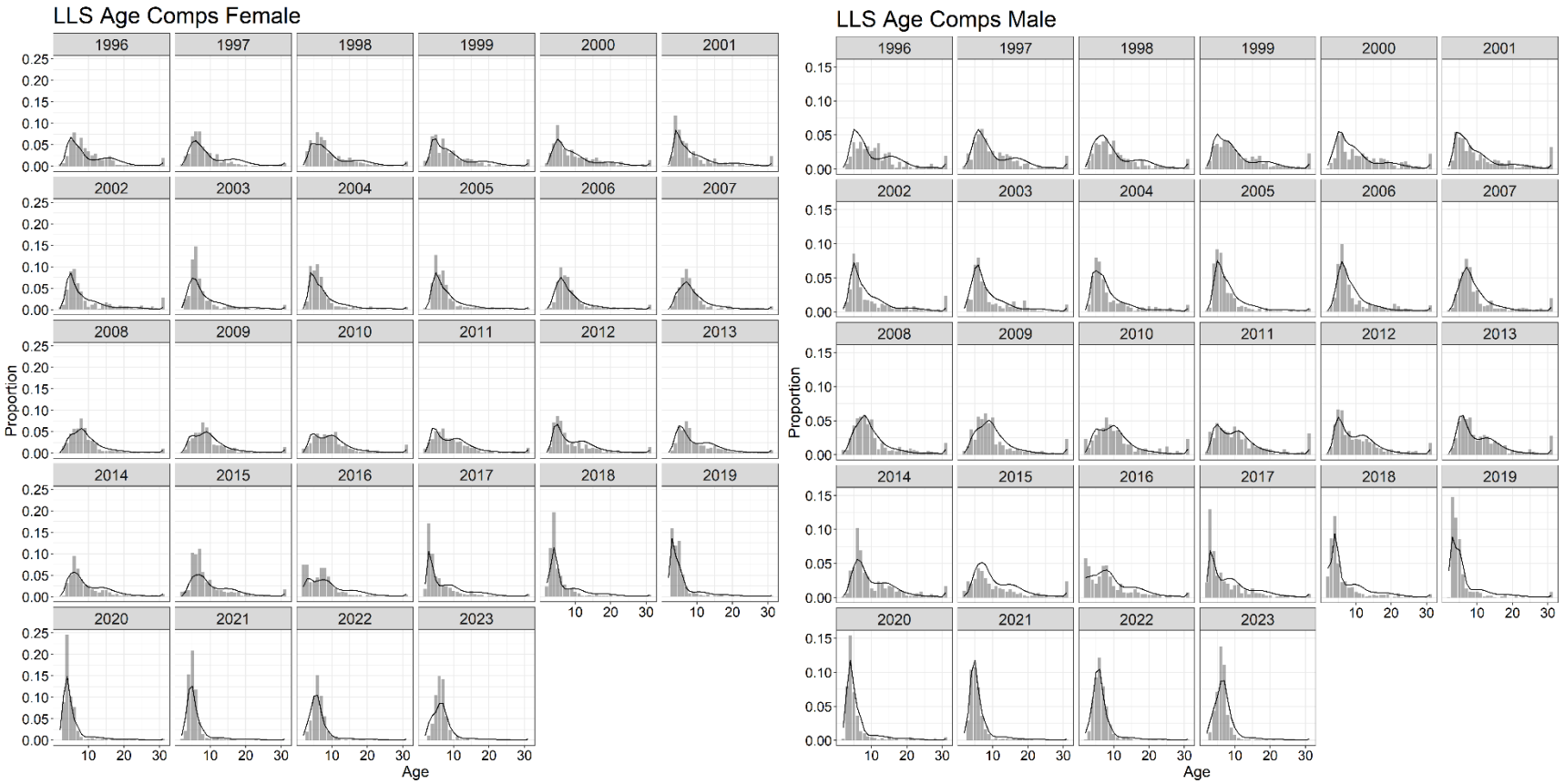


Figure 3.35. Fits to NOAA longline survey age composition data by year and sex (females left and males right) for Model *26.1_1Reg*.

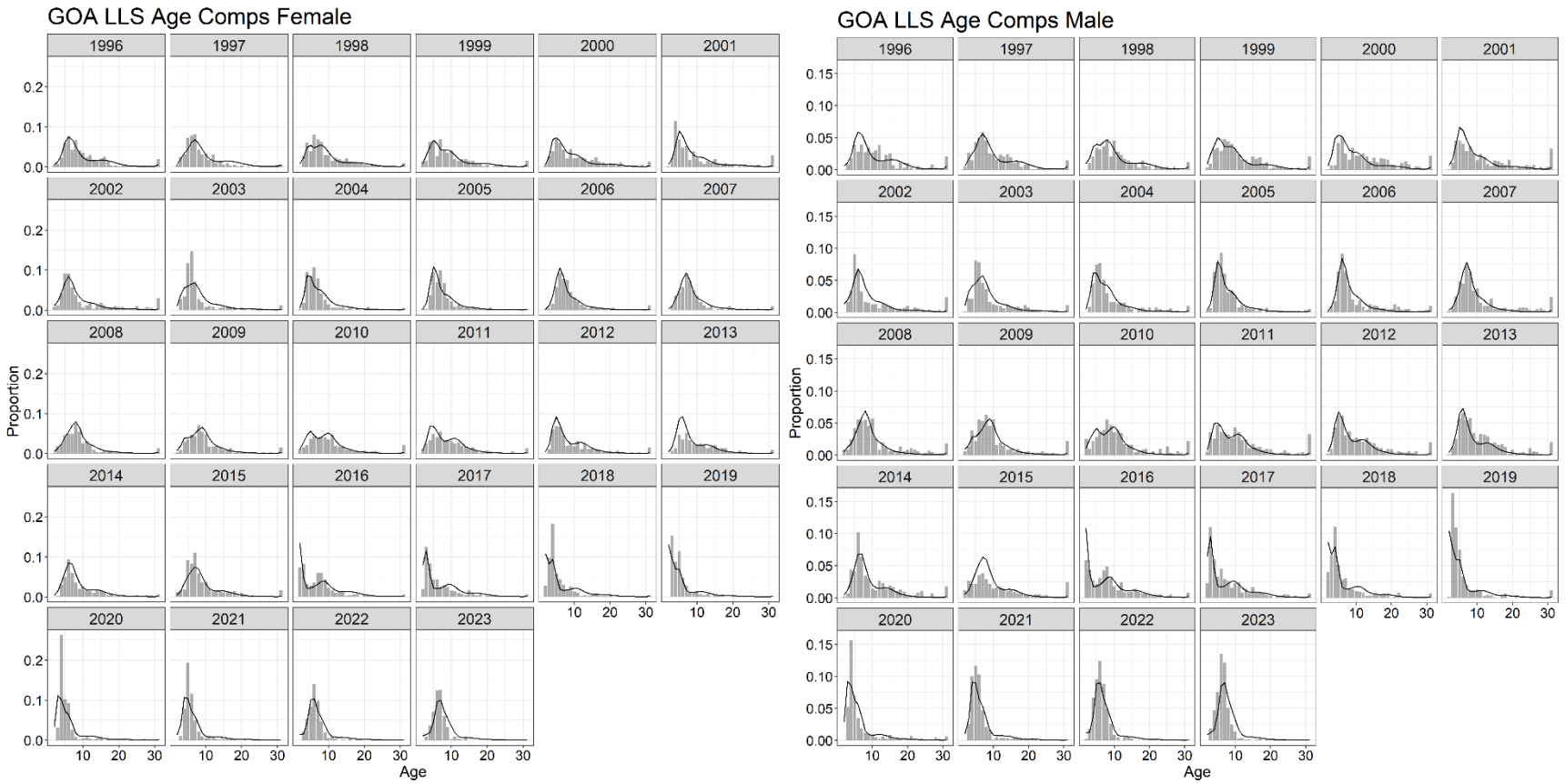


Figure 3.36. Fits to GOA NOAA Longline Survey age composition data by year and sex (females left and males right) for Model *26.2_FAA*.

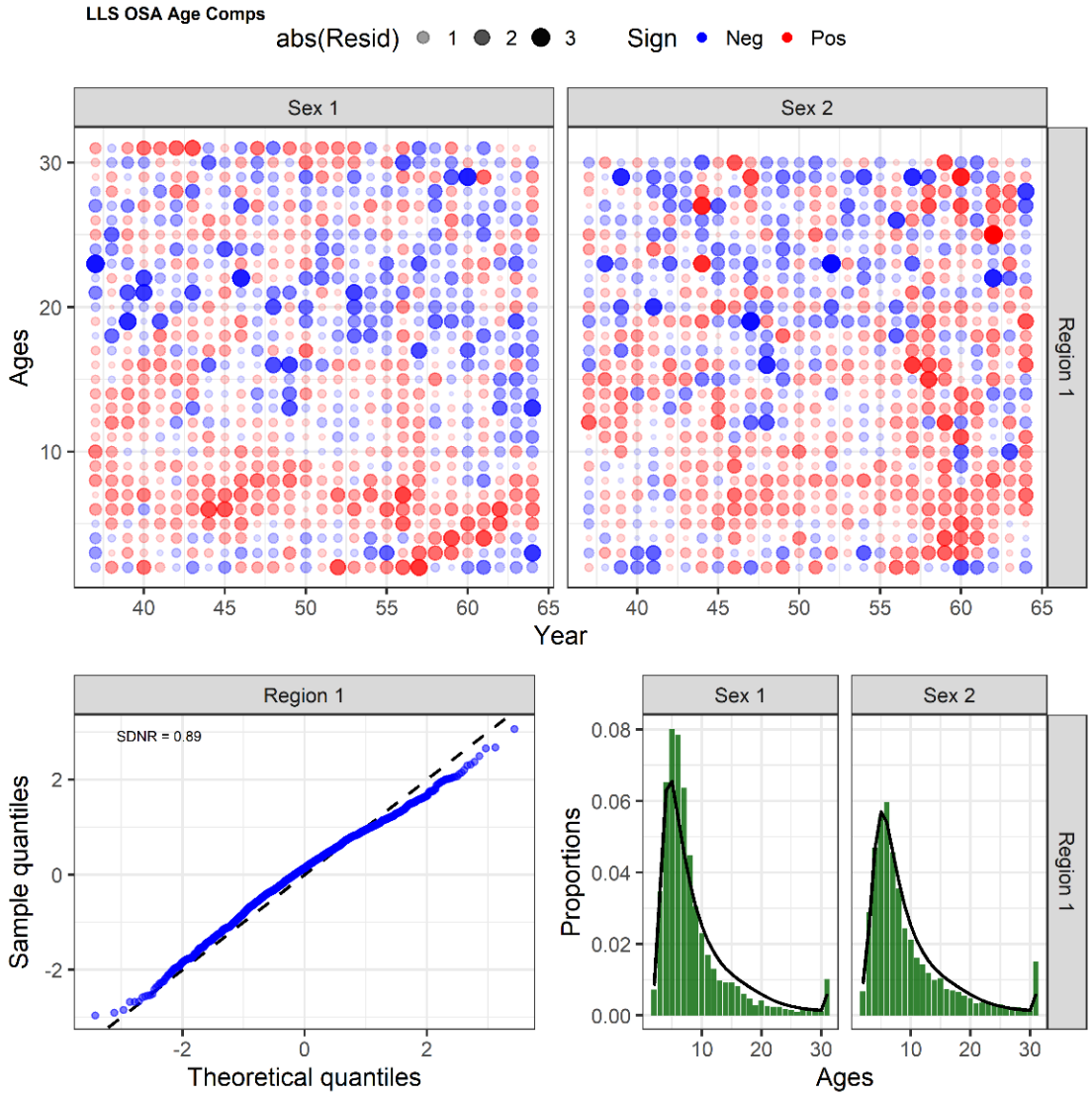


Figure 3.37. Fits to NOAA longline survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (mean across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg.*

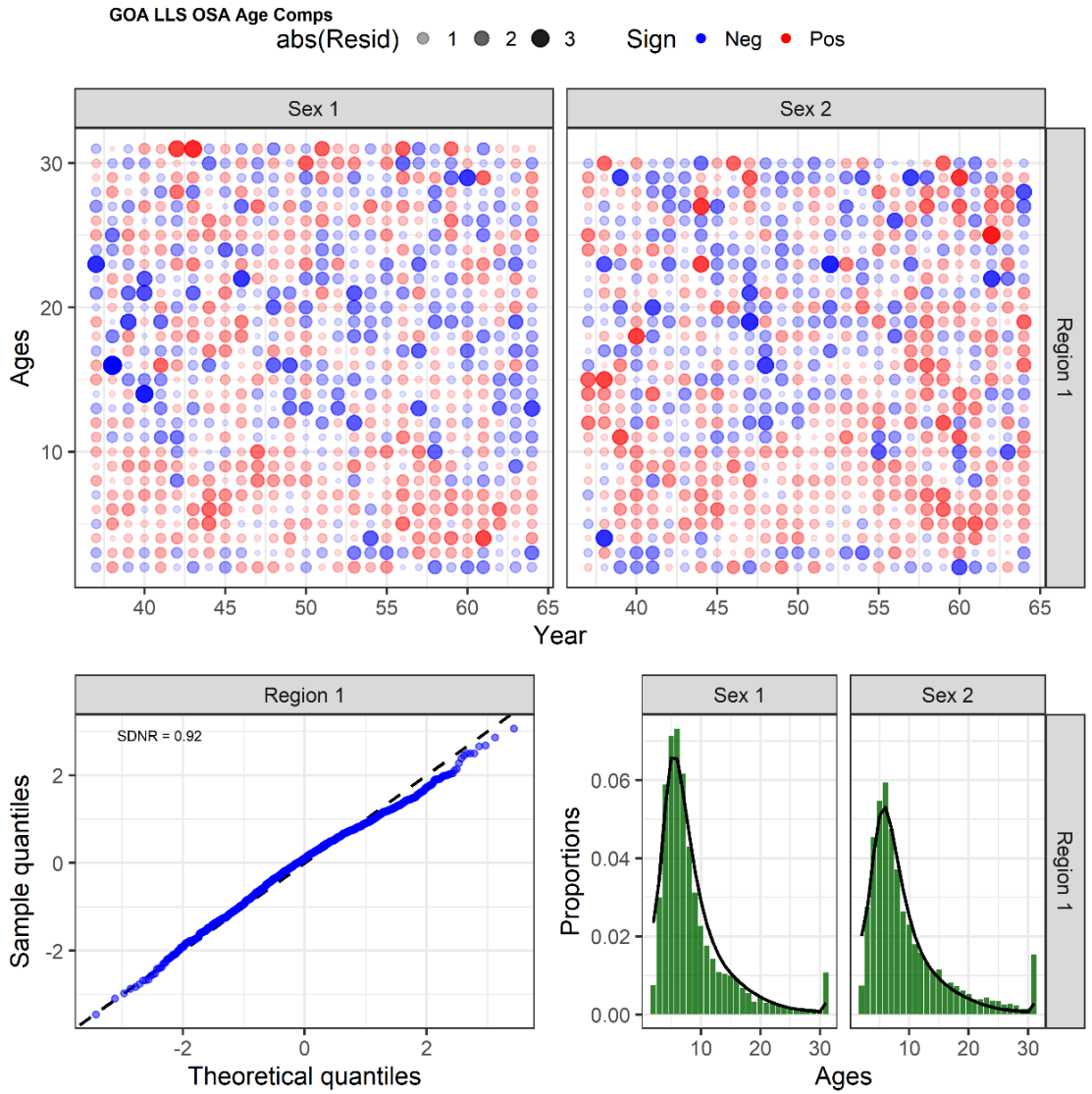


Figure 3.38. Fits to GOA NOAA Longline Survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (mean across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA*.

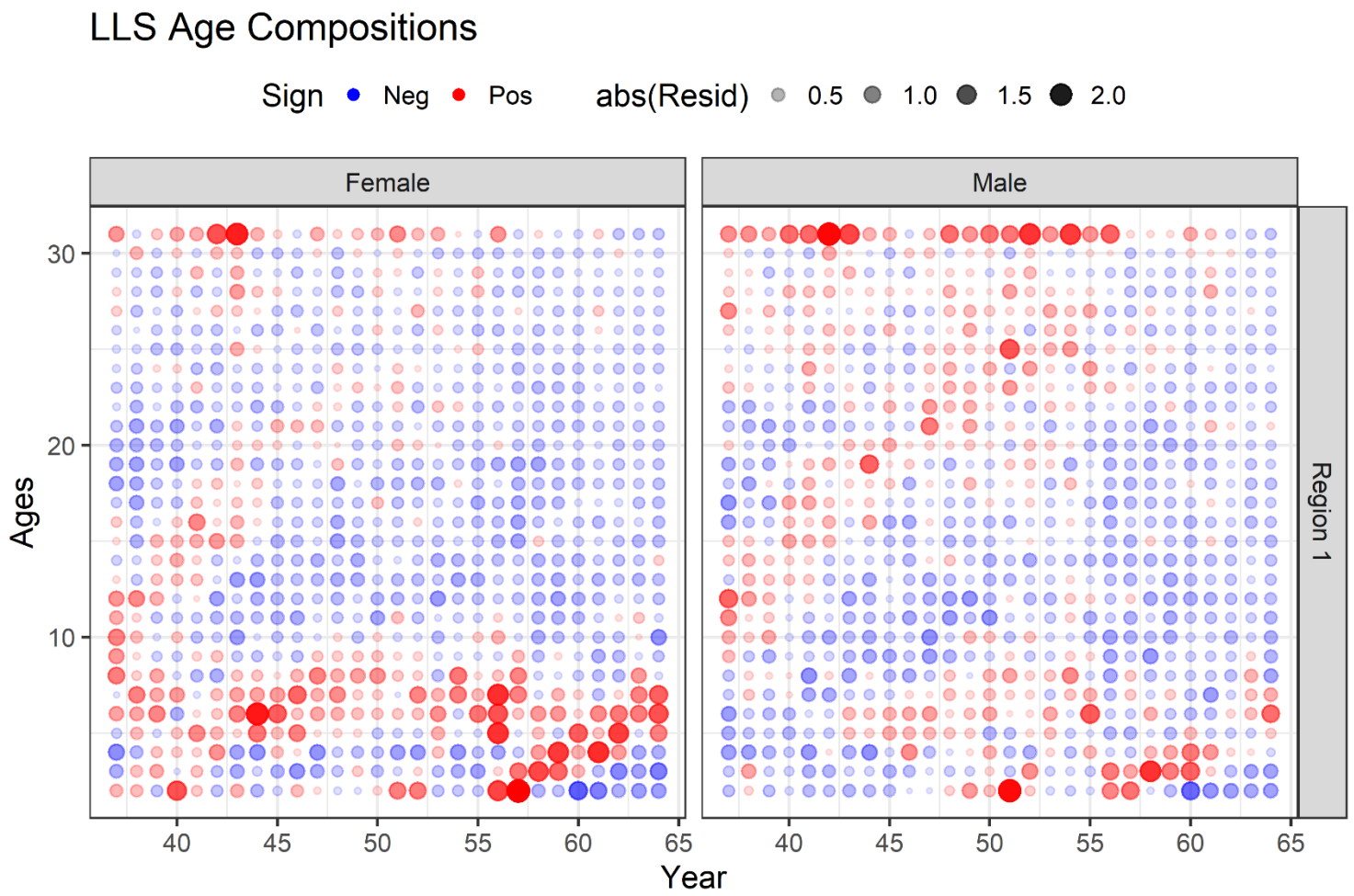


Figure 3.39. Fits to NOAA longline survey age composition data based on Pearson residuals for Model *26.1_1Reg*.

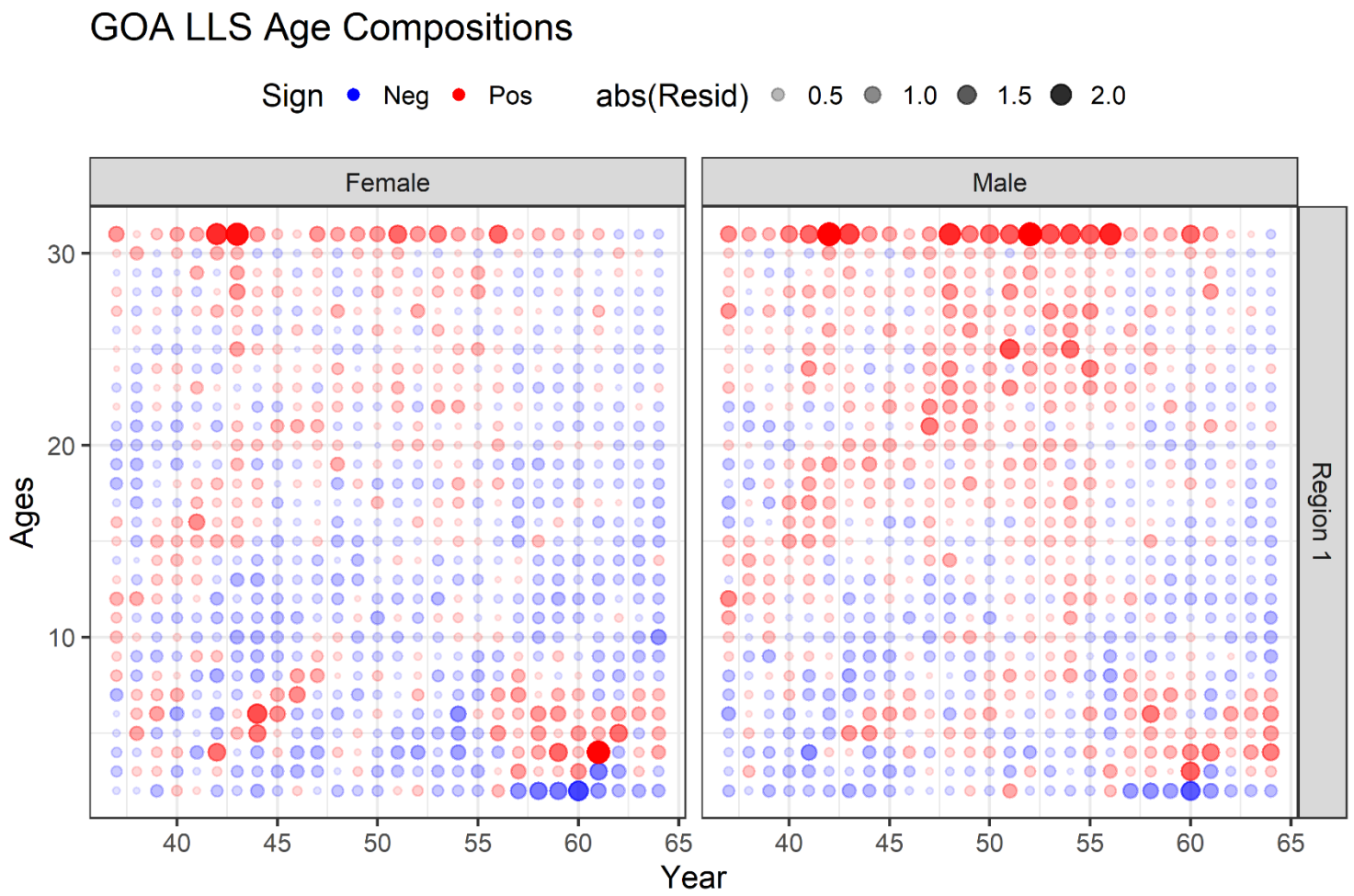


Figure 3.40. Fits to GOA NOAA Longline Survey age composition data based on Pearson residuals for Model *26.2_FAA*.

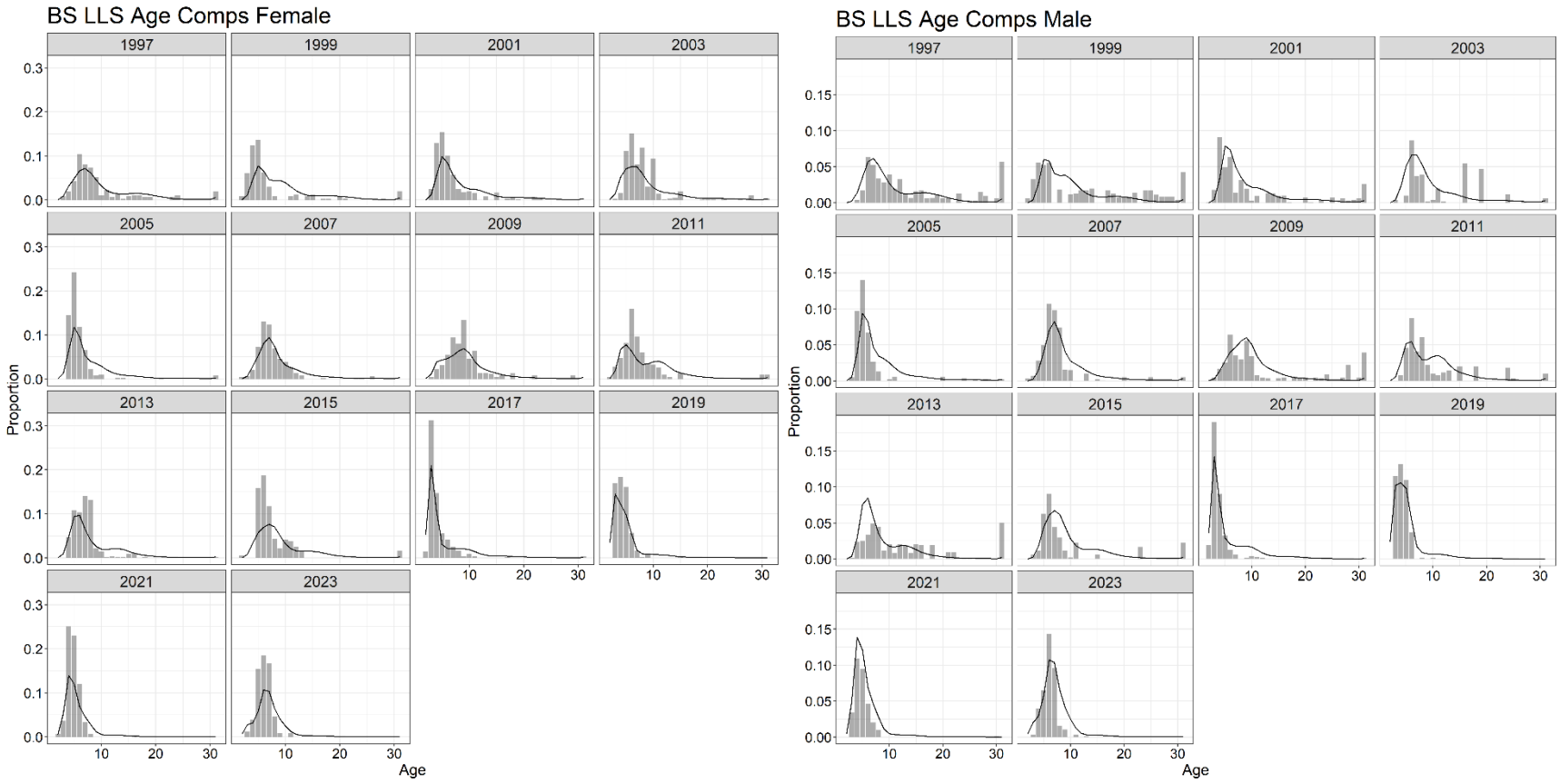


Figure 3.41. Fits to BS NOAA Longline Survey age composition data by year and sex (females left and males right) for Model *26.2_FAA*.

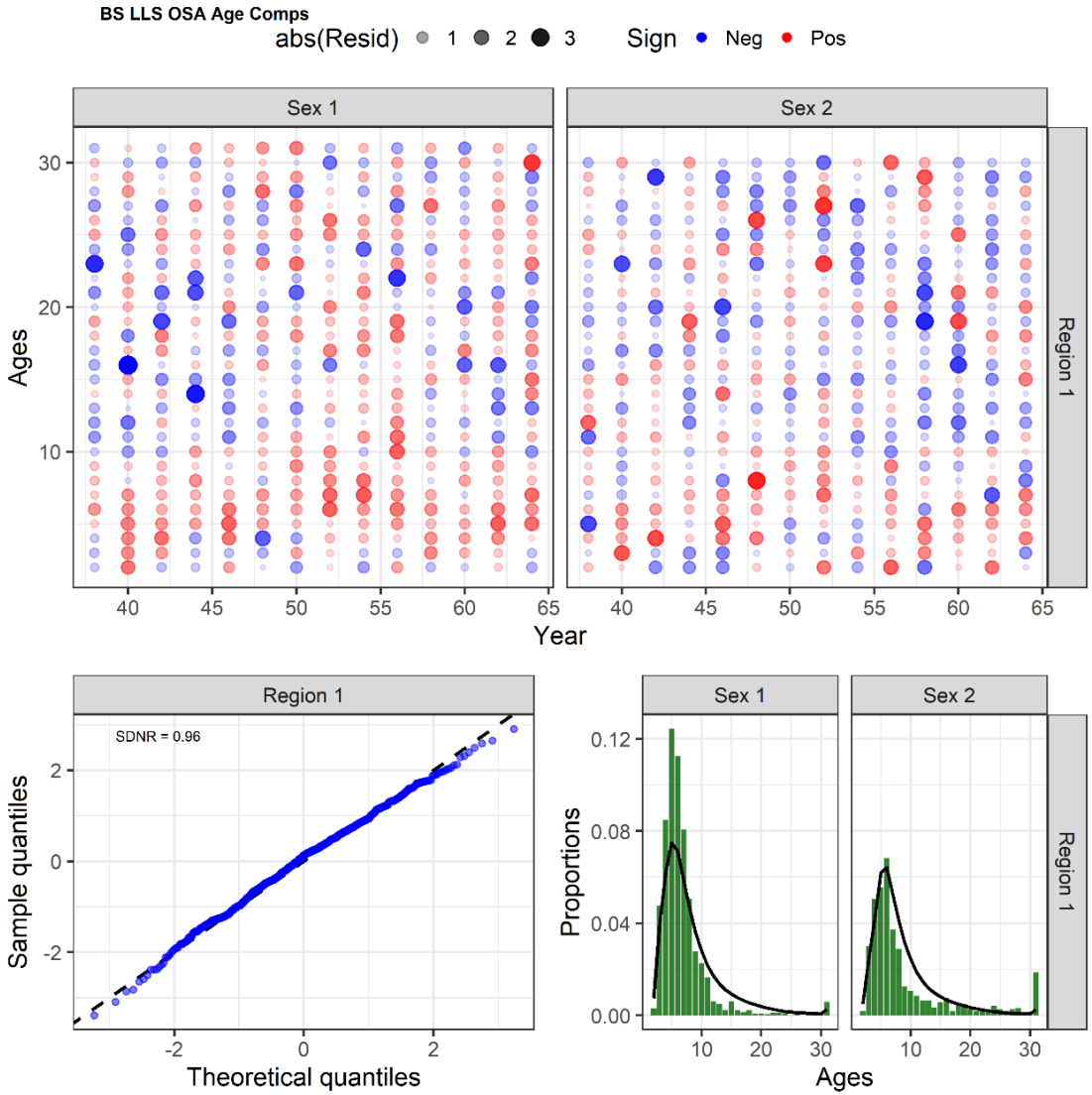


Figure 3.42. Fits to BS NOAA Longline Survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (mean across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA*.

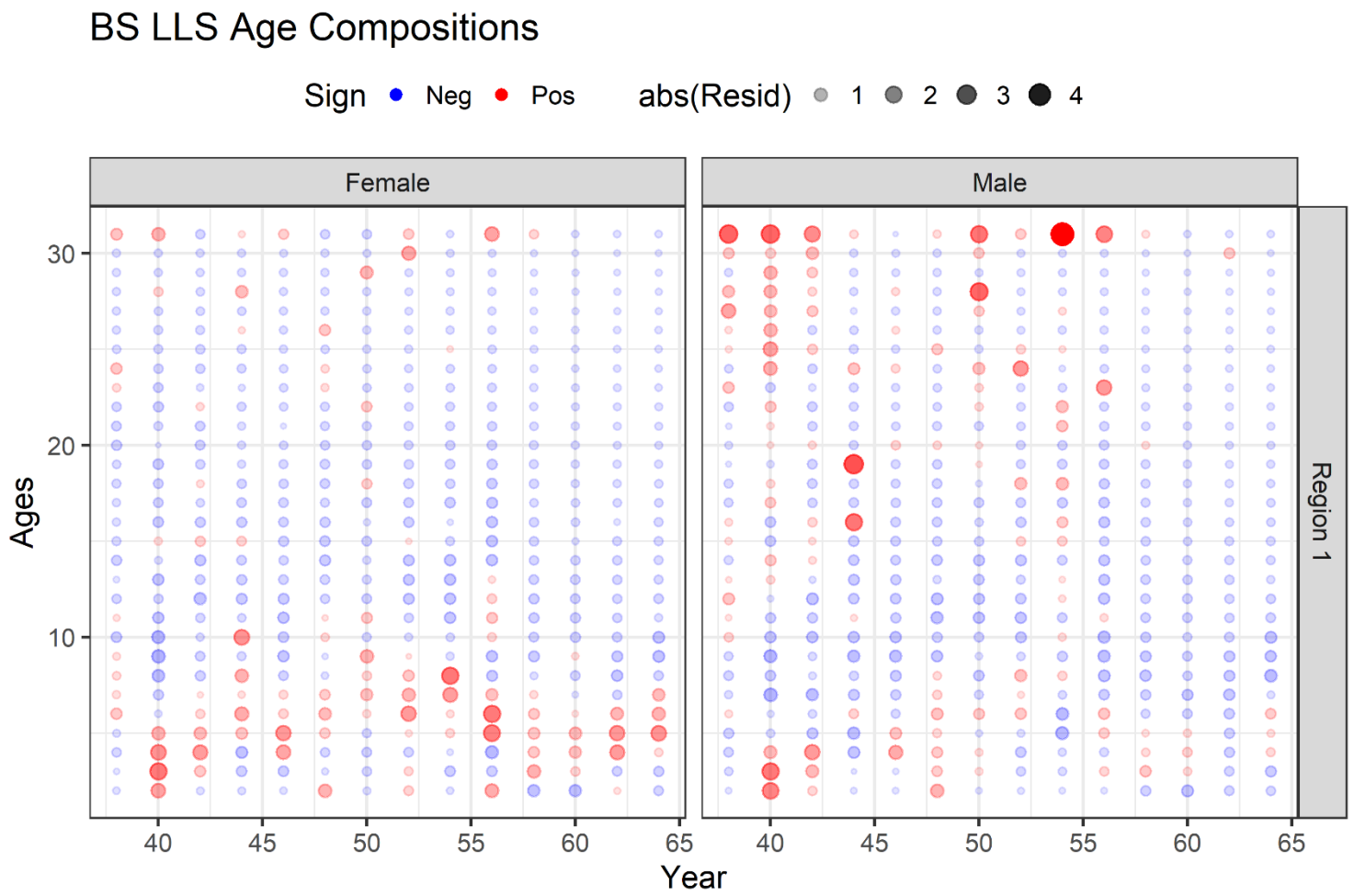


Figure 3.43. Fits to BS NOAA Longline Survey age composition data based on Pearson residuals for Model *26.2_FAA*.

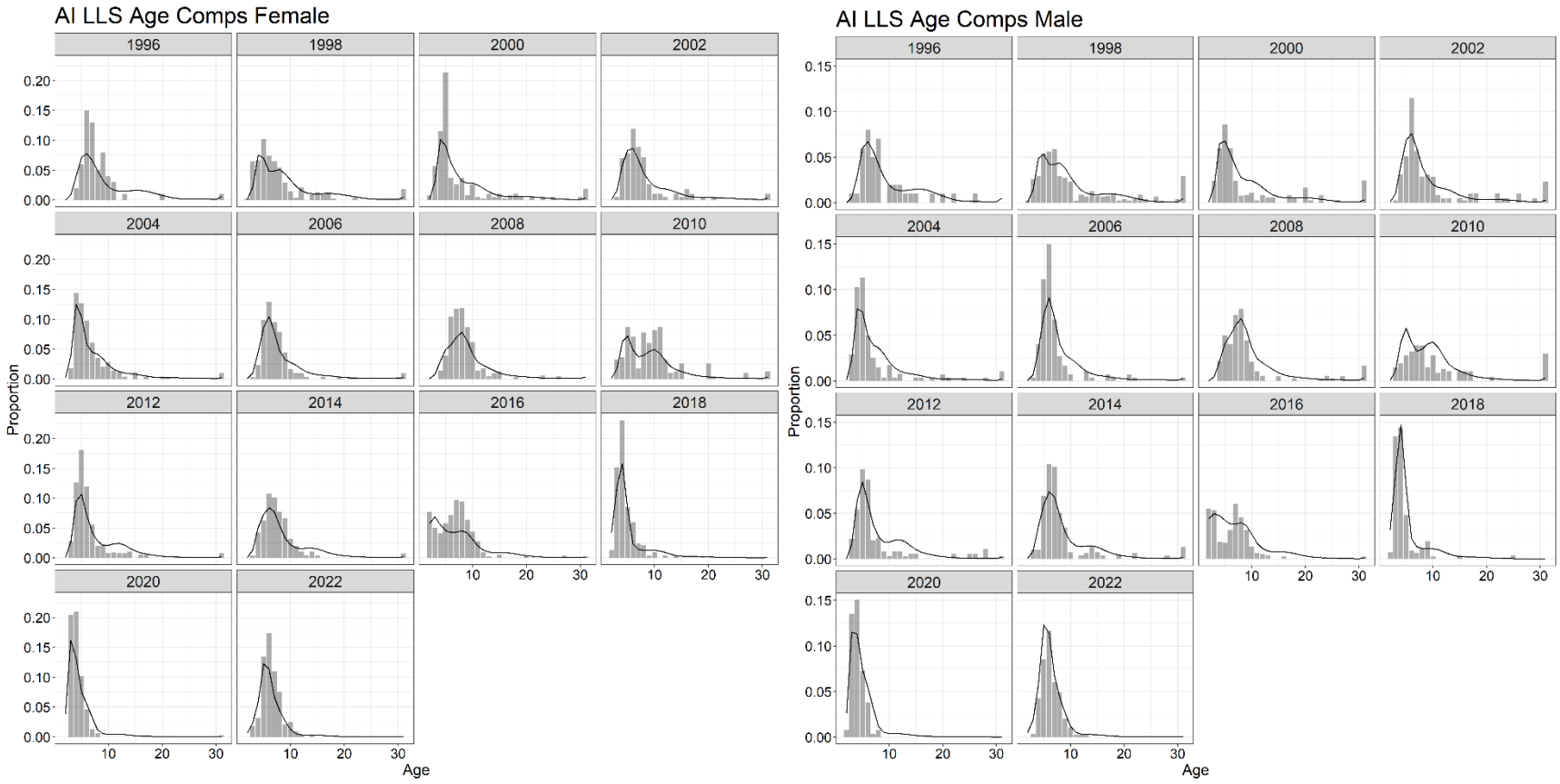


Figure 3.44. Fits to AI NOAA Longline Survey age composition data by year and sex (females left and males right) for Model *26.2_FAA*.

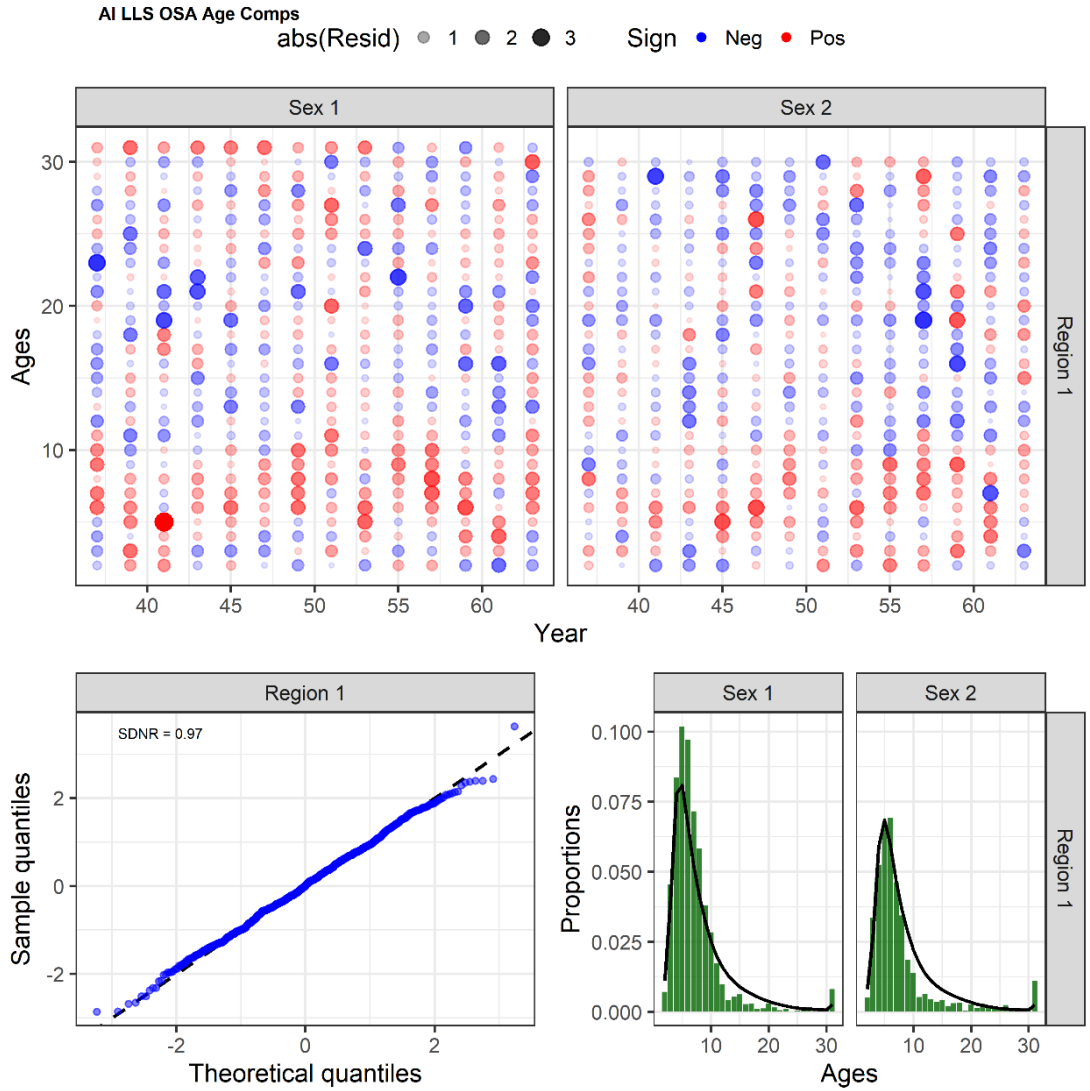


Figure 3.45. Fits to AI NOAA Longline Survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (mean across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA*.

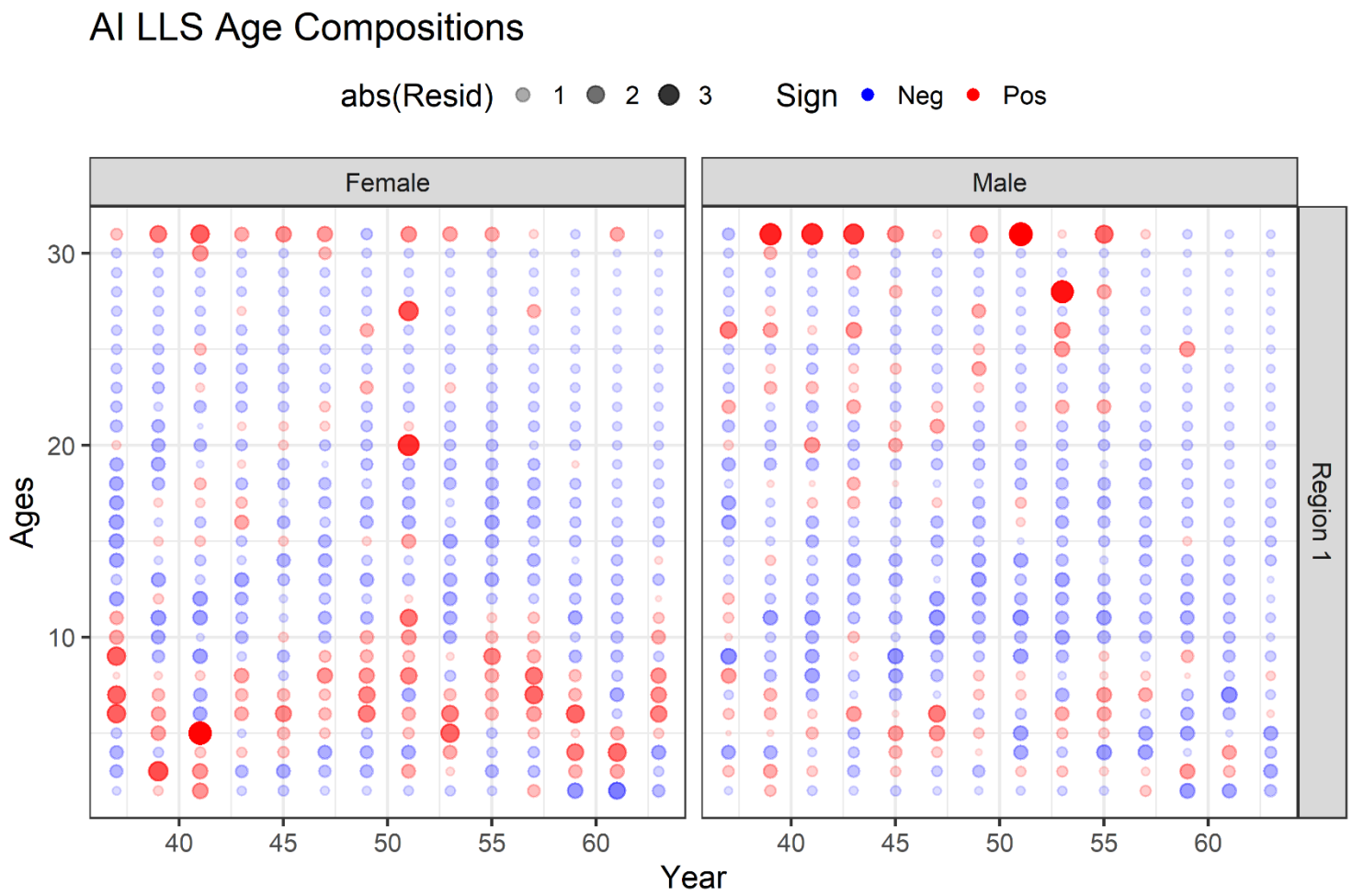


Figure 3.46. Fits to AI NOAA longline survey age composition data based on Pearson residuals for Model *26.2_FAA*.



Figure 3.47. Fits to fixed gear fishery age composition data by year and sex (females left and males right) for Model *26.1_1Reg*.

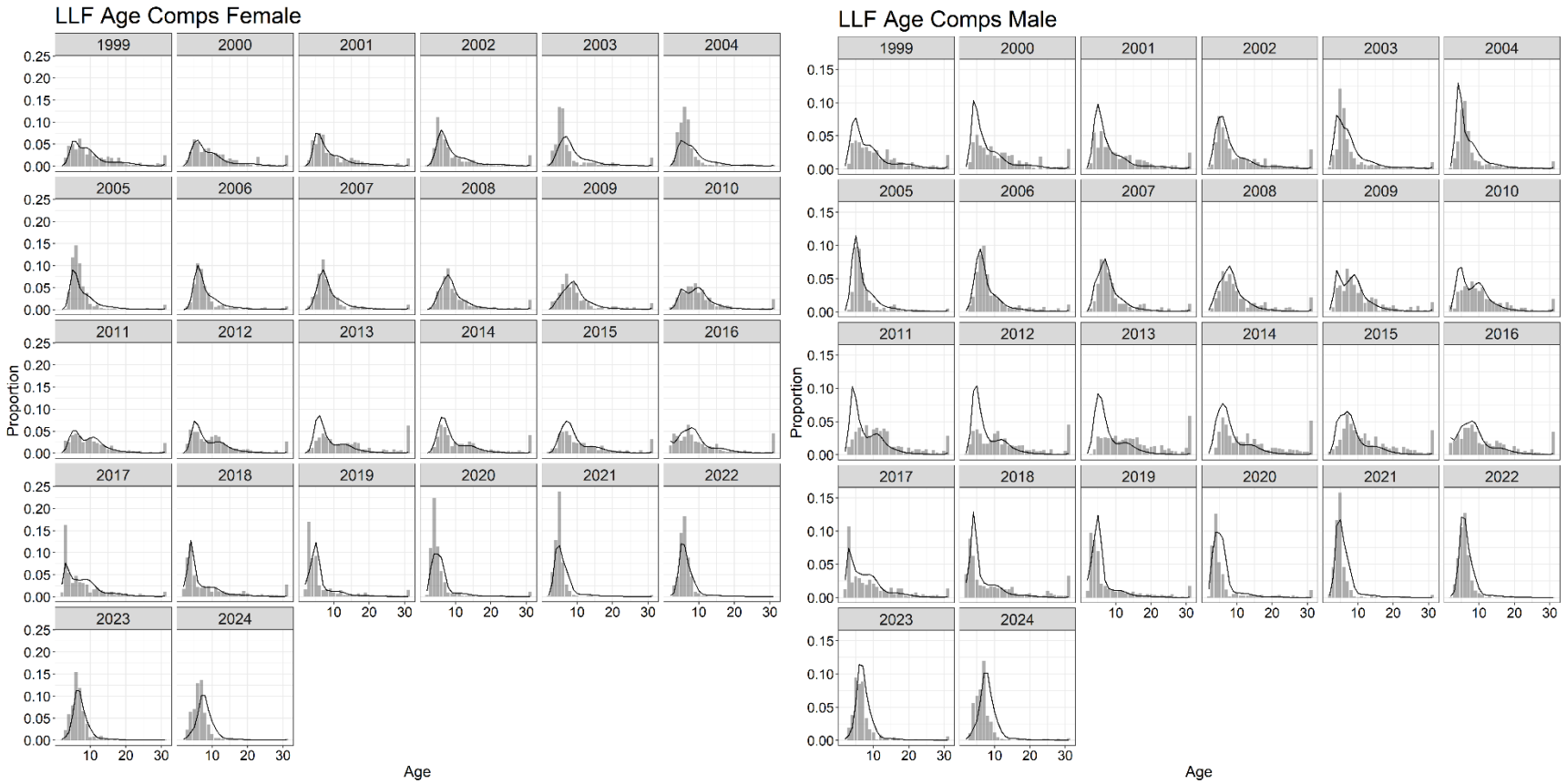


Figure 3.48. Fits to fixed gear fishery age composition data by year and sex (females left and males right) for Model *26.2_FAA*.

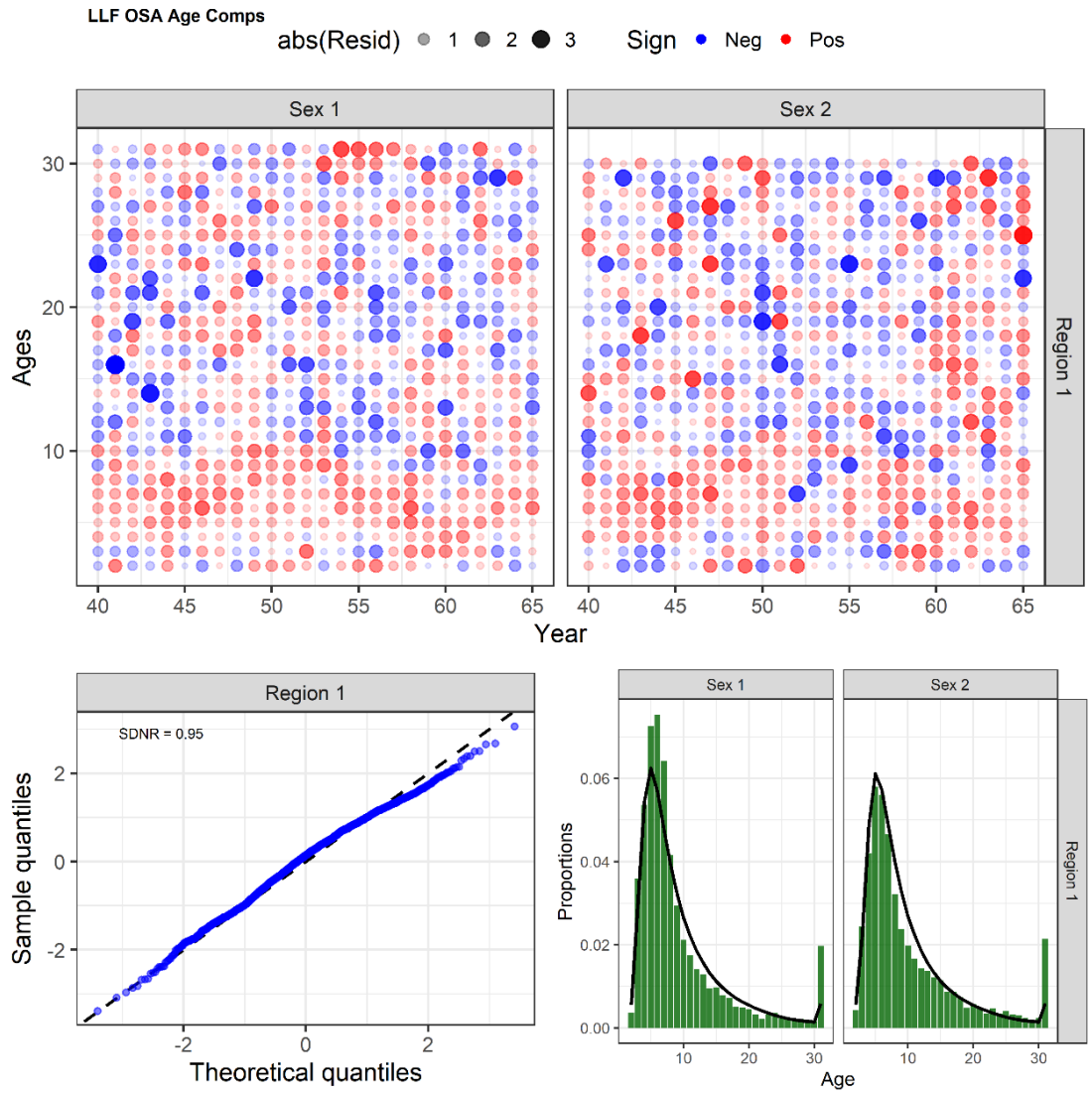


Figure 3.49. Fits to fixed gear fishery age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (mean across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg*.

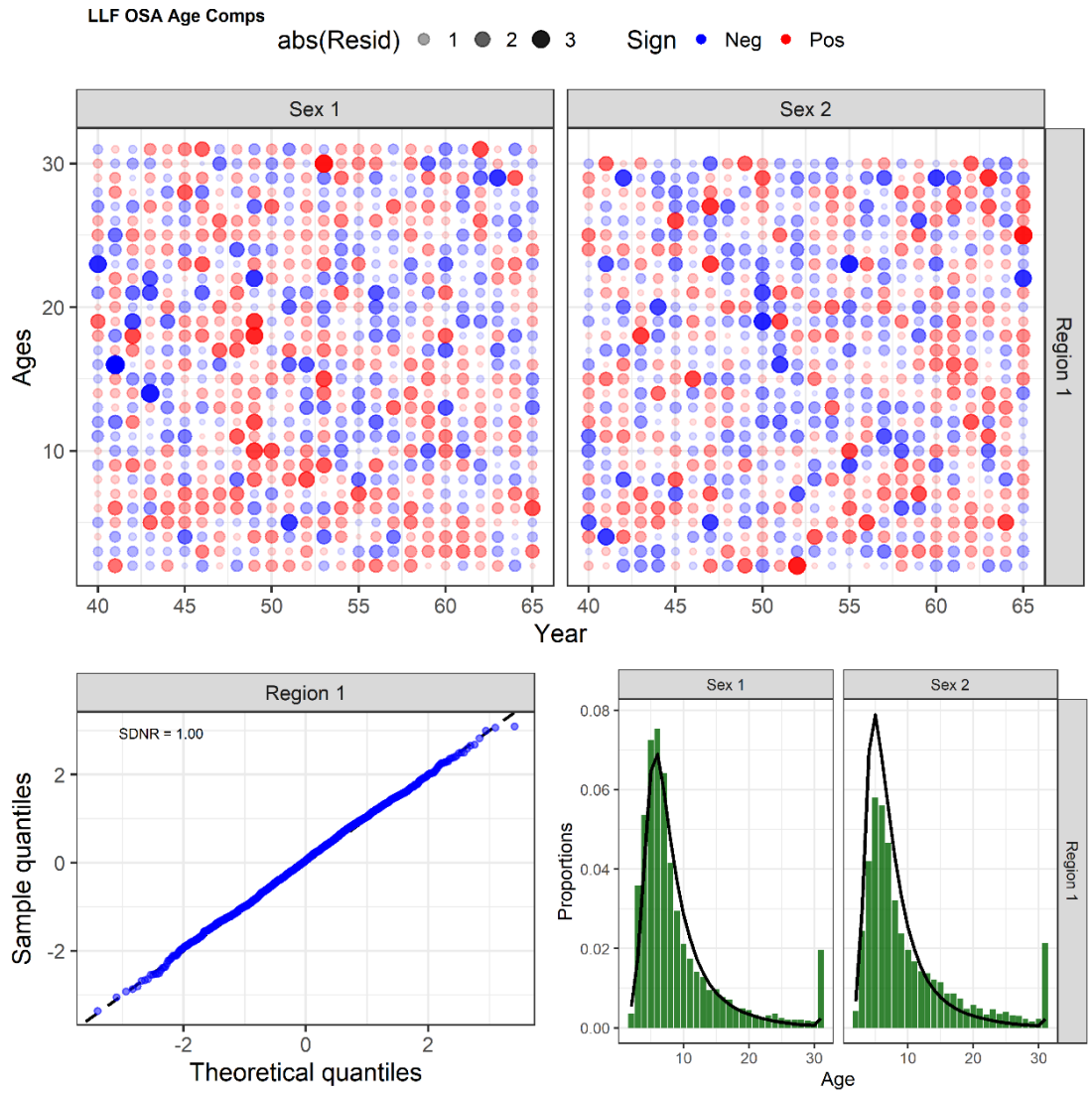


Figure 3.50. Fits to fixed gear fishery age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (mean across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA*.

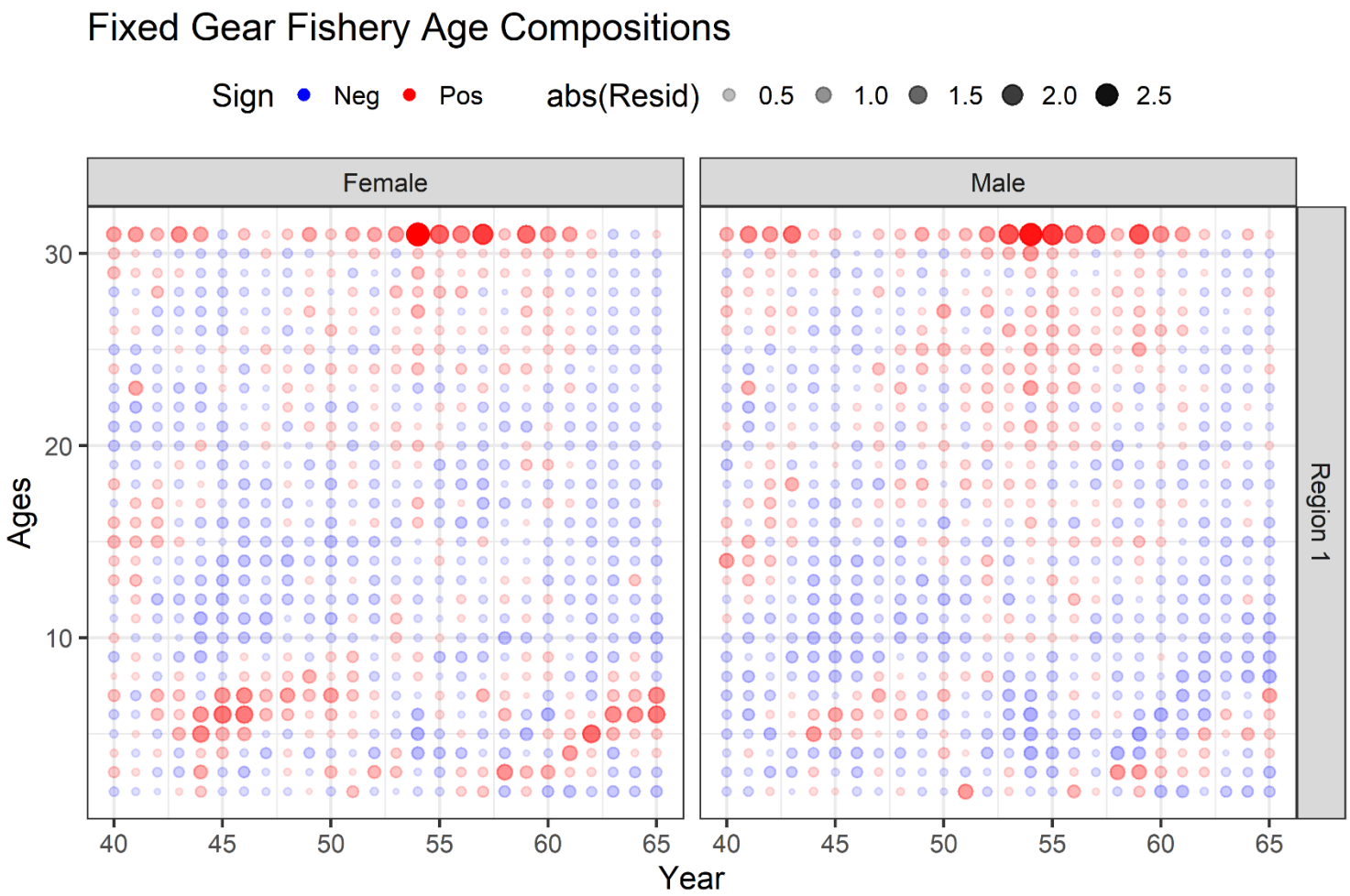


Figure 3.51. Fits to the fixed gear fishery age composition data based on Pearson residuals for Model *26.1_1Reg*.

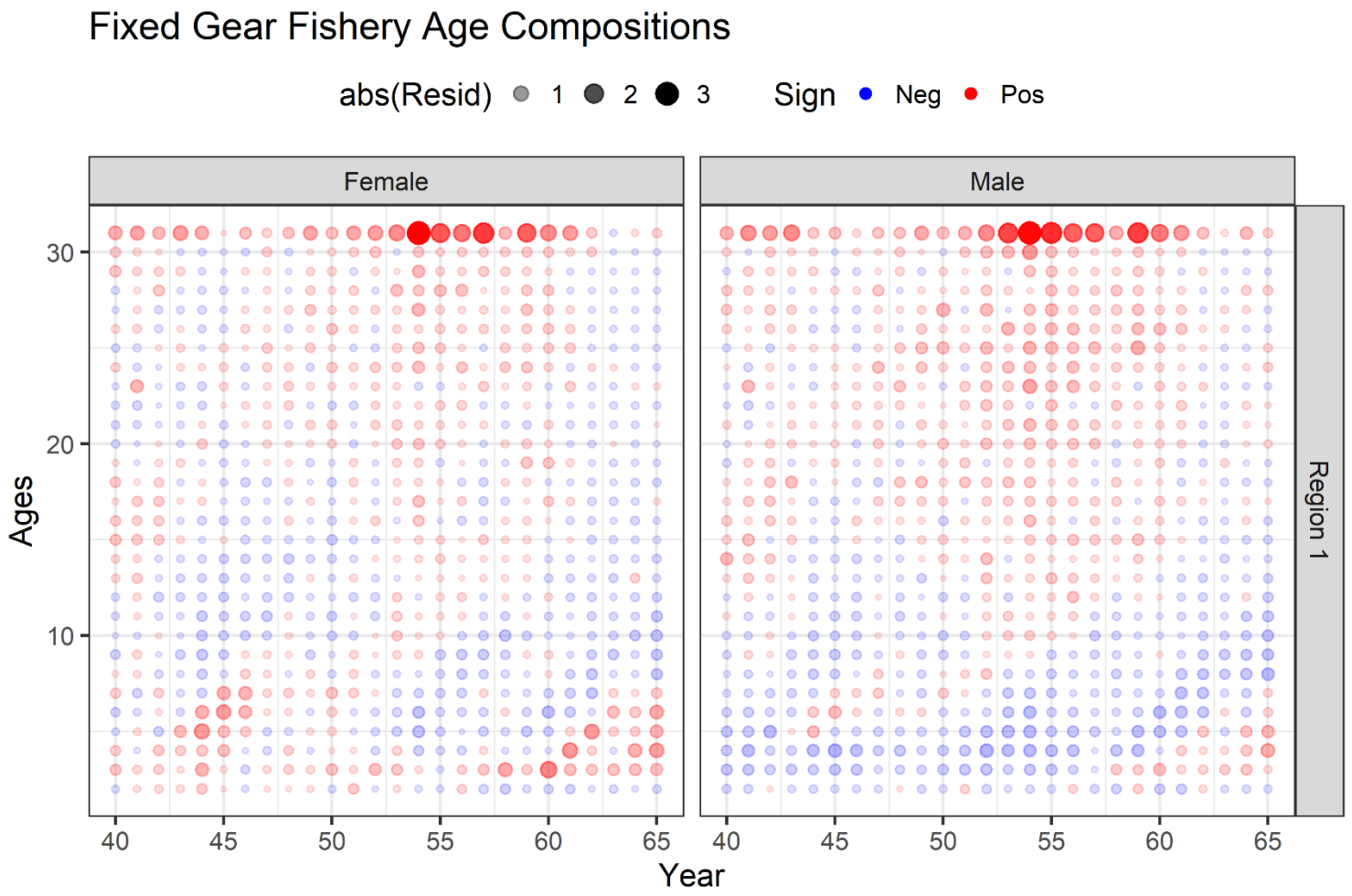


Figure 3.52. Fits to the fixed gear fishery age composition data based on Pearson residuals for Model *26. 2_FAA*.

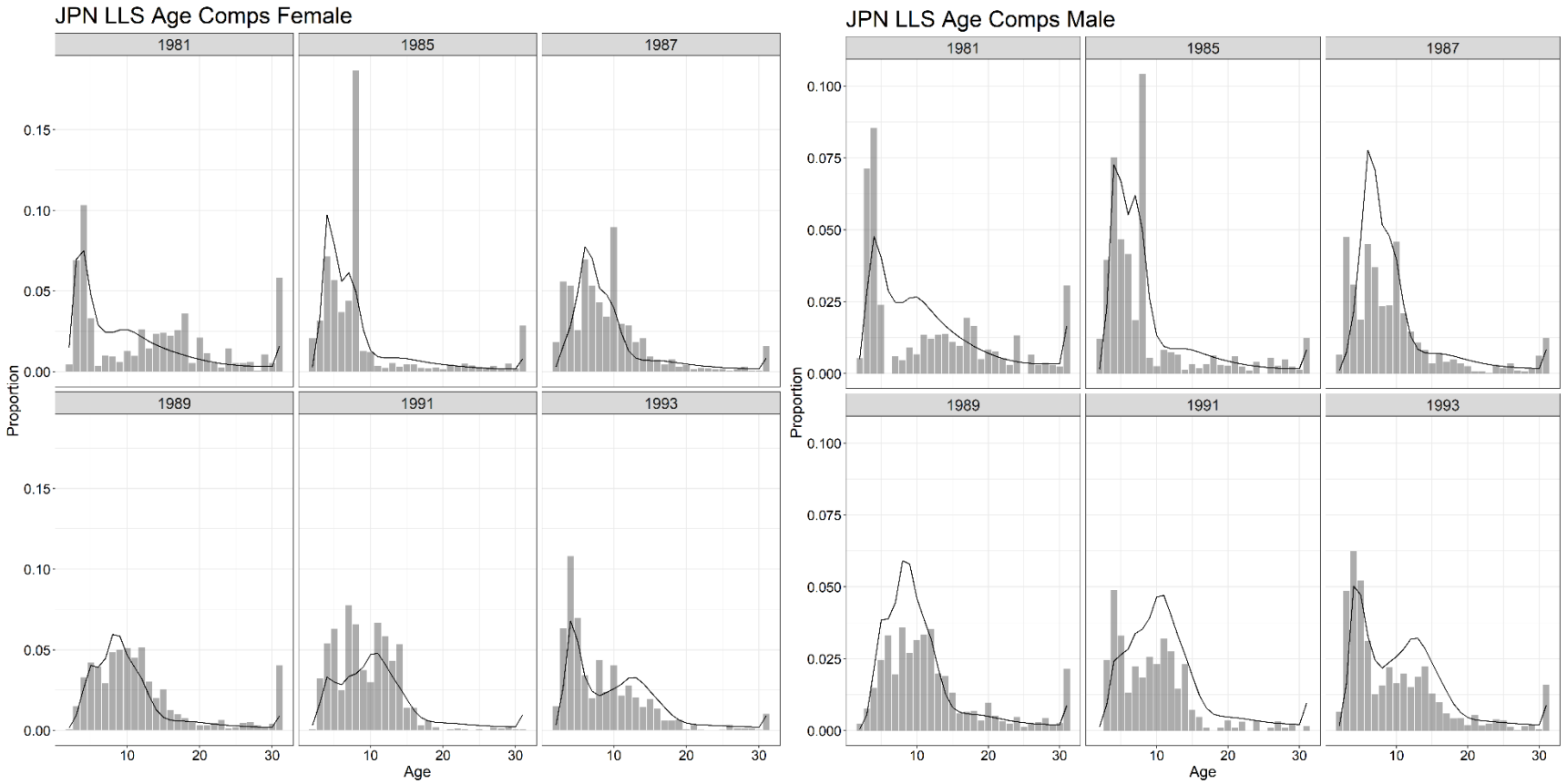


Figure 3.53. Fits to Japanese Longline Survey age composition data by year and sex (females left and males right) for model *26.1_1Reg*.

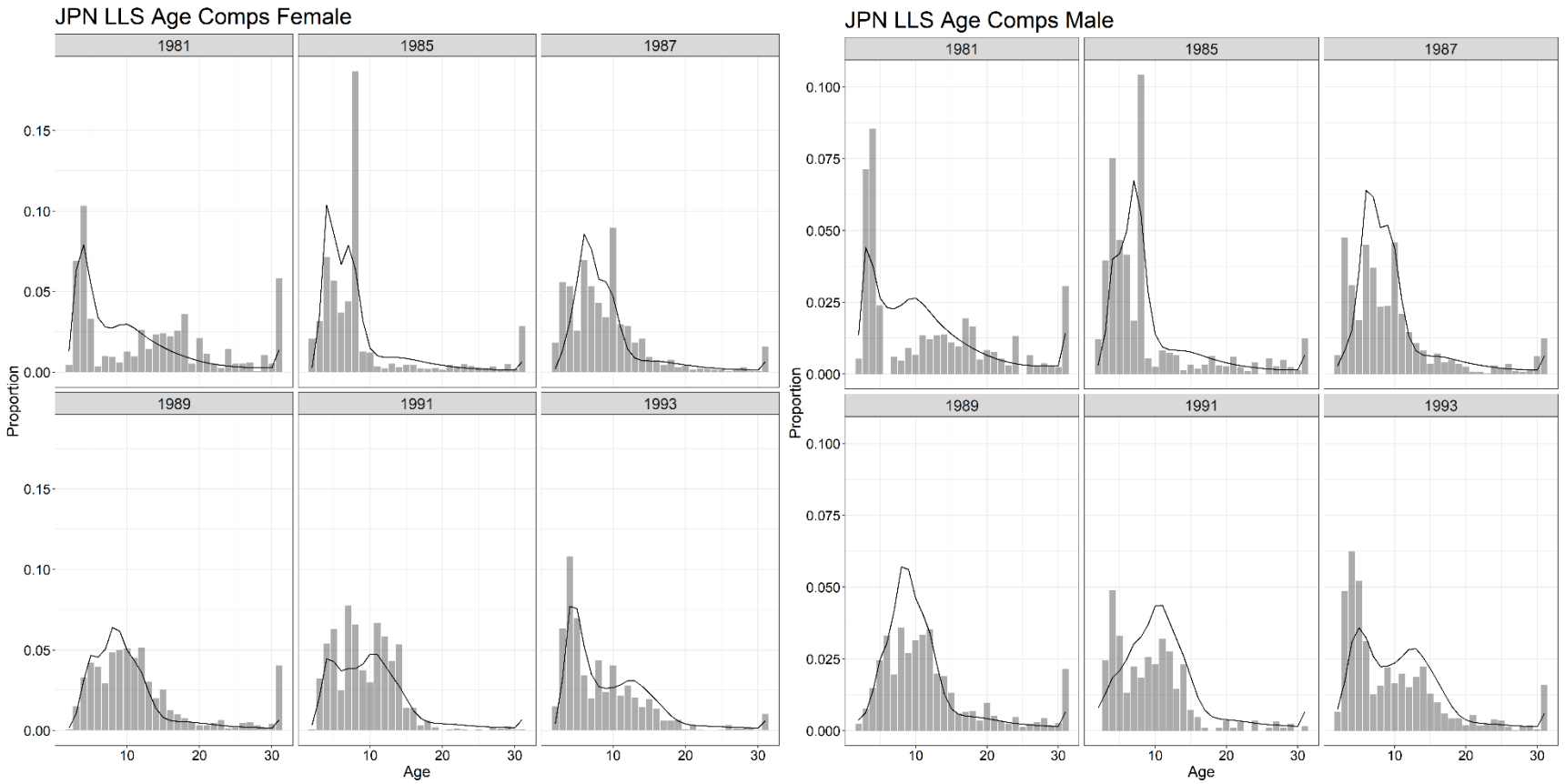


Figure 3.54. Fits to Japanese Longline Survey age composition data by year and sex (females left and males right) for Model *26.2_FAA*.

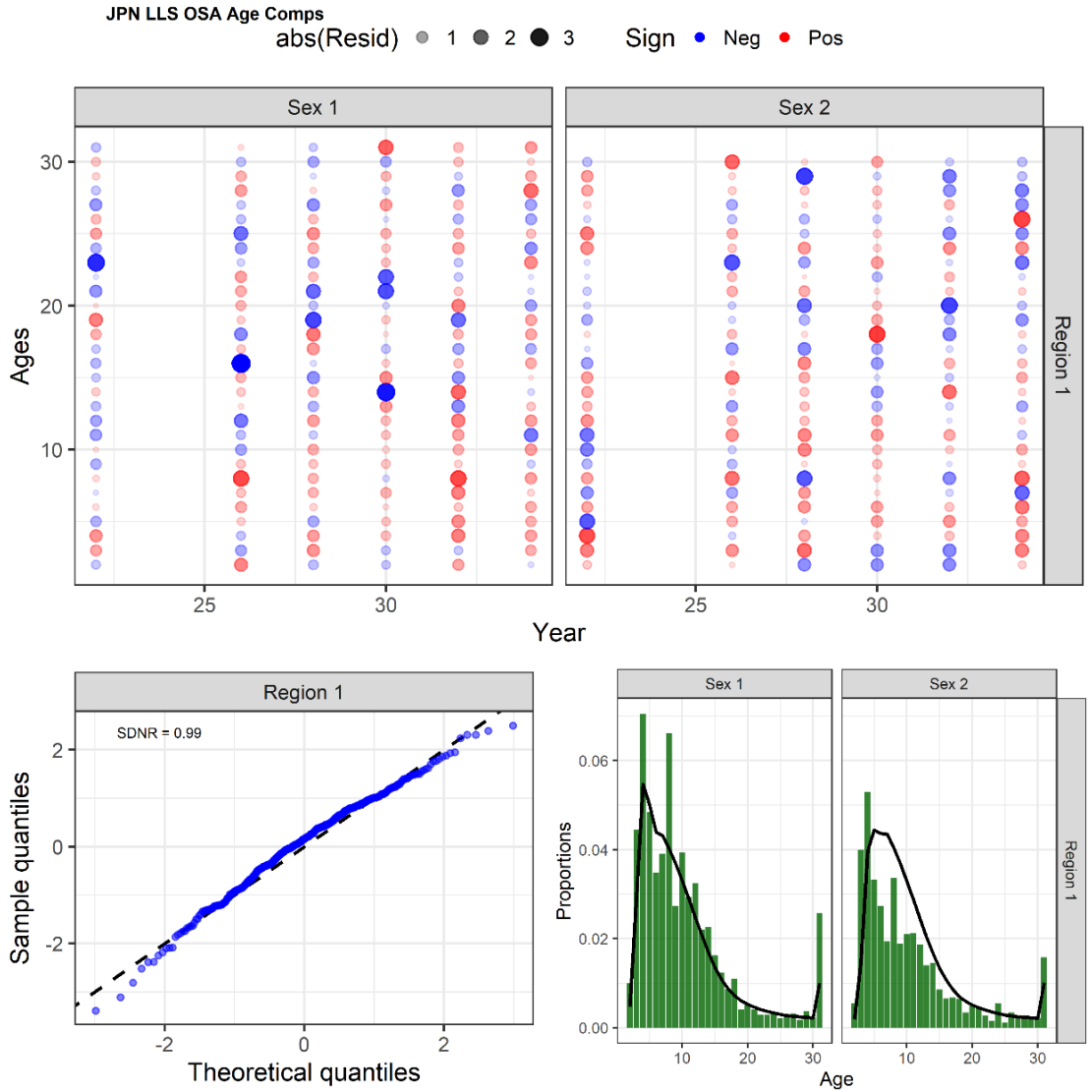


Figure 3.55. Fits to Japanese Longline Survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (mean across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg*.

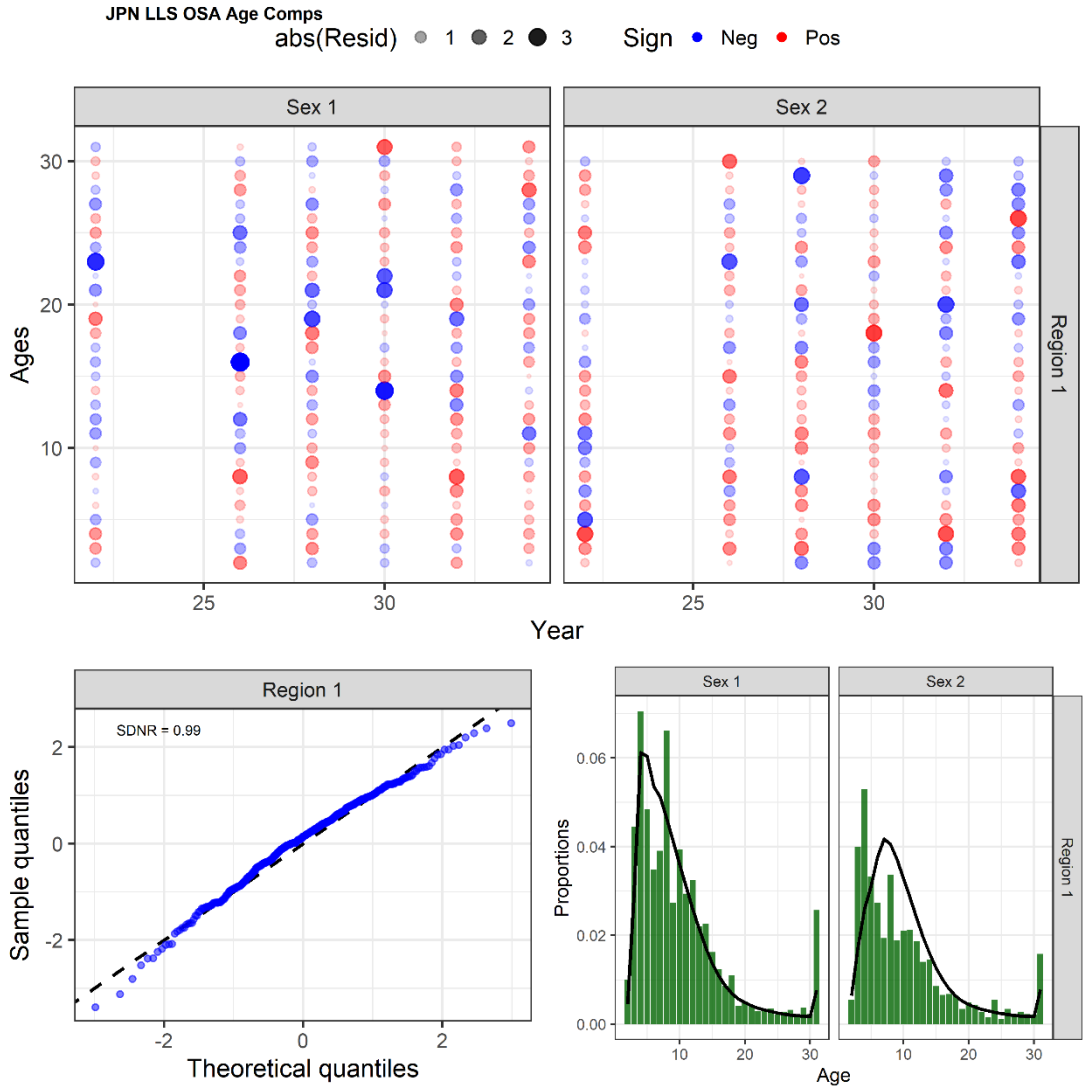


Figure 3.56. Fits to Japanese longline survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (mean across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA*.

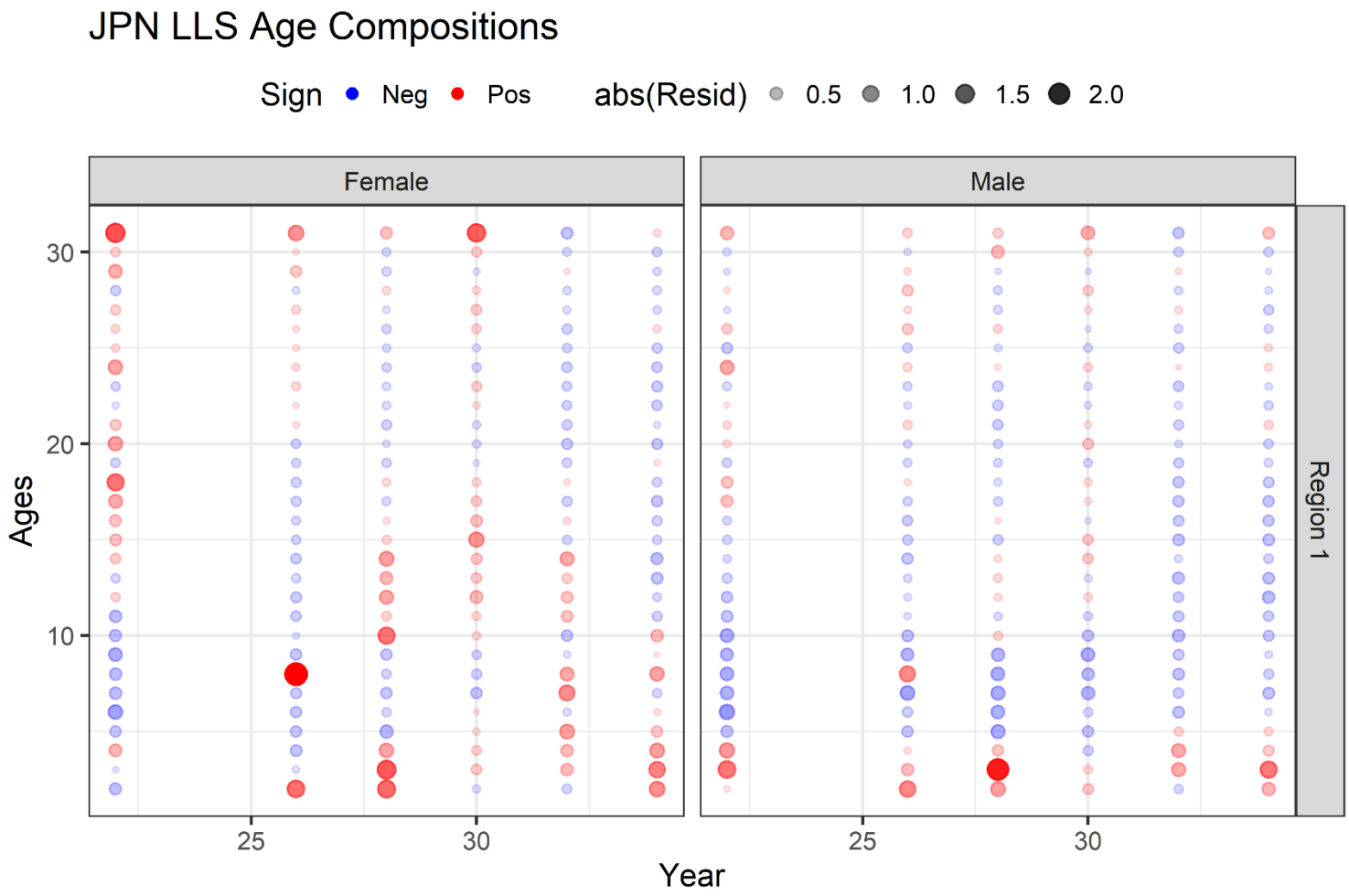


Figure 3.57. Fits to the Japanese Longline Survey age composition data based on Pearson residuals for Model *26.1_1Reg*.

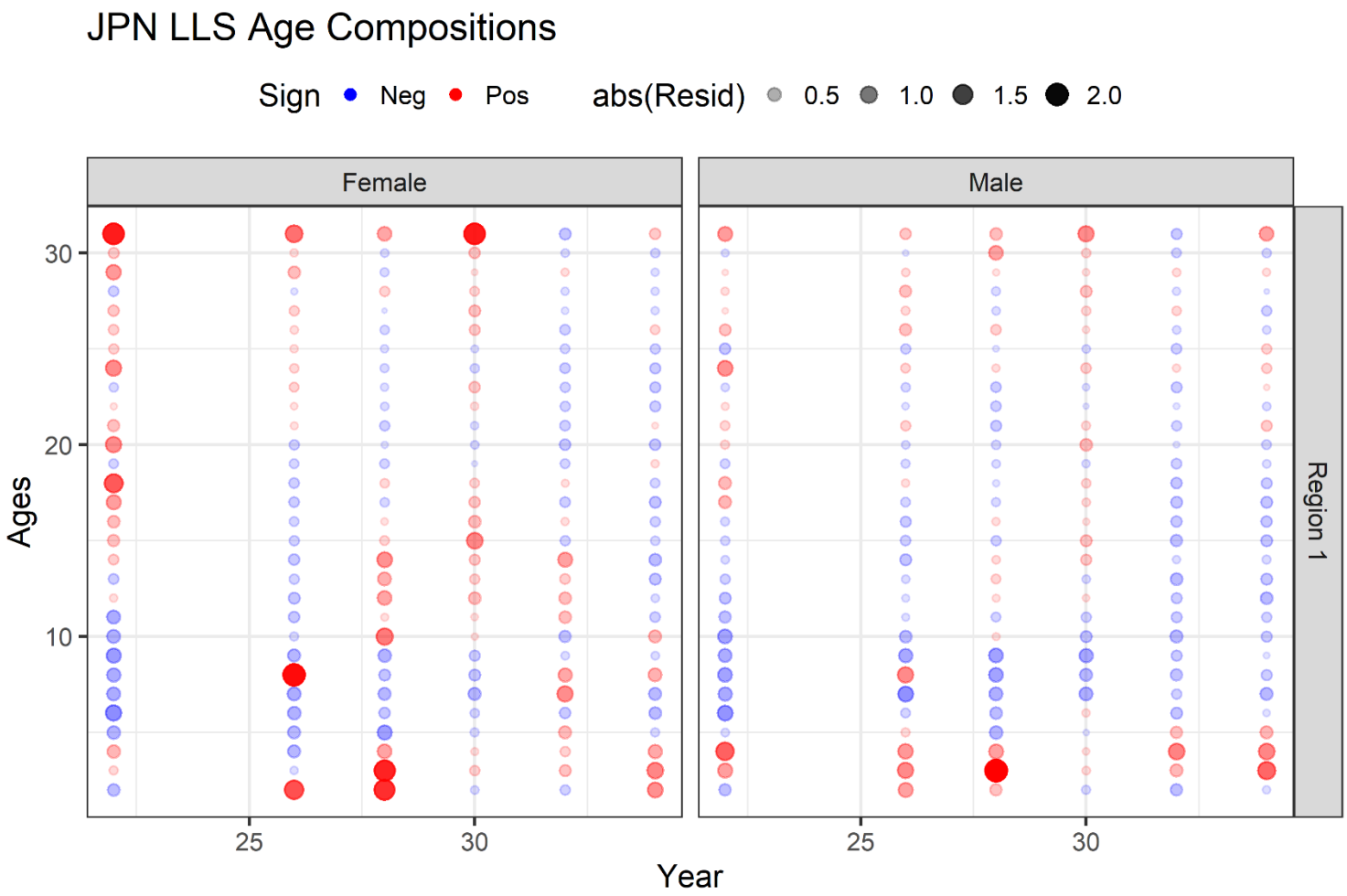


Figure 3.58. Fits to the Japanese Longline Survey age composition data based on Pearson residuals for Model *26.2_FAA*.

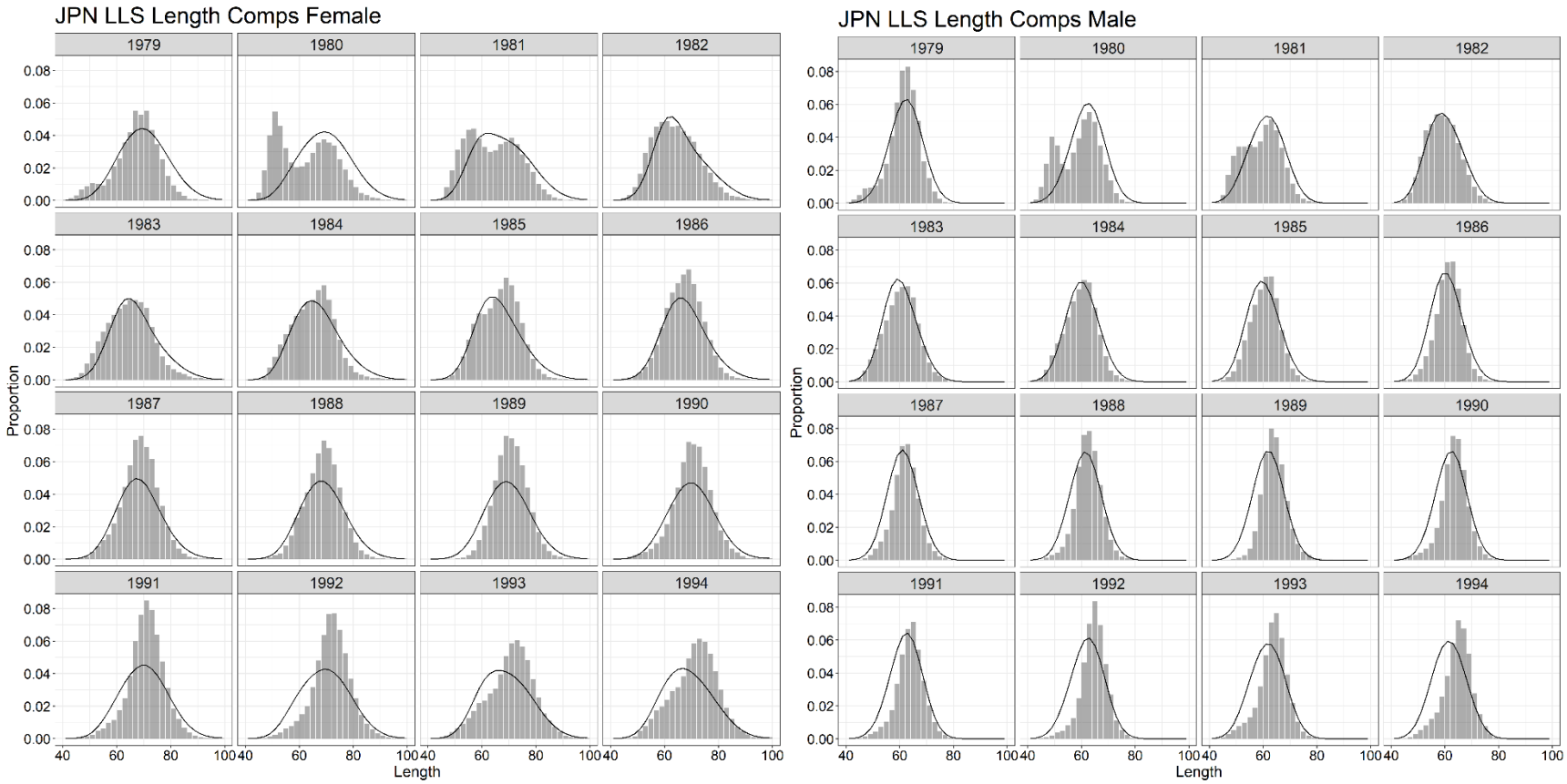


Figure 3.59. Fits to Japanese Longline Survey length composition data by year and sex (females left and males right) for Model *26.1_1Reg*.

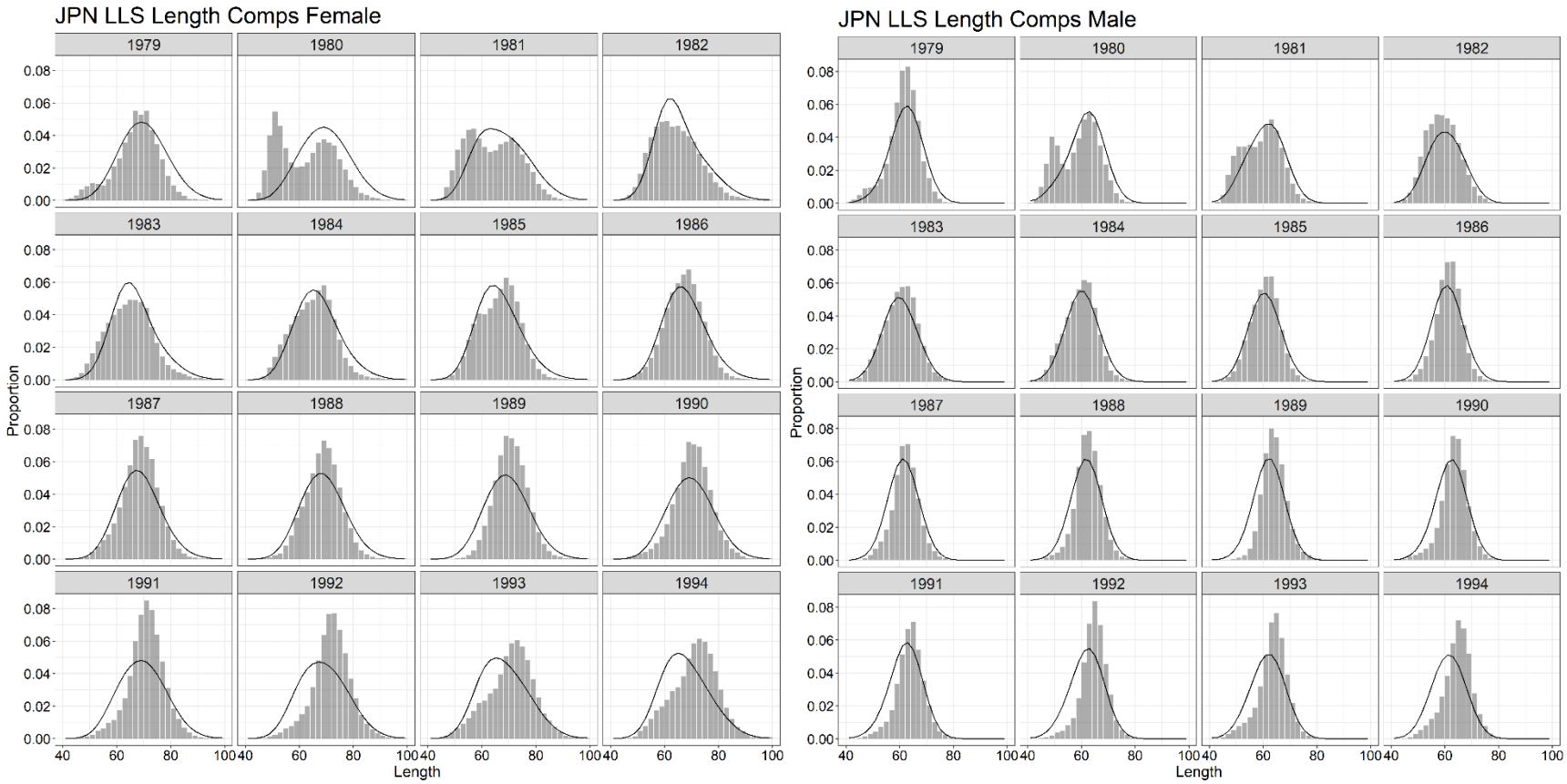


Figure 3.60. Fits to Japanese Longline Survey length composition data by year and sex (females left and males right) for Model *26.2_FAA*.

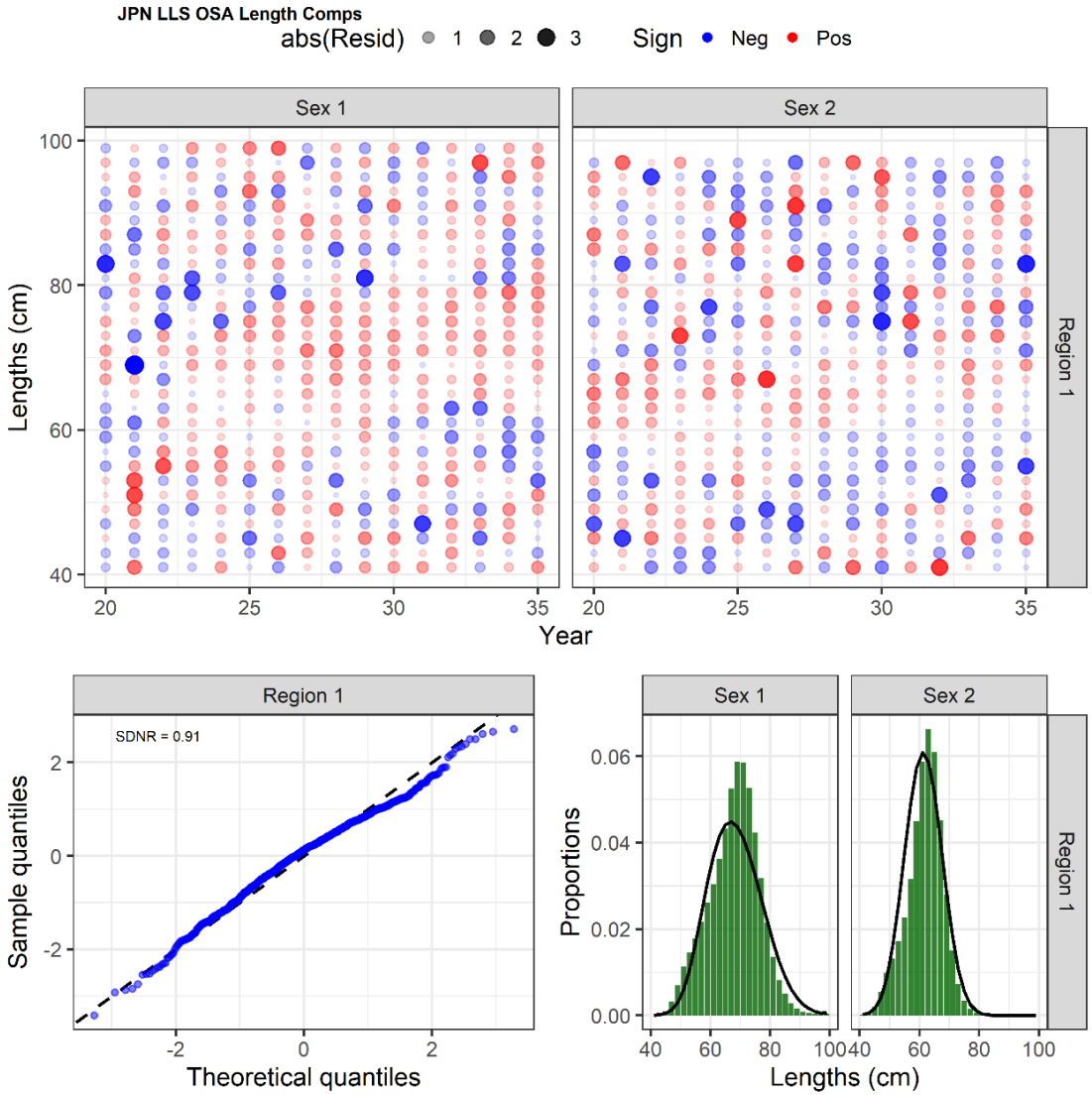


Figure 3.61. Fits to Japanese Longline Survey length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (mean across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg*.

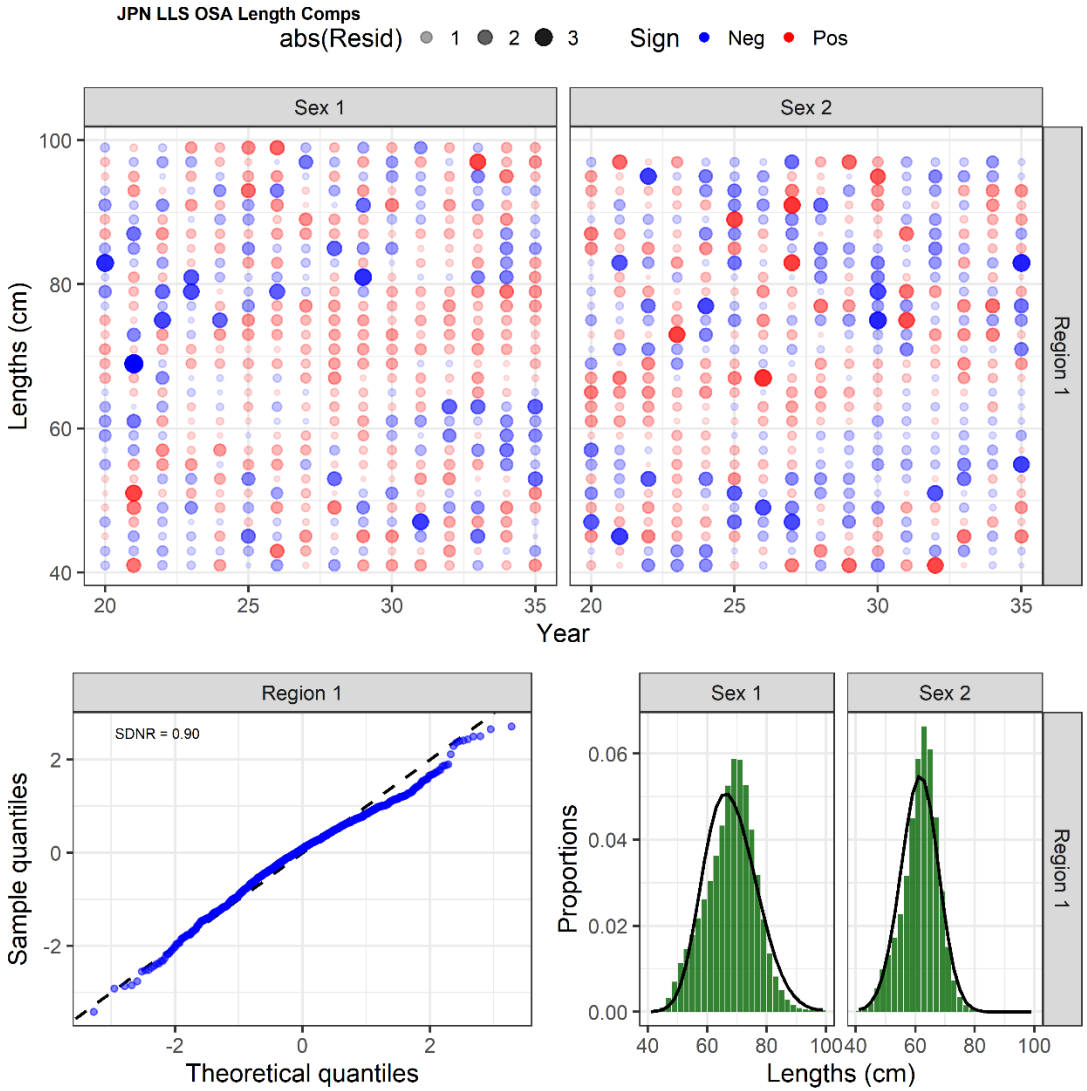


Figure 3.62. Fits to Japanese Longline Survey length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (mean across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA*.

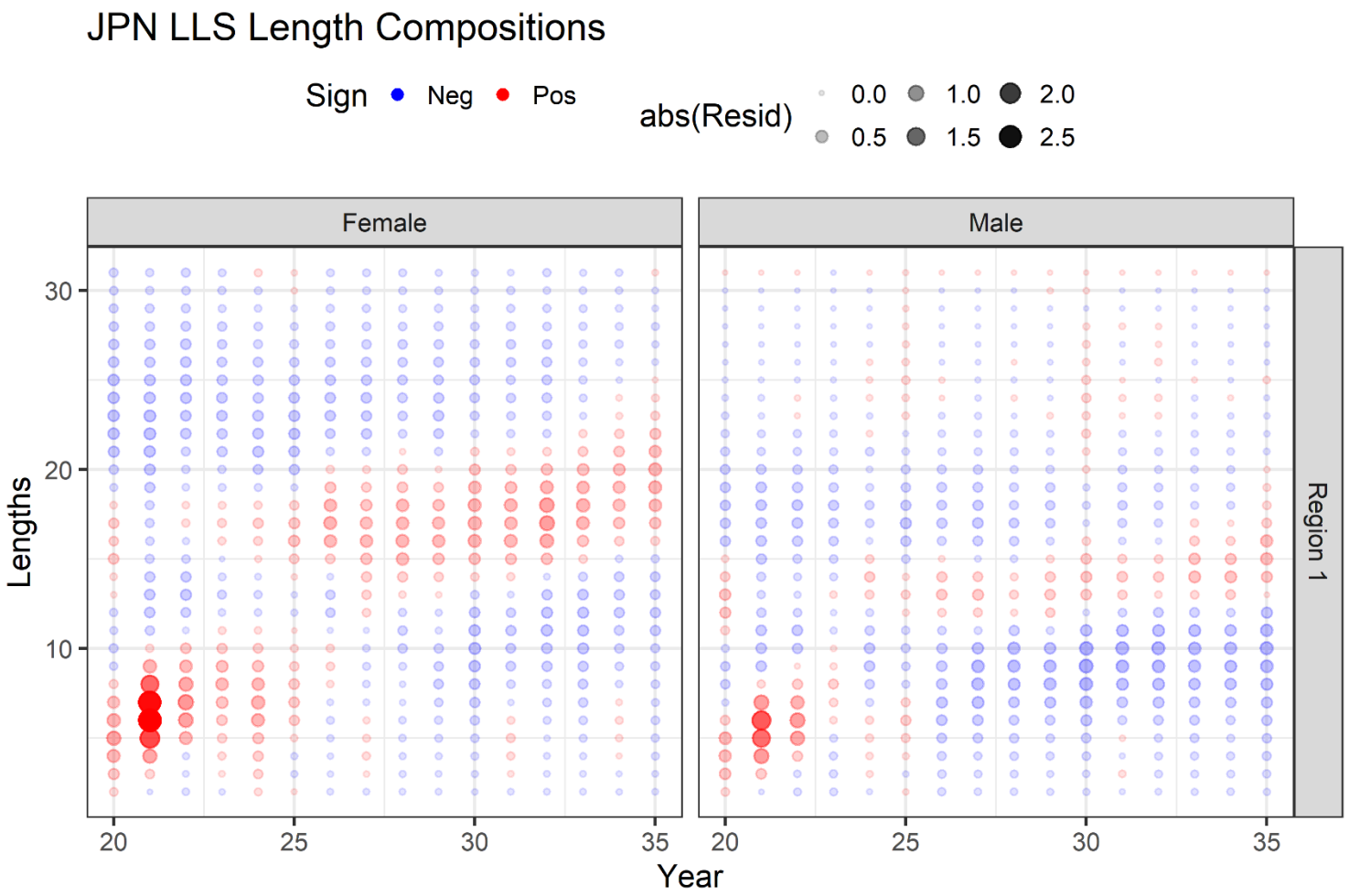


Figure 3.63. Fits to the Japanese Longline Survey length composition data based on Pearson residuals for Model *26.1_1Reg*.

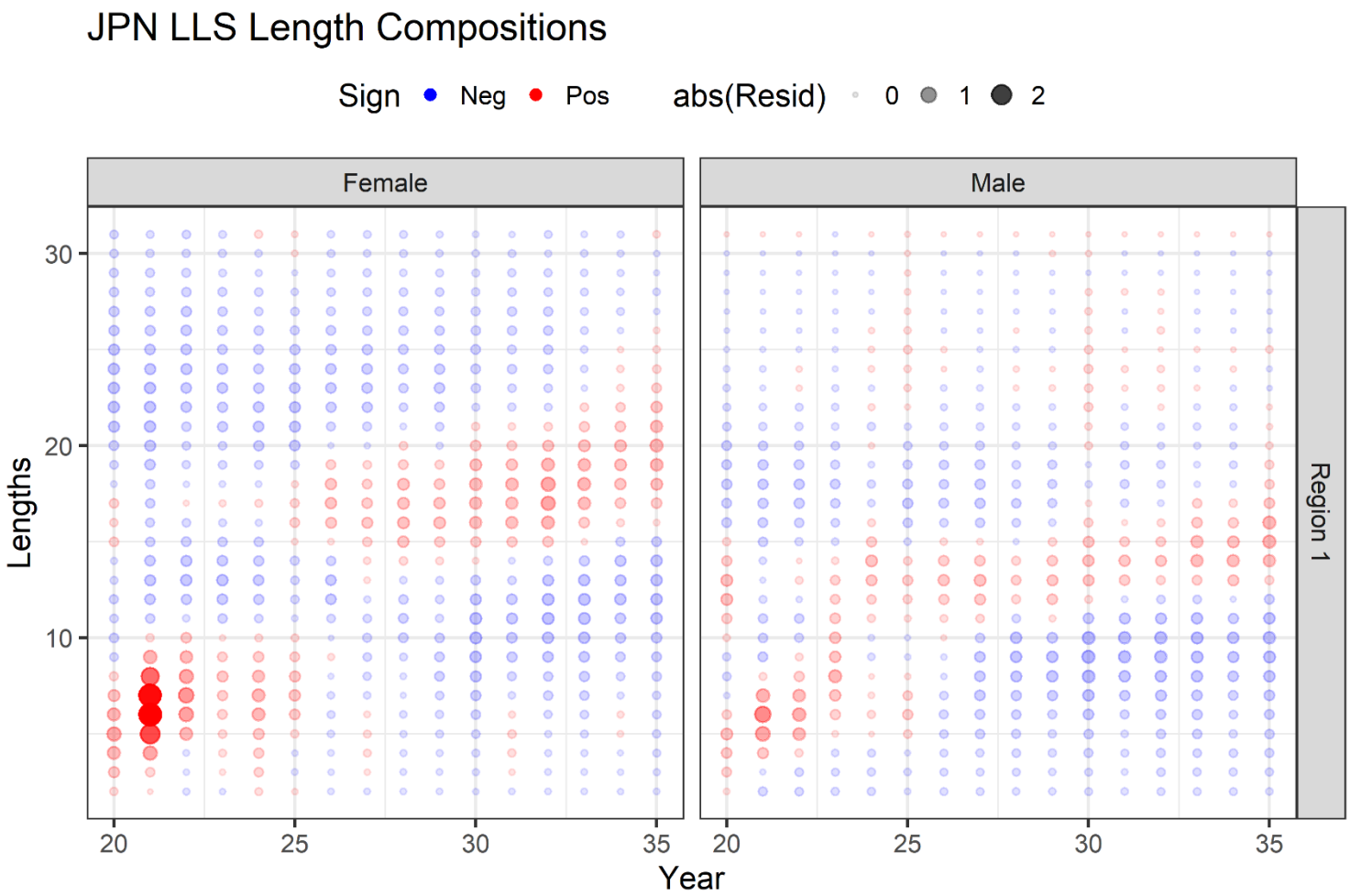


Figure 3.64. Fits to the Japanese Longline Survey length composition data based on Pearson residuals for Model *26.2_FAA*.

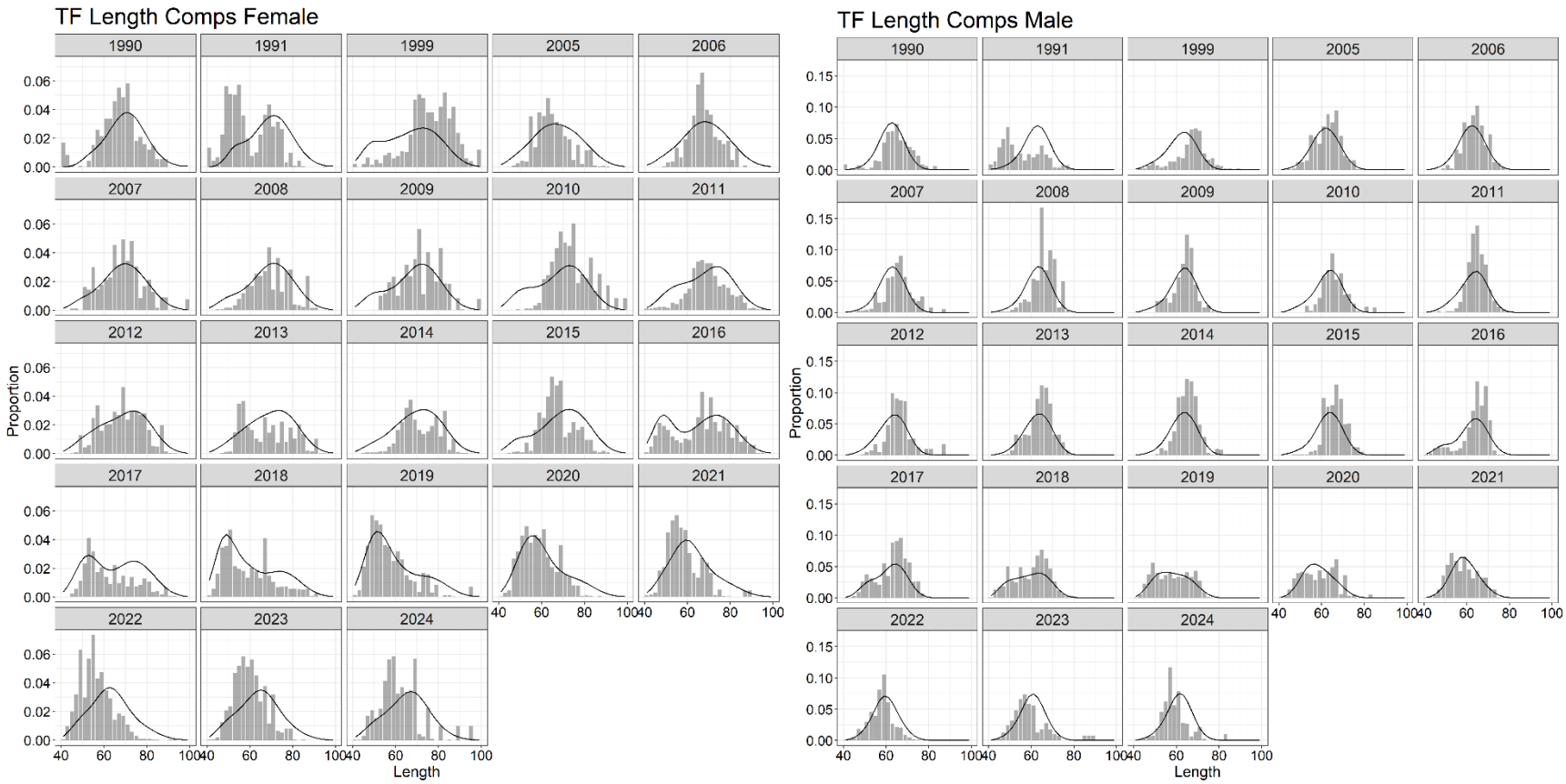


Figure 3.65. Fits to the trawl fishery length composition data by year and sex (females left and males right) for Model *26.1_1Reg*.

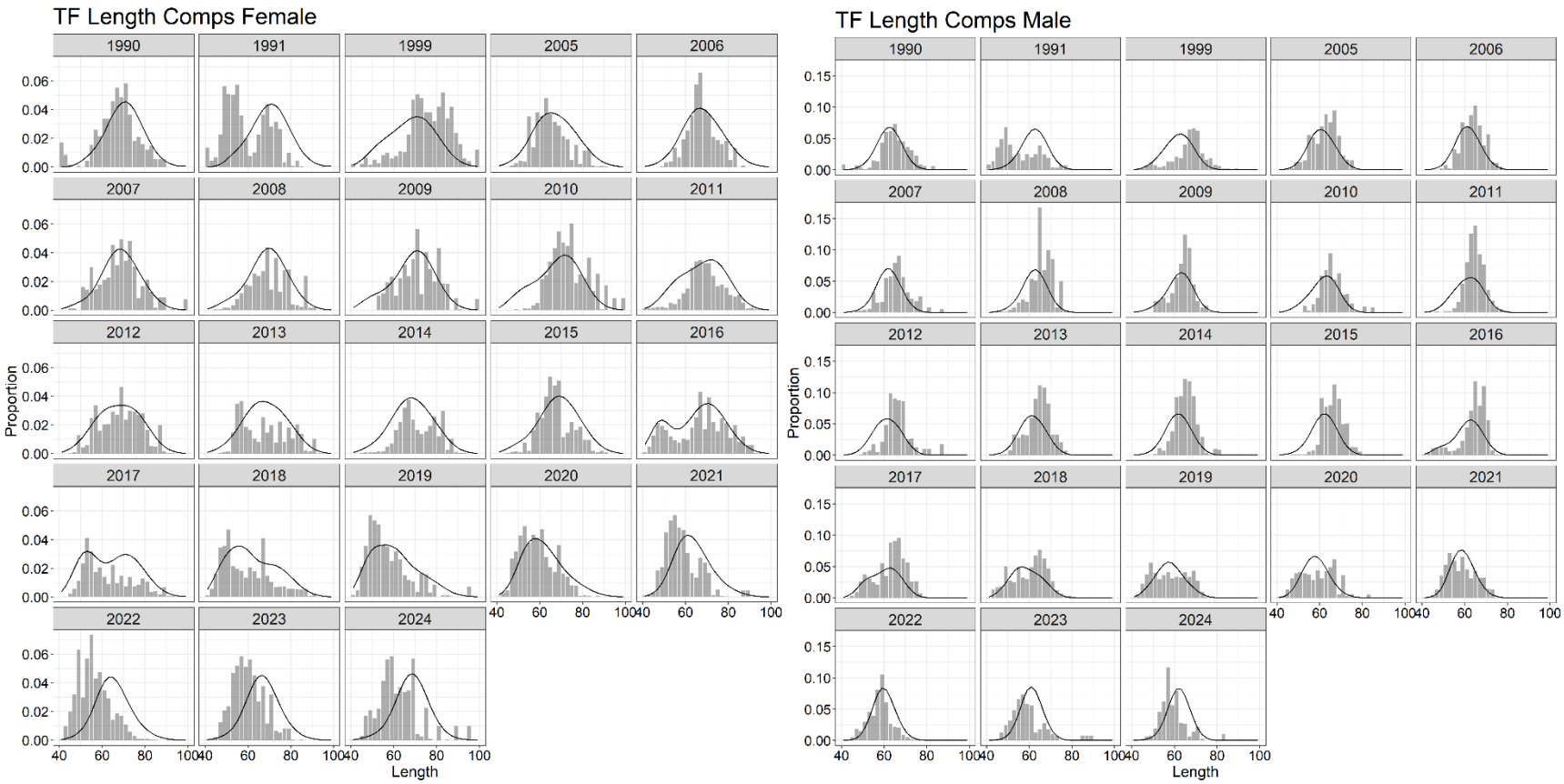


Figure 3.66. Fits to the trawl fishery length composition data by year and sex (females left and males right) for Model *26.2_FAA*.

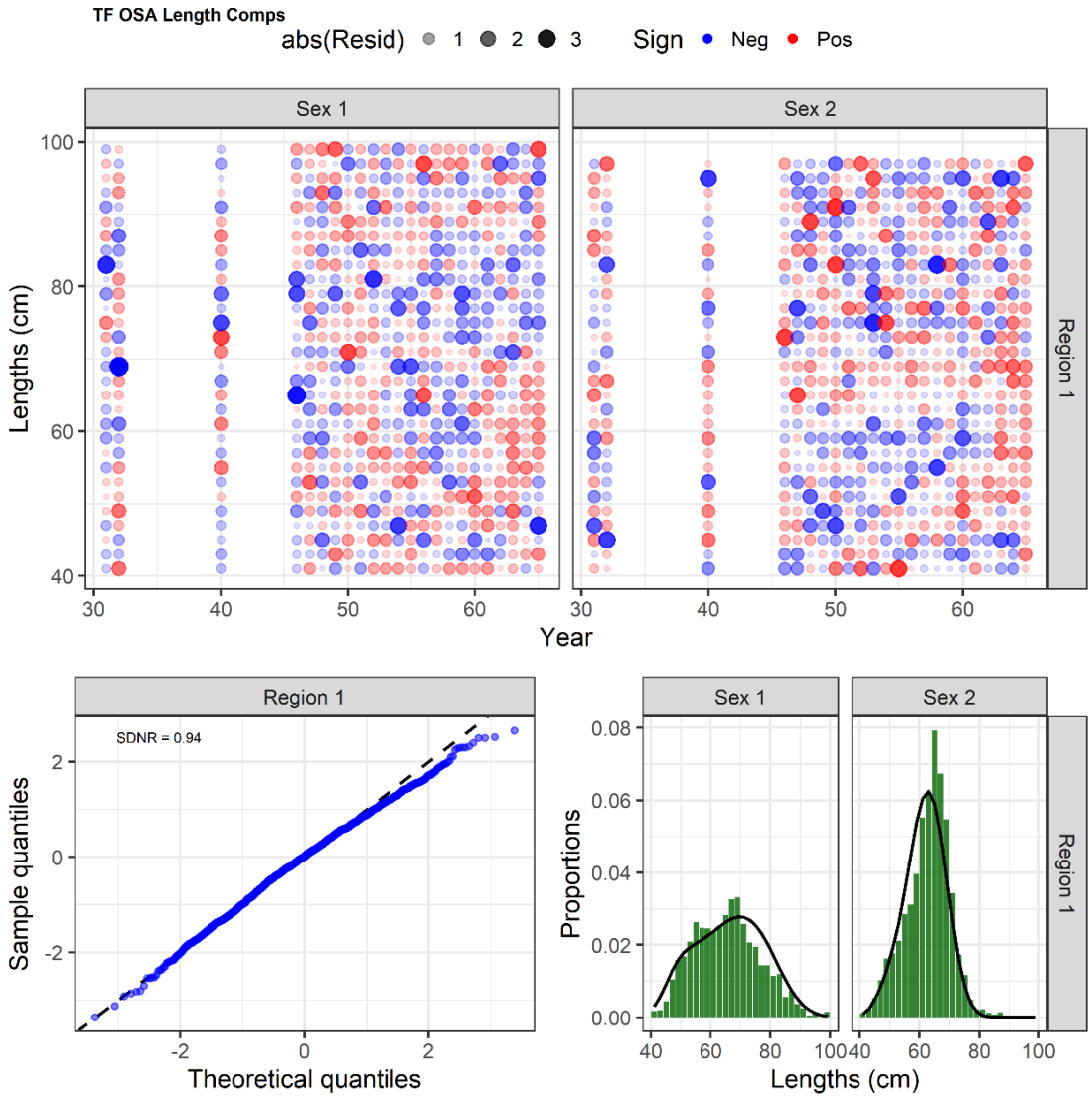


Figure 3.67. Fits to trawl fishery length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (mean across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg*.

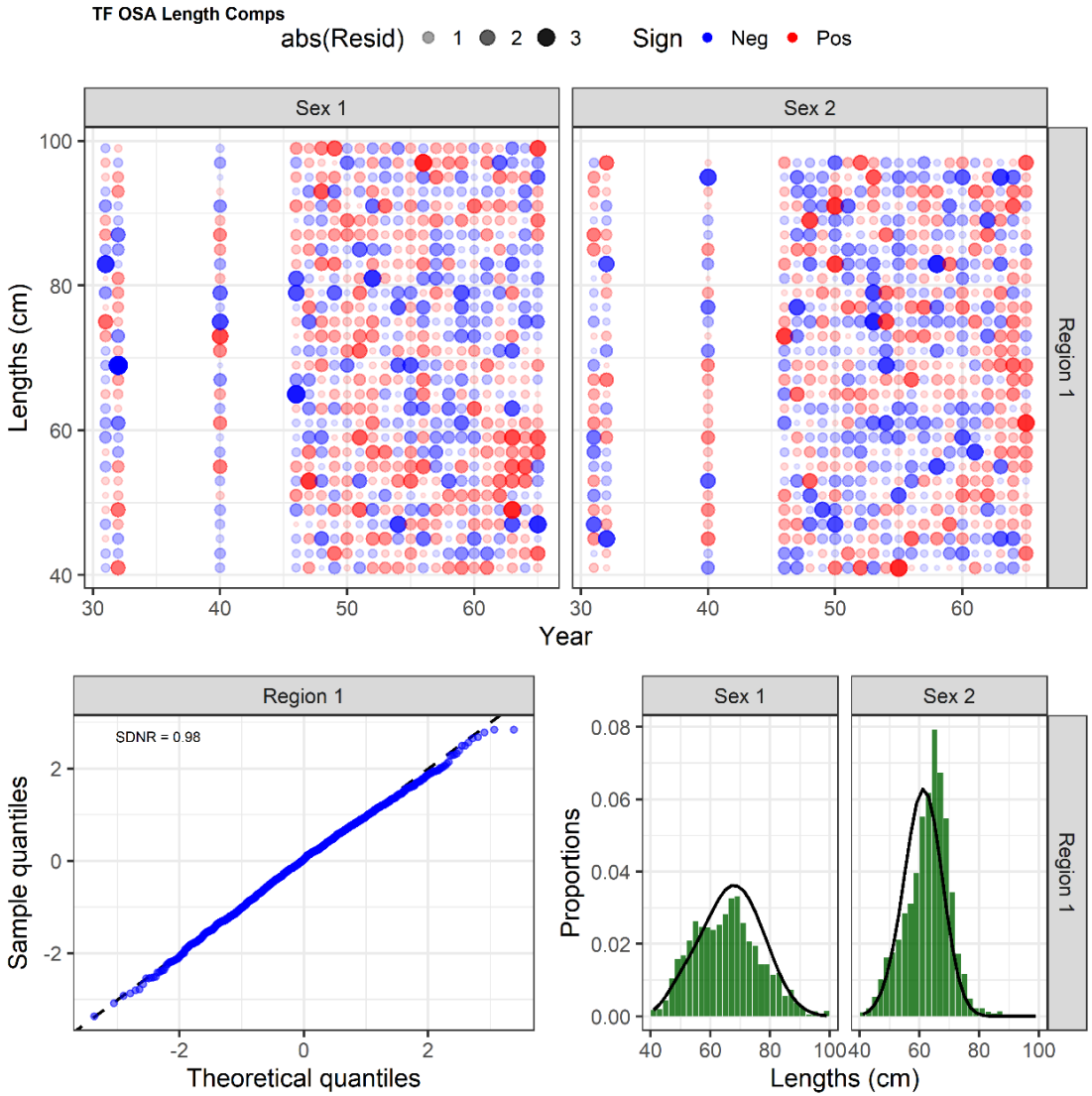


Figure 3.68. Fits to trawl fishery length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (mean across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA*.

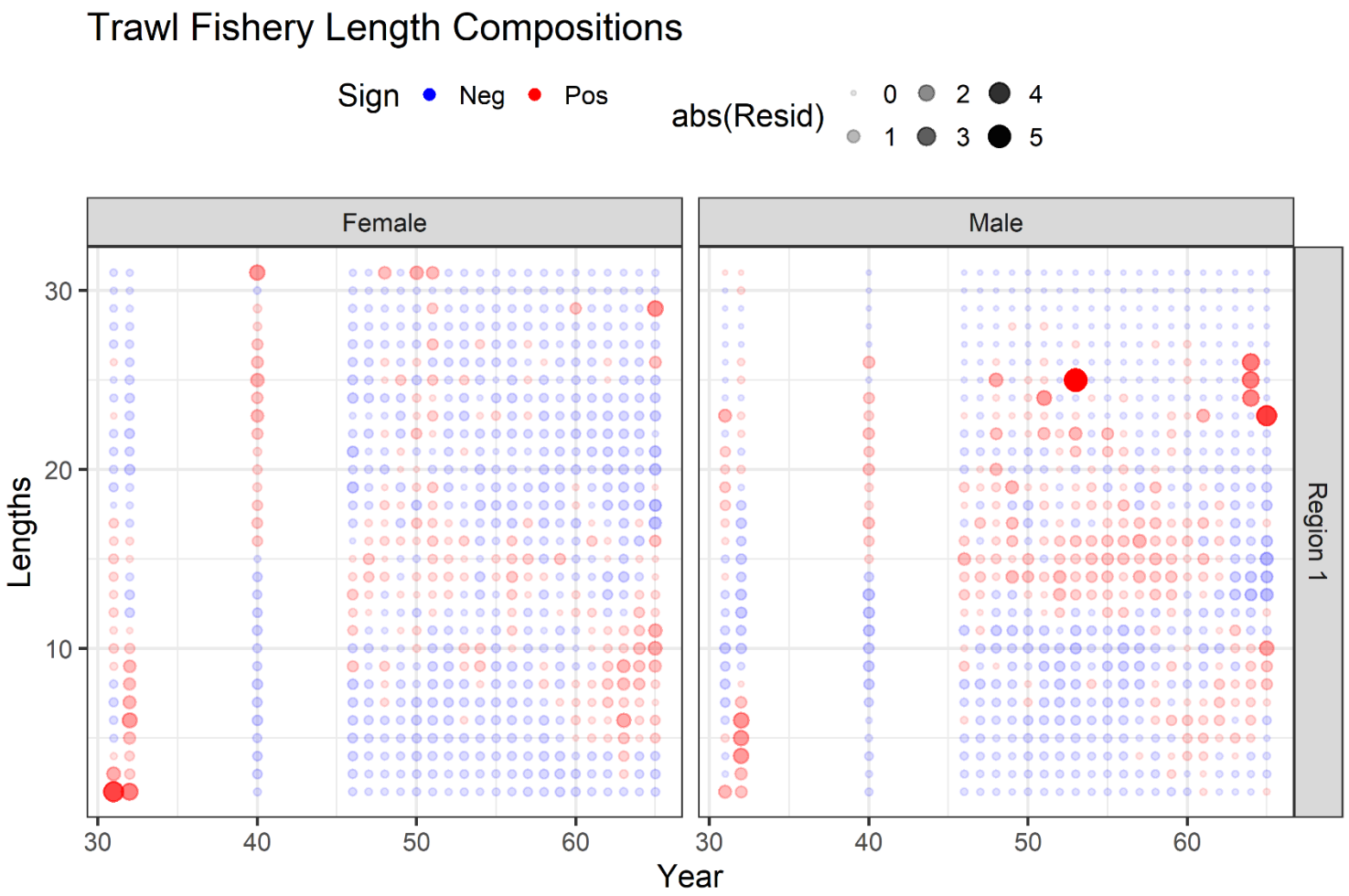


Figure 3.69. Fits to the trawl fishery length composition data based on Pearson residuals for Model *26.1_1Reg*.

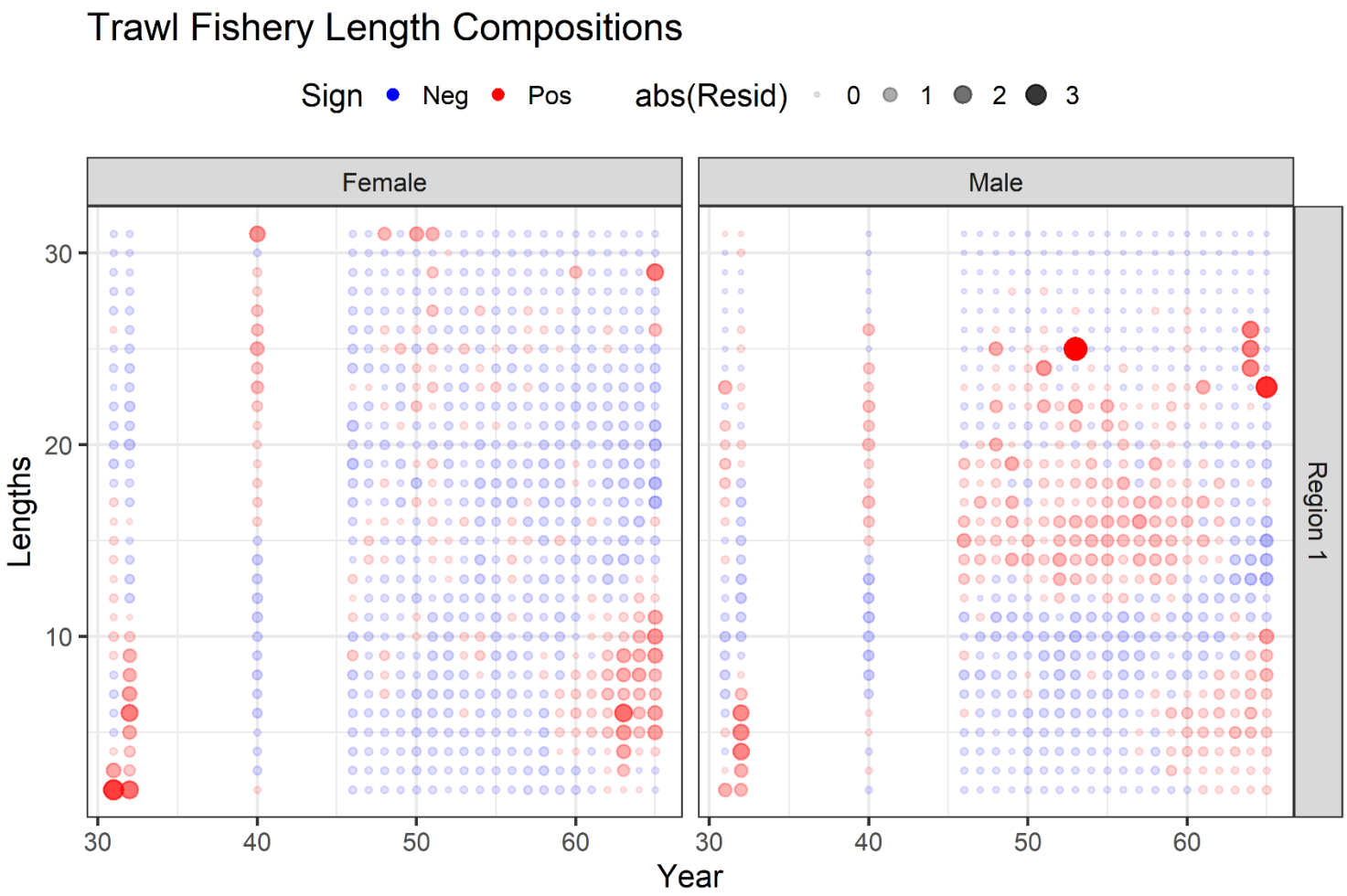


Figure 3.70. Fits to the trawl fishery length composition data based on Pearson residuals for Model *26.2_FAA*.

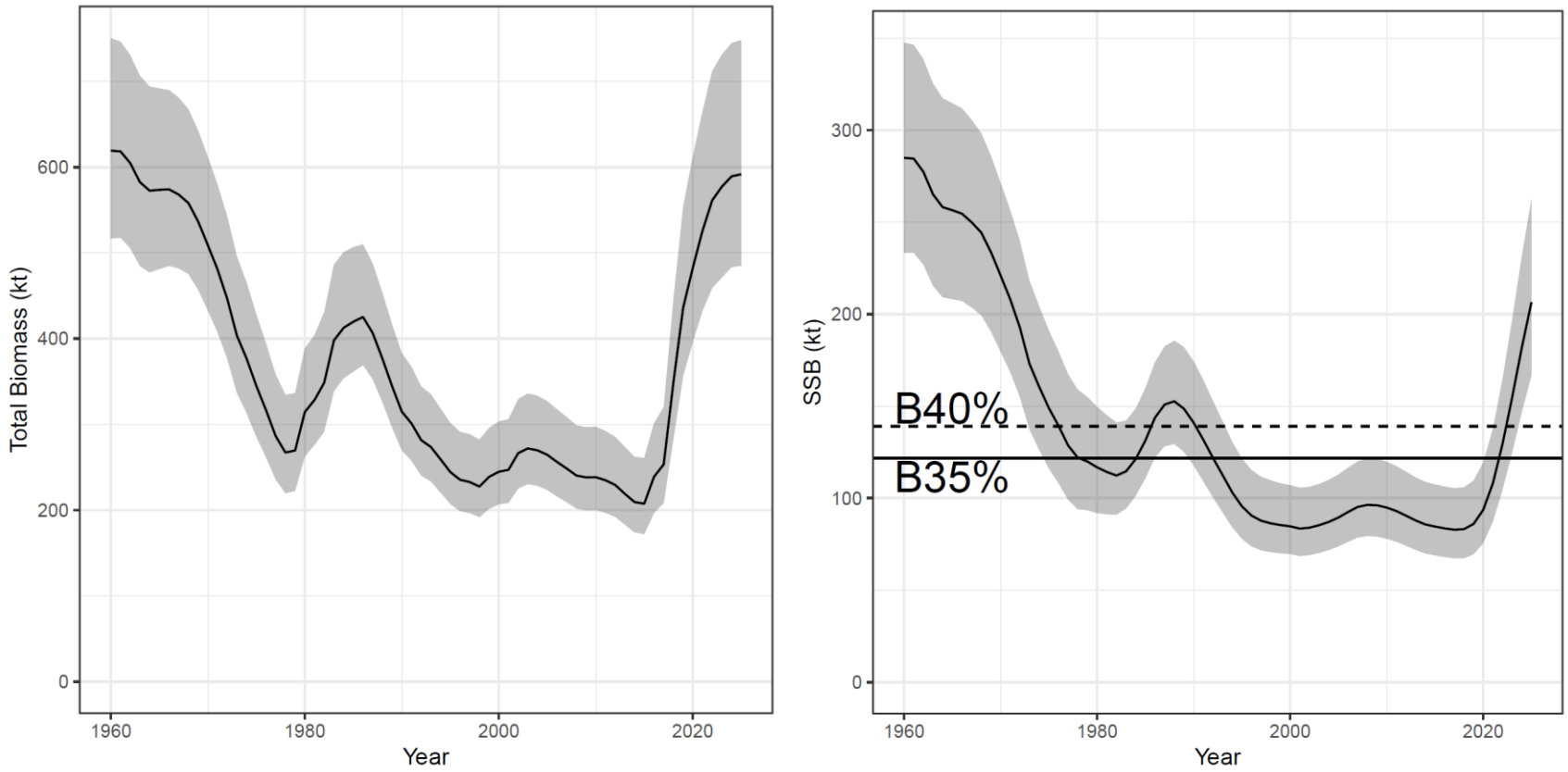


Figure 3.71. Estimated sablefish MCMC median (black line) total biomass (left panel) and spawning biomass (right panel) with 95% MCMC credible intervals (grey fill) for Model *26.1_1Reg*. Values are in kilotons. The $B_{35\%}$ (black solid line) and $B_{40\%}$ (black dashed line) reference points are shown on the SSB panel.

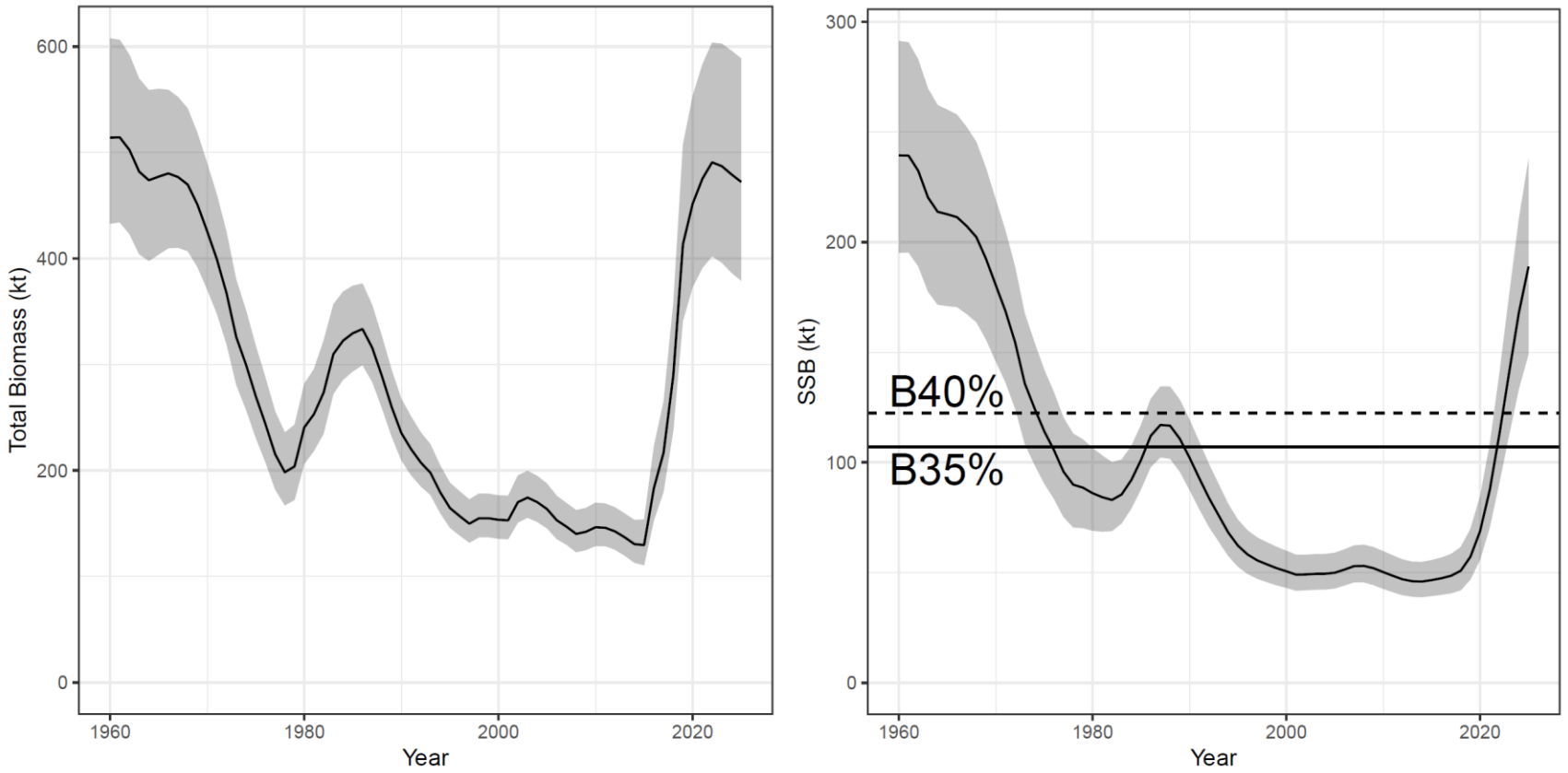


Figure 3.72. Estimated sablefish MCMC median (black line) total biomass (left panel) and spawning biomass (right panel) with 95% MCMC credible intervals (grey fill) for Model *26.2_FAA*. Values are in kilotons. The $B_{35\%}$ (black solid line) and $B_{40\%}$ (black dashed line) reference points are shown on the SSB panel.

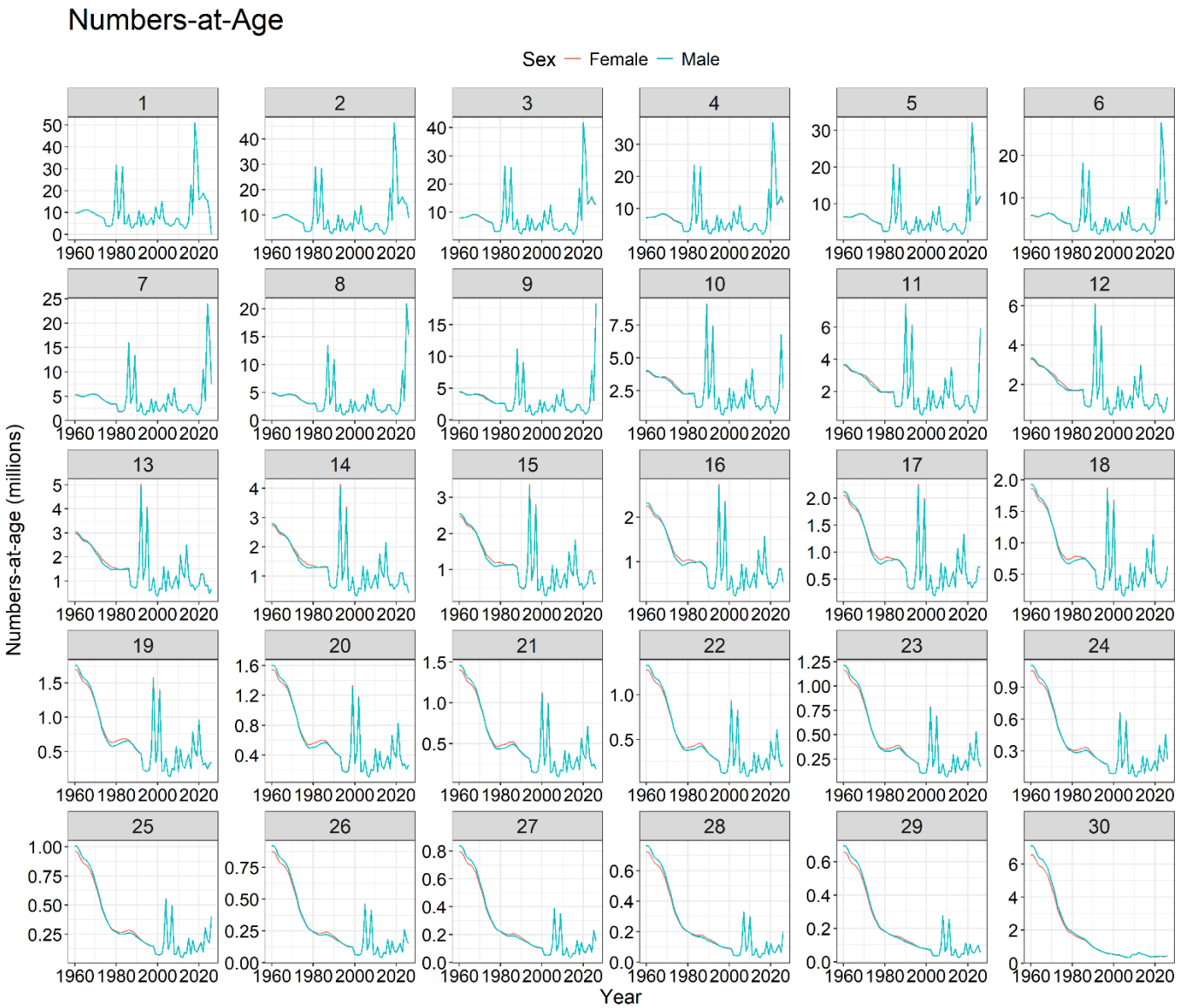


Figure 3.73. Model *26.1_1Reg* estimated population numbers (millions of fish) by sex (line color), age (facet), and year (x-axis).

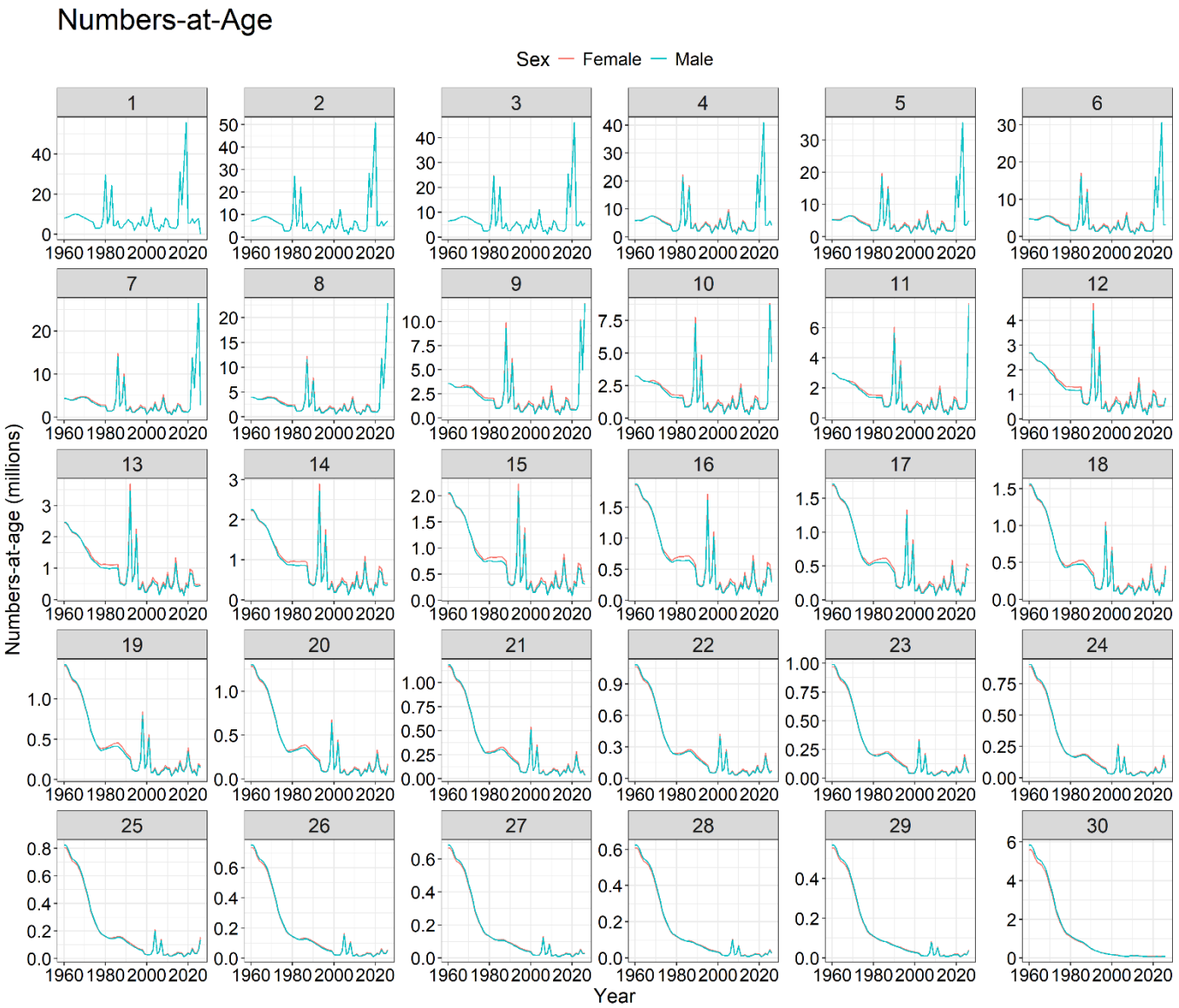


Figure 3.74. Model *26.2_FAA* estimated population numbers (millions of fish) by sex (line color), age (facet), and year (x-axis).

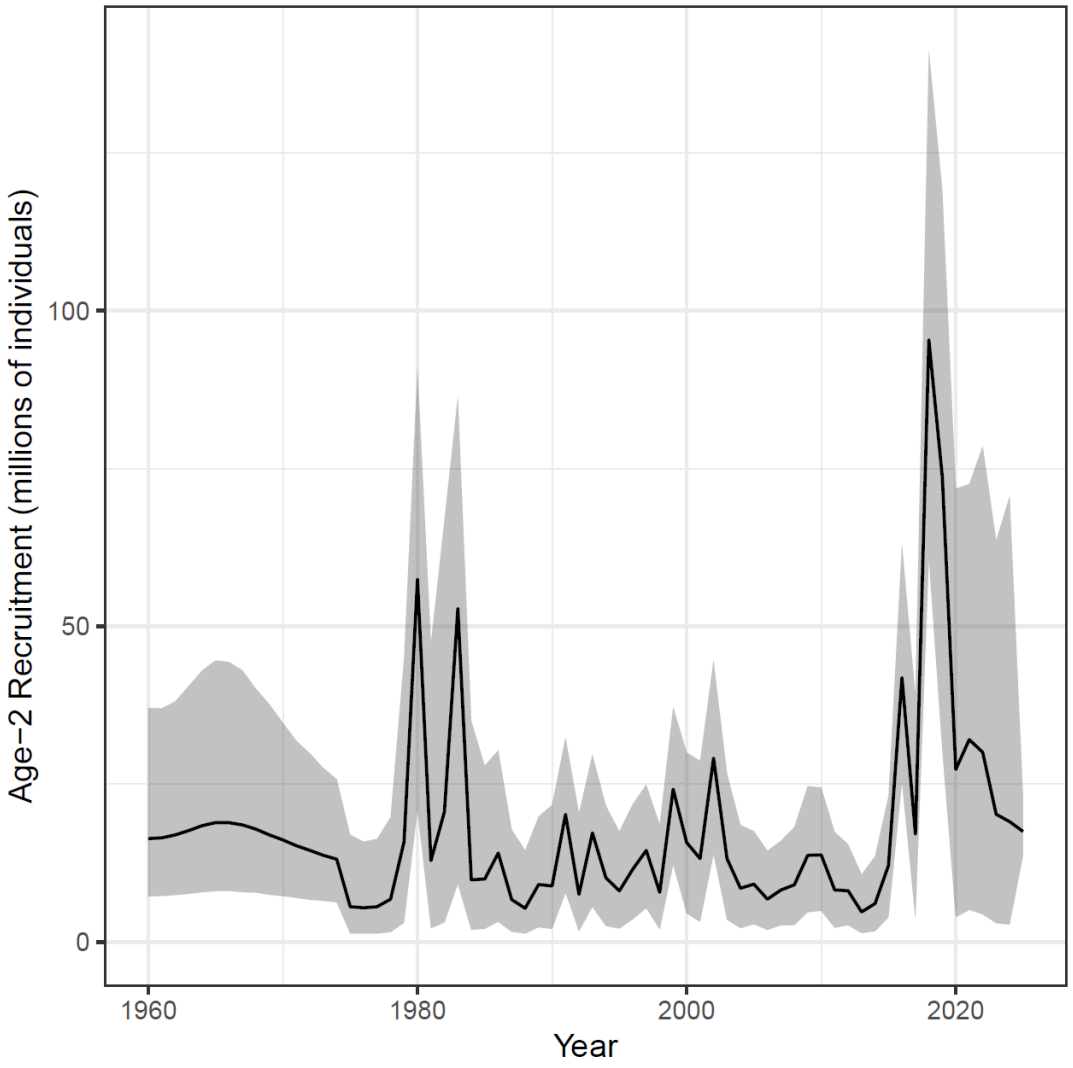


Figure 3.75. Estimated MCMC median (black line) recruitment of age-2 sablefish (millions of fish) with MCMC 95% credible intervals (grey fill) by recruitment year for Model *26.1_1Reg*. The estimate for the 2025 recruitment is fixed at the mean recruitment with no deviation parameter estimated.

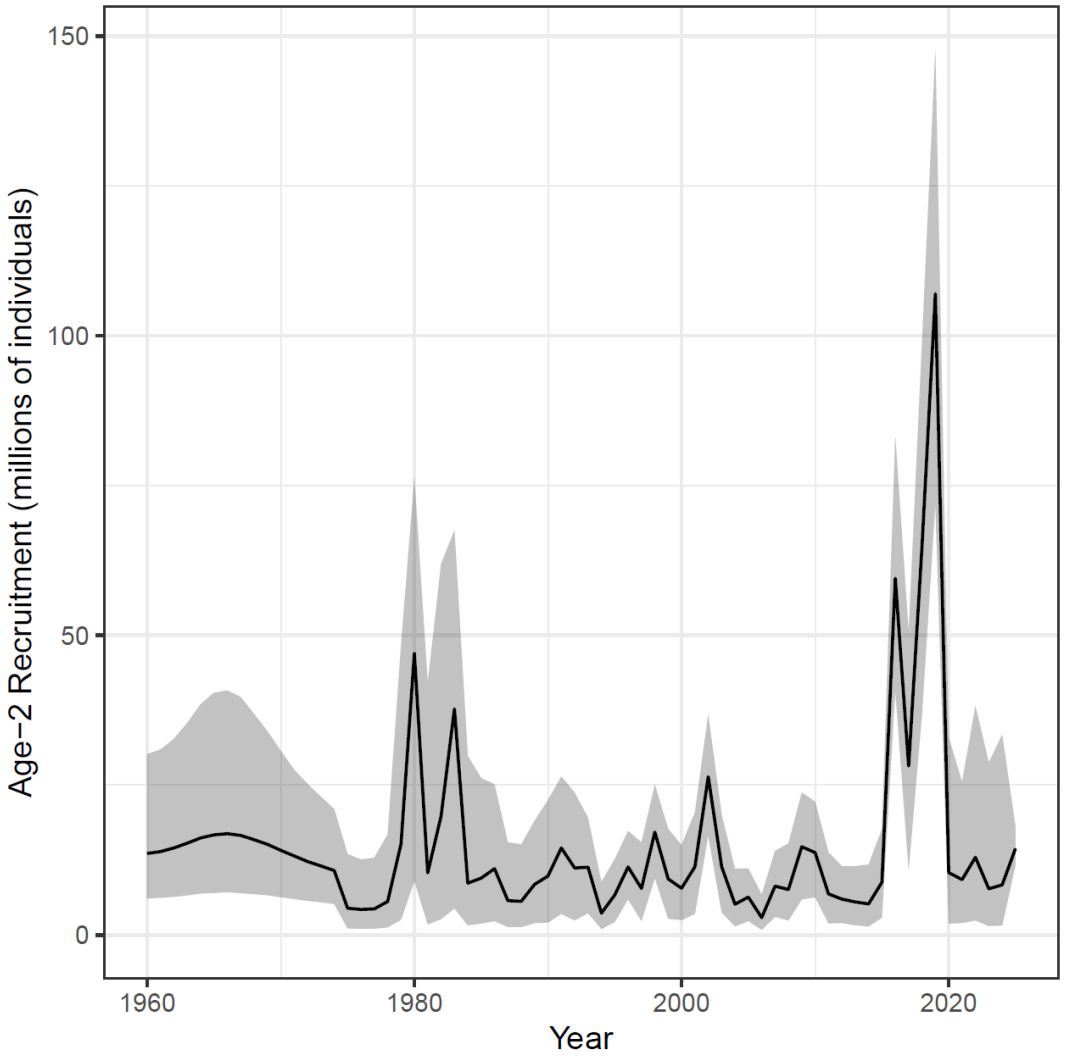


Figure 3.76. Estimated MCMC median (black line) recruitment of age-2 sablefish (millions of fish) with MCMC 95% credible intervals (grey fill) by recruitment year for Model *26.2_FAA*. The estimate for the 2025 recruitment is fixed at the mean recruitment with no deviation parameter estimated.

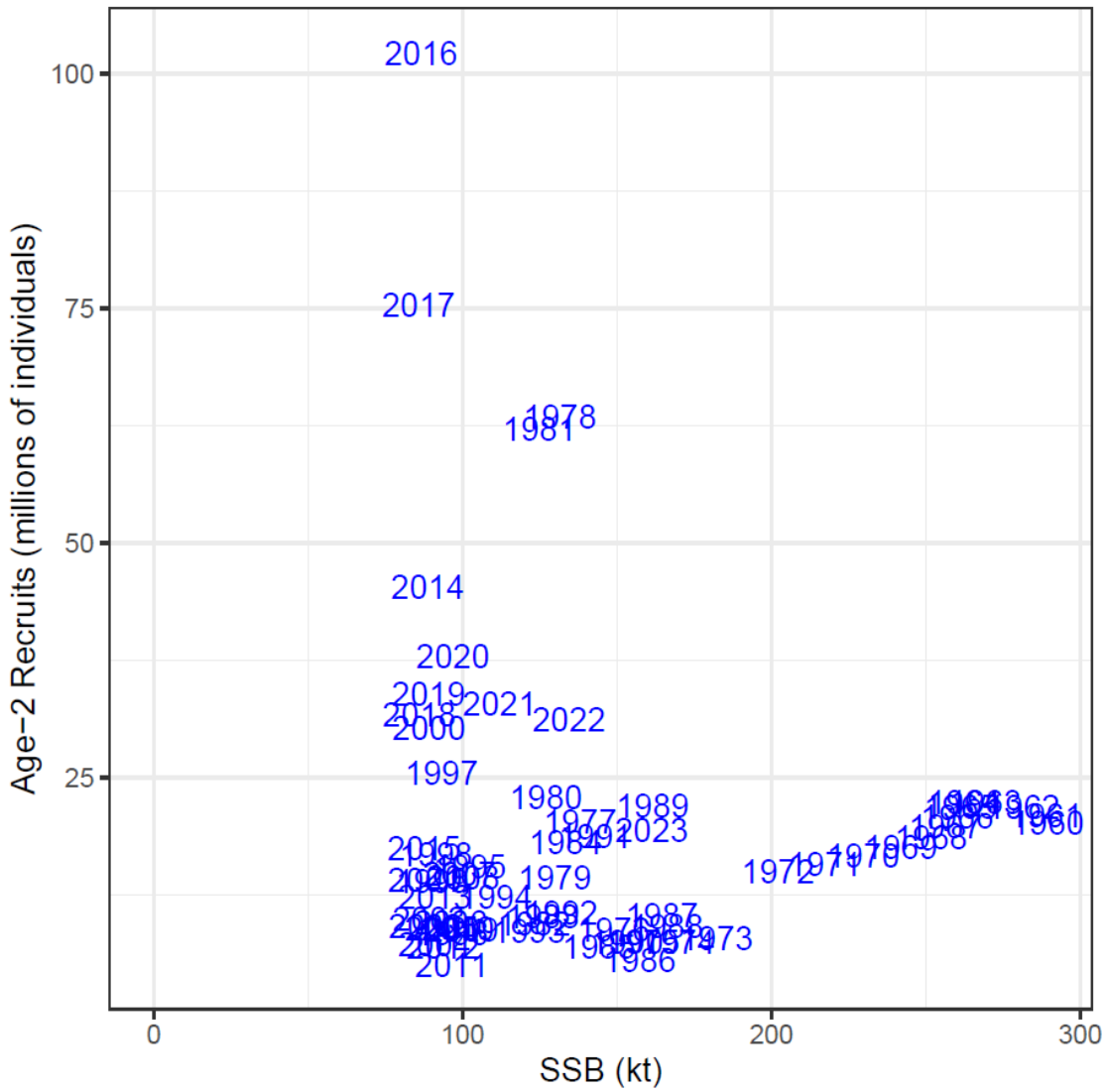


Figure 3.77. Age-2 recruits (millions of fish) and corresponding spawning stock biomass (kilotons) for each year class (identified by plotted year text) for Model *26.1_1Reg*.

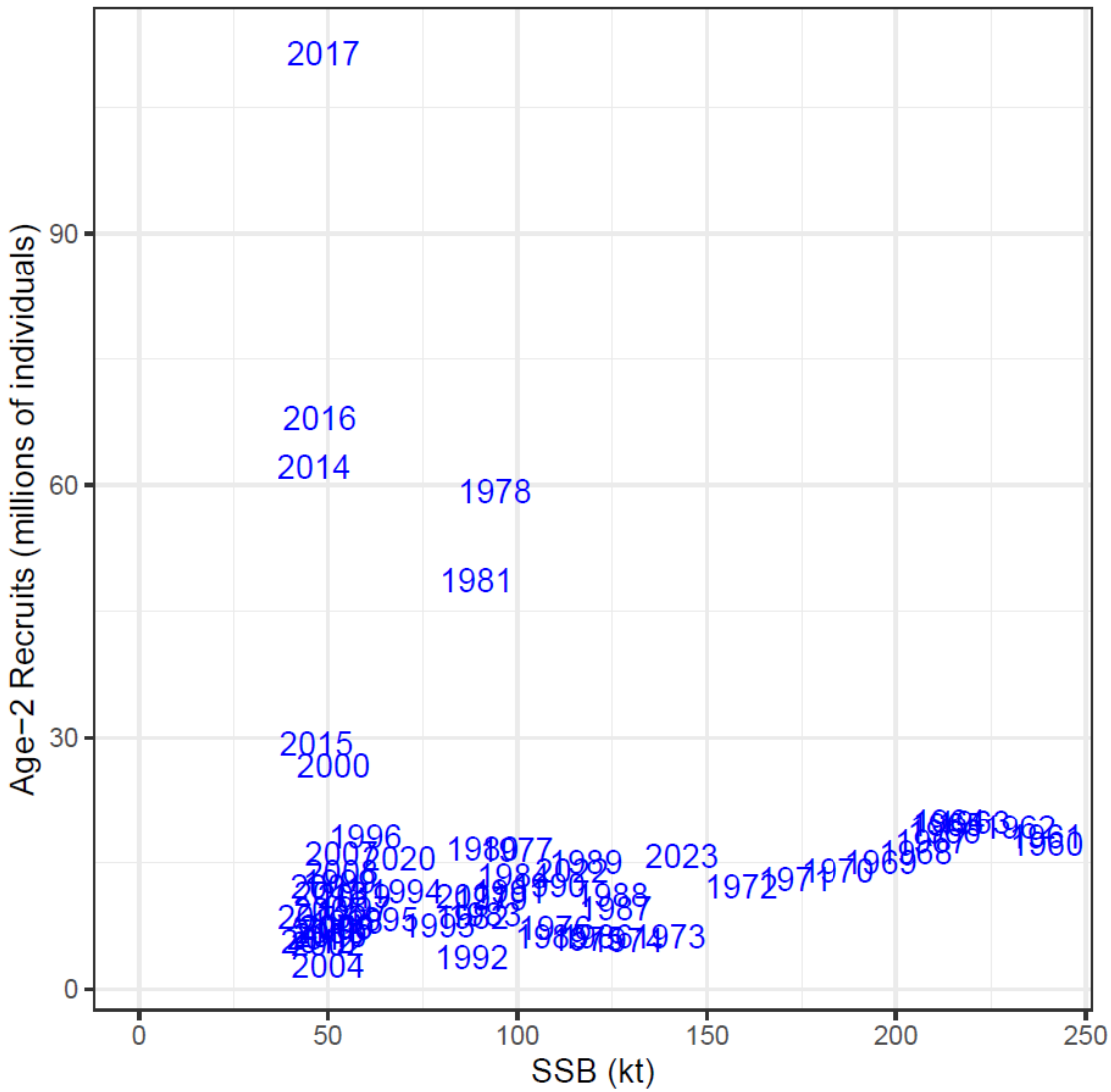


Figure 3.78. Age-2 recruits (millions of fish) and corresponding spawning stock biomass (kilotons) for each year class (identified by plotted year text) for Model *26.2_FAA*.

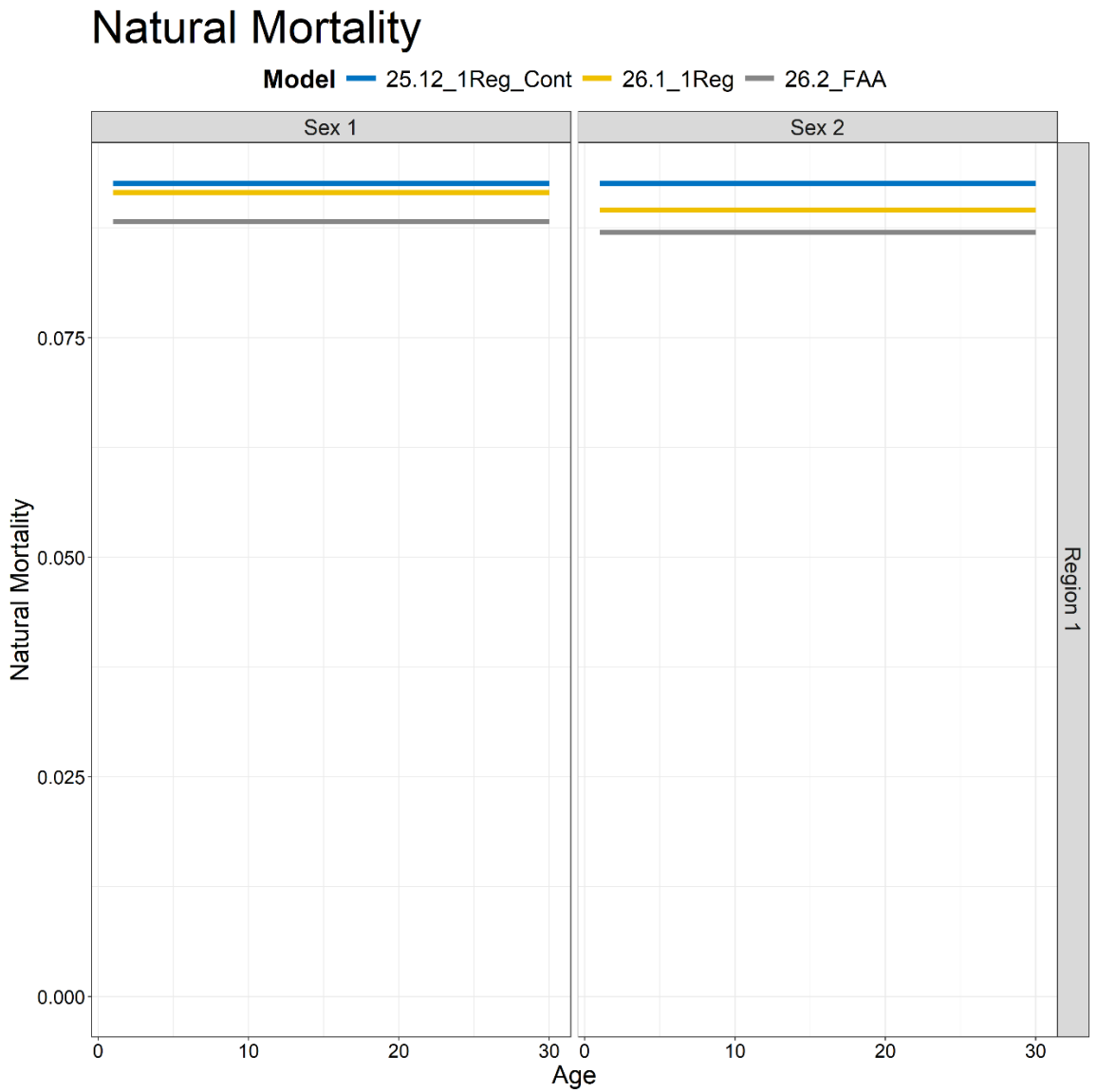


Figure 3.79. Natural mortality estimates from Model *25.12_1Reg_Cont* (blue line), Model *26.1_1Reg* (yellow line), and Model *26.2_FAA* (grey line). Sex 1 is females and sex 2 is males.

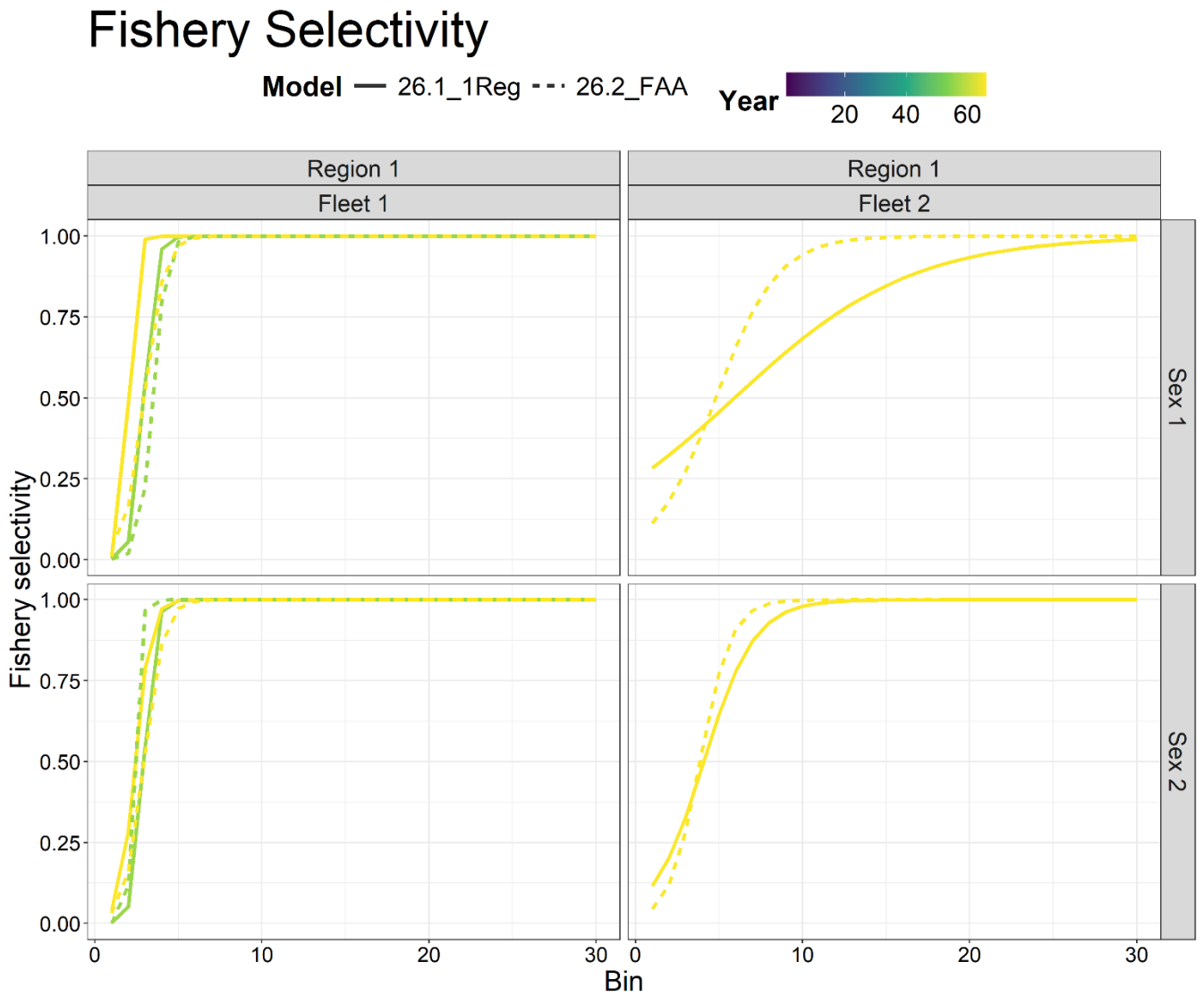


Figure 3.80. Fishery selectivity estimates from Model *26.1_1Reg* (solid line) and Model *26.2_FAA* (dashed line) where different colors represent the different time blocks. Fleet 1 is the fixed gear fleet (green is pre-2016 time block and yellow is post-2016 time block) and fleet 2 is the trawl fleet (single time block, yellow line). Sex 1 is females and sex 2 is males.

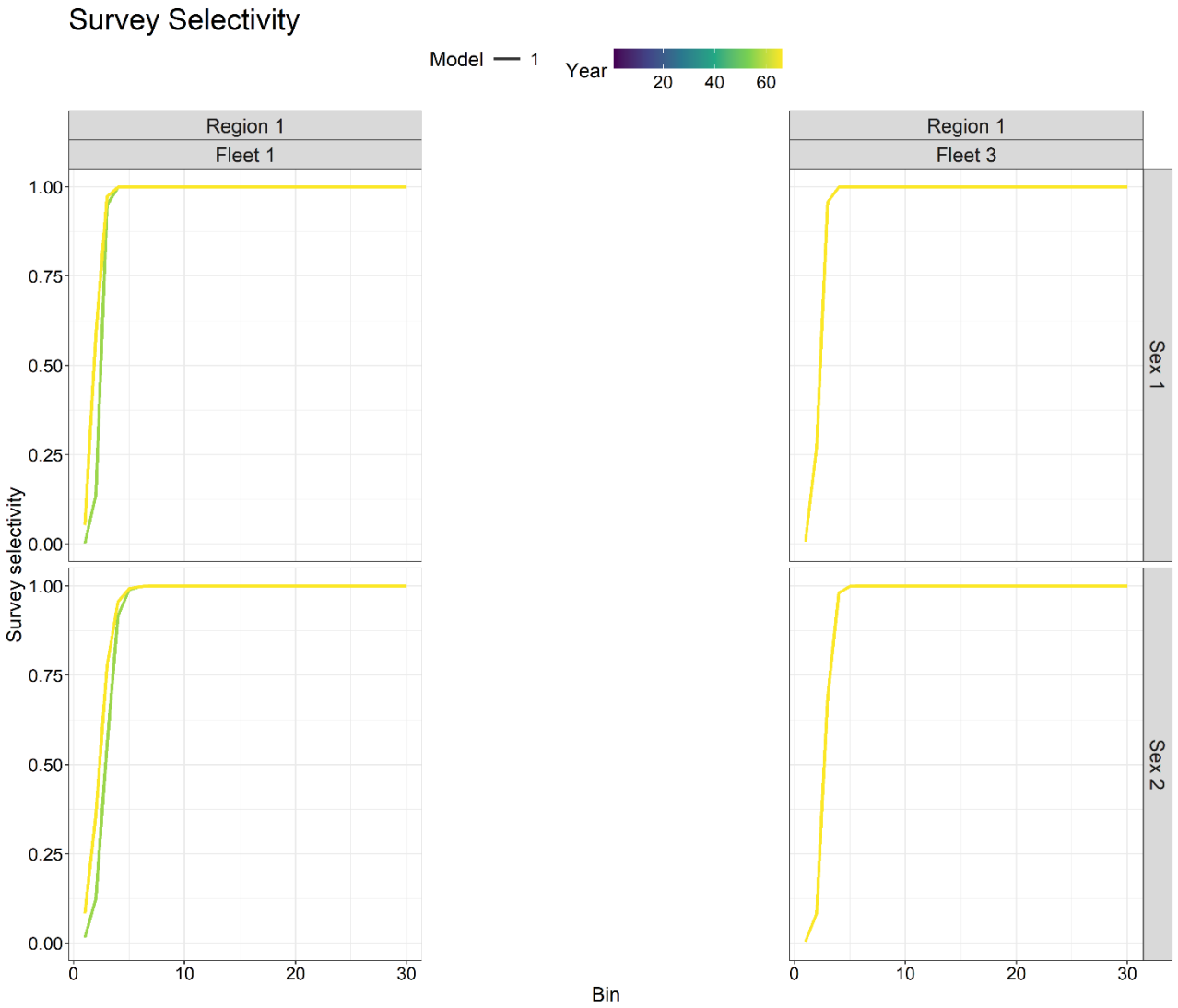


Figure 3.81. Survey selectivity estimates from Model *26.1_1Reg* where different colors represent the different time blocks. Fleet 1 is the NOAA Longline Survey (green is pre-2016 time block and yellow is post-2016 time block) and fleet 3 is the Japanese Longline Survey (single time block, yellow line). Sex 1 is females and sex 2 is males.

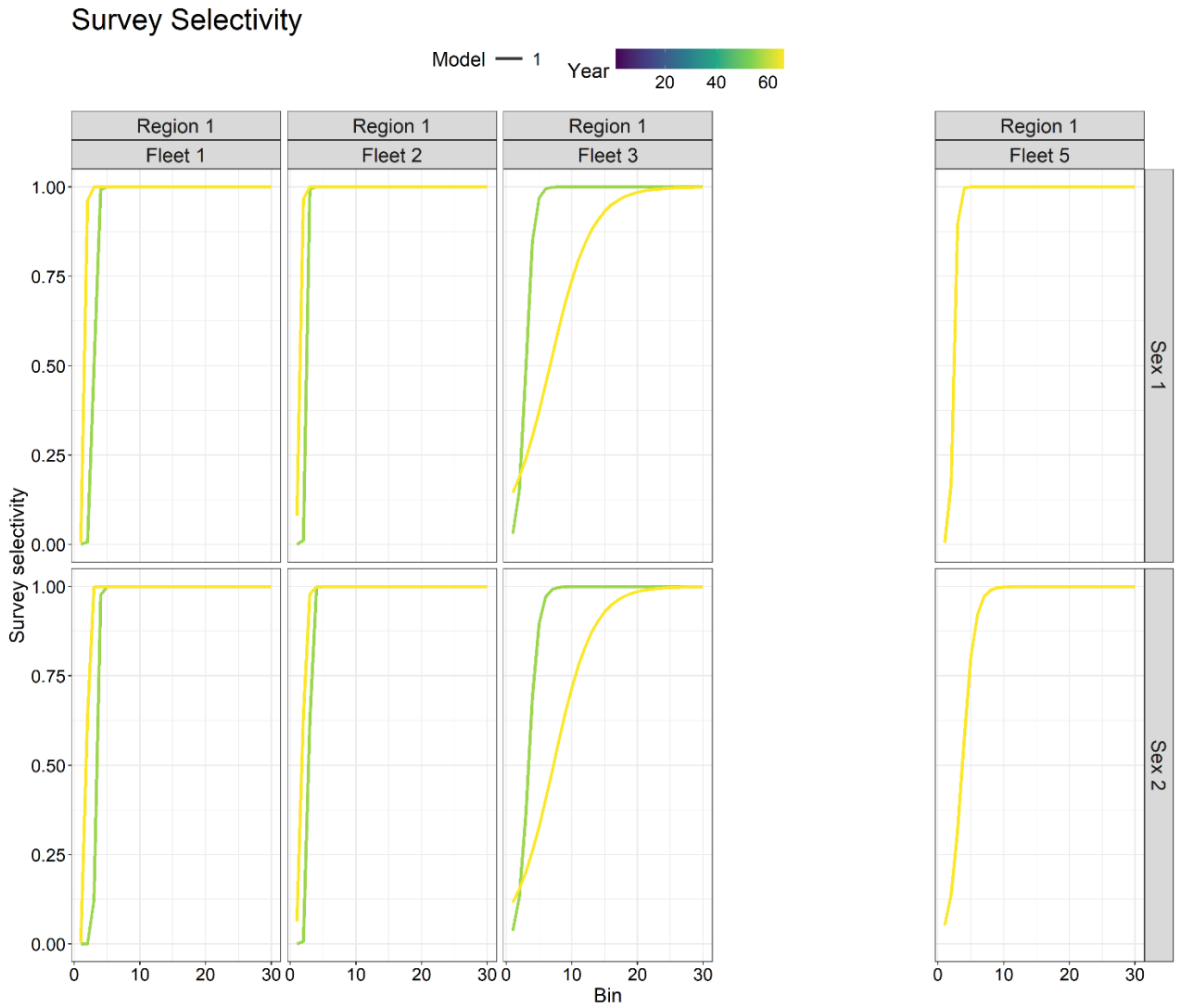


Figure 3.82. Survey selectivity estimates from Model *26.2_FAA* (solid lines) and Model *26.2a_FAA_Fsh+Srvy* (dashed lines) where different colors represent the different time blocks. Survey Fleet 1 is the BS NOAA Longline Survey, Survey Fleet 2 is the AI NOAA Longline Survey, Survey Fleet 3 is the GOA NOAA Longline Survey (green is pre-2016 time block and yellow is post-2016 time block). Survey Fleet 5 is the Japanese Longline Survey (single time block, yellow line). Sex 1 is females and sex 2 is males.

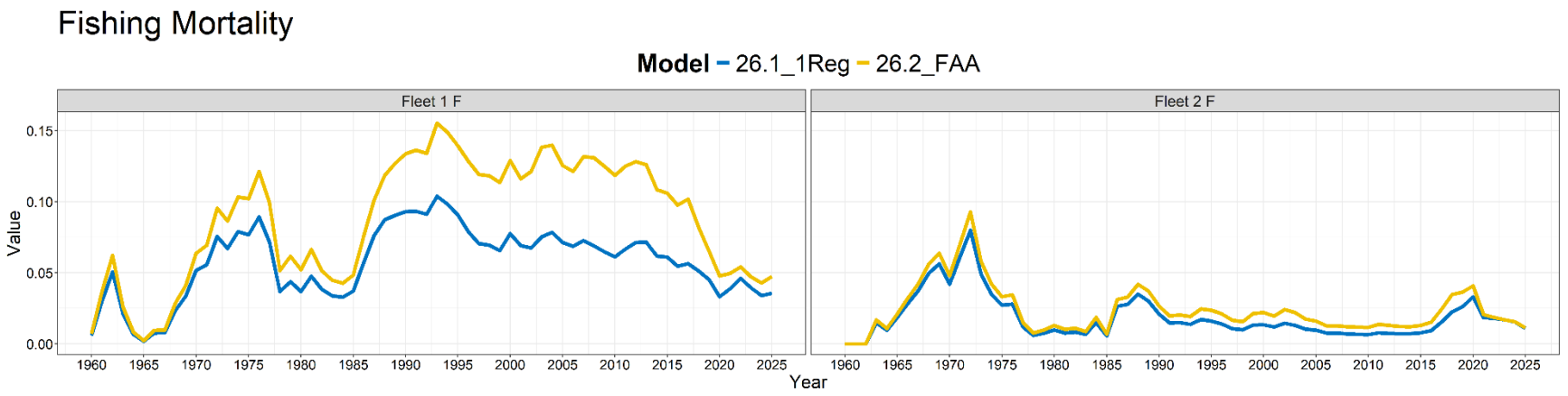


Figure 3.83. Time series estimates of fishing mortality (yr^{-1}) for Model *26.1_1Reg* and Model *26.2_FAA*. Fleet 1 is the fixed gear fishery and fleet 2 is the trawl fishery.

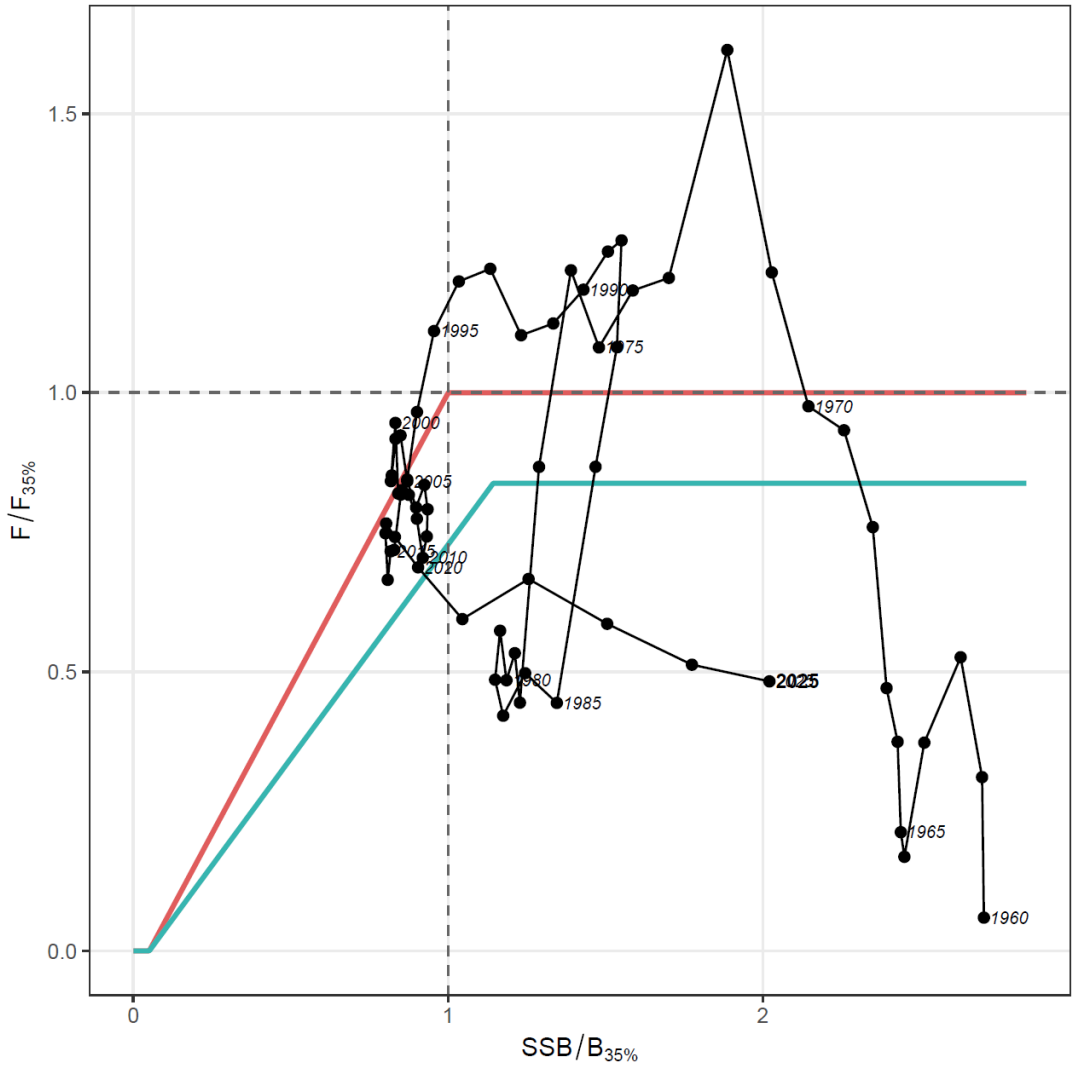


Figure 3.84. Phase-plane diagram illustrating the time series of sablefish estimated spawning biomass relative to the level at $B_{35\%}$ and fishing mortality relative to $F_{35\%}$ (equal to F_{OFL}) for Model *26.1_1Reg*. F_{ABC} for the max ABC is equivalent to $F_{40\%}$ and the associated threshold HCR is demonstrated by the blue line. The red line represents a threshold HCR associated with fishing at F_{OFL}.

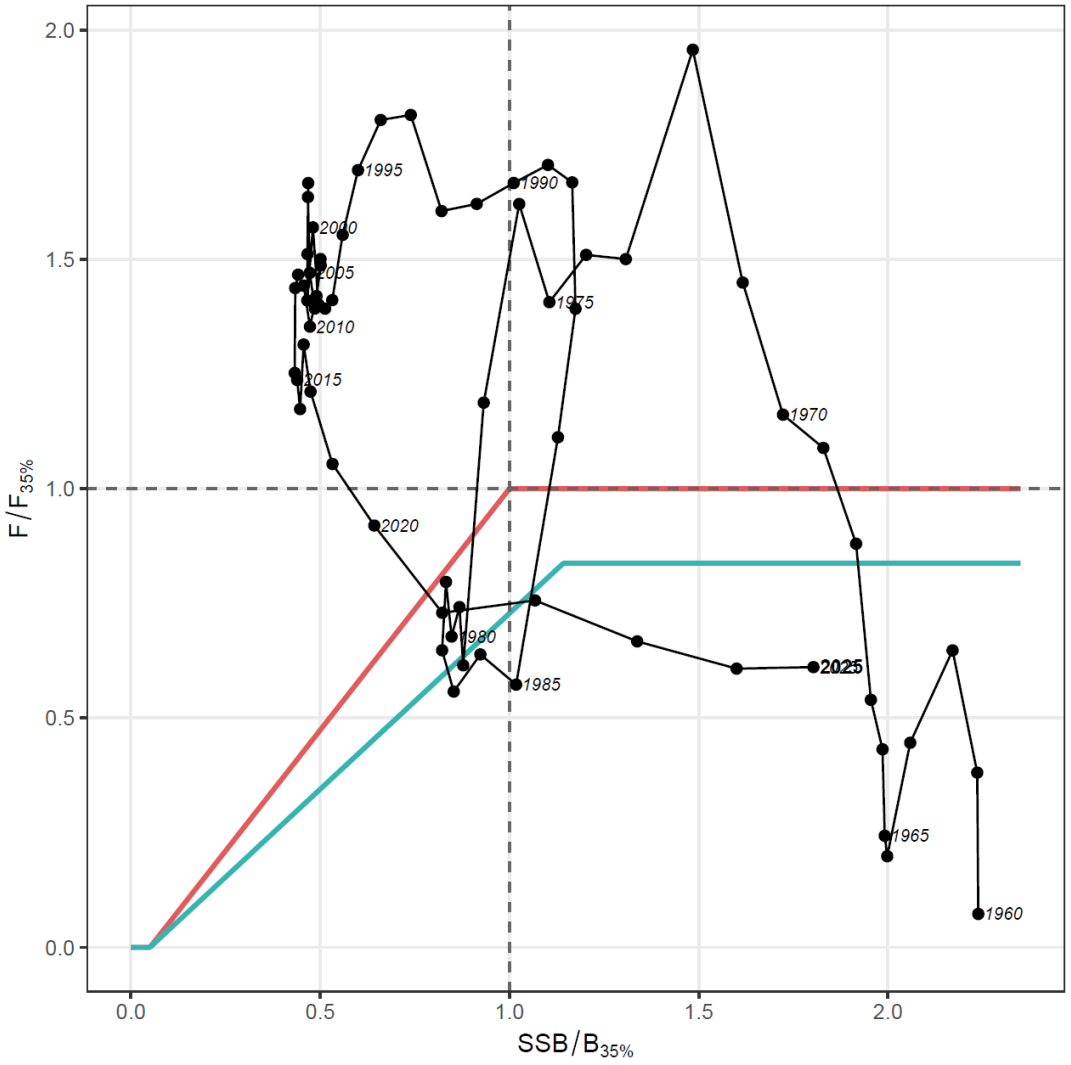


Figure 3.85. Phase-plane diagram illustrating the time series of sablefish estimated spawning biomass relative to the level at $B_{35\%}$ and fishing mortality relative to $F_{35\%}$ (equal to F_{OFL}) for Model *26.2_FAA*. F_{ABC} for the max ABC is equivalent to $F_{40\%}$ and the associated threshold HCR is demonstrated by the blue line. The red line represents a threshold HCR associated with fishing at F_{OFL}.

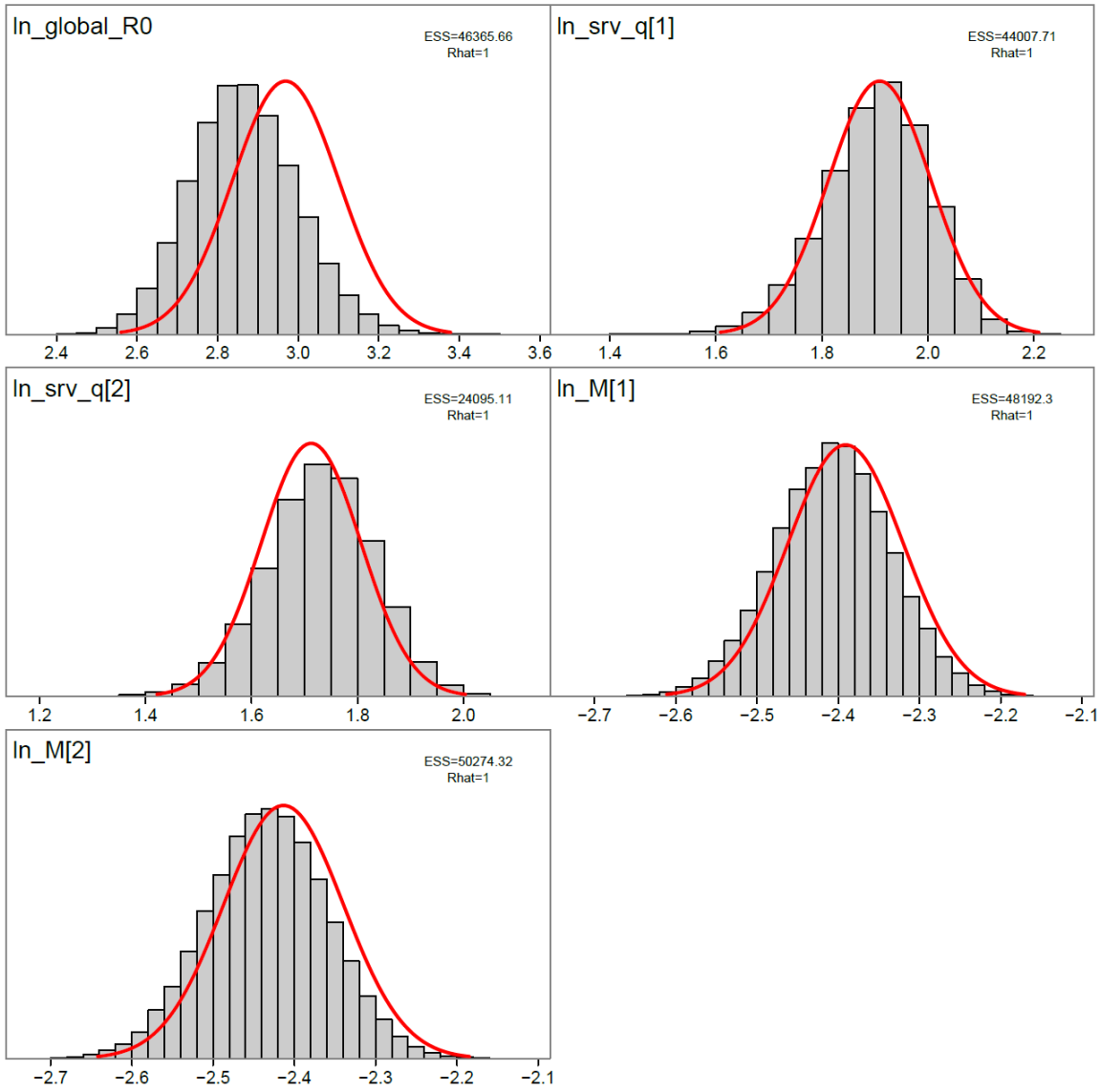


Figure 3.86. Key parameter marginal posterior histograms from MCMC and asymptotic normal distributions (red line) from the inverse Hessian (MLE) for Model *26.1_1Reg*. MCMC effective sample size (ESS) and R-hat estimates are printed in the top right corner.

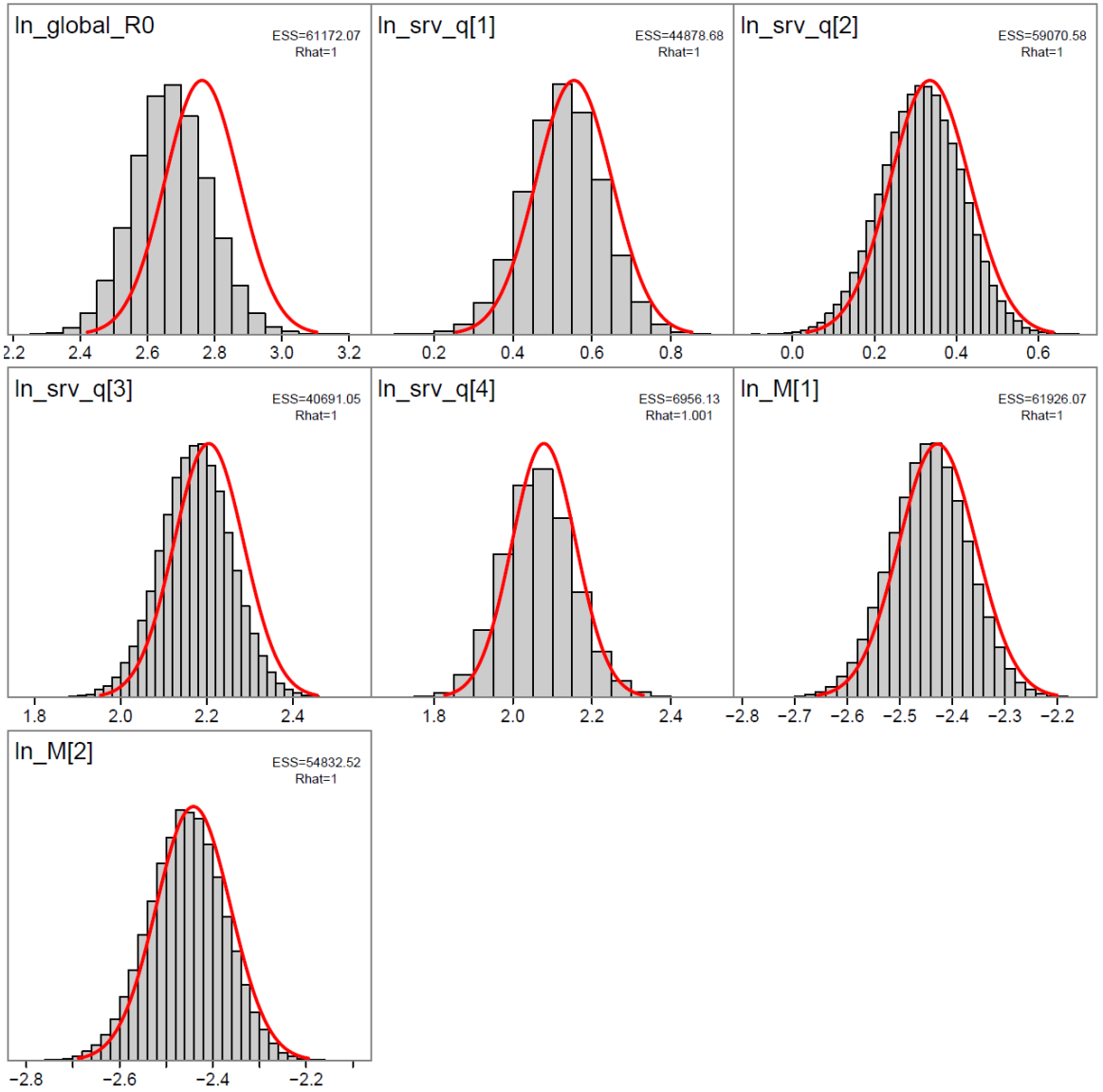


Figure 3.87. Key parameter marginal posterior histograms from MCMC and asymptotic normal distributions (red line) from the inverse Hessian (MLE) for Model *26.2_FAA*. MCMC effective sample size (ESS) and R-hat estimates are printed in the top right corner.

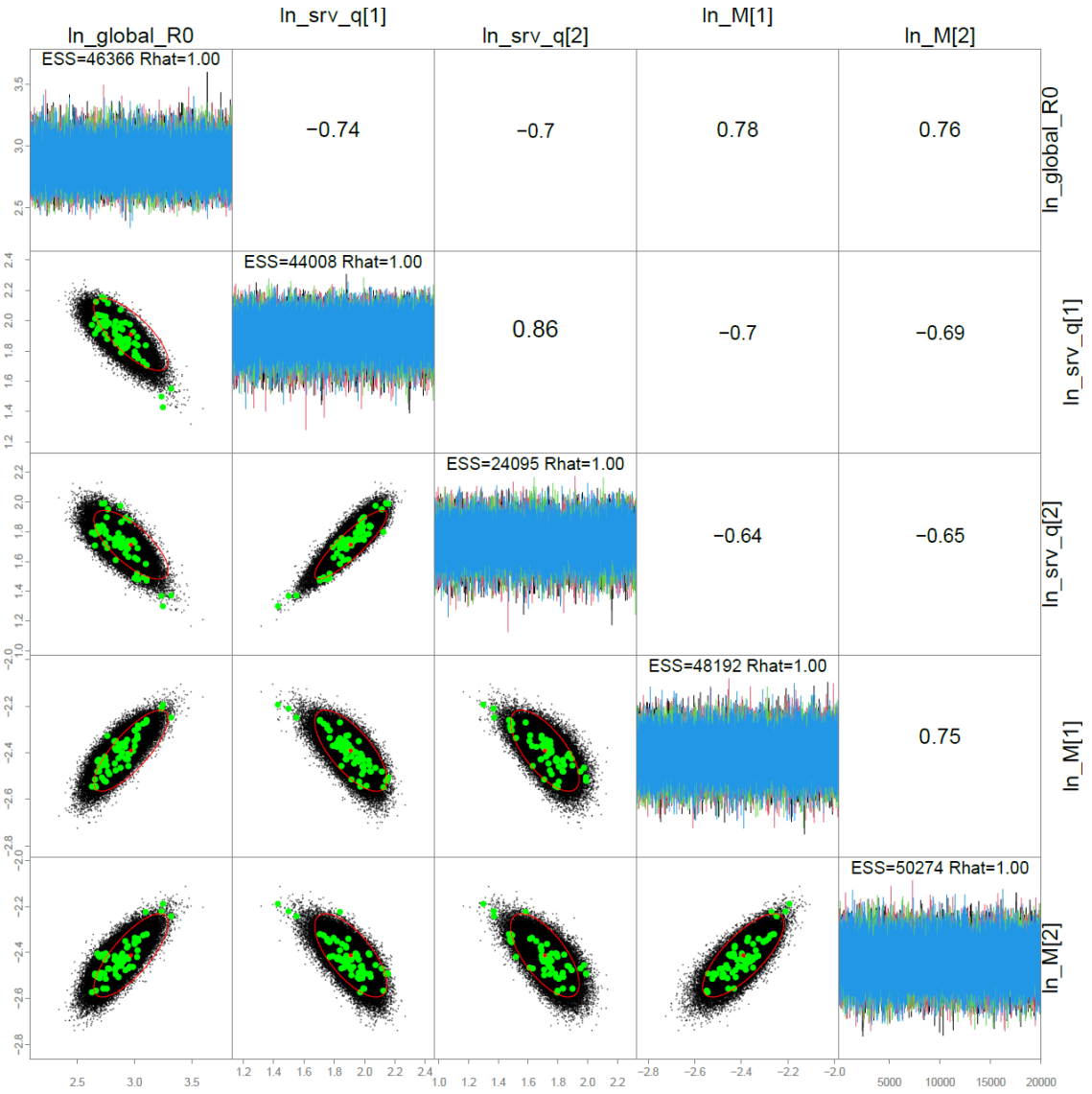


Figure 3.88. MCMC pairwise correlation plots among key model parameters for Model *26.1_1Reg*, where diagnostics are shown along the diagonal and parameter correlations are shown in the upper right quadrant.

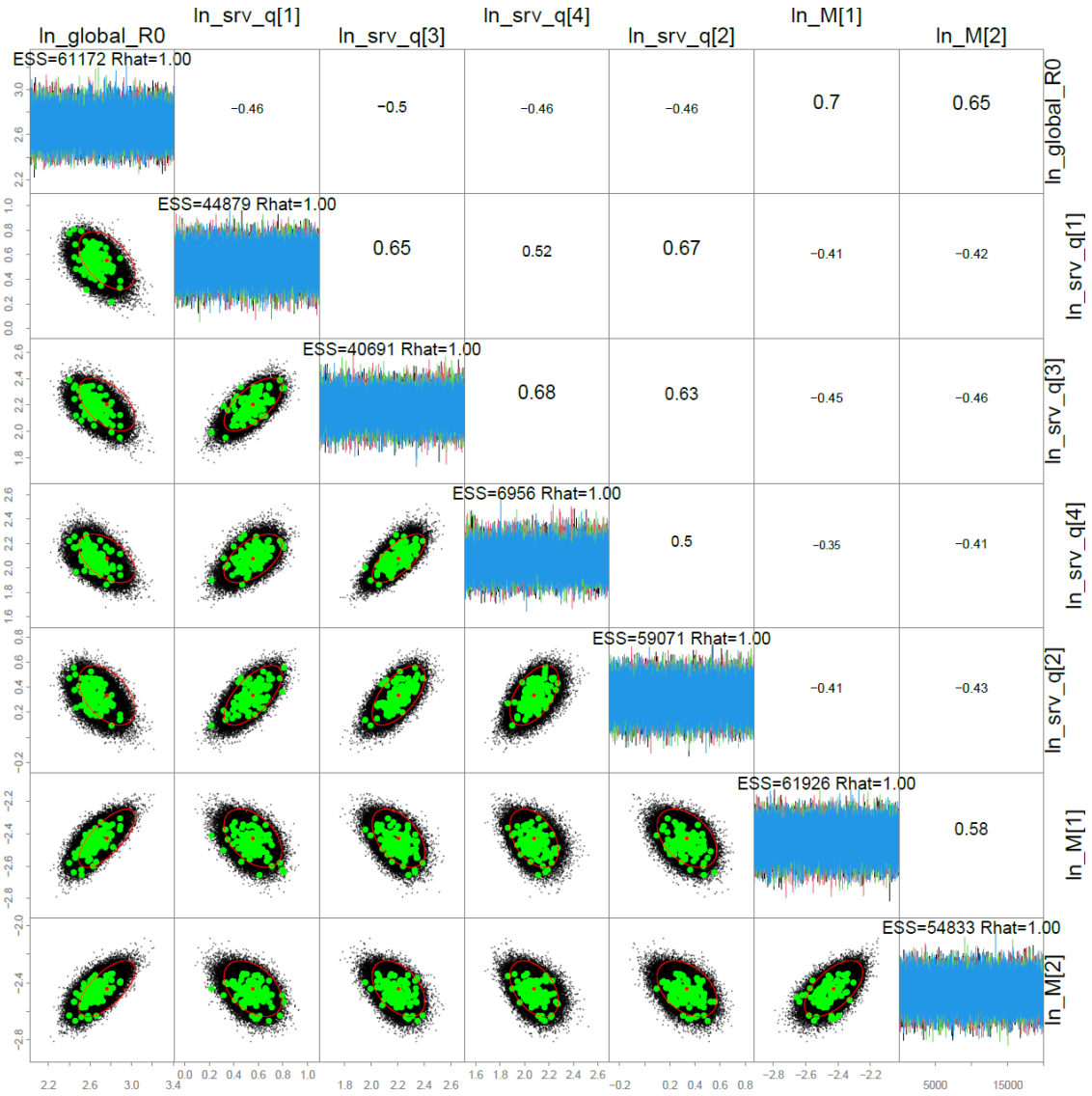


Figure 3.89. MCMC pairwise correlation plots among key model parameters for Model *26.2_FAA*, where diagnostics are shown along the diagonal and parameter correlations are shown in the upper right quadrant.

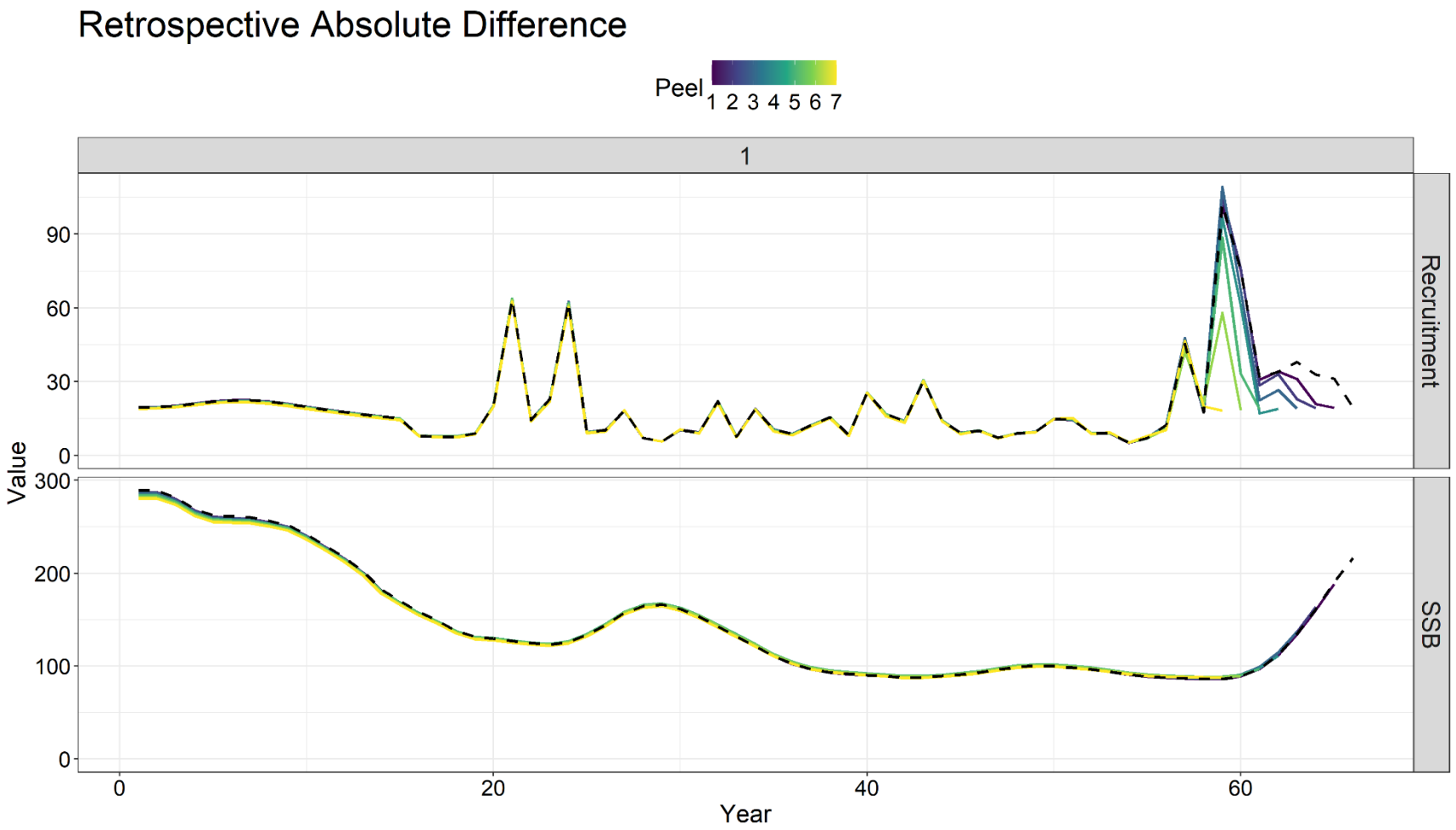


Figure 3.90. Comparison of recruitment (millions of fish; top panel) and spawning stock biomass (SSB; bottom panel) time series estimates from a seven year retrospective analysis for Model *26.1_1Reg*. Francis reweighting was undertaken for each peel to ensure consistency and to match how an assessment would be performed in a real-world situation. Mohn's rho for SSB was 0.0054.

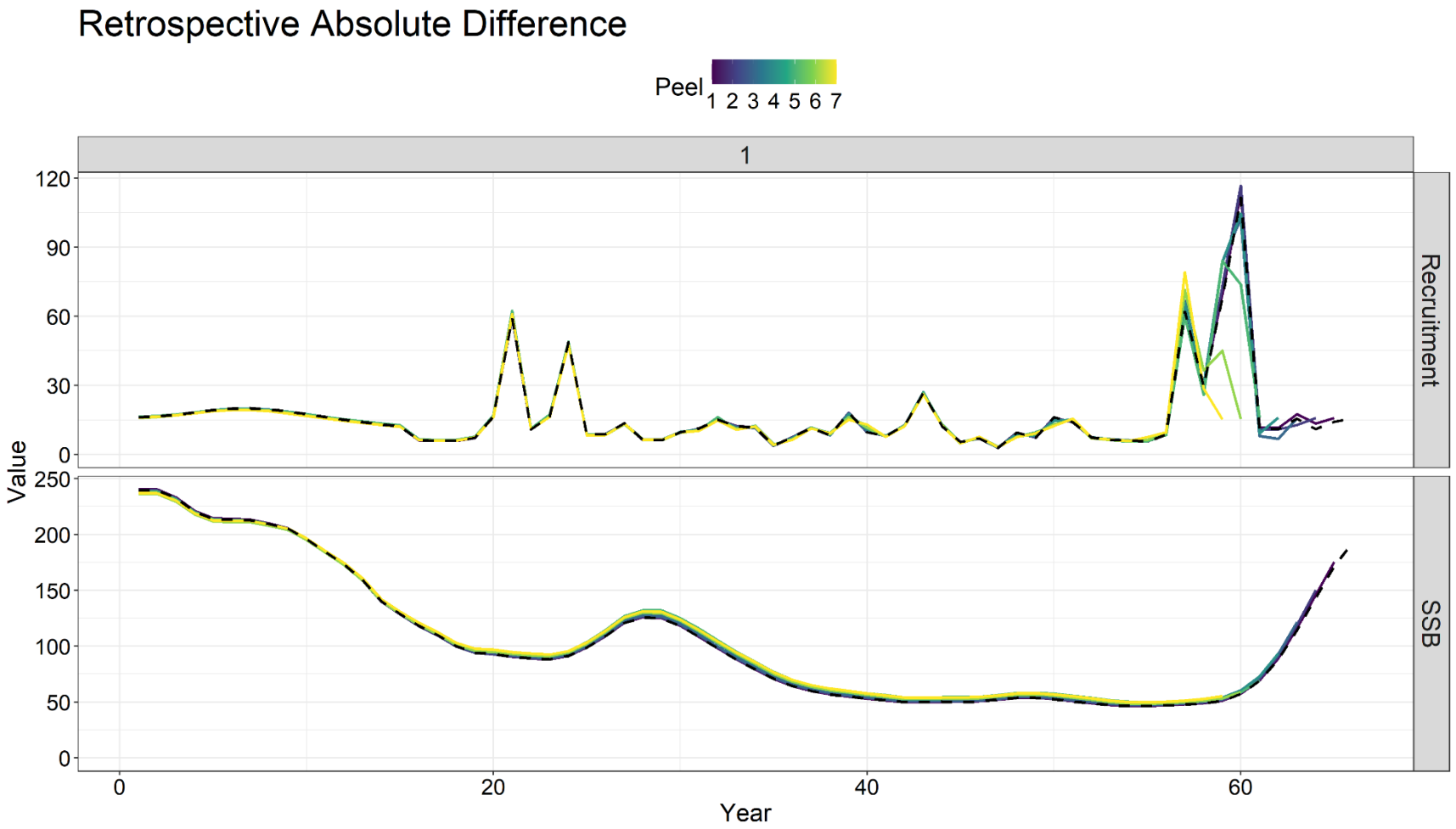


Figure 3.91. Comparison of recruitment (millions of fish; top panel) and spawning stock biomass (SSB; bottom panel) time series estimates from a seven-year retrospective analysis for Model *26.2_FAA*. Francis reweighting was undertaken for each peel to ensure consistency and to match how an assessment would be performed in a real-world situation. Mohn's rho for SSB was 0.046.



Figure 3.92. Comparison of a recruitment retrospective 'squid' plot representing how estimates of a given year class change as data are added in a seven-year retrospective analysis for model *26.1_1Reg*. Francis reweighting was undertaken for each peel to ensure consistency and to match how an assessment would be performed in a real-world situation. Cohort 56 represents the 2015 year class (2017 recruitment event).

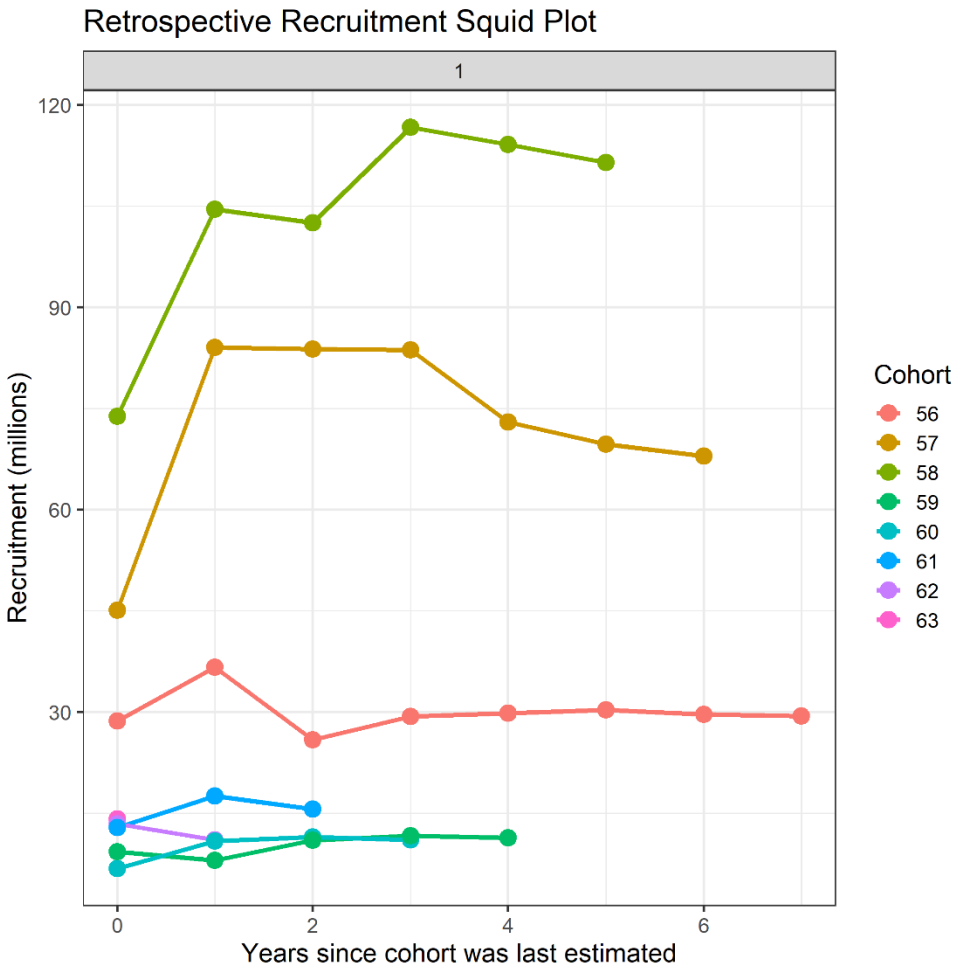


Figure 3.93. Comparison of a recruitment retrospective 'squid' plot representing how estimates of a given year class change as data are added in a seven-year retrospective analysis for model *26.2_FAA*. Francis reweighting was undertaken for each peel to ensure consistency and to match how an assessment would be performed in a real-world situation. Cohort 56 represents the 2015 year class (2017 recruitment event).

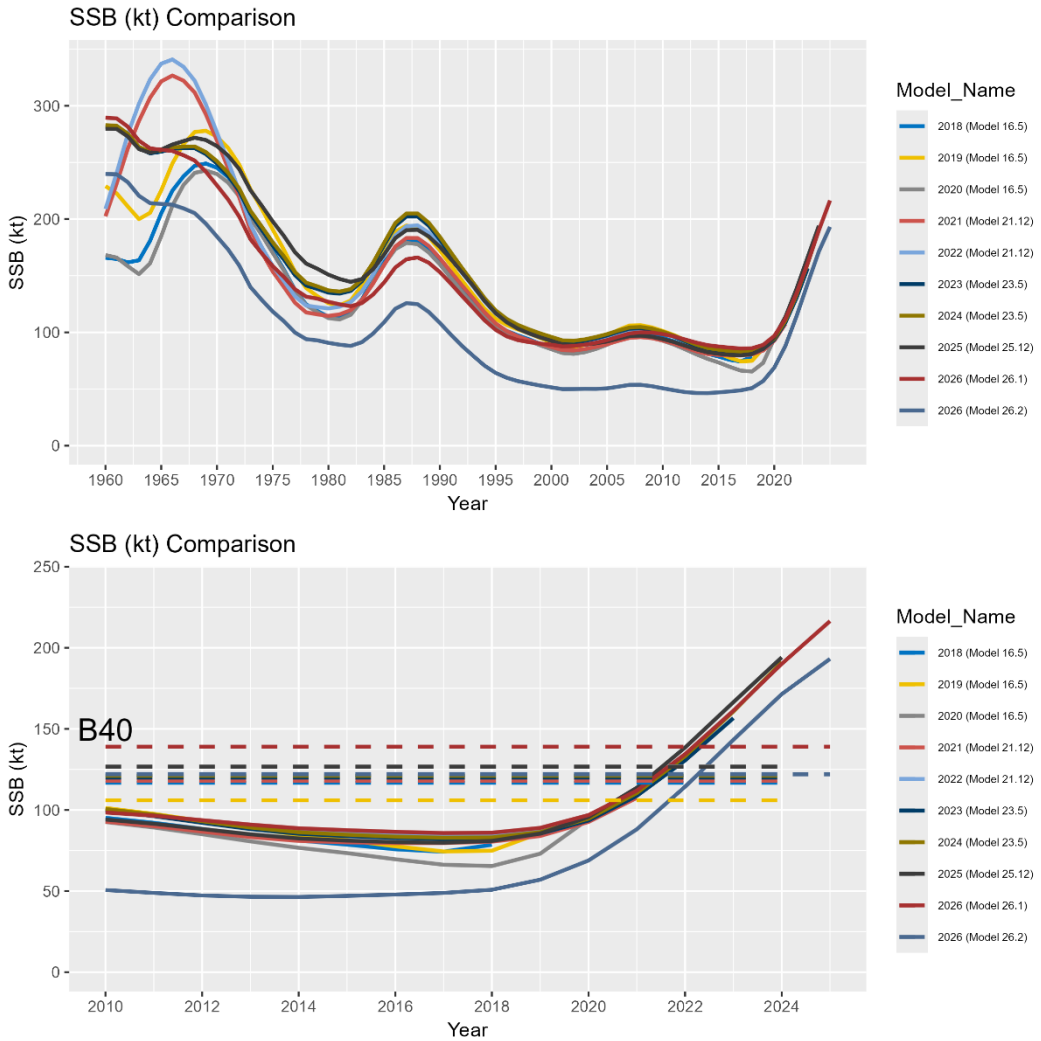


Figure 3.94. Results of the 'all model' historical retrospective illustrating estimated spawning stock biomass (in kilotons). Results are based on the accepted model in each terminal model year, including application of Model *25.12* for 2025 (although there was no SAFE), Model *23.5* for 2023 and 2024, Model *21.12* for 2021 and 2022, and Model *16.5* model for earlier model years. Models *26.1_1Reg* and Model *26.2_FAA* are shown for 2026. The top panel shows the entire time series of SSB from each assessment model, while the bottom panel shows the same results since 2010 overlaid with corresponding estimates of $B_{40\%}$.

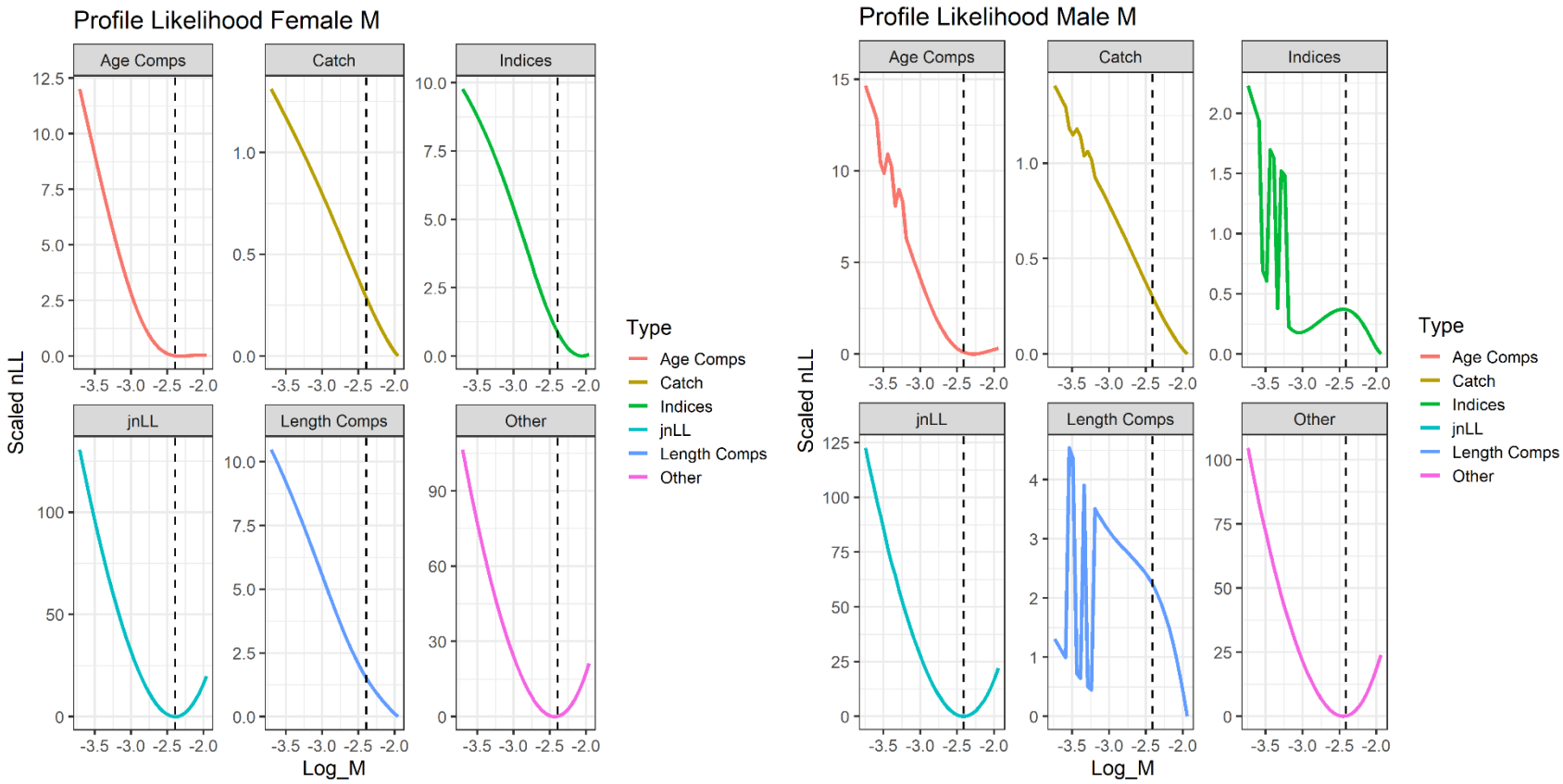


Figure 3.95. Likelihood profile for the natural mortality parameter (*M*) for females (left) and males (right), where panels represent the profile for each data type used in the model, for Model *26.1_1Reg*. The 'other' category represents the priors and penalties, which is particularly dominated by the *M* prior.

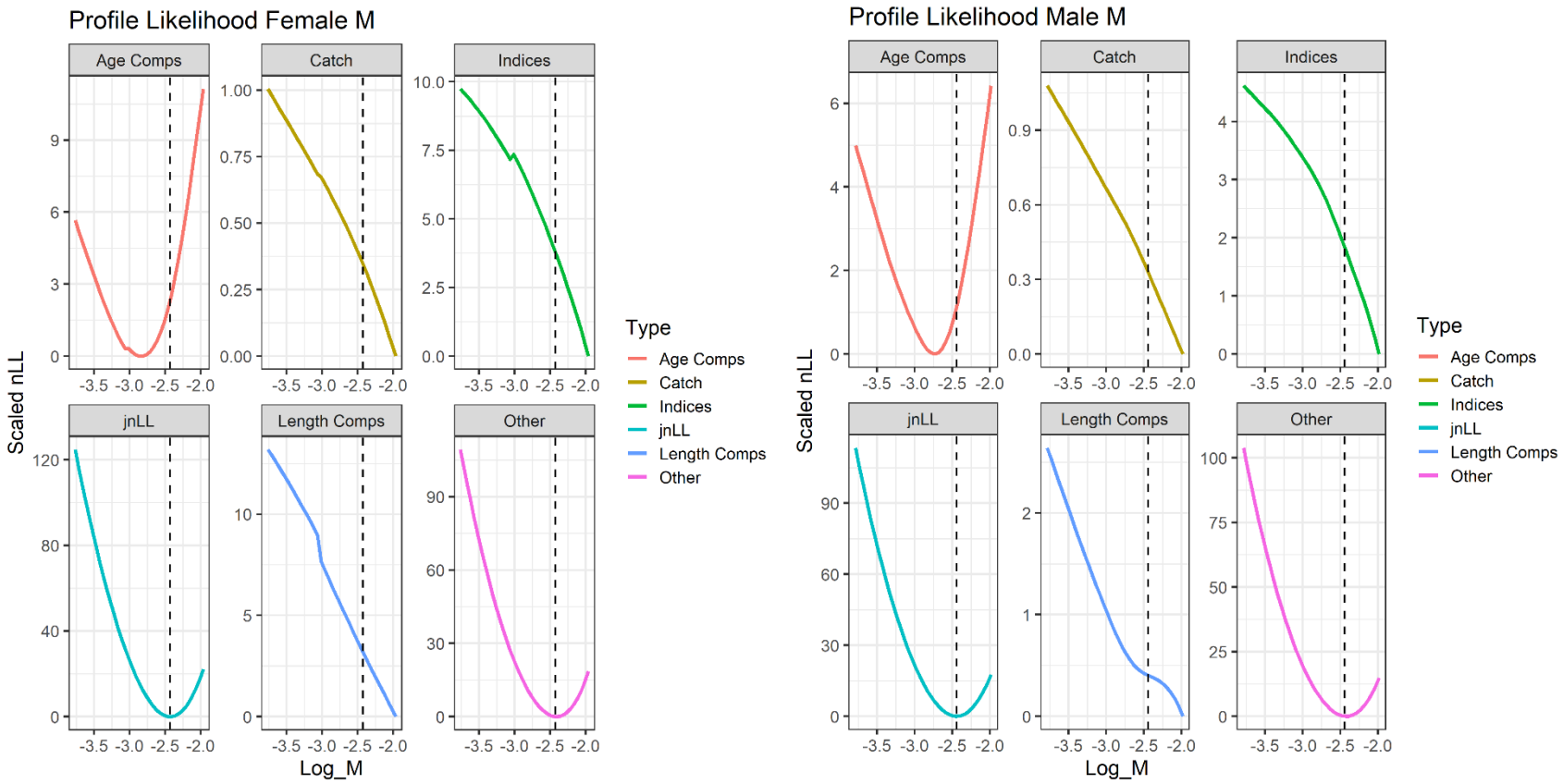


Figure 3.96. Likelihood profile for the natural mortality parameter (*M*) for females (left) and males (right), where panels represent the profile for each data type used in the model, for Model *26.2_FAA*. The 'other' category represents the priors and penalties, which is particularly dominated by the *M* prior.

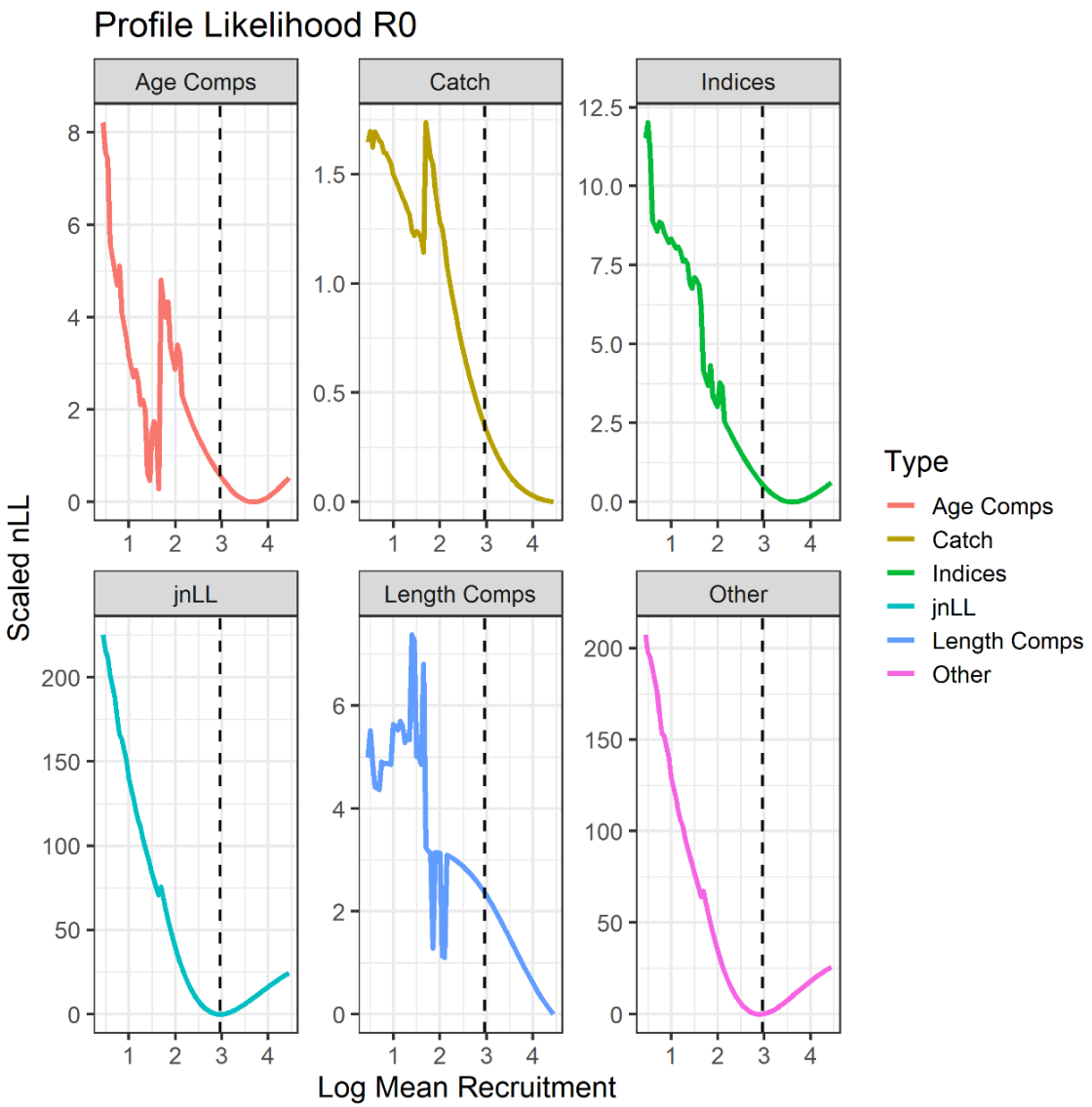


Figure 3.97. Likelihood profile for the mean recruitment (R_0), where panels represent the profile for each data type used in the model, for Model *26.1_1Reg*.

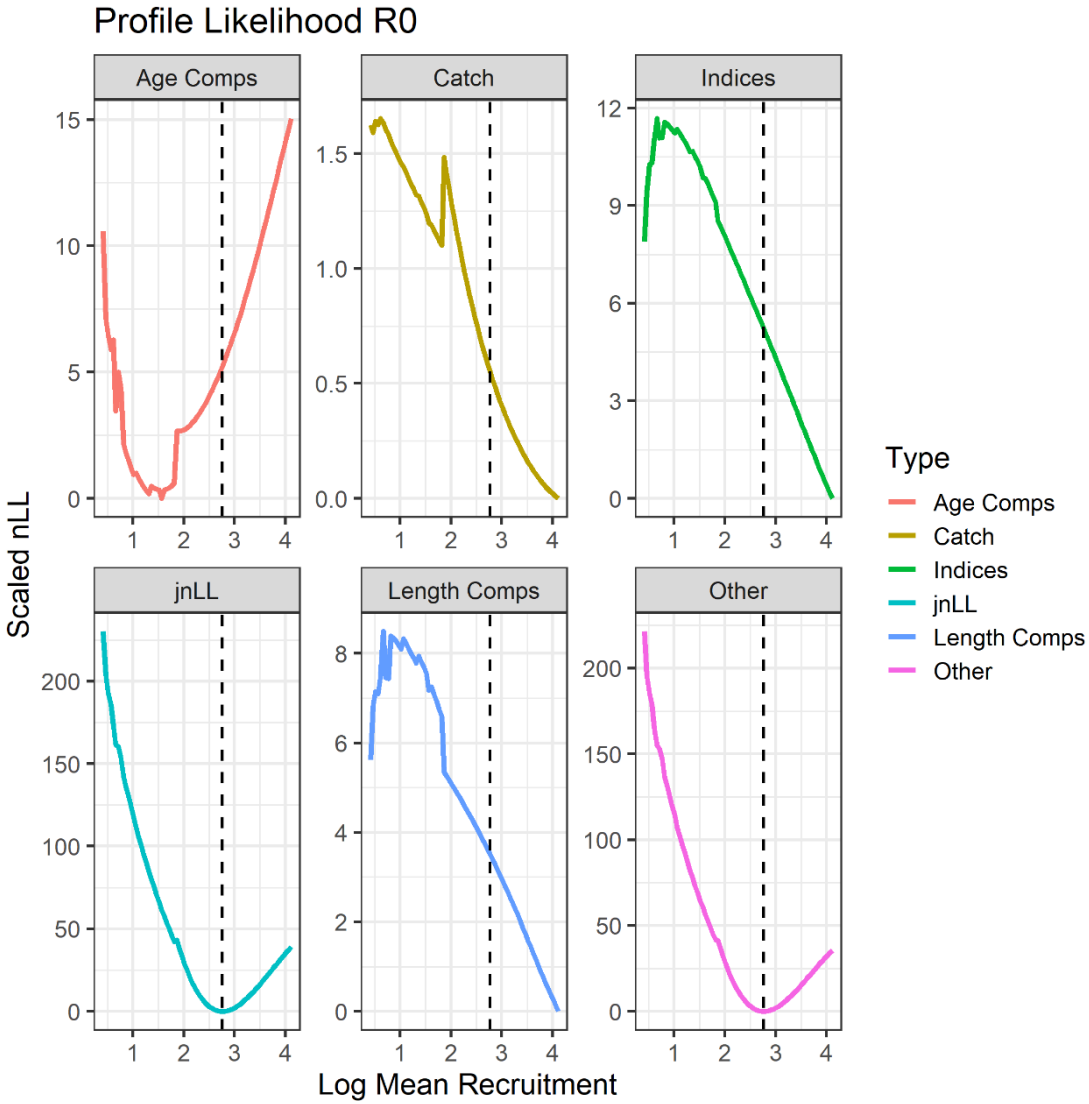


Figure 3.98. Likelihood profile for the mean recruitment (R_0), where panels represent the profile for each data type used in the model, for Model *26.2_FAA*.

Supplemental Material

Sablefish Longline Survey Fishery and Whale Interactions

NMFS has requested the assistance of the fishing fleet to avoid the annual longline survey stations since the inception of sablefish IFQ management in 1995. We request that fishermen stay at least five nm away from each survey station for 7 days before and 3 days after the planned sampling date (3 days allows for survey delays). Since 2021, survey calendars have been made available online (https://www.fisheries.noaa.gov/resource/document/alaska-sablefish-longline-survey-station-schedule). While the survey is being conducted, the skipper of the vessel makes announcements on the radio detailing the planned set locations for the upcoming days. Vessels encountered near survey stations are contacted by the survey vessel captain and interviewed to determine potential effects on survey catches. Typically, vessels have been aware of the survey and have not been fishing close to survey locations. Even with communication, there are some instances where survey gear was fished nearby commercial fishing gear or where commercial fishing had recently occurred. There are generally few interactions during the 90-day survey (Table SM3.1). In 2023 there were eight instances of vessel interactions that may have impacted survey catch or required the survey vessel to move the day's sets from their originally intended locations. In the GOA, there were 5 interactions with pot boats (3 in Southeast Outside, 1 in West Yakutat, and 1 in the Central GOA) and 3 interactions with longline vessels (1 in the western GOA and 2 in the central GOA). There were no vessel interactions in the Bering Sea. The number of sets impacted by whales is also tallied (Table SM3.2) and those sets are dropped (for orca interaction; 16 stations, mostly in the BS) or catch rates inflated using the survey whale inflation factor (for sperm whales; 12 stations, mostly in the eastern GOA; see 'Whale Depredation Estimation' section). No longline survey was conducted in 2024.

Table SM3.1. Count of longline survey and fishery vessel interactions by area, fishery gear type, and year.

	Longline		Trawl		Pot		Total	
Year	Stations	Vessels	Stations	Vessels	Stations	Vessels	Stations	Vessels
1995	8	7	9	15	0	0	17	22
1996	11	18	15	17	0	0	26	35
1997	8	8	8	7	0	0	16	15
1998	10	9	0	0	0	0	10	9
1999	4	4	2	6	0	0	6	10
2000	10	10	0	0	0	0	10	10
2001	1	1	1	1	0	0	2	2
2002	3	3	0	0	0	0	3	3
2003	4	4	2	2	0	0	6	6
2004	5	5	0	0	1	1	6	6
2005	1	1	1	1	0	0	2	2
2006	6	6	1	2	0	0	7	8
2007	8	6	2	2	0	0	10	8
2008	2	2	2	2	0	0	4	4
2009	3	3	0	0	0	0	3	3
2010	2	2	1	1	0	0	3	3
2011	3	3	0	0	0	0	3	3
2012	5	5	0	0	0	0	5	5
2013	5	5	0	0	0	0	5	5
2014	2	2	0	0	0	0	2	2
2015	3	3	1	1	0	0	6	6
2016	5	5	1	1	0	0	6	6
2017	8	10	3	3	3	3	13	16
2018	9	9	3	3	0	0	12	12
2019	4	4	1	1	4	4	9	9
2020	1	1	1	1	3	3	5	5
2021	0	0	0	0	4	4	4	4
2022	1	1	0	0	7	7	8	8
2023	2	3	0	0	4	5	6	8

Table SM3.2. Count of stations where sperm (S) or killer whale (K) depredation occurred and the number of stations sampled (in parentheses) by management area. Only stations used for RPN calculations are included. Areas not surveyed in a given year are left blank. If there were no whale depredation data taken, it is denoted with an "n/a". Killer whale depredation did not always occur on all skates of gear, and only those skates with depredation were removed from calculations of RPNs and RPWs.

	BS (16)		AI (14)		WG (10)		CG (16)		WY (8)		SEO (17)	
Year	**S**	**K**	**S**	**K**	**S**	**K**	**S**	**K**	**S**	**K**	**S**	**K**
1996			n/a	1	n/a	0	n/a	0	n/a	0	n/a	0
1997	n/a	2			n/a	0	n/a	0	n/a	0	n/a	0
1998			0	1	0	0	0	0	4	0	0	0
1999	0	7			0	0	3	0	6	0	4	0
2000			0	1	0	1	0	0	4	0	2	0
2001	0	5			0	0	3	0	0	0	2	0
2002			0	1	0	4	3	0	4	0	2	0
2003	0	7			0	3	2	0	1	0	2	0
2004			0	0	0	4	3	0	4	0	6	0
2005	0	2			0	4	0	0	2	0	8	0
2006			0	1	0	3	2	1	4	0	2	0
2007	0	7			0	5	1	1	5	0	6	0
2008			0	3	0	2	2	0	8	0	9	0
2009	0	10			0	2	5	1	3	0	2	0
2010			0	3	0	1	2	1	2	0	6	0
2011	0	7			0	5	1	1	4	0	9	0
2012			1	5	1	5	2	0	4	0	3	0
2013	0	11			0	2	2	2	3	0	7	0
2014			1	3	0	4	4	0	6	0	4	0
2015	0	9			0	5	4	0	6	0	7	0
2016			1	0	0	3	3	2	5	0	6	0
2017	0	11			1	2	4	0	3	0	9	0
2018			0	2	0	3	3	0	7	0	9	0
2019	0	10			1	4	6	3	6	0	4	0
2020			0	7	1	5	3	1	4	0	6	0
2021	0	10			0	1	5	0	1	0	2	0
2022			0	1	0	4	2	0	1	0	5	0
2023	0	12			0	3	2	1	2	0	8	0

Supplemental Catch Data

In order to address NS1 total accounting requirements, discards (Table SM3.3), FMP bycatch (Table SM3.4), nontarget bycatch (Table SM3.5), and prohibited species catch (PSC; Table SM3.6) are reported in this appendix. Note that all non-commercial catch is now accounted for in the assessment model catch, so no additional table is provided for these sources of removal.

Table SM3.3. Retained and discarded catch of sablefish (t) by year, fleet, and management (FMP) region. Abbreviations are: HAL = hook-and-line; POT = pot; TRW = trawl; and JIG = jig fishery. Source: NMFS Alaskan Regional Office Catch Accounting System via AKFIN (www.akfin.org), accessed on October 10, 2024. Discards are included in the assessment model catch assuming 100% mortality.

		Gulf of Alaska		Bering Sea/Aleutian Islands	
Year	**Gear Type**	**Retained**	**Discarded**	**Retained**	**Discarded**
2020	HAL	4,954	440	188	231
	POT	4,614	136	948	32
	TRW	1,088	1,243	2,239	2,924
2021	HAL	3,723	354	252	315
	POT	9,647	193	1,881	50
	TRW	1,142	460	1,374	1,875
2022	HAL	3,234	401	200	222
	POT	13,493	156	3,781	56
	TRW	1,354	527	2,209	1,276
2023	HAL	2,655	335	188	135
	POT	11,850	94	4,155	9
	TRW	1,173	291	3,648	485
2024	HAL	1,916	162	85	161
	JIG	0.01	NA	NA	NA
	POT	10,052	55	1,043	5
	TRW	993	229	3,864	48

Table SM3.4. Bycatch (t) of FMP groundfish species by year in the targeted sablefish fishery. Source: NMFS Alaskan Regional Office Catch Accounting System via AKFIN (www.akfin.org), accessed on October 10, 2024.

	2020		2021		2022		2023		2024	
Species	**R**	**D**	**R**	**D**	**R**	**D**	**R**	**D**	**R**	**D**
Arrowtooth Flounder	24.1	410.6	34.4	504.1	38.5	435.0	170.5	237.8	456.5	80.0
BSAI Kamchatka Flounder	9.0	7.4	9.7	3.9	60.3	18.9	235.6	26.6	497.3	12.9
BSAI Other Flatfish	0.7	0.2	5.0	0.7	22.4	14.6	133.8	6.3	374.8	19.9
BSAI Shortraker Rockfish	1.1	0.3	1.6	0.1	6.4	2.3	16.9	2.3	14.0	2.6
BSAI and GOA Skate	0.5	174.3	3.9	68.3	0.6	71.9	4.4	214.2	21.3	98.8
Flathead Sole	6.5	5.3	3.0	1.4	31.4	5.3	79.8	3.1	269.2	7.6
GOA Deep Water Flatfish	0.2	14.3	0.6	13.6	2.0	19.9	0.1	14.3	0.2	14.0
GOA Demersal Shelf Rockfish	9.8	1.0	17.5	2.1	20.7	2.2	17.9	0.5	20.9	0.3
GOA Rex Sole	1.5	6.8	0.3	2.2	0.7	3.3	2.1	0.9	3.8	1.4
GOA Skate, Big	0.7	28.0	2.4	31.0	1.5	27.7	1.6	22.2	0.4	2.4
GOA Skate, Longnose	7.9	54.1	4.8	122.3	4.7	212.8	4.6	91.7	3.4	29.6
GOA Thornyhead Rockfish	227.1	18.6	108.9	12.2	76.3	11.5	60.3	6.4	51.8	4.6
Greenland Turbot	10.4	1.0	1.9	1.7	91.3	7.2	117.9	23.3	237.7	5.3
Halibut	589.1	217.1	604.3	222.7	663.5	299.0	548.9	284.3	424.9	94.0
Octopus	0.0	2.6	0.0	1.3	0.2	2.1	0.2	25.3	0.0	3.1
Other Rockfish	17.6	12.8	22.6	27.8	119.6	26.1	106.6	52.9	395.5	82.8
Pacific Cod	39.7	19.9	64.6	16.4	23.6	13.2	44.3	4.4	135.7	11.2
Pacific Ocean Perch	58.5	1.7	23.7	0.4	40.7	12.4	104.7	18.0	124.2	8.3
Pollock	6.7	13.8	5.2	55.8	26.3	13.0	139.0	15.4	322.0	10.7
Rougheye Rockfish	125.2	49.2	96.6	62.5	98.7	35.2	105.2	19.0	66.4	12.0
Shortraker Rockfish	100.4	135.6	71.1	116.4	56.9	35.6	58.6	18.9	37.1	18.8

Table SM3.5. Bycatch of nontarget species and HAPC biota in the targeted sablefish fishery. Source: NMFS AKRO Blend/Catch Accounting System via AKFIN, October 10, 2024.

Species Group	2020	2021	2022	2023	2024
Benthic urochordata	0.01	0.07	0.04	0.18	0.11
Brittle star	0.29	0.33	2.56	0.60	1.93
Corals Bryozoans	1.08	1.48	2.91	0.43	0.27
Eelpouts	0.14	0.55	9.52	19.75	92.54
Giant Grenadier	1,408.95	795.91	589.45	955.23	700.49
Grenadier	482.61	167.89	143.37	148.39	15.12
Hermit crab	0.01	0.01	0.07	0.07	0.36
Invertebrate	0.06	0.13	0.24	0.04	0.12
Misc crabs	4.25	3.86	5.35	8.71	8.38
Misc fish	39.16	29.05	71.71	38.42	10.69
Pandalid shrimp	0.04	--	0.04	0.29	0.43
Sculpin	--	4.26	6.91	171.66	11.78
Scypho jellies	0.34	0.26	0.58	1.06	0.76
Sea anemone	1.01	2.58	3.90	8.57	14.17
Sea pens whips	0.54	0.22	0.04	0.27	0.65
Sea star	7.47	3.83	16.68	32.64	49.15
Snails	2.91	3.71	7.08	20.07	2.16
Sponge	0.33	0.30	0.68	1.31	1.47
Squid	0.73	0.47	10.26	24.08	55.49
Urchins, dollars, cucumbers	0.54	0.34	1.39	5.55	29.73

Table SM3.6. Prohibited Species Catch (PSC) estimates (in metric tons for halibut and numbers of animals for crab and salmon) by year and management (FMP) region for the sablefish target fishery. HAL is hook and line gear; POT is pot gear; and TRW is trawl gear. Source: NMFS Alaska Regional Office Catch Accounting System PSCNQ via AKFIN (www.akfin.org), accessed on October 10, 2024.

Gulf of Alaska

Species	Gear Type	2020	2021	2022	2023	2024
Bairdi Tanner Crab	HAL	-	10	16	29	16
	POT	96	359	481	71	25
	TRW	1,668	1,535	1,554	250	-
Blue King Crab	HAL	-	-	-	-	-
	POT	-	-	-	-	-
	TRW	-	-	-	-	-
Chinook Salmon	HAL	-	-	-	-	-
	POT	-	-	-	-	-
	TRW	-	711	-	-	-
Golden (Brown) King Crab	HAL	47	17	30	5	0
	POT	39	64	187	1,470	1,432
	TRW	-	-	-	16	-
Halibut	HAL	147	117	116	33	33
	POT	37	73	78	37	15
	TRW	21	35	18	3	2
Herring	HAL	-	-	-	-	-
	POT	-	-	-	-	-
	TRW	-	-	-	-	-
Non-Chinook Salmon	HAL	114	149	11	-	0
	POT	-	-	-	-	-
	TRW	-	-	-	-	-
Opilio Tanner (Snow) Crab	HAL	-	-	-	-	-
	POT	2	-	-	-	-
	TRW	-	-	-	-	-
Red King Crab	HAL	-	0	-	-	-
	POT	-	-	-	-	-
	TRW	-	-	-	-	-

Bering Sea/Aleutian Islands

Species	Gear Type	2020	2021	2022	2023	2024
Bairdi Tanner Crab	HAL	2	4	14	49	1
	POT	167	685	1,288	133	123
	TRW	-	-	69	1,223	1,469
Blue King Crab	HAL	1	0	9	6	0
	POT	-	-	-	-	-
	TRW	-	-	-	-	-
Chinook Salmon	HAL	0	0	0	0	0
	POT	-	-	-	-	-
	TRW	-	-	-	-	-
Golden (Brown) King Crab	HAL	0	42	2	1	46
	POT	5,789	28,750	10,340	22,792	6,992
	TRW	-	135	618	2,510	1,193
Halibut	HAL	4	11	0	0	-
	POT	7	10	13	8	8
	TRW	1	3	7	29	46
Herring	HAL	-	-	-	-	-
	POT	-	-	-	-	-
	TRW	-	0	-	0	1
Non-Chinook Salmon	HAL	0	0	0	0	0
	POT	-	-	-	-	-
	TRW	-	-	-	-	-
Opilio Tanner (Snow) Crab	HAL	14	29	44	204	8
	POT	374	846	1,675	1,274	65
	TRW	-	-	171	212	59
Red King Crab	HAL	0	0	2	3	0
	POT	18	-	-	25	-
	TRW	-	-	-	-	76

Additional Assessment Inputs and Outputs

Table SM3.7. Estimates of von Bertalanffy growth parameters and length-weight parameters for sablefish (top panel) and associated fork length (cm), weight (kg), and proportion mature by age and sex (bottom panel). The 2021 values represent estimates used in previous assessments as estimated in Echave (2021). The 2026 estimates represent the updated growth and weight estimates using all NOAA Longline Survey data from 1996 through 2023.

Parameter	Type	Sex	2021 Estimates	2026 Estimates
L_∞	Length-at-age	Female	81.2	81.5
L_∞	Length-at-age	Male	67.9	67.9
k	Length-at-age	Female	0.17	0.16
k	Length-at-age	Male	0.23	0.22
t_0	Length-at-age	Female	3.28	3.45
t_0	Length-at-age	Male	3.30	3.62
W_∞	Weight-at-age / Length-weight	Female	5.87	5.92
W_∞	Weight-at-age / Length-weight	Male	3.22	3.29
β	Weight-at-age / Length-weight	Female	3.02	3.23
β	Weight-at-age / Length-weight	Male	3.02	3.19

	Fork Length (cm)				Round Weight (kg)				Maturity
Age	2021 Male	2026 Male	2021 Female	2026 Female	2021 Male	2026 Male	2021 Female	2026 Female	Female
2	47.9	47.8	48.0	47.9	1.1	1.1	1.1	1.1	0.02
3	52.0	51.7	53.2	52.9	1.4	1.4	1.6	1.5	0.05
4	55.3	54.9	57.6	57.2	1.8	1.7	2.0	1.9	0.09
5	57.9	57.4	61.3	60.8	2.1	1.9	2.5	2.3	0.18
6	60.0	59.5	64.4	63.9	2.3	2.1	2.9	2.7	0.31
7	61.6	61.1	67.0	66.6	2.5	2.3	3.3	3.1	0.49
8	62.9	62.4	69.2	68.8	2.7	2.5	3.6	3.4	0.67
9	64.0	63.5	71.1	70.7	2.8	2.6	3.9	3.7	0.81
10	64.8	64.4	72.7	72.3	2.9	2.8	4.2	4.0	0.90
11	65.4	65.1	74.0	73.7	3.0	2.9	4.4	4.3	0.95
12	66.0	65.7	75.1	74.9	3.0	2.9	4.7	4.5	0.98
13	66.4	66.1	76.1	75.9	3.1	3.0	4.8	4.7	0.99
14	66.7	66.5	76.9	76.7	3.1	3.1	5.0	4.9	0.99
15	66.9	66.8	77.6	77.5	3.1	3.1	5.1	5.0	1.00
16	67.1	67.0	78.1	78.1	3.2	3.1	5.2	5.1	1.00
17	67.3	67.2	78.6	78.6	3.2	3.2	5.3	5.3	1.00
18	67.4	67.4	79.0	79.0	3.2	3.2	5.4	5.4	1.00
19	67.5	67.5	79.4	79.4	3.2	3.2	5.5	5.4	1.00
20	67.6	67.6	79.7	79.7	3.2	3.2	5.5	5.5	1.00
21	67.7	67.7	79.9	80.0	3.2	3.2	5.6	5.6	1.00
22	67.7	67.7	80.1	80.2	3.2	3.2	5.6	5.6	1.00
23	67.8	67.8	80.3	80.4	3.2	3.3	5.7	5.7	1.00
24	67.8	67.8	80.4	80.6	3.2	3.3	5.7	5.7	1.00
25	67.8	67.9	80.6	80.7	3.2	3.3	5.7	5.7	1.00
26	67.9	67.9	80.7	80.8	3.2	3.3	5.8	5.8	1.00
27	67.9	67.9	80.8	81.0	3.2	3.3	5.8	5.8	1.00
28	67.9	67.9	80.8	81.0	3.2	3.3	5.8	5.8	1.00
29	67.9	67.9	80.9	81.1	3.2	3.3	5.8	5.8	1.00
30	67.9	67.9	80.9	81.2	3.2	3.3	5.8	5.8	1.00
31+	67.9	68.0	81.0	81.2	3.2	3.3	5.8	5.9	1.00

Table SM3.8. Maximum likelihood estimation parameter values and standard errors from the inverse of the Hessian for all estimated parameters for Model *26.1_1Reg*.

Parameter	Estimate	Std. Error
ln_global_R0	2.97	0.13
ln_InitDevs	-0.01	0.40
ln_InitDevs	-0.03	0.39
ln_InitDevs	-0.05	0.39
ln_InitDevs	-0.05	0.39
ln_InitDevs	-0.05	0.39
ln_InitDevs	-0.05	0.39
ln_InitDevs	-0.05	0.39
ln_InitDevs	-0.05	0.39
ln_InitDevs	-0.04	0.39
ln_InitDevs	-0.04	0.39
ln_InitDevs	-0.04	0.39
ln_InitDevs	-0.04	0.39
ln_InitDevs	-0.04	0.39
ln_InitDevs	-0.03	0.39
ln_InitDevs	-0.03	0.39
ln_InitDevs	-0.03	0.39
ln_InitDevs	-0.03	0.39
ln_InitDevs	-0.02	0.40
ln_InitDevs	-0.02	0.40
ln_InitDevs	-0.02	0.40
ln_InitDevs	-0.02	0.40
ln_InitDevs	-0.02	0.40
ln_InitDevs	-0.02	0.40
ln_InitDevs	-0.01	0.40
ln_InitDevs	-0.01	0.40
ln_InitDevs	-0.01	0.40
ln_InitDevs	-0.01	0.40
ln_InitDevs	-0.01	0.40
ln_RecDevs	0.00	0.40
ln_RecDevs	0.01	0.40
ln_RecDevs	0.04	0.40
ln_RecDevs	0.08	0.41
ln_RecDevs	0.12	0.42
ln_RecDevs	0.14	0.42
ln_RecDevs	0.14	0.42
ln_RecDevs	0.12	0.42
ln_RecDevs	0.08	0.41
ln_RecDevs	0.03	0.40
ln_RecDevs	-0.03	0.39

ln_RecDevs	-0.08	0.38
ln_RecDevs	-0.13	0.37
ln_RecDevs	-0.18	0.36
ln_RecDevs	-0.23	0.35
ln_RecDevs	-0.72	0.66
ln_RecDevs	-0.74	0.64
ln_RecDevs	-0.72	0.64
ln_RecDevs	-0.53	0.65
ln_RecDevs	0.32	0.59
ln_RecDevs	1.48	0.27
ln_RecDevs	0.02	0.80
ln_RecDevs	0.50	0.75
ln_RecDevs	1.53	0.28
ln_RecDevs	-0.35	0.72
ln_RecDevs	-0.27	0.66
ln_RecDevs	0.30	0.46
ln_RecDevs	-0.65	0.61
ln_RecDevs	-0.88	0.60
ln_RecDevs	-0.27	0.49
ln_RecDevs	-0.38	0.58
ln_RecDevs	0.49	0.28
ln_RecDevs	-0.58	0.62
ln_RecDevs	0.33	0.34
ln_RecDevs	-0.24	0.50
ln_RecDevs	-0.44	0.50
ln_RecDevs	-0.09	0.40
ln_RecDevs	0.14	0.33
ln_RecDevs	-0.48	0.54
ln_RecDevs	0.63	0.25
ln_RecDevs	0.22	0.41
ln_RecDevs	0.03	0.51
ln_RecDevs	0.81	0.26
ln_RecDevs	0.05	0.45
ln_RecDevs	-0.39	0.51
ln_RecDevs	-0.30	0.42
ln_RecDevs	-0.62	0.49
ln_RecDevs	-0.42	0.42
ln_RecDevs	-0.34	0.45
ln_RecDevs	0.09	0.37
ln_RecDevs	0.06	0.37
ln_RecDevs	-0.42	0.48
ln_RecDevs	-0.42	0.40
ln_RecDevs	-0.97	0.51
ln_RecDevs	-0.65	0.48

ln_RecDevs	-0.09	0.41
ln_RecDevs	1.21	0.21
ln_RecDevs	0.26	0.55
ln_RecDevs	2.02	0.19
ln_RecDevs	1.72	0.30
ln_RecDevs	0.85	0.68
ln_RecDevs	0.92	0.65
ln_RecDevs	1.03	0.69
ln_RecDevs	0.71	0.87
ln_RecDevs	0.47	1.00
ln_M	-2.39	0.07
ln_M	-2.41	0.07
ln_F_mean	-3.03	0.13
ln_F_mean	-4.18	0.22
ln_F_devs	-2.13	0.13
ln_F_devs	-0.47	0.13
ln_F_devs	0.05	0.13
ln_F_devs	-0.83	0.13
ln_F_devs	-0.03	0.14
ln_F_devs	-1.98	0.13
ln_F_devs	-0.47	0.14
ln_F_devs	-3.27	0.13
ln_F_devs	0.20	0.14
ln_F_devs	-1.83	0.13
ln_F_devs	0.62	0.14
ln_F_devs	-1.80	0.13
ln_F_devs	0.89	0.13
ln_F_devs	-0.72	0.13
ln_F_devs	1.18	0.13
ln_F_devs	-0.36	0.12
ln_F_devs	1.31	0.13
ln_F_devs	0.07	0.12
ln_F_devs	1.02	0.13
ln_F_devs	0.14	0.12
ln_F_devs	1.39	0.13
ln_F_devs	0.45	0.12
ln_F_devs	1.65	0.12
ln_F_devs	0.33	0.12
ln_F_devs	1.16	0.12
ln_F_devs	0.49	0.12
ln_F_devs	0.83	0.12
ln_F_devs	0.47	0.12
ln_F_devs	0.58	0.12
ln_F_devs	0.62	0.12

ln_F_devs	0.60	0.12
ln_F_devs	0.40	0.12
ln_F_devs	-0.25	0.12
ln_F_devs	-0.27	0.12
ln_F_devs	-0.94	0.12
ln_F_devs	-0.10	0.12
ln_F_devs	-0.70	0.12
ln_F_devs	-0.27	0.11
ln_F_devs	-0.45	0.12
ln_F_devs	-0.01	0.12
ln_F_devs	-0.69	0.11
ln_F_devs	-0.22	0.12
ln_F_devs	-0.61	0.11
ln_F_devs	-0.35	0.11
ln_F_devs	-0.81	0.11
ln_F_devs	-0.38	0.11
ln_F_devs	-0.02	0.11
ln_F_devs	-0.26	0.11
ln_F_devs	-1.00	0.11
ln_F_devs	0.17	0.11
ln_F_devs	0.55	0.11
ln_F_devs	0.46	0.11
ln_F_devs	0.60	0.11
ln_F_devs	0.60	0.11
ln_F_devs	0.83	0.11
ln_F_devs	0.63	0.11
ln_F_devs	0.68	0.11
ln_F_devs	0.66	0.11
ln_F_devs	0.32	0.11
ln_F_devs	0.66	0.11
ln_F_devs	-0.03	0.11
ln_F_devs	0.64	0.11
ln_F_devs	-0.02	0.11
ln_F_devs	0.77	0.11
ln_F_devs	-0.11	0.11
ln_F_devs	0.71	0.11
ln_F_devs	0.11	0.11
ln_F_devs	0.63	0.11
ln_F_devs	0.05	0.11
ln_F_devs	0.49	0.11
ln_F_devs	-0.08	0.11
ln_F_devs	0.38	0.11
ln_F_devs	-0.35	0.11
ln_F_devs	0.37	0.11

ln_F_devs	-0.42	0.11
ln_F_devs	0.31	0.11
ln_F_devs	-0.14	0.11
ln_F_devs	0.48	0.11
ln_F_devs	-0.12	0.11
ln_F_devs	0.36	0.11
ln_F_devs	-0.25	0.11
ln_F_devs	0.34	0.11
ln_F_devs	-0.05	0.11
ln_F_devs	0.45	0.11
ln_F_devs	-0.16	0.11
ln_F_devs	0.49	0.11
ln_F_devs	-0.39	0.11
ln_F_devs	0.39	0.11
ln_F_devs	-0.46	0.11
ln_F_devs	0.36	0.11
ln_F_devs	-0.69	0.11
ln_F_devs	0.41	0.11
ln_F_devs	-0.69	0.11
ln_F_devs	0.36	0.11
ln_F_devs	-0.77	0.11
ln_F_devs	0.30	0.11
ln_F_devs	-0.81	0.11
ln_F_devs	0.24	0.11
ln_F_devs	-0.85	0.11
ln_F_devs	0.33	0.11
ln_F_devs	-0.68	0.11
ln_F_devs	0.39	0.11
ln_F_devs	-0.73	0.11
ln_F_devs	0.40	0.11
ln_F_devs	-0.76	0.11
ln_F_devs	0.25	0.11
ln_F_devs	-0.74	0.11
ln_F_devs	0.24	0.11
ln_F_devs	-0.66	0.11
ln_F_devs	0.13	0.11
ln_F_devs	-0.49	0.11
ln_F_devs	0.16	0.12
ln_F_devs	0.02	0.12
ln_F_devs	0.06	0.12
ln_F_devs	0.39	0.12
ln_F_devs	-0.06	0.13
ln_F_devs	0.54	0.13
ln_F_devs	-0.37	0.13

ln_F_devs	0.77	0.13
ln_F_devs	-0.22	0.12
ln_F_devs	0.19	0.13
ln_F_devs	-0.04	0.12
ln_F_devs	0.16	0.13
ln_F_devs	-0.20	0.12
ln_F_devs	0.11	0.13
ln_F_devs	-0.35	0.12
ln_F_devs	0.01	0.13
ln_F_devs	-0.30	0.12
ln_F_devs	-0.34	0.12
ln_fish_fixed_sel_pars	1.08	0.08
ln_fish_fixed_sel_pars	1.10	0.51
ln_fish_fixed_sel_pars	0.70	0.09
ln_fish_fixed_sel_pars	1.54	0.79
ln_fish_fixed_sel_pars	1.08	0.09
ln_fish_fixed_sel_pars	1.12	0.62
ln_fish_fixed_sel_pars	0.88	0.16
ln_fish_fixed_sel_pars	0.81	0.51
ln_fish_fixed_sel_pars	1.78	0.58
ln_fish_fixed_sel_pars	-1.67	0.69
ln_fish_fixed_sel_pars	1.41	0.29
ln_fish_fixed_sel_pars	-0.42	0.43
ln_srv_fixed_sel_pars	0.87	0.09
ln_srv_fixed_sel_pars	1.56	0.70
ln_srv_fixed_sel_pars	0.64	0.07
ln_srv_fixed_sel_pars	1.17	0.30
ln_srv_fixed_sel_pars	1.06	0.06
ln_srv_fixed_sel_pars	0.78	0.27
ln_srv_fixed_sel_pars	0.84	0.11
ln_srv_fixed_sel_pars	0.60	0.34
ln_srv_fixed_sel_pars	0.81	0.09
ln_srv_fixed_sel_pars	1.41	1.18
ln_srv_fixed_sel_pars	1.01	0.12
ln_srv_fixed_sel_pars	1.16	0.87
ln_srv_q	1.91	0.10
ln_srv_q	1.71	0.09

Table SM3.9. Maximum likelihood estimation parameter values and standard errors from the inverse of the Hessian for all estimated parameters for Model *26.2_FAA*.

Parameter	Estimate	Std. Error
ln_global_R0	2.76	0.11
ln_InitDevs	0.00	0.40
ln_InitDevs	-0.03	0.39
ln_InitDevs	-0.05	0.39
ln_InitDevs	-0.06	0.39
ln_InitDevs	-0.06	0.39
ln_InitDevs	-0.06	0.39
ln_InitDevs	-0.06	0.39
ln_InitDevs	-0.06	0.39
ln_InitDevs	-0.06	0.39
ln_InitDevs	-0.06	0.39
ln_InitDevs	-0.05	0.39
ln_InitDevs	-0.05	0.39
ln_InitDevs	-0.05	0.39
ln_InitDevs	-0.04	0.39
ln_InitDevs	-0.04	0.39
ln_InitDevs	-0.04	0.39
ln_InitDevs	-0.04	0.39
ln_InitDevs	-0.03	0.39
ln_InitDevs	-0.03	0.39
ln_InitDevs	-0.03	0.39
ln_InitDevs	-0.03	0.39
ln_InitDevs	-0.02	0.40
ln_InitDevs	-0.02	0.40
ln_InitDevs	-0.02	0.40
ln_InitDevs	-0.02	0.40
ln_InitDevs	-0.02	0.40
ln_InitDevs	-0.01	0.40
ln_InitDevs	-0.01	0.40
ln_RecDevs	0.02	0.40
ln_RecDevs	0.04	0.41
ln_RecDevs	0.08	0.41
ln_RecDevs	0.14	0.43
ln_RecDevs	0.20	0.44
ln_RecDevs	0.23	0.45
ln_RecDevs	0.24	0.45
ln_RecDevs	0.21	0.44
ln_RecDevs	0.17	0.43
ln_RecDevs	0.11	0.42
ln_RecDevs	0.04	0.40

ln_RecDevs	-0.03	0.39
ln_RecDevs	-0.10	0.37
ln_RecDevs	-0.17	0.36
ln_RecDevs	-0.23	0.35
ln_RecDevs	-0.75	0.65
ln_RecDevs	-0.79	0.64
ln_RecDevs	-0.75	0.64
ln_RecDevs	-0.52	0.66
ln_RecDevs	0.32	0.66
ln_RecDevs	1.61	0.28
ln_RecDevs	-0.06	0.80
ln_RecDevs	0.39	0.83
ln_RecDevs	1.49	0.31
ln_RecDevs	-0.25	0.75
ln_RecDevs	-0.21	0.68
ln_RecDevs	0.21	0.52
ln_RecDevs	-0.56	0.63
ln_RecDevs	-0.56	0.62
ln_RecDevs	-0.13	0.56
ln_RecDevs	0.02	0.59
ln_RecDevs	0.31	0.46
ln_RecDevs	0.11	0.52
ln_RecDevs	0.05	0.39
ln_RecDevs	-1.06	0.54
ln_RecDevs	-0.37	0.39
ln_RecDevs	0.06	0.27
ln_RecDevs	-0.28	0.44
ln_RecDevs	0.50	0.24
ln_RecDevs	-0.13	0.44
ln_RecDevs	-0.31	0.43
ln_RecDevs	0.15	0.37
ln_RecDevs	0.89	0.20
ln_RecDevs	0.10	0.37
ln_RecDevs	-0.72	0.50
ln_RecDevs	-0.47	0.35
ln_RecDevs	-1.35	0.52
ln_RecDevs	-0.14	0.32
ln_RecDevs	-0.42	0.45
ln_RecDevs	0.38	0.29
ln_RecDevs	0.24	0.30
ln_RecDevs	-0.40	0.45
ln_RecDevs	-0.55	0.42
ln_RecDevs	-0.58	0.47
ln_RecDevs	-0.67	0.51

ln_RecDevs	-0.24	0.42
ln_RecDevs	1.73	0.17
ln_RecDevs	0.98	0.34
ln_RecDevs	1.82	0.23
ln_RecDevs	2.32	0.17
ln_RecDevs	0.03	0.75
ln_RecDevs	0.00	0.64
ln_RecDevs	0.35	0.70
ln_RecDevs	-0.18	0.78
ln_RecDevs	-0.12	0.81
ln_M	-2.43	0.07
ln_M	-2.44	0.08
ln_F_mean	-2.65	0.11
ln_F_mean	-3.88	0.15
ln_F_devs	-2.31	0.13
ln_F_devs	-0.66	0.13
ln_F_devs	-0.13	0.13
ln_F_devs	-1.00	0.13
ln_F_devs	-0.21	0.15
ln_F_devs	-2.15	0.13
ln_F_devs	-0.64	0.14
ln_F_devs	-3.45	0.13
ln_F_devs	0.02	0.14
ln_F_devs	-2.01	0.13
ln_F_devs	0.44	0.14
ln_F_devs	-1.98	0.13
ln_F_devs	0.71	0.14
ln_F_devs	-0.91	0.13
ln_F_devs	1.00	0.13
ln_F_devs	-0.54	0.13
ln_F_devs	1.12	0.13
ln_F_devs	-0.10	0.12
ln_F_devs	0.84	0.13
ln_F_devs	-0.02	0.12
ln_F_devs	1.22	0.13
ln_F_devs	0.30	0.12
ln_F_devs	1.50	0.13
ln_F_devs	0.20	0.12
ln_F_devs	1.03	0.13
ln_F_devs	0.38	0.12
ln_F_devs	0.71	0.13
ln_F_devs	0.37	0.12
ln_F_devs	0.47	0.13
ln_F_devs	0.54	0.12

ln_F_devs	0.51	0.13
ln_F_devs	0.34	0.12
ln_F_devs	-0.32	0.13
ln_F_devs	-0.31	0.12
ln_F_devs	-1.01	0.13
ln_F_devs	-0.14	0.12
ln_F_devs	-0.76	0.12
ln_F_devs	-0.30	0.12
ln_F_devs	-0.47	0.12
ln_F_devs	-0.06	0.13
ln_F_devs	-0.70	0.12
ln_F_devs	-0.32	0.12
ln_F_devs	-0.64	0.12
ln_F_devs	-0.45	0.12
ln_F_devs	-0.86	0.12
ln_F_devs	-0.50	0.12
ln_F_devs	-0.10	0.12
ln_F_devs	-0.38	0.12
ln_F_devs	-1.11	0.12
ln_F_devs	0.07	0.12
ln_F_devs	0.41	0.12
ln_F_devs	0.36	0.11
ln_F_devs	0.46	0.12
ln_F_devs	0.52	0.11
ln_F_devs	0.70	0.11
ln_F_devs	0.59	0.11
ln_F_devs	0.58	0.12
ln_F_devs	0.64	0.11
ln_F_devs	0.25	0.12
ln_F_devs	0.66	0.11
ln_F_devs	-0.05	0.12
ln_F_devs	0.64	0.11
ln_F_devs	-0.02	0.12
ln_F_devs	0.79	0.11
ln_F_devs	-0.07	0.12
ln_F_devs	0.74	0.11
ln_F_devs	0.17	0.11
ln_F_devs	0.68	0.11
ln_F_devs	0.13	0.11
ln_F_devs	0.60	0.11
ln_F_devs	0.02	0.11
ln_F_devs	0.52	0.11
ln_F_devs	-0.22	0.11
ln_F_devs	0.52	0.11

ln_F_devs	-0.28	0.11
ln_F_devs	0.47	0.11
ln_F_devs	0.03	0.11
ln_F_devs	0.60	0.11
ln_F_devs	0.06	0.11
ln_F_devs	0.50	0.11
ln_F_devs	-0.06	0.11
ln_F_devs	0.54	0.11
ln_F_devs	0.15	0.11
ln_F_devs	0.67	0.12
ln_F_devs	0.06	0.11
ln_F_devs	0.68	0.12
ln_F_devs	-0.17	0.12
ln_F_devs	0.57	0.12
ln_F_devs	-0.26	0.12
ln_F_devs	0.54	0.11
ln_F_devs	-0.50	0.12
ln_F_devs	0.62	0.11
ln_F_devs	-0.49	0.12
ln_F_devs	0.62	0.11
ln_F_devs	-0.54	0.11
ln_F_devs	0.57	0.11
ln_F_devs	-0.56	0.11
ln_F_devs	0.52	0.12
ln_F_devs	-0.58	0.11
ln_F_devs	0.57	0.12
ln_F_devs	-0.41	0.11
ln_F_devs	0.60	0.12
ln_F_devs	-0.47	0.11
ln_F_devs	0.58	0.12
ln_F_devs	-0.53	0.12
ln_F_devs	0.43	0.12
ln_F_devs	-0.54	0.12
ln_F_devs	0.41	0.12
ln_F_devs	-0.47	0.12
ln_F_devs	0.32	0.12
ln_F_devs	-0.30	0.12
ln_F_devs	0.37	0.15
ln_F_devs	0.17	0.12
ln_F_devs	0.15	0.24
ln_F_devs	0.52	0.14
ln_F_devs	-0.08	0.22
ln_F_devs	0.56	0.15
ln_F_devs	-0.39	0.23

ln_F_devs	0.68	0.16
ln_F_devs	-0.35	0.24
ln_F_devs	-0.01	0.16
ln_F_devs	-0.27	0.17
ln_F_devs	-0.11	0.16
ln_F_devs	-0.41	0.14
ln_F_devs	-0.20	0.15
ln_F_devs	-0.50	0.13
ln_F_devs	-0.28	0.14
ln_F_devs	-0.40	0.14
ln_F_devs	-0.60	0.14
ln_fish_fixed_sel_pars	1.25	0.10
ln_fish_fixed_sel_pars	0.97	0.61
ln_fish_fixed_sel_pars	1.08	0.27
ln_fish_fixed_sel_pars	0.54	0.70
ln_fish_fixed_sel_pars	0.86	0.18
ln_fish_fixed_sel_pars	1.68	1.00
ln_fish_fixed_sel_pars	1.08	0.29
ln_fish_fixed_sel_pars	0.56	0.75
ln_fish_fixed_sel_pars	1.57	0.25
ln_fish_fixed_sel_pars	-0.61	0.43
ln_fish_fixed_sel_pars	1.35	0.20
ln_fish_fixed_sel_pars	0.07	0.43
ln_srv_fixed_sel_pars	1.10	0.07
ln_srv_fixed_sel_pars	1.60	0.79
ln_srv_fixed_sel_pars	0.48	0.21
ln_srv_fixed_sel_pars	2.14	0.70
ln_srv_fixed_sel_pars	1.21	0.09
ln_srv_fixed_sel_pars	1.74	0.86
ln_srv_fixed_sel_pars	0.64	0.10
ln_srv_fixed_sel_pars	1.77	0.76
ln_srv_fixed_sel_pars	0.90	0.12
ln_srv_fixed_sel_pars	2.26	0.61
ln_srv_fixed_sel_pars	0.35	0.18
ln_srv_fixed_sel_pars	1.76	0.62
ln_srv_fixed_sel_pars	1.07	0.04
ln_srv_fixed_sel_pars	1.69	0.72
ln_srv_fixed_sel_pars	0.60	0.11
ln_srv_fixed_sel_pars	1.18	0.36
ln_srv_fixed_sel_pars	1.10	0.09
ln_srv_fixed_sel_pars	0.55	0.30
ln_srv_fixed_sel_pars	1.90	0.16
ln_srv_fixed_sel_pars	-1.16	0.26
ln_srv_fixed_sel_pars	1.23	0.10

ln_srv_fixed_sel_pars	0.30	0.28
ln_srv_fixed_sel_pars	1.97	0.15
ln_srv_fixed_sel_pars	-1.11	0.24
ln_srv_fixed_sel_pars	0.89	0.11
ln_srv_fixed_sel_pars	1.32	0.76
ln_srv_fixed_sel_pars	1.31	0.16
ln_srv_fixed_sel_pars	0.07	0.36
ln_srv_q	0.55	0.10
ln_srv_q	0.34	0.10
ln_srv_q	2.20	0.08
ln_srv_q	2.08	0.08

Table SM3.10. Number (in millions of fish) of sablefish estimated by Model *26.1_1Reg* by age and year. Columns are age with age-31+ being the plus group. The terminal year plus one (2026) numbers are incremented based on the terminal year (2025) mortality and recruitment is set at 0.

Year	Sex	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31+
1960	Female	9.7	8.8	7.9	7.0	6.4	5.8	5.3	4.8	4.3	4.0	3.6	3.3	3.0	2.7	2.5	2.3	2.0	1.9	1.7	1.5	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.7	0.7	6.6
1961	Female	9.8	8.9	8.0	7.2	6.4	5.8	5.2	4.8	4.3	3.9	3.6	3.3	3.0	2.7	2.5	2.2	2.0	1.9	1.7	1.5	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.7	0.7	6.5
1962	Female	10.1	9.0	8.1	7.2	6.3	5.7	5.1	4.6	4.2	3.8	3.5	3.2	2.9	2.6	2.4	2.2	2.0	1.8	1.6	1.5	1.4	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.6	6.4
1963	Female	10.5	9.2	8.2	7.2	6.3	5.5	4.9	4.4	4.0	3.7	3.3	3.0	2.8	2.5	2.3	2.1	1.9	1.7	1.6	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.7	0.7	0.6	6.1
1964	Female	11.0	9.6	8.3	7.3	6.4	5.6	4.9	4.4	3.9	3.6	3.2	2.9	2.7	2.4	2.2	2.0	1.8	1.7	1.5	1.4	1.3	1.1	1.0	0.9	0.9	0.8	0.7	0.6	0.6	5.9
1965	Female	11.2	10.0	8.7	7.6	6.6	5.8	5.0	4.4	3.9	3.5	3.2	2.9	2.7	2.4	2.2	2.0	1.8	1.6	1.5	1.4	1.2	1.1	1.0	0.9	0.9	0.8	0.7	0.6	0.6	5.8
1966	Female	11.2	10.2	9.0	7.9	6.8	6.0	5.2	4.5	4.0	3.5	3.2	2.9	2.6	2.4	2.2	2.0	1.8	1.6	1.5	1.3	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.6	0.6	5.7
1967	Female	10.9	10.1	9.2	8.1	7.0	6.1	5.3	4.6	4.0	3.5	3.1	2.8	2.6	2.3	2.1	1.9	1.7	1.6	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.7	0.7	0.6	0.6	5.5
1968	Female	10.5	9.9	9.1	8.2	7.3	6.3	5.4	4.7	4.1	3.6	3.1	2.8	2.5	2.3	2.0	1.8	1.7	1.5	1.4	1.3	1.1	1.0	0.9	0.9	0.8	0.7	0.6	0.6	0.5	5.3
1969	Female	9.9	9.4	8.9	8.1	7.2	6.3	5.5	4.7	4.1	3.5	3.1	2.7	2.4	2.1	1.9	1.7	1.6	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.7	0.7	0.6	0.5	0.5	5.0
1970	Female	9.3	8.9	8.4	7.8	7.0	6.2	5.4	4.7	4.0	3.5	3.0	2.6	2.3	2.0	1.8	1.6	1.5	1.3	1.2	1.1	1.0	0.9	0.8	0.7	0.7	0.6	0.6	0.5	0.5	4.6
1971	Female	8.8	8.4	8.0	7.3	6.6	5.9	5.2	4.6	3.9	3.4	2.9	2.5	2.2	1.9	1.7	1.5	1.4	1.2	1.1	1.0	0.9	0.8	0.7	0.7	0.6	0.6	0.5	0.5	0.4	4.2
1972	Female	8.3	7.9	7.5	6.9	6.2	5.6	5.0	4.4	3.8	3.3	2.8	2.4	2.1	1.8	1.6	1.4	1.2	1.1	1.0	0.9	0.8	0.7	0.7	0.6	0.5	0.5	0.5	0.4	0.4	3.7
1973	Female	7.9	7.4	7.0	6.4	5.7	5.1	4.5	4.0	3.5	3.1	2.6	2.2	1.9	1.7	1.4	1.2	1.1	1.0	0.9	0.8	0.7	0.6	0.6	0.5	0.5	0.4	0.4	0.4	0.3	3.2
1974	Female	7.5	7.1	6.7	6.0	5.4	4.7	4.2	3.8	3.3	2.9	2.5	2.2	1.8	1.6	1.4	1.2	1.0	0.9	0.8	0.7	0.6	0.6	0.5	0.5	0.4	0.4	0.3	0.3	0.3	2.9
1975	Female	3.9	6.8	6.4	5.7	5.0	4.4	3.9	3.5	3.1	2.7	2.4	2.1	1.8	1.5	1.3	1.1	1.0	0.8	0.7	0.6	0.6	0.5	0.5	0.4	0.4	0.3	0.3	0.3	0.3	2.6
1976	Female	3.8	3.6	6.1	5.5	4.8	4.2	3.7	3.3	2.9	2.6	2.3	2.0	1.7	1.5	1.2	1.1	0.9	0.8	0.7	0.6	0.5	0.5	0.4	0.4	0.3	0.3	0.3	0.3	0.2	2.3
1977	Female	3.8	3.4	3.2	5.3	4.6	4.0	3.5	3.0	2.7	2.4	2.1	1.9	1.6	1.4	1.2	1.0	0.9	0.7	0.6	0.6	0.5	0.4	0.4	0.3	0.3	0.3	0.3	0.2	0.2	2.1
1978	Female	4.4	3.4	3.1	2.8	4.5	3.9	3.3	2.9	2.6	2.3	2.0	1.8	1.6	1.4	1.2	1.0	0.9	0.7	0.6	0.5	0.5	0.4	0.4	0.3	0.3	0.3	0.2	0.2	0.2	1.9
1979	Female	10.2	4.0	3.1	2.8	2.5	3.9	3.4	2.9	2.6	2.3	2.0	1.8	1.6	1.4	1.2	1.0	0.9	0.7	0.6	0.5	0.5	0.4	0.4	0.3	0.3	0.3	0.2	0.2	0.2	1.9
1980	Female	31.7	9.3	3.7	2.8	2.4	2.2	3.4	2.9	2.6	2.2	2.0	1.7	1.5	1.4	1.2	1.0	0.9	0.8	0.7	0.6	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.2	1.8
1981	Female	7.2	28.9	8.4	3.3	2.4	2.1	1.9	3.0	2.6	2.2	1.9	1.7	1.5	1.3	1.2	1.0	0.9	0.8	0.7	0.6	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.2	1.7
1982	Female	11.5	6.6	26.2	7.4	2.8	2.1	1.8	1.6	2.6	2.2	1.9	1.7	1.5	1.3	1.1	1.0	0.9	0.8	0.7	0.6	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.2	1.6
1983	Female	31.1	10.4	5.9	23.4	6.5	2.5	1.8	1.6	1.4	2.3	1.9	1.7	1.5	1.3	1.1	1.0	0.9	0.8	0.7	0.6	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.2	1.6
1984	Female	4.8	28.3	9.5	5.3	20.6	5.7	2.2	1.6	1.4	1.3	2.0	1.7	1.5	1.3	1.1	1.0	0.9	0.8	0.7	0.6	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.2	1.5
1985	Female	5.1	4.3	25.7	8.4	4.7	18.0	5.0	1.9	1.4	1.2	1.1	1.7	1.5	1.3	1.1	1.0	0.9	0.8	0.7	0.6	0.5	0.5	0.4	0.3	0.3	0.2	0.2	0.2	0.2	1.4
1986	Female	9.1	4.7	3.9	22.9	7.4	4.1	15.8	4.4	1.7	1.2	1.1	1.0	1.5	1.3	1.1	1.0	0.9	0.8	0.7	0.6	0.5	0.5	0.4	0.3	0.3	0.2	0.2	0.2	0.2	1.4
1987	Female	3.5	8.2	4.2	3.4	19.6	6.3	3.5	13.4	3.7	1.4	1.0	0.9	0.8	1.3	1.1	1.0	0.8	0.7	0.6	0.6	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.1	1.3
1988	Female	2.8	3.2	7.4	3.7	2.9	16.4	5.3	2.9	11.2	3.1	1.2	0.9	0.8	0.7	1.1	0.9	0.8	0.7	0.6	0.5	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.1	1.2
1989	Female	5.2	2.5	2.9	6.4	3.0	2.4	13.4	4.3	2.4	9.1	2.5	1.0	0.7	0.6	0.5	0.9	0.7	0.6	0.6	0.5	0.4	0.4	0.3	0.3	0.3	0.2	0.2	0.2	0.1	1.1
1990	Female	4.6	4.7	2.3	2.5	5.3	2.5	2.0	11.0	3.5	1.9	7.5	2.1	0.8	0.6	0.5	0.4	0.7	0.6	0.5	0.4	0.4	0.3	0.3	0.3	0.2	0.2	0.2	0.2	0.1	1.0
1991	Female	11.0	4.2	4.2	2.0	2.0	4.3	2.0	1.6	9.0	2.9	1.6	6.1	1.7	0.6	0.5	0.4	0.4	0.6	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.2	0.1	0.1	0.9
1992	Female	3.8	10.0	3.8	3.6	1.6	1.7	3.6	1.7	1.3	7.4	2.4	1.3	5.0	1.4	0.5	0.4	0.3	0.3	0.5	0.4	0.3	0.3	0.3	0.2	0.2	0.2	0.2	0.1	0.1	0.8
1993	Female	9.4	3.5	9.0	3.3	3.0	1.4	1.4	3.0	1.4	1.1	6.1	2.0	1.1	4.1	1.1	0.4	0.3	0.3	0.2	0.4	0.3	0.3	0.2	0.2	0.2	0.2	0.1	0.1	0.1	0.8
1994	Female	5.3	8.5	3.1	7.8	2.7	2.5	1.1	1.1	2.4	1.1	0.9	5.0	1.6	0.9	3.4	0.9	0.4	0.3	0.2	0.2	0.3	0.3	0.2	0.2	0.2	0.2	0.1	0.1	0.1	0.7
1995	Female	4.4	4.8	7.7	2.7	6.4	2.2	2.0	0.9	0.9	2.0	0.9	0.7	4.1	1.3	0.7	2.7	0.8	0.3	0.2	0.2	0.2	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.7
1996	Female	6.2	4.0	4.4	6.7	2.2	5.3	1.8	1.7	0.7	0.8	1.6	0.8	0.6	3.4	1.1	0.6	2.3	0.6	0.2	0.2	0.1	0.1	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.6
1997	Female	7.8	5.6	3.6	3.8	5.6	1.9	4.4	1.5	1.4	0.6	0.6	1.4	0.6	0.5	2.8	0.9	0.5	1.9	0.5	0.2	0.1	0.1	0.1	0.2	0.1	0.1	0.1	0.1	0.1	0.6
1998	Female	4.2	7.1	5.1	3.1	3.2	4.7	1.6	3.8	1.3	1.2	0.5	0.5	1.1	0.5	0.4	2.4	0.8	0.4	1.6	0.4	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.6
1999	Female	12.7	3.8	6.4	4.4	2.7	2.7	4.0	1.3	3.2	1.1	1.0	0.4	0.5	1.0	0.5	0.4	2.0	0.6	0.3	1.3	0.4	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.6
2000	Female	8.4	11.6	3.5	5.6	3.8	2.3	2.3	3.4	1.1	2.7	0.9	0.8	0.4	0.4	0.8	0.4	0.3	1.7	0.5	0.3	1.1	0.3	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.5
2001	Female	7.0	7.6	10.5	3.0	4.7	3.2	1.9	1.9	2.9	0.9	2.3	0.8	0.7	0.3	0.3	0.7	0.3	0.3	1.4	0.4	0.2	0.9	0.3	0.1	0.1	0.1	0.1	0.1	0.1	0.5
2002	Female	15.2	6.4	6.9	9.2	2.6	4.0	2.7	1.6	1.6	2.4	0.8	1.9	0.7	0.6	0.3	0.3	0.6	0.3	0.2	1.2	0.4	0.2	0.8	0.2	0.1	0.1	0.1	0.0	0.1	0.5
2003	Female	7.1	13.8	5.7	6.1	7.8	2.2	3.4	2.3	1.4	1.4	2.0	0.7	1.6	0.5	0.5	0.2	0.2	0.5	0.2	0.2	1.0	0.3	0.2	0.7	0.2	0.1	0.1	0.0	0.0	0.5
2004	Female	4.6	6.4	12.5	5.0	5.1	6.6	1.8	2.8	1.9	1.1	1.2	1.7	0.6	1.3	0.5	0.4	0.2	0.2	0.4	0.2	0.1	0.8	0.3	0.1	0.6	0.2	0.1	0.0	0.0	0.4
2005	Female	5.0	4.2	5.8	10.9	4.2	4.3	5.5	1.5	2.4	1.6	1.0	1.0	1.4	0.5	1.1	0.4	0.4	0.2	0.2	0.3	0.2	0.1	0.7	0.2	0.1	0.5	0.1	0.0	0.0	0.4
2006	Female	3.6	4.5	3.8	5.1	9.2	3.6	3.6	4.7	1.3	2.0	1.4	0.8	0.8	1.2	0.4	0.9	0.3	0.3	0.1	0.1	0.3	0.1	0.1	0.6	0.2	0.1	0.4	0.1	0.0	0.4
2007	Female	4.5	3.3	4.1	3.3	4.3	7.8	3.0	3.1	3.9	1.1	1.7	1.1	0.7	0.7	1.0	0.3	0.8	0.3	0.3	0.1	0.1	0.2	0.1	0.1	0.5	0.2	0.1	0.3	0.1	0.3
2008	Female	4.8	4.1	3.0	3.6	2.8	3.7	6.6	2.6	2.6	3.3	0.9	1.4	1.0	0.6	0.6	0.9	0.3	0.7	0.2	0.2	0.1	0.1	0.2	0.1	0.1	0.4	0.1	0.1	0.3	0.4
2009	Female	7.4	4.4	3.7	2.6	3.1	2.4	3.1	5.6	2.2	2.2	2.8	0.8	1.2	0.8	0.5	0.5	0.7	0.2	0.6	0.2	0.2	0.1	0.1	0.2	0.1	0.1	0.4	0.1	0.1	0.5
2010	Female	7.2	6.7	4.0	3.2	2.3	2.6	2.0	2.7	4.8	1.9	1.9	2.4	0.7	1.0	0.7	0.4	0.4	0.6	0.2	0.5	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.3	0.1	0.5
2011	Female	4.4	6.5	6.1	3.5	2.8	1.9	2.2	1.7	2.3	4.1	1.6	1.6	2.1	0.6	0.9	0.6	0.4	0.4	0.5	0.2	0.4	0.1	0.1	0.1	0.1	0.1	0.1	0.0	0.3	0.5
2012	Female	4.5	4.0	5.9	5.4	3.0	2.4	1.6	1.9	1.5	1.9	3.5	1.3	1.4	1.7	0.5	0.8	0.5	0.3	0.3	0.4	0.1	0.4	0.1	0.1	0.0	0.0	0.1	0.0	0.0	0.7
2013	Female	2.6	4.1	3.7	5.2	4.6	2.5	2.0	1.4	1.6	1.3	1.6	2.9	1.1	1.2	1.5	0.4	0.6	0.4	0.3	0.3	0.4	0.1	0.3	0.1	0.1	0.0	0.0	0.1	0.0	0.6
2014	Female	3.5	2.3	3.7	3.2	4.4	3.9	2.1	1.7	1.2	1.4	1.1	1.4	2.5	1.0	1.0	1.2	0.3	0.5	0.4	0.2	0.2	0.3	0.1	0.3	0.1	0.1	0.0	0.0	0.1	0.5
2015	Female	6.2	3.2	2.1	3.2	2.8	3.8	3.3	1.8	1.4	1.0	1.2	0.9	1.2	2.1	0.8	0.8	1.1	0.3	0.5	0.3	0.2	0.2	0.3	0.1	0.2	0.1	0.1	0.0	0.0	0.5
2016	Female	22.7	5.6	2.9	1.9	2.8	2.4	3.2	2.8	1.6	1.2	0.9	1.0	0.8	1.0	1.8	0.7	0.7	0.9	0.3	0.4	0.3	0.2	0.2	0.2	0.1	0.2	0.1	0.1	0.0	0.5
2017	Female	8.7	20.6	5.0	2.5	1.6	2.4	2.0	2.8	2.4	1.3	1.1	0.7	0.9	0.7	0.9	1.5	0.6	0.6	0.8	0.2	0.3	0.2	0.1	0.1	0.2	0.1	0.2	0.1	0.0	0.4
2018	Female	51.1	7.9	18.2	4.3	2.1	1.4	2.0	1.7	2.4	2.1	1.1	0.9	0.6	0.7	0.6	0.7	1.3	0.5	0.5	0.7	0.2	0.3	0.2	0.1	0.1	0.2	0.1	0.1	0.0	0.4
2019	Female	37.7	46.3	7.0	15.7	3.7	1.8	1.2	1.8	1.5	2.0	1.8	1.0	0.8	0.5	0.6	0.5	0.6	1.1	0.4	0.4	0.6	0.2	0.2	0.2	0.1	0.1	0.1	0.0	0.1	0.4

2020	Female	15.8	34.1	41.0	6.1	13.5	3.2	1.6	1.0	1.5	1.3	1.7	1.5	0.8	0.7	0.5	0.5	0.4	0.5	1.0	0.4	0.4	0.5	0.1	0.2	0.1	0.1	0.1	0.1	0.0	0.4
2021	Female	17.0	14.3	30.3	35.8	5.3	11.8	2.7	1.4	0.9	1.3	1.1	1.5	1.3	0.7	0.6	0.4	0.5	0.3	0.5	0.8	0.3	0.3	0.4	0.1	0.2	0.1	0.1	0.1	0.1	0.4
2022	Female	19.0	15.4	12.8	26.4	31.2	4.6	10.2	2.4	1.2	0.8	1.1	0.9	1.3	1.1	0.6	0.5	0.3	0.4	0.3	0.4	0.7	0.3	0.3	0.4	0.1	0.2	0.1	0.1	0.1	0.4
2023	Female	16.4	17.2	13.7	11.0	22.9	26.9	4.0	8.8	2.1	1.0	0.7	1.0	0.8	1.1	1.0	0.5	0.4	0.3	0.3	0.3	0.3	0.6	0.2	0.2	0.3	0.1	0.1	0.1	0.1	0.4
2024	Female	15.6	14.9	15.3	11.9	9.6	19.9	23.4	3.4	7.7	1.8	0.9	0.6	0.8	0.7	1.0	0.8	0.5	0.4	0.2	0.3	0.2	0.3	0.5	0.2	0.2	0.3	0.1	0.1	0.1	0.4
2025	Female	9.7	14.2	13.3	13.5	10.5	8.4	17.4	20.5	3.0	6.7	1.6	0.8	0.5	0.7	0.6	0.8	0.7	0.4	0.3	0.2	0.3	0.2	0.3	0.5	0.2	0.2	0.2	0.1	0.1	0.4
2026	Female	0.0	8.9	12.7	11.7	11.8	9.2	7.4	15.3	17.9	2.6	5.8	1.4	0.7	0.4	0.6	0.5	0.7	0.6	0.4	0.3	0.2	0.2	0.2	0.2	0.4	0.2	0.2	0.2	0.1	0.4
1960	Male	9.7	8.8	7.9	7.1	6.4	5.8	5.3	4.8	4.4	4.0	3.7	3.4	3.1	2.8	2.5	2.3	2.1	1.9	1.8	1.6	1.5	1.3	1.2	1.1	1.0	0.9	0.8	0.8	0.7	7.1
1961	Male	9.8	8.9	8.1	7.2	6.4	5.8	5.3	4.8	4.4	4.0	3.7	3.3	3.0	2.8	2.5	2.3	2.1	1.9	1.8	1.6	1.5	1.3	1.2	1.1	1.0	0.9	0.8	0.8	0.7	7.1
1962	Male	10.1	9.0	8.1	7.2	6.4	5.7	5.2	4.7	4.3	3.9	3.6	3.3	3.0	2.7	2.5	2.3	2.1	1.9	1.7	1.6	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.7	0.7	6.9
1963	Male	10.5	9.2	8.2	7.2	6.3	5.6	5.0	4.5	4.1	3.7	3.4	3.1	2.8	2.6	2.4	2.1	2.0	1.8	1.6	1.5	1.4	1.2	1.1	1.0	0.9	0.9	0.8	0.7	0.6	6.6
1964	Male	11.0	9.6	8.4	7.4	6.4	5.6	4.9	4.4	4.0	3.6	3.3	3.0	2.7	2.5	2.3	2.1	1.9	1.7	1.6	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.6	6.4
1965	Male	11.2	10.0	8.8	7.6	6.7	5.8	5.0	4.4	4.0	3.6	3.3	3.0	2.7	2.5	2.2	2.0	1.9	1.7	1.6	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.7	0.7	0.6	6.3
1966	Male	11.2	10.2	9.1	8.0	6.9	6.0	5.2	4.5	4.0	3.5	3.2	2.9	2.7	2.4	2.2	2.0	1.8	1.7	1.5	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.7	0.7	0.6	6.2
1967	Male	10.9	10.2	9.3	8.2	7.1	6.1	5.3	4.6	4.0	3.5	3.1	2.8	2.6	2.3	2.1	1.9	1.8	1.6	1.5	1.3	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.6	0.6	6.0
1968	Male	10.5	10.0	9.3	8.3	7.3	6.3	5.4	4.7	4.0	3.5	3.1	2.7	2.5	2.2	2.0	1.9	1.7	1.5	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.7	0.7	0.6	0.6	5.8
1969	Male	9.9	9.5	9.0	8.2	7.3	6.3	5.4	4.6	4.0	3.4	3.0	2.6	2.3	2.1	1.9	1.7	1.6	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.6	0.6	0.5	5.4
1970	Male	9.3	9.0	8.6	7.9	7.1	6.2	5.4	4.6	3.9	3.4	2.9	2.5	2.2	1.9	1.8	1.6	1.5	1.3	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.6	0.6	0.5	0.5	4.9
1971	Male	8.8	8.5	8.1	7.5	6.8	6.0	5.2	4.5	3.8	3.2	2.8	2.4	2.1	1.8	1.6	1.5	1.3	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.6	0.6	0.5	0.5	0.4	4.5
1972	Male	8.3	8.0	7.6	7.1	6.3	5.6	4.9	4.3	3.7	3.1	2.6	2.3	2.0	1.7	1.5	1.3	1.2	1.1	1.0	0.9	0.8	0.7	0.7	0.6	0.6	0.5	0.5	0.4	0.4	4.0
1973	Male	7.9	7.6	7.2	6.5	5.8	5.1	4.5	3.9	3.4	2.9	2.4	2.1	1.8	1.5	1.3	1.2	1.0	0.9	0.8	0.8	0.7	0.6	0.6	0.5	0.5	0.4	0.4	0.4	0.3	3.5
1974	Male	7.5	7.2	6.8	6.2	5.5	4.8	4.2	3.7	3.2	2.7	2.3	2.0	1.7	1.4	1.2	1.1	0.9	0.8	0.8	0.7	0.6	0.6	0.5	0.5	0.4	0.4	0.4	0.3	0.3	3.1
1975	Male	3.9	6.8	6.5	5.9	5.2	4.5	3.9	3.4	3.0	2.6	2.2	1.9	1.6	1.4	1.2	1.0	0.9	0.8	0.7	0.6	0.6	0.5	0.5	0.4	0.4	0.4	0.3	0.3	0.3	2.8
1976	Male	3.8	3.6	6.2	5.6	4.9	4.3	3.7	3.3	2.8	2.5	2.1	1.9	1.6	1.3	1.1	1.0	0.8	0.7	0.6	0.6	0.5	0.5	0.4	0.4	0.4	0.3	0.3	0.3	0.2	2.5
1977	Male	3.8	3.4	3.3	5.3	4.7	4.1	3.5	3.1	2.7	2.3	2.0	1.7	1.5	1.3	1.1	0.9	0.8	0.7	0.6	0.5	0.5	0.4	0.4	0.3	0.3	0.3	0.3	0.2	0.2	2.2
1978	Male	4.4	3.4	3.1	2.9	4.5	3.9	3.4	3.0	2.6	2.2	1.9	1.7	1.5	1.3	1.1	0.9	0.8	0.7	0.6	0.5	0.4	0.4	0.3	0.3	0.3	0.3	0.2	0.2	0.2	2.1
1979	Male	10.2	4.1	3.1	2.8	2.5	4.0	3.5	3.0	2.6	2.3	2.0	1.7	1.5	1.3	1.1	0.9	0.8	0.7	0.6	0.5	0.4	0.4	0.3	0.3	0.3	0.3	0.2	0.2	0.2	2.0
1980	Male	31.7	9.3	3.7	2.8	2.4	2.2	3.5	3.0	2.6	2.3	2.0	1.7	1.5	1.3	1.1	1.0	0.8	0.7	0.6	0.5	0.4	0.4	0.3	0.3	0.3	0.2	0.2	0.2	0.2	1.9
1981	Male	7.2	29.0	8.5	3.3	2.4	2.1	1.9	3.0	2.6	2.3	2.0	1.7	1.5	1.3	1.1	1.0	0.8	0.7	0.6	0.5	0.4	0.4	0.3	0.3	0.3	0.2	0.2	0.2	0.2	1.8
1982	Male	11.5	6.6	26.4	7.5	2.9	2.1	1.9	1.7	2.6	2.3	2.0	1.7	1.5	1.3	1.1	1.0	0.8	0.7	0.6	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.2	0.2	1.7
1983	Male	31.1	10.5	6.0	23.6	6.6	2.5	1.9	1.6	1.4	2.3	2.0	1.7	1.5	1.3	1.1	1.0	0.9	0.7	0.6	0.5	0.5	0.4	0.3	0.3	0.2	0.2	0.2	0.2	0.2	1.6
1984	Male	4.8	28.4	9.5	5.4	20.8	5.8	2.2	1.6	1.4	1.3	2.0	1.7	1.5	1.3	1.1	1.0	0.9	0.7	0.6	0.6	0.5	0.4	0.3	0.3	0.3	0.2	0.2	0.2	0.2	1.6
1985	Male	5.1	4.4	25.9	8.5	4.7	18.2	5.1	1.9	1.4	1.2	1.1	1.8	1.5	1.3	1.1	1.0	0.9	0.7	0.7	0.6	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.1	1.5
1986	Male	9.1	4.7	4.0	23.1	7.5	4.1	16.0	4.5	1.7	1.2	1.1	1.0	1.5	1.3	1.2	1.0	0.9	0.8	0.7	0.6	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.1	1.5
1987	Male	3.5	8.3	4.3	3.5	19.8	6.4	3.5	13.5	3.8	1.4	1.1	0.9	0.8	1.3	1.1	1.0	0.8	0.7	0.6	0.5	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.1	1.3
1988	Male	2.8	3.2	7.5	3.7	2.9	16.5	5.3	2.9	11.1	3.1	1.2	0.9	0.8	0.7	1.1	0.9	0.8	0.7	0.6	0.5	0.5	0.4	0.3	0.3	0.3	0.2	0.2	0.2	0.1	1.2
1989	Male	5.2	2.6	2.9	6.5	3.1	2.4	13.4	4.3	2.4	9.0	2.5	1.0	0.7	0.6	0.5	0.9	0.7	0.6	0.6	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.1	0.1	1.1
1990	Male	4.6	4.7	2.3	2.5	5.3	2.5	2.0	10.9	3.5	1.9	7.3	2.0	0.8	0.6	0.5	0.4	0.7	0.6	0.5	0.5	0.4	0.3	0.3	0.3	0.2	0.2	0.2	0.1	0.1	1.0
1991	Male	11.0	4.2	4.3	2.0	2.1	4.4	2.1	1.6	8.9	2.8	1.6	6.0	1.7	0.6	0.5	0.4	0.4	0.6	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.2	0.1	0.1	0.9
1992	Male	3.8	10.1	3.8	3.7	1.7	1.7	3.6	1.7	1.3	7.3	2.3	1.3	4.9	1.4	0.5	0.4	0.3	0.3	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.2	0.1	0.1	0.8
1993	Male	9.4	3.5	9.1	3.3	3.1	1.4	1.4	3.0	1.4	1.1	6.0	1.9	1.1	4.0	1.1	0.4	0.3	0.3	0.2	0.4	0.3	0.3	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.8
1994	Male	5.3	8.6	3.1	7.9	2.7	2.5	1.1	1.1	2.4	1.1	0.9	4.9	1.6	0.9	3.3	0.9	0.3	0.3	0.2	0.2	0.3	0.3	0.2	0.2	0.2	0.2	0.1	0.1	0.1	0.7
1995	Male	4.4	4.9	7.8	2.7	6.5	2.2	2.1	0.9	0.9	2.0	0.9	0.7	4.0	1.3	0.7	2.7	0.7	0.3	0.2	0.2	0.2	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.7
1996	Male	6.2	4.0	4.4	6.7	2.3	5.4	1.8	1.7	0.8	0.8	1.6	0.8	0.6	3.3	1.0	0.6	2.2	0.6	0.2	0.2	0.1	0.1	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.6
1997	Male	7.8	5.6	3.6	3.8	5.7	1.9	4.5	1.5	1.4	0.6	0.6	1.4	0.6	0.5	2.7	0.9	0.5	1.8	0.5	0.2	0.1	0.1	0.1	0.2	0.2	0.1	0.1	0.1	0.1	0.6
1998	Male	4.2	7.1	5.1	3.2	3.3	4.8	1.6	3.8	1.3	1.2	0.5	0.5	1.1	0.5	0.4	2.3	0.7	0.4	1.5	0.4	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.6
1999	Male	12.7	3.8	6.4	4.5	2.7	2.8	4.1	1.3	3.2	1.1	1.0	0.4	0.5	1.0	0.4	0.3	1.9	0.6	0.3	1.3	0.4	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.6
2000	Male	8.4	11.6	3.5	5.7	3.8	2.3	2.3	3.4	1.1	2.7	0.9	0.9	0.4	0.4	0.8	0.4	0.3	1.6	0.5	0.3	1.1	0.3	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.5
2001	Male	7.0	7.7	10.6	3.0	4.8	3.2	1.9	2.0	2.9	1.0	2.3	0.8	0.7	0.3	0.3	0.7	0.3	0.2	1.4	0.4	0.2	0.9	0.3	0.1	0.1	0.1	0.1	0.1	0.1	0.5
2002	Male	15.2	6.4	7.0	9.3	2.6	4.0	2.7	1.6	1.7	2.4	0.8	1.9	0.7	0.6	0.3	0.3	0.6	0.3	0.2	1.2	0.4	0.2	0.8	0.2	0.1	0.1	0.1	0.0	0.1	0.5
2003	Male	7.1	13.8	5.8	6.1	7.9	2.2	3.4	2.3	1.4	1.4	2.0	0.7	1.6	0.5	0.5	0.2	0.2	0.5	0.2	0.2	1.0	0.3	0.2	0.7	0.2	0.1	0.1	0.0	0.0	0.5
2004	Male	4.6	6.5	12.6	5.1	5.2	6.6	1.8	2.9	1.9	1.1	1.2	1.7	0.6	1.3	0.5	0.4	0.2	0.2	0.4	0.2	0.1	0.8	0.3	0.1	0.5	0.2	0.1	0.0	0.0	0.4
2005	Male	5.0	4.2	5.9	11.0	4.3	4.3	5.6	1.5	2.4	1.6	1.0	1.0	1.4	0.5	1.1	0.4	0.4	0.2	0.2	0.3	0.2	0.1	0.7	0.2	0.1	0.5	0.1	0.0	0.0	0.4
2006	Male	3.6	4.6	3.8	5.2	9.3	3.6	3.7	4.7	1.3	2.0	1.4	0.8	0.8	1.2	0.4	0.9	0.3	0.3	0.1	0.1	0.3	0.1	0.1	0.6	0.2	0.1	0.4	0.1	0.0	0.4
2007	Male	4.5	3.3	4.1	3.3	4.4	7.9	3.1	3.1	4.0	1.1	1.7	1.1	0.7	0.7	1.0	0.3	0.8	0.3	0.3	0.1	0.1	0.2	0.1	0.1	0.5	0.2	0.1	0.3	0.1	0.3
2008	Male	4.8	4.1	3.0	3.6	2.8	3.7	6.7	2.6	2.6	3.4	0.9	1.4	1.0	0.6	0.6	0.9	0.3	0.7	0.2	0.2	0.1	0.1	0.2	0.1	0.1	0.4	0.1	0.1	0.3	0.4
2009	Male	7.4	4.4	3.7	2.7	3.1	2.4	3.2	5.7	2.2	2.2	2.8	0.8	1.2	0.8	0.5	0.5	0.7	0.2	0.6	0.2	0.2	0.1	0.1	0.2	0.1	0.1	0.3	0.1	0.1	0.5
2010	Male	7.2	6.8	4.0	3.3	2.3	2.6	2.1	2.7	4.8	1.9	1.9	2.4	0.7	1.0	0.7	0.4	0.4	0.6	0.2	0.5	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.3	0.1	0.5
2011	Male	4.4	6.6	6.1	3.5	2.8	1.9	2.3	1.8	2.3	4.1	1.6	1.6	2.1	0.6	0.9	0.6	0.4	0.4	0.5	0.2	0.4	0.1	0.1	0.1	0.1	0.1	0.1	0.0	0.3	0.5
2012	Male	4.5	4.1	6.0	5.4	3.0	2.4	1.7	1.9	1.5	2.0	3.5	1.4	1.4	1.8	0.5	0.8	0.5	0.3	0.3	0.5	0.1	0.4	0.1	0.1	0.0	0.1	0.1	0.0	0.0	0.7
2013	Male	2.6	4.1	3.7	5.2	4.6	2.6	2.0	1.4	1.6	1.3	1.7	3.0	1.1	1.2	1.5	0.4	0.6	0.4	0.3	0.3	0.4	0.1	0.3	0.1	0.1	0.0	0.0	0.1	0.0	0.6
2014	Male	3.5	2.3	3.7	3.2	4.5	3.9	2.2	1.7	1.2	1.4	1.1	1.4	2.5	1.0	1.0	1.3	0.3	0.5	0.4	0.2	0.2	0.3	0.1	0.3	0.1	0.1	0.0	0.0	0.1	0.5
2015	Male	6.2	3.2	2.1	3.3	2.8	3.8	3.3	1.9	1.5	1.0	1.2	0.9	1.2	2.1	0.8	0.8	1.1	0.3	0.5	0.3	0.2	0.2	0.3	0.1	0.2	0.1	0.1	0.0	0.0	0.5
2016	Male	22.7	5.6	2.9	1.9	2.8	2.4	3.3	2.8	1.6	1.2	0.9	1.0	0.8	1.0	1.8	0.7	0.7	0.9	0.3	0.4	0.3	0.2	0.2	0.2	0.1	0.2	0.1	0.1	0.0	0.5
2017	Male	8.7	20.6	5.1	2.6	1.6	2.4	2.0	2.8	2.4	1.4	1.1	0.7	0.9	0.7	0.9	1.6	0.6	0.6	0.8	0.2	0.3	0.2	0.1	0.1	0.2	0.1	0.2	0.1	0.0	0.4
2018	Male	51.1	7.9	18.5	4.4	2.2	1.4	2.1	1.7	2.4	2.1	1.2	0.9	0.6	0.7	0.6	0.7	1.3	0.5	0.5	0.7	0.2	0.3	0.2	0.1	0.1	0.2	0.1	0.1	0.0	0.4
2019	Male	37.7	46.5	7.1	16.1	3.8	1.9	1.2	1.8	1.5	2.0	1.8	1.0	0.8	0.5	0.6	0.5	0.6	1.1	0.4	0.4	0.6	0.2	0.2	0.2	0.1	0.1	0.1	0.0	0.1	0.4

2020	Male	15.8	34.3	41.7	6.2	13.9	3.3	1.6	1.0	1.5	1.3	1.7	1.5	0.8	0.7	0.5	0.5	0.4	0.5	1.0	0.4	0.4	0.5	0.1	0.2	0.1	0.1	0.1	0.1	0.0	0.4
2021	Male	17.0	14.4	30.9	36.8	5.4	12.1	2.8	1.4	0.9	1.3	1.1	1.5	1.3	0.7	0.6	0.4	0.5	0.4	0.5	0.8	0.3	0.3	0.4	0.1	0.2	0.1	0.1	0.1	0.1	0.4
2022	Male	19.0	15.5	13.0	27.2	32.1	4.7	10.5	2.4	1.2	0.8	1.1	0.9	1.3	1.1	0.6	0.5	0.3	0.4	0.3	0.4	0.7	0.3	0.3	0.4	0.1	0.2	0.1	0.1	0.1	0.4
2023	Male	16.4	17.3	13.9	11.4	23.6	27.7	4.1	9.0	2.1	1.0	0.6	1.0	0.8	1.1	1.0	0.5	0.4	0.3	0.3	0.3	0.3	0.6	0.2	0.2	0.3	0.1	0.1	0.1	0.1	0.4
2024	Male	15.6	15.0	15.6	12.3	9.9	20.5	24.0	3.5	7.8	1.8	0.9	0.6	0.8	0.7	0.9	0.8	0.5	0.4	0.2	0.3	0.2	0.3	0.5	0.2	0.2	0.3	0.1	0.1	0.1	0.4
2025	Male	9.7	14.2	13.5	13.8	10.8	8.7	17.9	21.0	3.1	6.8	1.6	0.8	0.5	0.7	0.6	0.8	0.7	0.4	0.3	0.2	0.3	0.2	0.3	0.5	0.2	0.2	0.2	0.1	0.1	0.4
2026	Male	0.0	8.9	12.8	12.0	12.1	9.4	7.6	15.7	18.3	2.7	5.9	1.4	0.7	0.4	0.6	0.5	0.7	0.6	0.3	0.3	0.2	0.2	0.2	0.2	0.4	0.2	0.2	0.2	0.1	0.5

Table SM3.11. Number (in millions of fish) of sablefish estimated by Model *26.2_FAA* by age and year. Columns are age with age-31+ being the plus group. The terminal year plus one (2026) numbers are incremented based on the terminal year (2025) mortality and recruitment is set at 0.

Year	Sex	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31+
1960	Female	8.1	7.2	6.4	5.8	5.2	4.7	4.3	3.9	3.6	3.2	2.9	2.7	2.4	2.2	2.0	1.9	1.7	1.5	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.7	0.7	0.6	0.6	5.6
1961	Female	8.3	7.4	6.6	5.9	5.2	4.7	4.3	3.9	3.5	3.2	2.9	2.7	2.4	2.2	2.0	1.9	1.7	1.5	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.7	0.7	0.6	0.6	5.6
1962	Female	8.6	7.6	6.7	6.0	5.2	4.6	4.2	3.8	3.4	3.1	2.8	2.6	2.4	2.2	2.0	1.8	1.6	1.5	1.4	1.2	1.1	1.0	0.9	0.9	0.8	0.7	0.6	0.6	0.5	5.4
1963	Female	9.1	7.9	6.9	6.1	5.2	4.5	4.0	3.6	3.3	3.0	2.7	2.5	2.2	2.0	1.9	1.7	1.5	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.7	0.7	0.6	0.6	0.5	5.1
1964	Female	9.6	8.3	7.2	6.3	5.4	4.6	4.0	3.5	3.2	2.9	2.6	2.4	2.2	2.0	1.8	1.6	1.5	1.4	1.2	1.1	1.0	0.9	0.9	0.8	0.7	0.6	0.6	0.5	0.5	5.0
1965	Female	9.9	8.8	7.6	6.6	5.7	4.9	4.2	3.6	3.2	2.8	2.6	2.3	2.1	1.9	1.8	1.6	1.5	1.3	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.6	0.6	0.5	0.5	4.9
1966	Female	10.0	9.1	8.0	6.9	5.9	5.1	4.4	3.7	3.2	2.8	2.5	2.3	2.1	1.9	1.7	1.6	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.6	0.6	0.5	0.5	4.8
1967	Female	9.8	9.1	8.3	7.3	6.2	5.3	4.6	3.9	3.3	2.8	2.5	2.2	2.0	1.8	1.7	1.5	1.4	1.3	1.1	1.0	1.0	0.9	0.8	0.7	0.7	0.6	0.5	0.5	0.5	4.6
1968	Female	9.3	8.9	8.3	7.5	6.5	5.5	4.7	4.0	3.4	2.9	2.5	2.2	1.9	1.8	1.6	1.5	1.3	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.6	0.6	0.5	0.5	0.4	4.4
1969	Female	8.8	8.5	8.1	7.4	6.5	5.6	4.7	4.0	3.4	2.9	2.4	2.1	1.8	1.6	1.5	1.3	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.6	0.6	0.5	0.5	0.4	0.4	4.1
1970	Female	8.1	8.0	7.7	7.2	6.4	5.6	4.7	3.9	3.3	2.8	2.4	2.0	1.7	1.5	1.4	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.6	0.6	0.5	0.5	0.4	0.4	0.4	3.7
1971	Female	7.6	7.4	7.2	6.8	6.2	5.4	4.6	3.9	3.3	2.7	2.3	2.0	1.7	1.4	1.2	1.1	1.0	0.9	0.8	0.7	0.7	0.6	0.6	0.5	0.5	0.4	0.4	0.4	0.3	3.3
1972	Female	7.0	6.9	6.7	6.4	5.8	5.1	4.4	3.7	3.2	2.6	2.2	1.8	1.6	1.3	1.1	1.0	0.9	0.8	0.7	0.7	0.6	0.5	0.5	0.5	0.4	0.4	0.3	0.3	0.3	2.9
1973	Female	6.5	6.4	6.2	5.8	5.2	4.6	4.0	3.4	2.9	2.4	2.0	1.7	1.4	1.2	1.0	0.8	0.7	0.7	0.6	0.5	0.5	0.5	0.4	0.4	0.3	0.3	0.3	0.3	0.2	2.4
1974	Female	6.1	5.9	5.7	5.5	4.9	4.3	3.7	3.2	2.7	2.3	1.9	1.6	1.3	1.1	0.9	0.8	0.7	0.6	0.5	0.5	0.4	0.4	0.4	0.3	0.3	0.3	0.2	0.2	0.2	2.1
1975	Female	3.1	5.6	5.4	5.1	4.5	4.0	3.4	3.0	2.5	2.2	1.8	1.5	1.3	1.0	0.9	0.7	0.6	0.5	0.5	0.4	0.4	0.3	0.3	0.3	0.3	0.2	0.2	0.2	0.2	1.8
1976	Female	2.9	2.8	5.0	4.8	4.2	3.7	3.2	2.8	2.4	2.0	1.7	1.5	1.2	1.0	0.8	0.7	0.6	0.5	0.4	0.4	0.3	0.3	0.3	0.2	0.2	0.2	0.2	0.2	0.2	1.6
1977	Female	3.0	2.7	2.6	4.5	3.9	3.4	2.9	2.5	2.2	1.9	1.6	1.4	1.1	1.0	0.8	0.7	0.6	0.5	0.4	0.3	0.3	0.3	0.2	0.2	0.2	0.2	0.2	0.1	0.1	1.4
1978	Female	3.7	2.7	2.4	2.3	3.7	3.2	2.8	2.4	2.1	1.8	1.5	1.3	1.1	0.9	0.8	0.6	0.5	0.5	0.4	0.3	0.3	0.2	0.2	0.2	0.2	0.2	0.1	0.1	0.1	1.2
1979	Female	8.3	3.3	2.5	2.2	2.0	3.2	2.8	2.4	2.1	1.8	1.5	1.3	1.1	1.0	0.8	0.7	0.6	0.5	0.4	0.3	0.3	0.2	0.2	0.2	0.2	0.2	0.1	0.1	0.1	1.2
1980	Female	29.6	7.6	3.1	2.2	1.9	1.7	2.8	2.4	2.0	1.8	1.5	1.3	1.1	1.0	0.8	0.7	0.6	0.5	0.4	0.3	0.3	0.2	0.2	0.2	0.2	0.2	0.1	0.1	0.1	1.1
1981	Female	5.4	27.1	6.9	2.8	1.9	1.6	1.5	2.4	2.0	1.8	1.5	1.3	1.1	1.0	0.8	0.7	0.6	0.5	0.4	0.3	0.3	0.2	0.2	0.2	0.2	0.2	0.1	0.1	0.1	1.0
1982	Female	8.3	4.9	24.7	6.2	2.4	1.7	1.4	1.3	2.0	1.7	1.5	1.3	1.1	1.0	0.8	0.7	0.6	0.5	0.4	0.3	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.1	1.0
1983	Female	24.3	7.6	4.5	22.3	5.4	2.1	1.4	1.2	1.1	1.7	1.5	1.3	1.1	1.0	0.8	0.7	0.6	0.5	0.4	0.4	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.9
1984	Female	4.3	22.2	7.0	4.1	19.6	4.7	1.8	1.2	1.1	0.9	1.5	1.3	1.1	1.0	0.8	0.7	0.6	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.9
1985	Female	4.4	3.9	20.3	6.3	3.6	17.1	4.1	1.5	1.1	0.9	0.8	1.3	1.1	1.0	0.8	0.7	0.6	0.5	0.5	0.4	0.3	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.8
1986	Female	6.8	4.1	3.6	18.3	5.5	3.1	14.8	3.6	1.3	0.9	0.8	0.7	1.1	1.0	0.8	0.7	0.6	0.5	0.5	0.4	0.3	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.8
1987	Female	3.1	6.2	3.7	3.2	15.6	4.6	2.6	12.3	2.9	1.1	0.8	0.6	0.6	0.9	0.8	0.7	0.6	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.7
1988	Female	3.1	2.9	5.6	3.3	2.7	12.7	3.7	2.1	9.9	2.4	0.9	0.6	0.5	0.5	0.7	0.6	0.5	0.5	0.4	0.3	0.3	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.6
1989	Female	4.8	2.9	2.6	4.9	2.7	2.1	10.1	2.9	1.6	7.7	1.8	0.7	0.5	0.4	0.4	0.6	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.6
1990	Female	5.6	4.4	2.6	2.3	4.0	2.1	1.7	7.9	2.3	1.3	6.0	1.4	0.5	0.4	0.3	0.3	0.5	0.4	0.3	0.3	0.2	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.5
1991	Female	7.5	5.1	4.0	2.3	1.9	3.2	1.7	1.3	6.2	1.8	1.0	4.7	1.1	0.4	0.3	0.2	0.2	0.4	0.3	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.4
1992	Female	6.2	6.9	4.7	3.5	1.9	1.5	2.5	1.3	1.0	4.9	1.4	0.8	3.7	0.9	0.3	0.2	0.2	0.2	0.3	0.2	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.4
1993	Female	5.8	5.6	6.3	4.1	2.9	1.5	1.2	2.0	1.0	0.8	3.8	1.1	0.6	2.9	0.7	0.3	0.2	0.2	0.1	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.3
1994	Female	1.9	5.3	5.1	5.5	3.3	2.2	1.1	0.9	1.5	0.8	0.6	2.9	0.9	0.5	2.2	0.5	0.2	0.1	0.1	0.1	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.0	0.3
1995	Female	3.8	1.7	4.8	4.5	4.4	2.6	1.7	0.9	0.7	1.2	0.6	0.5	2.3	0.7	0.4	1.7	0.4	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.0	0.3
1996	Female	5.8	3.5	1.6	4.2	3.6	3.5	2.0	1.4	0.7	0.5	0.9	0.5	0.4	1.8	0.5	0.3	1.3	0.3	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.0	0.0	0.2
1997	Female	4.2	5.3	3.2	1.4	3.5	2.9	2.8	1.6	1.1	0.5	0.4	0.7	0.4	0.3	1.4	0.4	0.2	1.1	0.3	0.1	0.1	0.1	0.0	0.1	0.1	0.1	0.0	0.0	0.0	0.2
1998	Female	9.1	3.8	4.9	2.8	1.2	2.8	2.3	2.2	1.3	0.9	0.4	0.3	0.6	0.3	0.2	1.1	0.3	0.2	0.8	0.2	0.1	0.1	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.2
1999	Female	4.8	8.3	3.5	4.3	2.3	0.9	2.3	1.9	1.8	1.0	0.7	0.4	0.3	0.5	0.2	0.2	0.9	0.3	0.1	0.7	0.2	0.1	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.2
2000	Female	4.0	4.4	7.5	3.1	3.6	1.9	0.8	1.8	1.5	1.4	0.8	0.6	0.3	0.2	0.4	0.2	0.2	0.7	0.2	0.1	0.5	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2
2001	Female	6.4	3.7	4.0	6.7	2.5	2.8	1.5	0.6	1.4	1.2	1.1	0.7	0.4	0.2	0.2	0.3	0.2	0.1	0.6	0.2	0.1	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.2
2002	Female	13.4	5.8	3.3	3.6	5.5	2.0	2.3	1.2	0.5	1.1	1.0	0.9	0.5	0.3	0.2	0.1	0.2	0.1	0.1	0.4	0.1	0.1	0.3	0.1	0.0	0.0	0.0	0.0	0.0	0.2
2003	Female	6.1	12.2	5.3	3.0	2.9	4.4	1.6	1.8	1.0	0.4	0.9	0.8	0.7	0.4	0.3	0.1	0.1	0.2	0.1	0.1	0.4	0.1	0.1	0.3	0.1	0.0	0.0	0.0	0.0	0.1
2004	Female	2.7	5.6	11.1	4.7	2.4	2.3	3.5	1.3	1.4	0.7	0.3	0.7	0.6	0.6	0.3	0.2	0.1	0.1	0.1	0.1	0.1	0.3	0.1	0.0	0.2	0.0	0.0	0.0	0.0	0.1
2005	Female	3.4	2.5	5.1	9.8	3.8	1.9	1.8	2.7	1.0	1.1	0.6	0.2	0.6	0.5	0.4	0.3	0.2	0.1	0.1	0.1	0.1	0.0	0.2	0.1	0.0	0.2	0.0	0.0	0.0	0.1
2006	Female	1.4	3.1	2.2	4.5	8.1	3.0	1.5	1.5	2.2	0.8	0.9	0.5	0.2	0.4	0.4	0.3	0.2	0.1	0.1	0.1	0.1	0.0	0.0	0.2	0.0	0.0	0.1	0.0	0.0	0.1
2007	Female	4.8	1.3	2.9	2.0	3.7	6.5	2.4	1.2	1.2	1.7	0.6	0.7	0.4	0.1	0.4	0.3	0.3	0.2	0.1	0.1	0.0	0.1	0.0	0.0	0.1	0.0	0.0	0.1	0.0	0.1
2008	Female	3.6	4.4	1.2	2.5	1.6	3.0	5.2	1.9	1.0	0.9	1.4	0.5	0.6	0.3	0.1	0.3	0.2	0.2	0.1	0.1	0.0	0.0	0.1	0.0	0.0	0.1	0.0	0.0	0.1	0.1
2009	Female	8.0	3.3	4.0	1.1	2.1	1.3	2.4	4.1	1.5	0.8	0.7	1.1	0.4	0.4	0.2	0.1	0.2	0.2	0.2	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.1
2010	Female	7.0	7.3	3.0	3.5	0.9	1.7	1.0	1.9	3.3	1.2	0.6	0.6	0.9	0.3	0.4	0.2	0.1	0.2	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.1
2011	Female	3.7	6.4	6.7	2.7	2.9	0.7	1.4	0.8	1.5	2.7	1.0	0.5	0.5	0.7	0.3	0.3	0.2	0.1	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1
2012	Female	3.2	3.4	5.8	5.9	2.2	2.3	0.6	1.1	0.7	1.2	2.1	0.8	0.4	0.4	0.6	0.2	0.2	0.1	0.0	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1
2013	Female	3.1	2.9	3.1	5.2	4.9	1.8	1.9	0.4	0.9	0.5	1.0	1.7	0.6	0.3	0.3	0.4	0.2	0.2	0.1	0.0	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1
2014	Female	2.8	2.8	2.7	2.7	4.2	3.9	1.4	1.5	0.4	0.7	0.4	0.8	1.3	0.5	0.3	0.2	0.4	0.1	0.1	0.1	0.0	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.1
2015	Female	4.3	2.6	2.6	2.4	2.3	3.5	3.2	1.2	1.2	0.3	0.6	0.3	0.6	1.1	0.4	0.2	0.2	0.3	0.1	0.1	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1
2016	Female	31.1	4.0	2.4	2.3	2.0	1.9	2.8	2.6	0.9	1.0	0.2	0.5	0.3	0.5	0.9	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1
2017	Female	14.7	28.3	3.6	2.0	1.9	1.6	1.5	2.3	2.1	0.8	0.8	0.2	0.4	0.2	0.4	0.7	0.3	0.1	0.1	0.2	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1
2018	Female	34.0	13.4	25.4	3.1	1.7	1.6	1.3	1.2	1.9	1.7	0.6	0.7	0.2	0.3	0.2	0.3	0.6	0.2	0.1	0.1	0.2	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1
2019	Female	55.7	30.9	12.0	22.1	2.6	1.4	1.3	1.1	1.0	1.5	1.4	0.5	0.5	0.1	0.2	0.2	0.3	0.5	0.2	0.1	0.1	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.1

2020	Female	5.7	50.7	27.8	10.5	18.8	2.2	1.2	1.1	0.9	0.8	1.3	1.2	0.4	0.4	0.1	0.2	0.1	0.2	0.4	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1
2021	Female	5.5	5.2	45.7	24.5	9.1	16.1	1.8	1.0	0.9	0.8	0.7	1.1	1.0	0.4	0.4	0.1	0.2	0.1	0.2	0.3	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.1
2022	Female	7.8	5.0	4.7	40.5	21.4	7.8	13.9	1.6	0.9	0.8	0.7	0.6	0.9	0.8	0.3	0.3	0.1	0.1	0.1	0.2	0.3	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.1
2023	Female	5.5	7.1	4.6	4.1	35.2	18.4	6.7	11.8	1.4	0.7	0.7	0.6	0.5	0.8	0.7	0.3	0.3	0.1	0.1	0.1	0.2	0.1	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.1
2024	Female	7.1	5.0	6.4	4.1	3.6	30.5	15.9	5.8	10.2	1.2	0.6	0.6	0.5	0.4	0.7	0.6	0.2	0.2	0.1	0.1	0.1	0.1	0.2	0.1	0.0	0.0	0.1	0.0	0.0	0.1
2025	Female	7.9	6.4	4.6	5.7	3.6	3.2	26.5	13.8	5.0	8.8	1.0	0.5	0.5	0.4	0.4	0.6	0.5	0.2	0.2	0.0	0.1	0.1	0.1	0.2	0.1	0.0	0.0	0.0	0.0	0.1
2026	Female	0.0	7.2	5.8	4.1	5.0	3.1	2.7	22.9	11.9	4.3	7.6	0.9	0.5	0.4	0.4	0.3	0.5	0.5	0.2	0.2	0.0	0.1	0.0	0.1	0.2	0.1	0.0	0.0	0.0	0.1
1960	Male	8.1	7.2	6.4	5.8	5.2	4.7	4.3	3.9	3.6	3.2	3.0	2.7	2.5	2.3	2.1	1.9	1.7	1.6	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.6	0.6	5.8
1961	Male	8.3	7.4	6.6	5.9	5.2	4.7	4.3	3.9	3.6	3.2	3.0	2.7	2.5	2.2	2.1	1.9	1.7	1.6	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.7	0.7	0.6	0.6	5.8
1962	Male	8.6	7.6	6.7	5.9	5.2	4.6	4.2	3.8	3.4	3.1	2.9	2.6	2.4	2.2	2.0	1.8	1.7	1.5	1.4	1.3	1.1	1.0	1.0	0.9	0.8	0.7	0.7	0.6	0.6	5.7
1963	Male	9.1	7.9	6.9	5.8	5.0	4.5	4.0	3.6	3.3	3.0	2.7	2.5	2.2	2.1	1.9	1.7	1.6	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.6	0.6	0.5	5.3
1964	Male	9.6	8.3	7.2	6.1	5.2	4.4	3.9	3.5	3.2	2.9	2.6	2.4	2.2	2.0	1.8	1.6	1.5	1.4	1.3	1.1	1.0	1.0	0.9	0.8	0.7	0.7	0.6	0.5	0.5	5.2
1965	Male	9.9	8.8	7.6	6.5	5.5	4.6	4.0	3.5	3.1	2.8	2.6	2.3	2.1	1.9	1.8	1.6	1.5	1.3	1.2	1.1	1.0	0.9	0.9	0.8	0.7	0.7	0.6	0.5	0.5	5.1
1966	Male	10.0	9.1	8.1	6.9	5.9	5.0	4.2	3.6	3.2	2.8	2.5	2.3	2.1	1.9	1.7	1.6	1.5	1.3	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.6	0.6	0.5	0.5	5.0
1967	Male	9.8	9.2	8.3	7.3	6.2	5.2	4.4	3.7	3.2	2.8	2.5	2.2	2.0	1.8	1.7	1.5	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.7	0.7	0.6	0.6	0.5	0.5	4.8
1968	Male	9.3	9.0	8.4	7.5	6.4	5.4	4.6	3.8	3.2	2.7	2.4	2.2	1.9	1.8	1.6	1.5	1.3	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.6	0.6	0.5	0.5	0.4	4.6
1969	Male	8.8	8.5	8.1	7.3	6.4	5.5	4.6	3.9	3.2	2.7	2.3	2.0	1.8	1.6	1.5	1.4	1.2	1.1	1.0	0.9	0.9	0.8	0.7	0.6	0.6	0.5	0.5	0.5	0.4	4.2
1970	Male	8.1	8.0	7.7	7.0	6.2	5.4	4.6	3.8	3.2	2.7	2.2	1.9	1.7	1.5	1.4	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.6	0.6	0.5	0.5	0.4	0.4	0.4	3.8
1971	Male	7.6	7.4	7.2	6.6	5.9	5.2	4.4	3.7	3.1	2.6	2.2	1.8	1.6	1.4	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.6	0.6	0.5	0.5	0.4	0.4	0.4	0.3	3.5
1972	Male	7.0	6.9	6.7	6.1	5.4	4.8	4.1	3.6	3.0	2.5	2.1	1.7	1.5	1.2	1.1	1.0	0.9	0.8	0.7	0.7	0.6	0.6	0.5	0.5	0.4	0.4	0.4	0.3	0.3	3.0
1973	Male	6.5	6.4	6.2	5.5	4.8	4.2	3.7	3.2	2.7	2.3	1.9	1.6	1.3	1.1	0.9	0.8	0.7	0.7	0.6	0.6	0.5	0.5	0.4	0.4	0.3	0.3	0.3	0.3	0.2	2.5
1974	Male	6.1	6.0	5.8	5.1	4.5	3.9	3.3	2.9	2.5	2.1	1.8	1.5	1.3	1.1	0.9	0.8	0.7	0.6	0.5	0.5	0.4	0.4	0.4	0.3	0.3	0.3	0.3	0.2	0.2	2.2
1975	Male	3.1	5.6	5.4	4.7	4.2	3.6	3.1	2.7	2.3	2.0	1.7	1.4	1.2	1.0	0.8	0.7	0.6	0.5	0.5	0.4	0.4	0.3	0.3	0.3	0.3	0.2	0.2	0.2	0.2	1.9
1976	Male	2.9	2.8	5.0	4.4	3.8	3.4	2.9	2.5	2.1	1.9	1.6	1.4	1.1	1.0	0.8	0.7	0.6	0.5	0.4	0.4	0.3	0.3	0.3	0.3	0.2	0.2	0.2	0.2	0.2	1.7
1977	Male	3.0	2.7	2.6	4.1	3.5	3.0	2.6	2.2	1.9	1.7	1.5	1.2	1.1	0.9	0.7	0.6	0.5	0.4	0.4	0.3	0.3	0.3	0.2	0.2	0.2	0.2	0.2	0.2	0.1	1.4
1978	Male	3.7	2.7	2.4	2.1	3.3	2.9	2.5	2.2	1.8	1.6	1.4	1.2	1.0	0.9	0.7	0.6	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.2	0.2	0.2	0.1	0.1	1.3
1979	Male	8.3	3.3	2.5	2.1	1.8	2.9	2.5	2.2	1.9	1.6	1.4	1.2	1.0	0.9	0.8	0.6	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.2	0.2	0.2	0.1	0.1	1.2
1980	Male	29.6	7.6	3.0	2.1	1.8	1.6	2.5	2.1	1.8	1.6	1.4	1.2	1.0	0.9	0.8	0.6	0.5	0.5	0.4	0.3	0.3	0.2	0.2	0.2	0.2	0.2	0.1	0.1	0.1	1.1
1981	Male	5.4	27.1	6.9	2.6	1.8	1.6	1.4	2.1	1.8	1.6	1.4	1.2	1.0	0.9	0.8	0.6	0.6	0.5	0.4	0.3	0.3	0.2	0.2	0.2	0.2	0.2	0.1	0.1	0.1	1.1
1982	Male	8.3	5.0	24.6	5.9	2.3	1.6	1.3	1.2	1.8	1.6	1.3	1.2	1.0	0.9	0.7	0.6	0.5	0.5	0.4	0.3	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.1	1.0
1983	Male	24.3	7.6	4.5	21.4	5.1	1.9	1.4	1.1	1.0	1.6	1.3	1.2	1.0	0.9	0.7	0.6	0.5	0.5	0.4	0.3	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.9
1984	Male	4.3	22.3	6.9	3.9	18.7	4.5	1.7	1.2	1.0	0.9	1.3	1.2	1.0	0.9	0.7	0.6	0.5	0.5	0.4	0.4	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.9
1985	Male	4.4	3.9	20.3	6.1	3.4	16.2	3.8	1.5	1.0	0.9	0.7	1.2	1.0	0.9	0.7	0.6	0.6	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.9
1986	Male	6.8	4.1	3.6	17.7	5.3	3.0	14.0	3.3	1.3	0.9	0.7	0.6	1.0	0.9	0.7	0.6	0.6	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.8
1987	Male	3.1	6.2	3.7	3.0	14.8	4.4	2.5	11.6	2.7	1.0	0.7	0.6	0.5	0.8	0.7	0.6	0.5	0.5	0.4	0.3	0.3	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.7
1988	Male	3.1	2.9	5.6	3.0	2.5	12.0	3.5	2.0	9.3	2.2	0.8	0.6	0.5	0.4	0.7	0.6	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.7
1989	Male	4.8	2.9	2.6	4.5	2.4	1.9	9.4	2.8	1.5	7.3	1.7	0.7	0.5	0.4	0.3	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.6
1990	Male	5.6	4.4	2.6	2.1	3.6	1.9	1.5	7.3	2.1	1.2	5.6	1.3	0.5	0.4	0.3	0.3	0.4	0.3	0.3	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.5
1991	Male	7.5	5.1	4.0	2.1	1.6	2.8	1.5	1.2	5.7	1.7	0.9	4.4	1.0	0.4	0.3	0.2	0.2	0.3	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.4
1992	Male	6.2	6.9	4.6	3.2	1.6	1.3	2.2	1.2	0.9	4.5	1.3	0.7	3.5	0.8	0.3	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.4
1993	Male	5.8	5.6	6.2	3.7	2.5	1.3	1.0	1.7	0.9	0.7	3.5	1.0	0.6	2.7	0.6	0.2	0.2	0.1	0.1	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.3
1994	Male	1.9	5.3	5.1	4.9	2.9	1.9	1.0	0.8	1.3	0.7	0.6	2.7	0.8	0.4	2.1	0.5	0.2	0.1	0.1	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.0	0.3
1995	Male	3.8	1.7	4.8	4.0	3.8	2.2	1.5	0.8	0.6	1.0	0.5	0.4	2.1	0.6	0.3	1.6	0.4	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.0	0.3
1996	Male	5.8	3.5	1.6	3.8	3.1	3.0	1.7	1.2	0.6	0.5	0.8	0.4	0.3	1.6	0.5	0.3	1.3	0.3	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.0	0.0	0.2
1997	Male	4.2	5.3	3.1	1.3	3.0	2.5	2.4	1.4	0.9	0.5	0.4	0.6	0.3	0.3	1.3	0.4	0.2	1.0	0.2	0.1	0.1	0.1	0.0	0.1	0.1	0.1	0.0	0.0	0.0	0.2
1998	Male	9.1	3.8	4.8	2.6	1.0	2.4	2.0	1.9	1.1	0.7	0.4	0.3	0.5	0.3	0.2	1.0	0.3	0.2	0.8	0.2	0.1	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.2
1999	Male	4.8	8.3	3.4	3.9	2.1	0.8	1.9	1.6	1.5	0.9	0.6	0.3	0.2	0.4	0.2	0.2	0.8	0.2	0.1	0.6	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2
2000	Male	4.0	4.4	7.5	2.8	3.2	1.7	0.7	1.6	1.3	1.2	0.7	0.5	0.2	0.2	0.3	0.2	0.1	0.7	0.2	0.1	0.5	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2
2001	Male	6.4	3.7	4.0	6.0	2.2	2.5	1.3	0.5	1.2	1.0	1.0	0.6	0.4	0.2	0.1	0.3	0.1	0.1	0.5	0.2	0.1	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.2
2002	Male	13.4	5.8	3.3	3.2	4.9	1.8	2.0	1.1	0.4	1.0	0.8	0.8	0.4	0.3	0.2	0.1	0.2	0.1	0.1	0.4	0.1	0.1	0.3	0.1	0.0	0.0	0.0	0.0	0.0	0.1
2003	Male	6.1	12.2	5.2	2.7	2.6	3.9	1.4	1.6	0.8	0.3	0.8	0.6	0.6	0.4	0.2	0.1	0.1	0.2	0.1	0.1	0.3	0.1	0.1	0.3	0.1	0.0	0.0	0.0	0.0	0.1
2004	Male	2.7	5.6	11.0	4.2	2.1	2.0	3.0	1.1	1.2	0.7	0.3	0.6	0.5	0.5	0.3	0.2	0.1	0.1	0.1	0.1	0.1	0.3	0.1	0.0	0.2	0.0	0.0	0.0	0.0	0.1
2005	Male	3.4	2.5	5.0	8.8	3.3	1.7	1.6	2.4	0.9	1.0	0.5	0.2	0.5	0.4	0.4	0.2	0.1	0.1	0.1	0.1	0.1	0.0	0.2	0.1	0.0	0.2	0.0	0.0	0.0	0.1
2006	Male	1.4	3.2	2.2	4.1	7.0	2.6	1.3	1.3	1.9	0.7	0.8	0.4	0.2	0.4	0.3	0.3	0.2	0.1	0.1	0.0	0.1	0.0	0.0	0.2	0.0	0.0	0.1	0.0	0.0	0.1
2007	Male	4.8	1.3	2.8	1.8	3.3	5.7	2.1	1.1	1.0	1.5	0.6	0.6	0.3	0.1	0.3	0.3	0.2	0.1	0.1	0.0	0.0	0.1	0.0	0.0	0.1	0.0	0.0	0.1	0.0	0.1
2008	Male	3.6	4.4	1.2	2.3	1.4	2.6	4.5	1.7	0.8	0.8	1.2	0.4	0.5	0.3	0.1	0.2	0.2	0.2	0.1	0.1	0.0	0.0	0.1	0.0	0.0	0.1	0.0	0.0	0.1	0.1
2009	Male	8.0	3.3	3.9	0.9	1.8	1.1	2.1	3.6	1.3	0.7	0.6	1.0	0.4	0.4	0.2	0.1	0.2	0.2	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.1
2010	Male	7.0	7.4	3.0	3.2	0.8	1.5	0.9	1.7	2.9	1.1	0.5	0.5	0.8	0.3	0.3	0.2	0.1	0.2	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.1
2011	Male	3.7	6.4	6.6	2.4	2.6	0.6	1.2	0.7	1.3	2.3	0.9	0.4	0.4	0.6	0.2	0.3	0.1	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1
2012	Male	3.2	3.4	5.8	5.4	2.0	2.1	0.5	0.9	0.6	1.1	1.8	0.7	0.3	0.3	0.5	0.2	0.2	0.1	0.0	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1
2013	Male	3.1	2.9	3.0	4.7	4.3	1.6	1.6	0.4	0.8	0.5	0.8	1.5	0.5	0.3	0.3	0.4	0.1	0.2	0.1	0.0	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1
2014	Male	2.8	2.8	2.6	2.5	3.7	3.4	1.3	1.3	0.3	0.6	0.4	0.7	1.2	0.4	0.2	0.2	0.3	0.1	0.1	0.1	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1
2015	Male	4.3	2.6	2.6	2.2	2.0	3.0	2.8	1.0	1.1	0.3	0.5	0.3	0.6	0.9	0.4	0.2	0.2	0.3	0.1	0.1	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1
2016	Male	31.1	4.0	2.3	2.1	1.8	1.6	2.5	2.3	0.8	0.9	0.2	0.4	0.2	0.4	0.8	0.3	0.1	0.1	0.2	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1
2017	Male	14.7	28.4	3.6	2.0	1.8	1.5	1.3	2.0	1.9	0.7	0.7	0.2	0.3	0.2	0.4	0.6	0.2	0.1	0.1	0.2	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1
2018	Male	34.0	13.4	25.5	3.1	1.7	1.4	1.2	1.1	1.6	1.5	0.5	0.6	0.1	0.3	0.2	0.3	0.5	0.2	0.1	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1
2019	Male	55.7	31.0	12.1	22.2	2.6	1.4	1.2	1.0	0.9	1.3	1.2	0.4	0.5	0.1	0.2	0.1	0.2	0.4	0.2	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1

2020	Male	5.7	50.9	28.0	10.6	18.9	2.2	1.2	1.0	0.8	0.7	1.1	1.0	0.4	0.4	0.1	0.2	0.1	0.2	0.3	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1
2021	Male	5.5	5.2	46.1	24.7	9.1	16.0	1.8	1.0	0.8	0.7	0.6	0.9	0.9	0.3	0.3	0.1	0.1	0.1	0.2	0.3	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.1
2022	Male	7.8	5.1	4.7	40.9	21.5	7.8	13.7	1.6	0.8	0.7	0.6	0.5	0.8	0.7	0.3	0.3	0.1	0.1	0.1	0.1	0.2	0.1	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.1
2023	Male	5.5	7.1	4.6	4.2	35.4	18.4	6.7	11.7	1.3	0.7	0.6	0.5	0.5	0.7	0.6	0.2	0.2	0.1	0.1	0.1	0.1	0.2	0.1	0.0	0.0	0.1	0.0	0.0	0.0	0.1
2024	Male	7.1	5.0	6.5	4.1	3.6	30.6	15.9	5.8	10.1	1.1	0.6	0.5	0.4	0.4	0.6	0.5	0.2	0.2	0.0	0.1	0.1	0.1	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.1
2025	Male	7.9	6.5	4.6	5.8	3.6	3.2	26.5	13.7	5.0	8.7	1.0	0.5	0.4	0.4	0.3	0.5	0.5	0.2	0.2	0.0	0.1	0.1	0.1	0.2	0.1	0.0	0.0	0.0	0.0	0.1
2026	Male	0.0	7.2	5.9	4.1	5.0	3.1	2.7	22.9	11.9	4.3	7.5	0.9	0.5	0.4	0.3	0.3	0.4	0.4	0.1	0.2	0.0	0.1	0.0	0.1	0.1	0.1	0.0	0.0	0.0	0.1